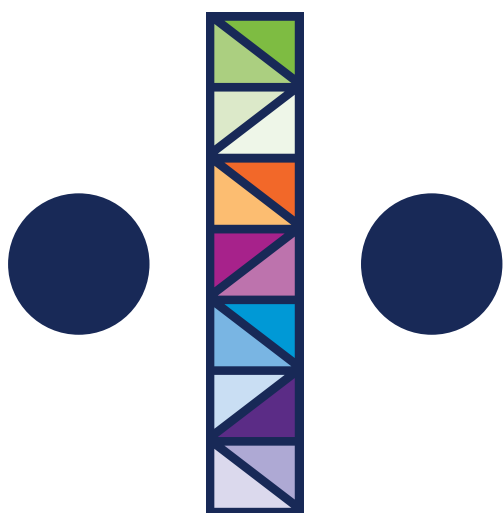
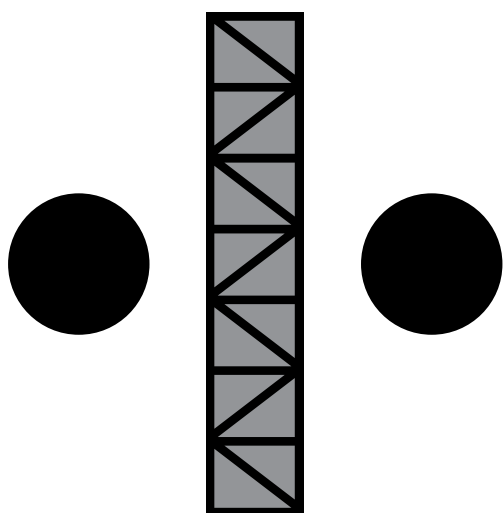




World report on social determinants of health equity



World Health
Organization



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Foreword

Our world is an unequal one. Where we are born, grow, live, work and age significantly influences our health and well-being. These social determinants of health are shaped by policy choices, social values, and the systems we create and sustain. So, despite current inequalities and inequities, change for the better is possible.

This comprehensive *World report on social determinants of health equity* underscores that achieving more equitable health outcomes requires a concerted effort to address the complex web of social, economic, environmental and political factors that impact health. These determinants are rooted in the structures of our societies, from educational access and income distribution, to living conditions and social protection.

The COVID-19 pandemic exposed and intensified health inequities, dramatically illustrating how systemic inequalities place marginalized and vulnerable groups at disproportionate risk. The impacts reverberated beyond physical health to affect mental, social and economic well-being, highlighting the need for holistic and inclusive approaches to health equity. We must foster environments that promote health and help protect against adverse outcomes across all facets of society.

This report presents evidence-based strategies and policy recommendations to guide governments, civil society and international organizations in creating just and equitable health systems. Drawing on evidence, examples and country case studies from around the world, it illustrates the positive impacts of interventions that address social determinants – whether through universal access to health and social protection, fair labour practices or inclusive education.

I invite leaders and communities worldwide to engage with this report's findings, as we work together to improve health equity through addressing social inequities. Together, let us commit to building a world where everyone, regardless of circumstance, has the opportunity to enjoy the highest attainable standard of health and well-being.



A blue ink signature of Dr. Tedros Adhanom Ghebreyesus, written in a cursive style.

Dr Tedros Adhanom Ghebreyesus
Director-General
World Health Organization

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Key technical writers and reviewers

The primary writers of the report from WHO were, in alphabetical order: Julia Berenson, Kumanan Rasanathan, Sudhvir Singh and Nicole Valentine (WHO Department of Social Determinants of Health). Monika Kosinska, Nathalie Roebbel and Tamitza Toroyan from the WHO Department of Social Determinants of Health served as internal reviewers. Julia Berenson coordinated the development and review of the country examples. Data on the progress against the Commission on Social Determinants of Health targets was prepared by Ahmad Reza Hosseinpour, Katherine Kirkby and Nicole Bergen from the WHO Department of Data and Analytics.

The report was edited by Michael Bartos (Australia). Jessica Allen, Antia Castedo, Angela Donkin and Michael Marmot from the University College London Institute of Health Equity in the United Kingdom of Great Britain and Northern Ireland (United Kingdom) reviewed evidence and contributed to drafting sections of the report. Mike Kalmus Elias (Honorary Clinical Fellow, University of Liverpool, United Kingdom) and Pooja Yerramilli (Health Specialist, the World Bank Group, United States of America) made important technical contributions to the report.

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Policy Expert Group

Rowaida Al-Maaitah (Faculty of Nursing, Jordan University of Science and Technology, Jordan), Paulo Buss (Emeritus Professor, Oswaldo Cruz Foundation, Brazil), Helen Clark (Former Prime Minister, New Zealand and Former Administrator, United Nations Development Programme, New Zealand), Tim Evans (Inaugural Director and Associate Dean, School of Population and Global Health, McGill University, Canada), Michael Marmot (Professor and Director, Institute of Health Equity, University College London, United Kingdom), Anders Nordström (Advisor Karolinska

Institutet and Stockholm School of Economics and Former Ambassador for Global Health, Sweden), Loyce Pace (Former Assistant Secretary for Global Affairs, US Department of Health and Human Services, United States), Joy Phumaphi (Executive Secretary, African Leaders Malaria Alliance, Botswana), Erika Placella (Head of Health, Swiss Agency for Development and Cooperation, Switzerland), Dr K. Srinath Reddy (Honorary Distinguished Professor and Former President, Public Health Foundation of India, India), John-Arne Røttingen (Chief Executive Officer, Wellcome Trust and Former Ambassador for Global Health, Norway), Viroj Tangcharoensathien (Senior Advisor, International Health Policy Foundation, Thailand), Jeanette Vega (Senior Adviser to the Minister of Health, Chile), Guo Yan (School of Public Health, Peking University, China).

Scientific Expert Group

Debashis Basu (Academic and Clinical Head of Public Health Medicine, Steve Biko Academic Hospital and University of Pretoria, South Africa), Fran Baum (Professor of Health Equity, The Stretton Institute, The University of Adelaide, Australia), Mickey Chopra (Global Solutions Lead for Service Delivery, Health, Nutrition and Population global practice, World Bank), Mandeep Dhaliwal (Director, HIV and Health Group, Bureau for Policy and Programme Support, United Nations Development Programme), Carlos Dora (Visiting Professor of Public Policies, Federal University of Rio Grande do Sul, Porto Alegre, Brazil), Sharon Friel (Professor of Health Equity; Director, Planetary Health Equity Hothouse and Australian Research Centre for Health Equity, School of Regulation and Global Governance, Australian National University, Australia), Johanna Hanefeld (Vice President (Acting), Robert Koch Institute; Director, Centre for International Health Protection, Robert Koch Institute, Germany), Meri Koivusalo (Professor of Global Health Policy, Faculty of Social Sciences, Tampere University, Finland), Vivian Kwang-wen Lin (Honorary Professor, Li Ka Shing Faculty of Medicine, University of Hong Kong), Rene Loewenson (Director, Training and Research Support Centre, Zimbabwe), David McCoy (Research Lead, United Nations University International Institute for Global Health, Malaysia), Iman Nuwayhid (Professor, Faculty of Health Sciences, American University of Beirut, Lebanon), Hoda Rashad (Professor and Director, Social Research Center, The American University in Cairo, Egypt), Sabina Rashid (Professor, Mushtaque Chowdhury Chair in Health and Poverty; Director, Centre for Gender and Sexual and Reproductive Health and Rights, BRAC James P Grant School of Public Health, Bangladesh), Papaarangi Reid (Professor, Māori Health, Faculty of Medical and Health Sciences, University of Auckland, New Zealand), Gita Sen (Professor Emeritus, Indian Institute of Management Bangalore; Former Director, Honorary Distinguished Professor, Ramalingaswami Centre on Equity & Social Determinants of Health, Public Health Foundation of India).

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WHO contributors and reviewers

WHO Regional Office for Africa: Doris Kirigia, Guy Mbayo and Peter Phori (Division of Universal Health Coverage/Healthier Population).

WHO Regional Office for the Americas and Pan American Health Organization: Francisco Armada Perez, Pamela Bernales Baksai, Pilar Campos Esteban, Gerry Eijkemans, Fernanda Lanzagorta Cerecer, Fabrizio Mendez, Orielle Solar Hormazabal, Soto Santiago and Hugo Vasquez. (Department of Social and Environmental Determinants for Health Equity).

WHO Regional Office for the Eastern Mediterranean: Zahra Ahmed, Maha El-Adawy and Samar Elfeky (Division of Health Promotion); Hafid Hachri (WHO Office, Morocco).

WHO Regional Office for Europe: Matthias Braubach, Kira Fortune and Sinaia Netanyahu (Department of Environment and Communicable Diseases); Christine Brown, Yannish Naik and Dorota Sienkiewicz (Office for Investment for Health & Development); Kathrin Hetz and Kremlin Wickramasinghe (Special Initiative on Noncommunicable Diseases and Innovation).

WHO Regional Office for South-East Asia: Suvajee Good (Department of Family Health, Gender and Life Course).

WHO Regional Office for the Western Pacific: Emma Callon, Isabel Espinosa, Joana Madureira Lima and Antonio Palazuelos Prieto (Division of Healthy Environments and Populations); Josaia Tiko (WHO Office, Papua New Guinea); April Joy David (WHO Office, Philippines).

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Abbreviations

AI	Artificial Intelligence
CONNECT	Community Network Engagement for Essential Healthcare and COVID-19 Responses through Trust
COVID-19	Coronavirus Disease 2019
GDP	Gross Domestic Product
Global Fund	Global Fund to Fight AIDS, Tuberculosis and Malaria
HiAP	Health in All Policies
HIV	Human Immunodeficiency Virus
HIV/AIDS	Human Immunodeficiency Virus/Acquired Immune Deficiency Syndrome
ILO	International Labour Organization
LEB	Life expectancy at birth
Mpox	Monkeypox (now known as Mpox)
OECD	Organisation for Economic Co-operation and Development
PHC	Primary Health Care
PROGRESS	Place of residence (urban/rural), Race/ethnicity, Occupation, Gender, Religion, Education, Socioeconomic status, Social capital/resources
SDGs	Sustainable Development Goals
SDH Special Initiative	WHO Special Initiative for Action on the Social Determinants for Advancing Health Equity
UHC	Universal health coverage
UN	United Nations
WHO	World Health Organization
WTO	World Trade Organization

Executive summary

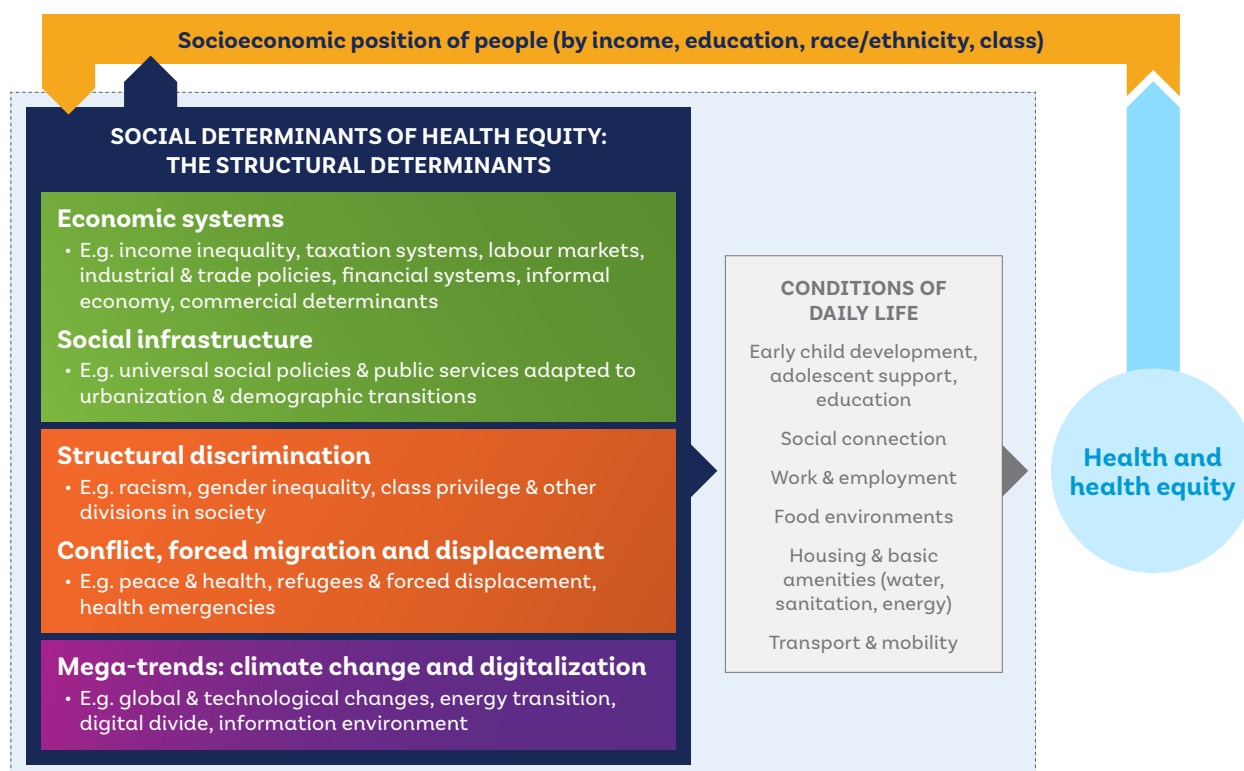
The place where you live, the communities you belong to, your education level, ethnicity, race, income and gender, and whether you have a disability, all make a huge difference to how long you can expect to live a healthy life. People in the country with the highest life expectancy will, on average, live for 33 years more than those born in the country with the lowest life expectancy. There are major differences in life expectancy between countries at very similar income levels: data shows that regardless of income level, some countries have managed to halve premature death over the past half-century, while in others, it has remained the same or even increased. Within countries, life expectancy varies by decades, depending on which area you live in and the social group to which you belong.

These avoidable differences in health are largely due to the social determinants of health equity. The conclusion of the WHO Commission on the Social Determinants of Health (“the Commission”) in 2008 that “social injustice kills on a grand scale” continues to resonate. Despite global consensus that these social determinants must be tackled, progress since 2008 has been insufficient. While health gaps between countries have reduced, as many countries have advanced their overall social and economic development, within many countries, inequalities are widening, in turn deepening health inequity. Widening inequity within countries, especially in lower-income countries, depletes their capacity to establish the inclusive institutions which are key to achieving prosperity, and increases the chance that countries will be trapped in cycles of conflict and under-development.

In resolution WHA74.16 (2021), the Seventy-fourth World Health Assembly requested the Director-General to prepare an updated report on the social determinants of health, their impact on health and health equity, progress made so far in addressing them, and recommendations for further action. This *World report on social determinants of health equity* responds to that request.

Fig. 1 provides an overview of the social determinants of health equity and areas for action addressed in this report.

FIG. 1: Social determinants of health equity as outlined in this report



Progress against the targets of the Commission on the Social Determinants of Health

The Commission set out the aspiration to close health gaps in a generation. At the current rate of progress, this goal will not be achieved. Improvements in the conditions of daily life – including in access to quality education, economic opportunities and employment, as well as water, sanitation, energy, housing and transport – have contributed greatly to improvements in population health. But this progress has been brought to a halt by a series of predictable and preventable global crises – financial, climate, and driven by conflict and pandemics.

The Commission had a target to halve the gap in life expectancy between social groups within countries by 2040. Where there is data available, such life expectancy gaps within countries have often widened. Globally, under-five mortality halved between 2000 and 2023, falling short of the improvement needed to meet the Commission’s target of a 90% decline by 2040. Low-income countries saw the fastest relative decline, but in 2023, the rate of under-five mortality in low-income countries was still 13 times higher than in high-income countries.

Maternal mortality fell by 40% globally between 2000 and 2023, from 328 to 197 deaths per 100 000 live births, but will need to fall to less than 16 deaths per 100 000 live births by 2040 to meet the Commission target. Women from disadvantaged or marginalized groups – measured by income, race and ethnicity, educational level, or place of residence – are still far more likely to die from pregnancy-related causes than their more advantaged counterparts in countries at all income levels.

Sustained progress in health equity requires political will and action to tackle the structural social determinants of health equity, as well as the daily conditions of life.

The inequitable impacts of the COVID-19 pandemic

The COVID-19 pandemic is a profound global example of the inequitable health outcomes driven by upstream social determinants. From its outset, socio-economic inequality was a predictor of higher mortality. Between countries, inequity in COVID-19 outcomes was exacerbated by inequitable access to diagnostics, vaccines and other countermeasures. At subnational level, higher COVID-19 incidence and mortality were associated with higher social and economic inequality, including alarming inequities due to race and ethnicity status in many countries. Beyond unequal access to pharmaceuticals, social and economic inequities hindered the effectiveness of public health and social measures at reducing the spread and impact of the pandemic. The pandemic's impact in accelerating inequity has persisted beyond the initial crisis. Many low- and middle-income countries emerged with large amounts of sovereign debt, reducing the fiscal space for investment in necessary social and physical infrastructure to improve health, and resulting in large backlogs in health services. Global progress towards the Sustainable Development Goals (SDGs) was already faltering before the additional shock of the pandemic; now it is severely off-track. COVID-19 will not be the last pandemic with severe health and health equity impacts unless enhanced efforts are taken to address social and economic inequities as part of pandemic prevention, preparedness and response.

Areas for action

Four interlocking strategies are needed to make lasting contributions to addressing the social determinants of health equity:



1. Addressing economic inequality and investing in social infrastructure and universal public services.
2. Overcoming structural discrimination.
3. Managing the challenges and opportunities posed by the mega-trends of climate change and the digital transformation in a way that systematically promotes health equity.
4. Putting in place governance arrangements that will maximize the coherence and impact of action.



Economic inequality and universal public services



Economic inequality and inadequate access to quality public services in areas such as health, education, social protection, transport and housing are major drivers of health inequities. Addressing health equity requires whole-of-government commitment and action through economic and social policies. Dedicated public investment is needed to underpin access to high-quality and affordable health, welfare and social services, whether provided by the public, private or third (not-for-profit) sectors. Increased spending on social services has been shown to result in measurable health equity improvements. Social protection programmes across the life course are investments in the necessary conditions to enable all people to prosper and flourish. Interventions such as unemployment benefits, pensions and child support provide a safety net during times of need. They reduce poverty and debt, and improve health outcomes. International cooperation on development financing and on minimum levels of taxation can be leveraged within the inclusive global forum of the United Nations, to support significant progress in health equity.

Many governments are starting to consider well-being frameworks, to go beyond gross domestic product (GDP) as a measure of progress. Well-being budgeting frames national development so that health in the broadest sense can become the focus of policy-making. Health equity is integral to well-being approaches, helping to ensure that financial resources are allocated to areas that maximize health outcomes for all citizens.

Growing levels of external debt observed in many countries are limiting fiscal space for essential investments in universal public services. In 2024, an estimated 3.3 billion people lived in countries that spent more on debt interest payments than on either education or health. Reducing external debt levels, creating fiscal space and enhancing commitment to social services are essential for increasing investments in these areas, ultimately leading to healthier and more equitable communities.

Commercial actors that produce goods and services that improve health, support good employment conditions and have positive impacts on communities can benefit health and reduce health inequities. Yet the impacts of business actors that are escalating avoidable levels of ill health, planetary damage and inequity need to be addressed through action by governments and agencies equipped to take measures in the

interests of human and planetary health, with the support of regional and international mechanisms. There is evidence that the public sector can be effective in mitigating negative impacts of health-harming commercial practices, including through enacting regulatory and legislative changes that better value human and planetary health. The public sector can also use its considerable leverage as a procurer of goods and services – worth more than US\$ 8 trillion globally – as a force for good, in shaping everything from sustainable agriculture to decent work.

Structural discrimination

Structural discrimination is discrimination which occurs systematically across society and is embedded in prevailing cultures, legal mechanisms and economic structures. Comprehensive strategies are required to dismantle inequities that are systemic in nature and often long-standing. Historic discrimination has left some population groups with fewer resources and diminished fulfilment of human rights, perpetuating ongoing intergenerational inequity, which manifests as health inequities. Racism, sexism, classism and ableism, for instance, often intersect and compound, acting across the life course and between generations to undermine health and people's agency to lead flourishing lives. Rectifying deep-seated inequities requires institutional and legal change, systematic realization of human rights, and redistribution of social and economic resources between and within countries. Restorative justice and reparation to affected communities are increasingly being considered to improve health equity and redress structural violence, and reduce intergenerational transmission of harm, stress and disadvantage.



Gender inequality compounds and intersects with other aspects of structural discrimination and social inequity. Women and girls often suffer from poorer health and quality of life, made worse by inadequate social and health protection. Policies grounded in human rights, supporting women's equitable participation and remuneration in the workforce, and reducing their burden of unpaid care and domestic labour, are associated with lower gender inequalities in health, and must accompany efforts to overcome women's poverty. (i)

The increasing volume and precariousness of migration, together with unprecedented levels of conflict with widening impacts on civilian populations, are a form of structural violence which decimates

i Throughout this publication, reference is made to human rights law and standards in international law, the applicability of which in a specific setting will depend on factors such as the State's ratification of relevant human rights instruments.



health equity. Robust social protection which adapts to the changing circumstances of health emergencies and of conflict is an important tool to protect migrants and those affected by conflict. So, too, are practices which embed the principle of the right to health regardless of migrancy or citizenship status.

Steer action on climate change and digitalization towards health equity

Climate change and digital transformation are two important existential forces that significantly impact health equity. The green and digital economies are key pillars of many countries' economic plans. Effective climate action must integrate a "just energy transition" and health equity considerations, together with effective approaches to the commercial determinants of health, ensuring that actions aimed at climate mitigation and adaptation do not make inequities worse, and prioritizing populations experiencing conditions leading to greater vulnerability.



Climate-related actions addressing the social determinants of health equity must be intersectoral. These include measures to reduce pollution through stricter environmental regulation, expanding public transport, and promoting sustainable agricultural practices that enhance food security and nutrition while benefiting rural populations. The positive frameworks that recognize the unequal burdens of global warming, including the initiation by COP29 of a Loss and Damage Fund, are important measures for creating the means to redress climate-related health inequities.

Digital technologies and big data can improve health and social care delivery and outcomes, but their effectiveness in supporting health equity depends on how they are deployed. To fulfil the potential of digitalization to improve health equity, a combination of approaches is needed. These include ensuring equitable access to services regardless of digital access and literacy; inclusive service design, with priorities set by end users; regulation of inequitable platform-based business models, and the implementation of human rights frameworks that mitigate risks to human dignity, autonomy and privacy. In the world of work and education, digitalization can be a force for good, but also for exclusion of large segments of the population from essential social protection and services.

Information is an important determinant of health in its own right, with misinformation and disinformation growing mediators of health outcomes. The proliferation of health information channels and the



complex interplay between trust, authority and social connection has created new challenges and opportunities which health institutions are barely able to keep up with. Frameworks which systematically promote health information equity, including with particular reference to information on the social determinants of health, are urgently needed.

In the context of rapid technological change, public regulation and the accountability of commercial actors have often lagged. Policies should ensure digital advancements are used for positive reasons, such as to harness the opportunities from telemedicine and distance learning, and are accessible to all. Platform operators and regulators should take responsibility for addressing the harms of addictive and antisocial online behaviours. Generative AI and other technological advances are being embraced in a range of health care delivery, but deliberate design and governance efforts must be employed to ensure that these changes help bridge, rather than widen, the digital divide.

Governance to maximize the coherence and impact of action

National governments and international organizations must support and collaborate with local governments, communities and civil society in their efforts to address the social determinants of health equity.

The international system supporting health equity needs reinforcement, in order to tackle conflicts of interest, promote action across sectors, and boost democratic accountability and the full realization of human rights.



Empowering local governments, civil society and communities is crucial for addressing health inequities. People on the ground and local entities understand their specific challenges, opportunities and context, making them better positioned to identify and implement actions on social determinants. Local initiatives also show promise in improving social connection and addressing isolation and loneliness. Supporting these local efforts with adequate resources, authority and policy backing is crucial for their success.

Creating platforms for community voices, and co-designing policies and programmes with those they aim to serve, ensures services are responsive to needs. By building a robust health and care sector that prioritizes equity, participation and primary health care approaches, as well as developing structures to promote intersectoral action, significant progress can be made in addressing the social determinants of health equity.

Primary health care approaches orient health services more directly to population needs, creating systemic feedback loops that generate opportunities from local to national levels to change the social determinants of health equity. Such approaches require a capable workforce across health and social sectors. Training health professionals on social determinants and delivery in culturally appropriate and inclusive ways can improve their ability to provide equitable care and to work intersectorally, engaging communities and promoting better responses to their needs, to improve their daily living conditions.

Robust and representative governance needs the support of comprehensive data systems. Reliable monitoring of conditions and actions affecting the social determinants is crucial for tracking progress, ensuring accountability and informing policy decisions. Data disaggregation by age, gender, race, ethnicity, migration and socioeconomic status offers valuable insights into health inequities,

providing the baseline understanding to drive comprehensive actions. Countries' experience has been that effective monitoring of progress is a necessary counterpart to action at all stages – in formulating policy goals and identifying needs; marshalling coalitions for action, including social movements, and maintaining the long-term political commitment needed for fundamental change.

A strategic framework for action

Proactive and comprehensive strategies that integrate health, social and environmental impacts into economic and development thinking can be powerful forces for change towards health equity. The most successful strategies will come from resolute leadership and whole-of-government approaches that generate and learn from evidence. Closing the inequality gap in health outcomes is not just a matter of ensuring aggregate outcomes are more fairly distributed; it also requires a commitment to “leave no one behind”, which is the central promise of the 2030 Agenda for Sustainable Development.

In taking forward this agenda, there are key roles across institutions and sectors:

- ▶ **National and local governments:** lead with strong policies, integrating action on the social determinants of health equity into strategies, supported by intersectoral governance structures and investments.
- ▶ **Civil society:** engage in advocacy, policy-making, community actions, knowledge-building and social mobilization on the social determinants of health equity.
- ▶ **Private sector:** commit to social contracts that support health equity, enhance corporate social responsibility and good citizenship, avoid exploiting regulatory weaknesses, and enhance accountability in employment and production practices.
- ▶ **Research institutions:** advance partnerships and expand the evidence base of complex adaptive policies and responses to address the social determinants of health equity.
- ▶ **The UN system and global financial institutions:** adopt health equity as a central goal, support countries with implementation of the key recommendations in this report, and ensure aid, investment and lending support to policies addressing the social determinants of health equity.

- ▶ **The World Health Organization:** continue to proactively advance the health equity agenda through evidence-gathering for norms and standard setting; supporting global and national policy coherence; convening diverse partners; monitoring and implementation, and building capacity.

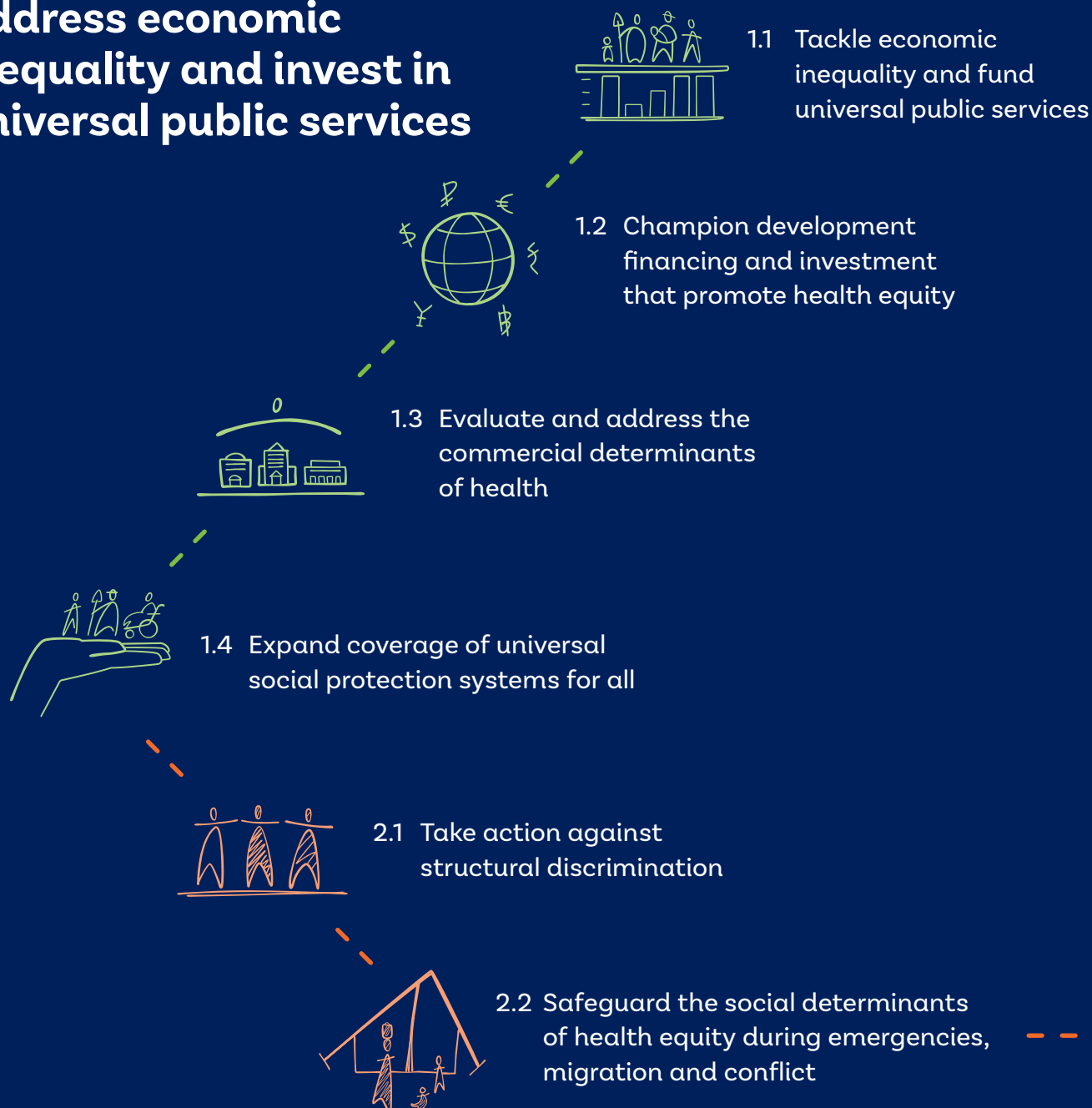
Much of the global disease and mortality burden is preventable. Perpetuating avoidable gaps in health outcomes is unfair. A renewed commitment to address the social determinants of health equity is a vital pathway to right these wrongs. This world report makes clear recommendations for tackling the social determinants at all levels (see Fig. 2 and Table 1). Leaders at all levels need to reinvigorate and purposefully pursue the actions and comprehensive strategies which show greatest promise in addressing these determinants, bridging different disciplines and sectors. Only through such collective and holistic efforts to address the root causes of health inequities will the world achieve long, healthy lives for all, as envisioned in the Sustainable Development Agenda.

FIG. 2:

Recommendations for actions on the social determinants of health equity

ACTION AREA 1

Address economic inequality and invest in universal public services



ACTION AREA 2

Tackle structural discrimination and the determinants and impacts of conflict, emergencies and migration

ACTION AREA 4

Bringing about change through new governance approaches



4.6 Monitor social determinants of health equity



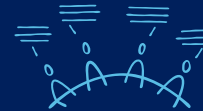
4.5 Strengthen focus on social determinants in health systems and policy platforms



4.4 Build and retain a health and care workforce capable of delivering equity



4.3 Reach universal health coverage through progressive health financing and primary health care



4.2 Support community engagement and civil society



4.1 Equip local governments to reduce health inequities



3.2 Ensure digital transformation promotes health equity



3.1 Articulate and accelerate the health equity benefits of climate action and the preservation of biodiversity

ACTION AREA 3

Steer mega-trends towards health equity: climate change and digitalization

TABLE 1. Recommendations for action on social determinants of health equity

Recommendation	Report section	Key actions
Action area 1: Address economic inequality and invest in universal public services (Part 2, Chapter 3)		
Address economic inequality and invest in universal public services. 	3.1	Use progressive taxation to expand fiscal space for income transfers and universal, equitable public services. Provide adequate public funding for good-quality, equitable infrastructure and services, and ensure that effective mechanisms are in place to safeguard quality and secure equity. Move towards well-being economies and consider well-being budgeting.
Promote the social determinants of health equity in development financing and investment. 	3.2	Ensure that the need for funding (fiscal space) to address the social determinants of health equity is included in approaches to development financing, debt relief and taxation. Deliver and monitor development financing to support public investment in policies, actions and infrastructure that address the social determinants of health equity.
Analyse and address the commercial determinants of health. 	3.3	Analyse, legislate on and regulate commercial activities that negatively affect health and health equity, while maximizing the health-promoting capacity of the private sector. Address and manage conflicts of interest in policy environments, to prevent negative impacts on population health and health equity. Use the scale of the public sector to incentivize commercial activities that positively affect health and health equity – for example, through mandating public procurement that requires sustainable, safe and healthy products, and safe and fair labour standards throughout supply chains. Strengthen health equity considerations in global and regional trade processes, including in relation to intellectual property.
Increase comprehensiveness and expand coverage of universal social protection systems across the life course. 	3.4	Ensure adequate income guarantees and care throughout the life course, including for people with disabilities or chronic health conditions. Build and expand paid leave benefits for sickness and parental leave, including for the precariously employed and informal workers.

Recommendation	Report section	Key actions
Action area 2: Tackle structural discrimination and the determinants and impacts of conflict, emergencies and migration (Part 2, Chapter 4)		
Address structural discrimination. 	4.1	Recognize and repair discrimination embedded in policies, laws, institutions and social norms. Redress the negative impacts of colonization by developing standards for reparative justice that measure impacts on health. Acknowledge Indigeneity as a determinant of health and health equity (shaped by discrimination and historical injustices, as well as protective factors), while upholding the provisions of the United Nations Declaration on the Rights of Indigenous Peoples.
Address and protect the social determinants of health equity in emergencies, migration and conflict. 	4.2	Recognize peace as a fundamental determinant of health equity, and health as a bridge to peace. Ensure preparedness and response efforts for health emergencies reflect social determinants of health equity considerations at their core, including the equitable and rapid rollout of social protection. Ensure the rights of, and support, displaced people and migrants to access health and social services without discrimination.
Action area 3: Steer action on climate change and digitalization towards health equity (Part 2, Chapter 5)		
Articulate and accelerate the health equity benefits of climate action and the preservation of biodiversity. 	5.1	Support the development and implementation of climate change mitigation and adaptation policies that maximize health equity and co-benefits. Enable the energy and food transitions to prioritize addressing energy poverty and food insecurity. Integrate health systems more centrally in climate and environment strategies, and ensure that local and equitable climate action is a core objective of health systems. Strengthen support for Indigenous communities in their stewardship of land and natural resources, and recognize and address the impacts of environmental degradation, land confiscation and climate-related migration.
Steer the digital transformation in favour of health equity and the public good. 	5.2	Address the digital divide. Prevent harmful impacts of digital and technological transformations and artificial intelligence on health and health equity, including in relation to labour markets and social cohesion.

Recommendation	Report section	Key actions
Action area 4: Bringing about change through new governance approaches (Part 3, Chapters 6–8)		
Empower local governments to address health inequities. 	6.1	Strengthen the role of local government with the functions and resources to implement community-centred actions for health equity. Ensure that urban, rural and territorial planning, transport and housing investments are underpinned by approaches that deliver healthy housing and built environments, and adopt universal design principles. Equip local government to address health equity across the life course, including in supporting age-friendly communities, and combating social isolation and loneliness.
Support community engagement and civil society. 	6.2	Incorporate representative community engagement and social participation in local policy processes, and in health and social service decision-making and delivery. Create the enabling conditions that maximize the capabilities of independent and inclusive civil society to address the social determinants of health equity.
Achieve universal health coverage through progressive health financing and primary health care approaches. 	7.1	Improve equitable access to a continuum of quality health services, addressing both physical and mental health across the life course, by strengthening primary health care-orientation within health systems. Minimize patients' out-of-pocket expenditure, and finance health services from pooled government resources. Increase the share of health and care sector funding in proportion to need, with special intensity for populations experiencing marginalization, and recognize and address discrimination within health systems. Facilitate equitable access to health technologies, from research and development, through to manufacturing and equitable delivery.
Build and retain a health and care workforce capable of delivering equity. 	7.2	Recognize and reward work in the care economy, including informal care, and demonstrate the role of the health and care sector as a driver of health equity and social inclusion, by ensuring decent and safe working conditions and addressing pay and representation gaps. Develop human capacity in health, social protection, education, labour, local government and service organizations to enhance intersectoral efforts to address the social determinants of health equity.
Strengthen focus on social determinants in health systems and policy platforms. 	8.1	Integrate addressing the social determinants of health equity in all health strategies, policies, emergency preparedness and response plans, and public health laws. Establish coordination and accountability mechanisms for intersectoral collaboration and community engagement for health (including Health in All Policies functions) at all levels of government.
Monitor social determinants of health equity. 	8.2	Strengthen statistical infrastructure, institutionalize and build capacity for the use of disaggregated data to measure progress on the social determinants of health equity, including the distribution of socially produced health risks across social groups and geographic areas; the mechanisms that drive inequities, and actions to address them. Utilize new technologies and novel data sources to fill data gaps and inform action on the social determinants of health equity.



PART 1:

The state of social
determinants of
health equity

PART 2:

Tackling the social
determinants of
health equity on
multiple fronts

PART 3:

Bringing about
change through
new governance
approaches

CHAPTER 1:

Inequities in today's world

1.1

This report

Social injustice continues to kill on a grand scale

Unacceptable gaps persist in how long people can hope to live healthy lives, depending on where they live, the community they belong to, their level of education, how much money they have, how they identify, their gender, their race and ethnicity, whether they have a disability, and other characteristics. Between countries, in 2021, the difference in life expectancy was as wide as 33 years (1). Inequalities of similar magnitude are seen within countries (2). These inequities are not explained only by a country's income level: of the 34 countries that over the past 20 years have halved premature death (before age 70), some are high-income, some middle- and some low-income. Similarly for those countries that have seen either minimal change or increases in premature deaths (1, 3). Inequities in life outcomes are closely linked to degrees of social disadvantage: health follows a social gradient whereby the more deprived the area in which people live, the lower their income, the fewer their years of education, the worse their health and the fewer healthy years they can expect to live. Independent of income, marked inequalities in health are also seen among minority racial and ethnic groups, underprivileged castes, Indigenous Peoples, refugees and displaced populations, people with disabilities, and other groups experiencing marginalization and exclusion (4, 5, 6, 7).

Health inequalities are the measurable differences in health across population subgroups, and health inequities are those differences that are avoidable and unjust. They are the result of political, cultural, social and economic systems that shape daily living conditions: the social determinants of health equity (for more information on definitions, see Box 1). Although groups are defined on the basis of characteristics of individuals, it is the operation of social determinants of health, not properties of individuals, that account for most of the health inequities between groups.

In 2005, WHO convened the Commission on the Social Determinants of Health (hereafter, "the Commission"). Its goal was to support countries in tackling social determinants of health to improve health equity. It aimed to do so by catalysing policy and institutional change to address social determinants of health within countries. The Commission's 2008 final report reviewed the evidence on social determinants of health and actions addressing them, and identified policy options and opportunities for action to close the health gap in a generation. The report concluded that "social injustice is killing people on a grand scale" and "a toxic combination of poor policies and programmes, unfair economic arrangements, and bad governance" contributes to the unjust life chances of children and adults around the world (6).

Since the Commission, there has been some promising progress in reducing health gaps between countries, as described in Section 1.2. This progress is largely due to improved conditions of daily life – greater access to education and employment, improved housing and environments, and increased access to effective primary health care (PHC). However, these actions can only achieve so much within the context of inequitable political, cultural, social and economic systems that reward the most powerful, to the detriment of the least powerful. The persistence of structural inequities has meant that action has not been at sufficient scale, nor sustained for long enough, to effect widespread and lasting improvements to health and health equity (8).

The social determinants of health – that is, the conditions in which people are born, grow, live, work and age, and people’s access to power, money and resources – have a powerful influence on these avoidable and unjust health gaps (6, 9, 10, 11). These determinants are the main ways health is shaped, more so than access to health care services, or genetic influences (6, 12, 13). Social determinants encompass both downstream “intermediary” determinants, representing the conditions of daily life, and upstream “structural” determinants, representing governance, political and economic systems, and societal norms and values that shape who has access to resources and opportunities, and who does not. The structural determinants create unequal distributions of money, power and resources according to a person’s social status, as defined by gender, age, education, income, race or ethnicity, disability status and other characteristics, which create inequities in the conditions of daily life that shape health. Therefore, structural determinants are the “root causes” of health outcomes (for more information on definitions, see Box 1).

This *World report on social determinants of health equity* reviews progress since the Commission. It draws on updated evidence to set out the rationale for more intensive action on the social determinants of health equity, and makes recommendations for action. These recommendations signpost key areas where stakeholders can implement strategies to address the social determinants of health equity. The report emphasizes the need to focus on equity and levelling up all people’s health, creating healthier and more equitable societies. While the phrases “social determinants of health” and “social determinants of health equity” are often used interchangeably, this report uses “social determinants of health equity”, to keep the focus on reducing health inequities. It emphasizes the need for action on structural determinants, because action on these “causes of the causes” is essential for achieving health equity.

1.1.1 Definitions

Box 1 provides definitions of key terms used in this world report.

BOX 1. Definitions of key terms used in this world report

The social determinants of health equity: These refer to the societal factors that give rise to social position and the association between social position; access to power, money and resources, and health. These include the conditions in which people are born, grow, live, work and age, together with **structural determinants**, such as the formal and informal rules of systems and institutions (including economic systems and commercial determinants), policies, culture and values (including classism, racism, sexism, ageism, ableism, xenophobia and homophobia). They are influenced by historical context, and operate over the lifespan (14, 15, 16). Societal factors are conditioned by their physical and technological environment, and changes in these environments are important constraints on or enablers of social change.

Studies suggest that social determinants account for at least 50% of health outcomes (12, 17, 18) and health inequities (13). While, for some, the phrase “social determination of health” better emphasizes the active exercise of power in creating health outcomes (19), this report uses “determinants”, although the term should be understood in this dynamic sense.

Health equity: The absence of unfair and avoidable or remediable differences in health among population groups defined socially, economically, demographically or geographically (14).

Health inequities: The unfair and avoidable or remediable systematic differences in health among population groups defined socially, economically, demographically or geographically (20). Health inequities arise fundamentally from the unequal allocation of power, money and resources, which manifest in the unequal distribution of social determinants (5, 6).

The social gradient in health: This means people and communities have progressively better health, the higher their socioeconomic position or conditions.

Health inequalities: The observable differences in health between population groups. Health inequality is the metric by which health inequity can be assessed, meaning characteristics that stratify people’s power and access to opportunities should most closely reflect likely areas of unfair health differences (21).

Characteristics that stratify people’s power and access to opportunities are summarized by the PROGRESS acronym: place of residence (urban/rural), race/ethnicity, occupation, gender, religion, education, socioeconomic status, and social capital/resources (20). Intersections of disadvantage can be analysed by applying multiple stratifications. The Campbell and Cochrane Equity Methods Group has expanded PROGRESS to include additional elements, termed PROGRESS-Plus (22):

- ▶ personal characteristics associated with discrimination (such as age or disability)
- ▶ features of relationships (for example, smoking parents; exclusion from school)
- ▶ time-dependent relationships (for example, leaving hospital, respite care).

A Health in All Policies (HiAP) approach: This is an approach to public policies across sectors that systematically takes into account the health implications of decisions, seeks synergies, and avoids harmful health impacts in order to improve population health and health equity. It improves accountability of policy-makers for health impacts at all levels of policy-making, and includes an emphasis on the consequences of public policies on health systems, determinants of health, and well-being (23).

The commercial determinants of health: These are the commercial dimension of the social determinants of health, comprising activities undertaken by commercial actors that affect people’s health, directly or indirectly, positively or negatively (24).

Social infrastructure: This refers to the policies, services, resources and related spaces that people have access to, which enable them to fully participate in social, civic and economic life without barriers (25, 26).

Structural discrimination: The rules, norms, routines, patterns of attitudes and behaviours in institutions and other societal structures that represent obstacles to groups or individuals in achieving the same rights and opportunities as those available to most of the population (27). Structural discrimination occurs systematically across society and is embedded in prevailing cultures, legal mechanisms and economic structures.

1.1.2 Goals and objectives of the report

In resolution WHA74.16 (2021), the Seventy-fourth World Health Assembly requested the Director-General to prepare an updated report on the social determinants of health, their impact on health and health equity, progress made so far in addressing them and recommendations for further action (28). This report responds to that request.

1.1.3 Purpose of the recommendations

The action areas and recommendations in this report build on the analysis and recommendations made by the Commission, and are attuned to a rapidly changed and changing global context.

The recommendations are based on evidence of which social determinants shape health inequities and what actions to address them work to improve health equity. Since the Commission, the evidence base has grown significantly, as a result of greater focus on the social determinants in academic literature, and in the development and implementation of policies and programmes to act on social determinants, which have led to greater expertise in how to deliver practical actions (29).

Drawing on this evidence, the recommendations signpost key areas that can be considered and adapted in the development and implementation of strategies to address the social determinants of health equity. Implementing these recommendations would lead to better health, greater health equity and progress on social, economic and development domains that have already been agreed globally, including the Sustainable Development Goals (SDGs). The recommendations are relevant to all regions and countries at all income levels, but implementation plans must be mindful of the specific patterns of inequity in each setting.

1.1.4 Audience for the report

This report primarily targets national decision-makers whose work in different sectors impacts health. Given the breadth of policies that influence health, these decision-makers include national leaders and officials across a wide range of sectors. The report is also relevant to decision-makers in local government, parliaments, health systems, multilateral organizations, civil society, academia and the private sector, and to researchers, practitioners and advocates seeking to advance health equity through their work.

1.1.5 Development of the report

The development of this world report followed a rigorous, inclusive and transparent process, adhering to WHO's standards for evidence synthesis, stakeholder engagement and ethical governance.

The WHO Secretariat commissioned 10 scoping reviews and background papers to identify the latest evidence on the social determinants of health equity, progress in tackling them and best practice (see Annex 1). Two of these have been synthesised and published in the peer reviewed literature (8, 30). Following these preliminary scoping reviews, an initial outline was proposed for review and advice by the policy and scientific expert groups. Once the thematic focus areas were identified, updated structured evidence reviews were carried out specifically for the report by the UCL Institute of Health Equity. Evidence synthesis combined systematic reviews of peer-reviewed literature, WHO reports and grey literature with country examples demonstrating diverse contexts and practical applications. Country examples were selected using criteria such as geographic diversity, relevance to social determinants of health, and innovative approaches to addressing inequities. Drafts were rigorously reviewed by expert group members and validated through consultations with stakeholders, including governments in featured countries. Twenty consultations were held with UN Partners, policy-makers and user audiences to discuss the report.

“Pathfinder countries’ have generated important experiences to support action on the social determinants of health in advancing health equity”

Beyond the literature reviews and expert groups, the report draws on learning from the multicountry Special Initiative for Action on the Social Determinants of Health for Advancing Health Equity (SDH Special Initiative), launched in 2021 (15) and from specific case studies identified in the reviewed literature. The initiative's participants – so-called “pathfinder countries” – in the Americas, Asia and the Eastern Mediterranean generated important experiences in successful governance approaches to support action on the social determinants of health in advancing health equity. Some of these experiences that have demonstrable results were recorded as country examples in the report, while other examples originated outside WHO. The report also draws on evidence generated for the WHO regional commissions on the social determinants of health in the Americas (31), European (32) and Eastern Mediterranean (5) regions.

“Implementing these recommendations would lead to progress on the Sustainable Development Goals”

In response to Resolution WHA74.16, a draft outline of the report was presented to the 152nd session of the WHO Executive Board, and received positive feedback. A summary of the report and its proposed recommendations, based on evidence reviews, was subsequently presented to the 154th session of the WHO Executive Board and the 77th World Health Assembly. Member State feedback during these sessions was positive and informed the iterative refinement and finalization of the report.

See the Acknowledgements for a list of the expert group members, and Annex 1 for further description of the approach and the methods used for the world report.

1.1.6 Report overview

The report consists of eight chapters in three parts:

Part 1 comprises two chapters which provide the rationale for the recommendations. Chapter 1 provides an overview of progress towards health equity since the Commission, and makes the case that while there has been progress on improving daily living conditions, there has been insufficient action on the structural determinants. Chapter 2 provides the framework for the analysis and the recommendations which follow, surveying the social determinants of health equity and summarizing the changing global context and new threats to health.

Part 2 considers recommendations on the main determinants identified in Chapter 2. Chapter 3 recommends approaches to address economic inequalities and invest in universal public services. Chapter 4 addresses structural discrimination and the inequitable determinants and impacts of conflict and forced migration. Chapter 5 sets out how to steer action on two major planetary transitions: climate change and digitalization.

Part 3 covers governance and addresses the ways in which new approaches to governance in a variety of sectors can enable more equitable outcomes. Chapter 6 makes recommendations for the greater involvement of local government, communities and civil society in prioritizing and taking action to address health inequities. Chapter 7 discusses the role of health systems in supporting action on the social determinants of health equity, including through PHC approaches, and Chapter 8 sets out recommendations for stronger governance to facilitate intersectoral action and enhanced monitoring of health inequities and social determinants.

1.2

Updates since the Commission on Social Determinants of Health, and progress against its targets

In 2008, the Commission set out an agenda for change with three overarching recommendations:

1. Improve daily living conditions
2. Tackle the inequitable distribution of power, money and resources
3. Measure and understand the problem and assess the impact of action.

Three years after the Commission report was published, governments met in 2011 for the World Conference on Social Determinants of Health, where the Rio Political Declaration was adopted by over 100 countries (33). The Declaration provides information to countries on how to advance the recommendations of the Commission's report, identifying five key areas considered critical to addressing health inequities:

1. Adopting better governance for health and development
2. Promoting participation in policy-making and implementation
3. Further reorienting the health sector towards reducing health inequities
4. Strengthening global governance and collaboration
5. Monitoring progress and increasing accountability.



1.2.1 Progress against the targets of the Commission on Social Determinants of Health

This report assesses progress towards the three guiding targets set out in the Commission's report, to be achieved by 2040:

1. Halve the gap in life expectancy between countries and between social groups within countries
2. Halve adult mortality rates in all countries (probability of dying between 15 and 60 years of age)
3. Achieve 90% and 95% reductions in child and maternal mortality, respectively.

Progress has been made against all three targets, but the current rates of improvement are insufficient to meet them by 2040 (see Fig. 3). Tracking progress towards the Commission's targets has been hampered by a lack of data in many countries, especially on within-country health outcomes and data disaggregated by population groups.

1.2.1.1 Progress against Commission Target 1:

Reduce by 10 years, between 2000 and 2040, the **life expectancy at birth (LEB) gap** between the third of countries with the highest LEB levels and the third of countries with the lowest, by levelling up countries with lower life expectancy at birth.

Halve, between 2000 and 2040, the **LEB gap between social groups within countries**, by levelling up the LEB of lower socioeconomic groups.

The gap in life expectancy at birth (in rounded figures) between the top and bottom third of countries declined by 5.6 years between 2000 and 2021, from a gap of 18.2 years in 2000 to one of 12.5 years in 2021, marking important progress (1). The greatest increase in life expectancy at birth occurred in the bottom third of countries, where it increased from 59 years in 2000 to 66 years in 2021 – a 12% increase, compared to a 2% increase in the top third of countries during the same period. However, progress has slowed significantly, with most of the increase in life expectancy occurring between 2000 and 2010, and a much slower rate of improvement in the subsequent decade. From 2019, life expectancy fell for all regions, except Oceania, due to the COVID-19 pandemic (1).

Within-country inequalities in life expectancy

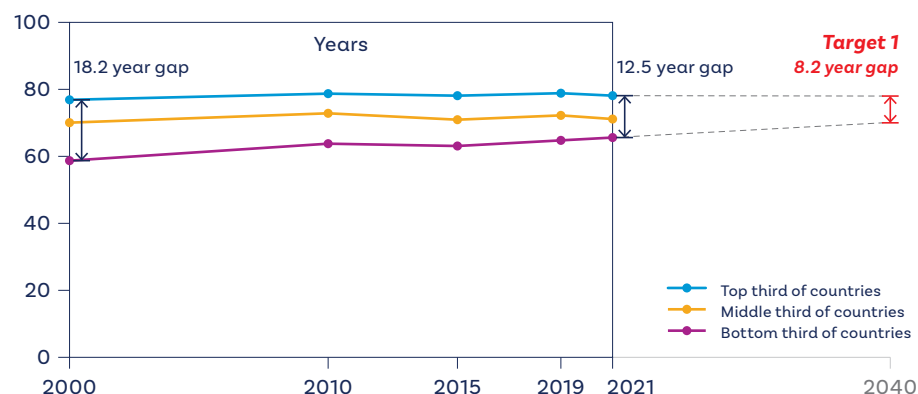
The available data shows that during the past two decades, the gap in life expectancy between the most and least advantaged groups within some countries has widened (see illustrative examples in Box 2). There is insufficient data to draw conclusions about the trend in low- and middle-income countries.

FIG. 3: Progress against the Commission on Social Determinants of Health's targets

Life expectancy at birth

Between 2000 and 2021, the gap between the top third and bottom third of countries declined by 5.6 years. **To achieve the 2040 target, an additional 4.4 years is needed.**

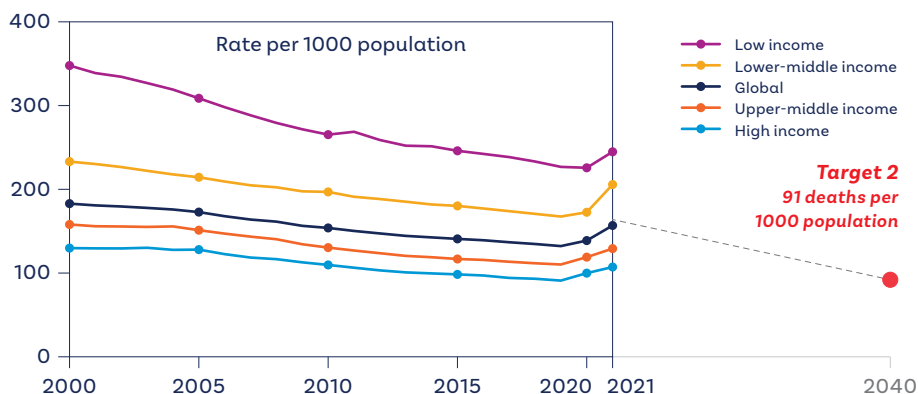
Life expectancy at birth (years) among the top and bottom thirds of countries (WHO estimates, 2000–2021)



Adult mortality

Globally, the adult mortality rate fell by 14% between 2000 and 2021. The COVID-19 pandemic slowed progress between 2019 and 2021. **To meet the 2040 target, the global adult mortality rate needs to decline by 56%.**

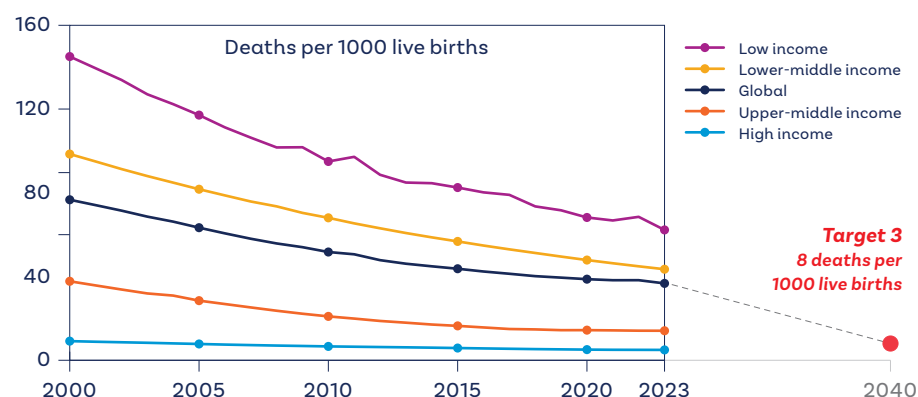
Adult mortality rate (probability of dying between 15 and 60 years per 1000 population) globally and by World Bank income group (WHO estimates, 2000–2021)



Under-five mortality

Between 2000 and 2023, the global rate of under-five mortality dramatically improved, halving from 77 to 37 deaths per 1000 live births. **Meeting the 2040 target will require a further reduction to 8 deaths per 1000 live births.**

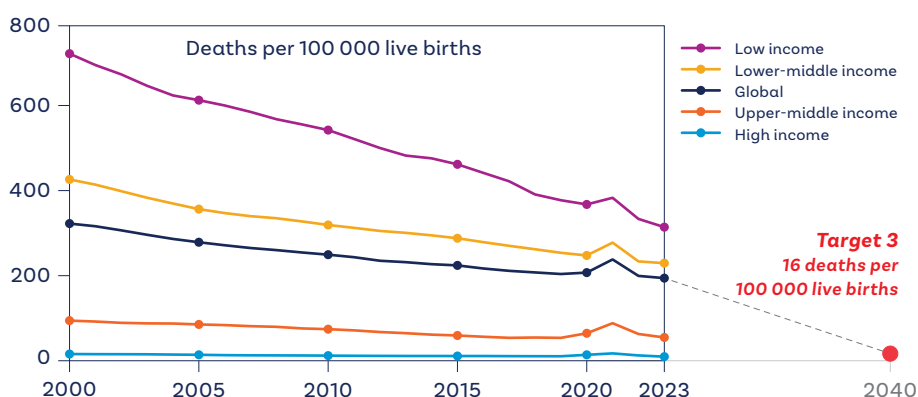
Under-five mortality rate (deaths per 1000 live births) globally and by World Bank income grouping (UN IGME estimates, 2000–2023)



Maternal mortality

Between 2000 and 2023, the global maternal mortality rate greatly improved, declining by 40%. **To achieve the 2040 target, it will need to reduce from 197 deaths to 16 deaths per 100 000 live births.**

Maternal mortality ratio (per 100 000 live births) globally and by World Bank income grouping (Maternal Mortality Estimation Inter-Agency Group estimates, 2000–2023)





BOX 2. Within-country trends in life expectancy by income or social group, sample countries

Where data is available, patterns emerge showing within-country inequalities in life expectancy across national populations.

Indigenous Peoples have lower life expectancy than non-Indigenous populations in rich and poor countries alike, although data is limited. The gap in life expectancy between the Indigenous and non-Indigenous population is 12.5 years for the Inuit in **Canada**, 10 years for Indigenous **Australians**, 21.5 years for the Baka in **Cameroon** and 13 years for the Maasai in **Kenya** (34).

In **Japan**, the gap in life expectancy between the most and least deprived areas is 2.5 years for men and 1.2 years for women between 2010 and 2014 (35).

In the **United States of America**, between 2001 and 2014, the 5% of men with the lowest incomes saw no gain in life expectancy, while the highest-earning 5% gained an extra 2.3 years of life expectancy. Between 2000 and 2019, life expectancy increased for the Black population, but still remains 3.6 years lower than that of the White population (36, 37). Since 2010, life expectancy has stagnated for White individuals in low-income counties in Appalachia and the Lower Mississippi Valley, and declined for American Indian and Alaska Native populations in the Western United States. Progress for Black populations in regions with low life expectancy stalled after 2010 (38).

In the **United Kingdom (England)** in 2001, the life expectancy gap between the richest and most deprived 20% of local areas was 7.4 years for men and 5.0 years for women, but by 2017, the gap had widened to 7.7 and 6.1 years for men and women, respectively. The social gradient in life expectancy closely tracks the level of neighbourhood deprivation (39).

In **Norway**, between 2005 and 2015, life expectancy of the richest quartile of both men and women grew, while for the lowest-income quartile it barely increased for men and fell for women (40).

In **Hungary, Poland, Latvia and Slovakia**, there are gaps in life expectancy of 10 years and more between men with high and low education levels (41).

In **Europe**, Roma people are one of the largest ethnic minorities, consisting of approximately 12 million people. The European Commission documents that Roma life expectancy in 2018 was between five and 20 years shorter than the general population (42).

Analysis of life expectancy in the 22 districts of Tehran, in the **Islamic Republic of Iran**, showed a gap of 5.5 years for females and 3.7 years for males between the most and least deprived districts (43).

1.2.1.2 Progress against Commission Target 2:

Halve between 2000 and 2040, adult mortality rates (probability of dying between the age of 15 and 60) in all countries and in all social groups within countries.

Globally, the adult mortality rate fell by 14% between 2000 and 2021, though the COVID-19 pandemic led to a rise in mortality between 2019 and 2021, slowing progress (44). There are substantial differences in adult mortality rates between country income groups. In 2021, the adult mortality rate in low-income countries (245 per 1000 population) was 2.3 times higher than that in high-income countries (107 per 1000 population).

Between 2000 and 2021, the steepest declines in adult mortality rates occurred in low-income countries, which experienced a 30% decline, compared with a decline of 12% in lower-middle-income countries, 18% in upper-middle-income countries and 17% in high-income countries. In 2021, lower-middle-income countries, representing approximately 43% of the total global population, accounted for over half (54%) of adult deaths globally (44).

To meet Target 2, between 2021 and 2040, the global adult mortality rate needs to decline by 58% to reach 91 per 1000 population.

Within-country inequalities in adult mortality

Within countries, marginalized or disadvantaged population subgroups in relation to race and ethnicity, education (see Box 3), income and immigrant status tend to have higher adult mortality rates than more advantaged populations. In some countries with available data, inequalities within countries have widened.



BOX 3. Trends in adult mortality by education level in sample high-income countries

Research in a range of high-income contexts shows a strong social gradient in adult mortality associated with people's level of education. For example:

In the **Republic of Korea**, mortality inequalities by education widened for men between 1995 and 2010, with those in the lowest education category experiencing a mortality risk 8.7 times higher than those with the highest level of education (45).

In **Australia**, men who had not completed secondary school had an age-adjusted mortality rate of 102 per 10 000 person-years – 2.2 times higher than the rate for men with university degrees (46.3 per 10 000 person-years). For Australian women, the difference in age-adjusted mortality was 60% between those with the highest and lowest education levels (46).

Within many **European** countries, large relative inequalities in all-cause mortality between educational groups have widened over recent decades (47). For example, in **Sweden**, men with the lowest educational attainment have an age-adjusted mortality rate 2.2 times higher than those with the highest educational levels. In **Norway**, overall mortality rates have declined, but inequalities related to level of education persist, and among women these inequities widened between 2014 and 2020 (48).

Cross-sectional data from **36 countries** indicates that, on average, older people's cognitive function (as measured by delayed word recall) is better among those with post-secondary or secondary education, compared to those with only primary or no formal education (49).

Within countries, there are sex-based inequalities in adult mortality rates: adult mortality is higher for males than females in almost all settings, and this gap has only slightly decreased globally between 2000 and 2021. To reach the 2040 target, accelerated improvements for males are required, particularly in low- and lower-middle-income countries (44).

Similarly, disability-related inequalities in adult mortality persist within countries. A systematic review of 70 studies from 22 low- and middle-income countries shows that people with disabilities have double the mortality rates of those without disabilities (50). In high-income countries, the mortality gap for people with psychosocial disabilities is 20 years for men and 15 years for women, due to a combination of lifestyle risk factors, social determinants and poorer health care (51, 52).

“People with disabilities have double the mortality rates of those without disabilities”

1.2.1.3 Progress against Commission Target 3:

Reduce by 90%, between 2000 and 2040, the under-five mortality rate in all countries and all social groups within countries, and reduce by 95%, between 2000 and 2040, the maternal mortality rate in all countries and all social groups within countries.

Over the past few decades, there have been reductions in maternal and child mortality rates. Between 2000 and 2023, the global rate of under-five mortality halved, declining from 77 to 37 deaths per 1000 live births (53). Meanwhile, between 2000 and 2023, maternal mortality improved globally from 328 to 197 deaths per 100 000 live births, representing a decline of 40% (54). However, global successes in reducing maternal mortality stagnated between 2016 and 2023 (and increased in 2021, due to impacts from the COVID-19 pandemic) (55).

Progress to date has been driven by substantial declines in low-income and lower-middle-income countries, but these countries still account for 92% of maternal deaths, and under-five mortality in low-income countries remained 13 times higher than in high-income countries in 2023 (62 compared with 5 deaths per 1000 live births) (53, 54).

Sixty countries are not on track to reach the SDG target of 25 or fewer under-five deaths per 1000 live births by 2030. Of these countries, three-quarters are in sub-Saharan Africa, 85% are low- or lower-middle-income, and almost half are fragile and conflict-affected (53).

“Maternal mortality ratios are higher among populations in fragile settings”

Likewise, maternal mortality ratios are higher among populations in fragile settings that are susceptible to economic, social and political instability. In 2023, the 10 countries that ranked most highly on the Fragile States Index elaborated by The Fund for Peace (56) had an average maternal mortality ratio of almost 500 deaths per 100 000 live births, whereas the ratio among the 10 lowest-ranked countries ranged from 1 (Norway) to 12 (Canada and Luxembourg) (54).

Meeting Target 3 for under-five mortality will require the global rate to further reduce to 8 deaths per 1000 live births, or lower. This will be achieved if the rate in all countries is at least as low as the mortality rate from the year 2003 in high-income countries. For maternal mortality, the global maternal mortality rate will need to reduce to 16 deaths per 100 000 live births, or lower, by 2040. If, by 2040, all countries reach the maternal mortality rate that high-income countries had in 2000, then the Commission’s target of a 95% overall reduction would be surpassed.

Within-country inequalities in under-five mortality

Within countries, under-five mortality is higher among the poorest quintile than the richest. Compared to their urban counterparts, children living in rural areas are more likely to die prematurely, especially in low- and middle-income countries (57, 58, 59). Across low- and lower-middle-income countries, where 90% of under-five deaths occur, the gap in under-five mortality between the richest and poorest 20% of the population narrowed between 2000 and 2023 from 61 to 29 deaths per 1000 live births, but still remained substantial (53). To achieve Target 3 in terms of reducing the gap between the richest and poorest 20% of the population, the rich-poor gap needs to narrow by at least 23 deaths per 1000 live births between 2023 and 2040. As noted by the Commission, mortality is at best a proxy measure for the overall goals of child well-being, but consistent data on more comprehensive measures is lacking.

Within-country inequalities in maternal mortality

Within countries, women who are from disadvantaged or marginalized groups are disproportionately more likely to die from pregnancy-related causes than their more advantaged counterparts, with inequalities related to socioeconomic position including level of education, area deprivation, and race and ethnicity (see Box 4). There are also strong associations between higher levels of gender inequality, including child marriage, and higher maternal mortality rates (60, 61).

In many high-income countries, racial and ethnic inequities in maternal death rates persist. For example, in Australia, Aboriginal and Torres Strait Island mothers are up to three times more likely to die during childbirth than non-Indigenous women (62).

In all countries, contributing factors to inequalities in mortality include barriers to contraception; abortion restrictions; an undersupply of maternity providers (especially midwives), and a lack of access to comprehensive postpartum support (including maternity care coverage and mandated paid maternity leave). Women from disadvantaged or marginalized groups often have more limited access to health care, particularly reproductive health services, contributing to higher rates of maternal mortality (see Box 4).

Promising progress on the Commission's targets in relation to under-five and maternal mortality reflects the focus that many countries and the international community have placed on improving the health of mothers and children. High levels of under-five and maternal mortality in the lowest-income countries and population groups have been seen as morally unacceptable, resulting in sustained efforts involving many partners to reduce mortality. Visible targets with clear data showing the extent of progress have contributed to increased accountability and corresponding resolve to continue improvements.

BOX 4. Evidence of unequal maternal health outcomes within sample countries

There are significant inequalities in maternal mortality within many countries, on measures of race, socio-economic status, rurality and disability, as recent research shows:

In the **United States** in 2020, the maternal mortality rate for non-Hispanic Black women was nearly three times higher than that of non-Hispanic White women (55.3 vs. 19.1 deaths per 100 000 live births). That gap widened over the previous year (63). In **Brazil**, while maternal mortality increased for all during the pandemic, Black women have 1.44 times the mortality rate of non-Black women (64). In the **United Kingdom of Great Britain and Northern Ireland** between 2020 and 2022, the risk of maternal death for women from Black ethnic minority backgrounds was almost three times higher than for White women (65).

Evidence from a number of countries shows a correlation between maternal mortality and socioeconomic status. In the **United Kingdom**, between 2009 and 2022, women from the most

deprived 20% of areas were more than twice as likely to suffer maternal mortality than those from the least deprived 20% (65).

Evidence from 27 **African countries** shows that women living in rural areas delivered their babies in a health facility 25 percentage points less than those in urban areas, with a significant proportion of this inequity explained by household wealth and maternal education (66). In the **United States**, the maternal mortality ratio in 2017 was estimated to be between 45% and 65% higher in rural than urban areas (67).

A population-based cohort study evaluated more than 1.8 million births in Ontario, **Canada**. Compared with women without a disability, the adjusted relative risk of severe maternal morbidity or death was 14% higher among women with a sensory disability, 29% higher among women with a physical disability, 57% higher among women with an intellectual/developmental disability, and 74% higher among women with two or more disabilities (68).

1.3

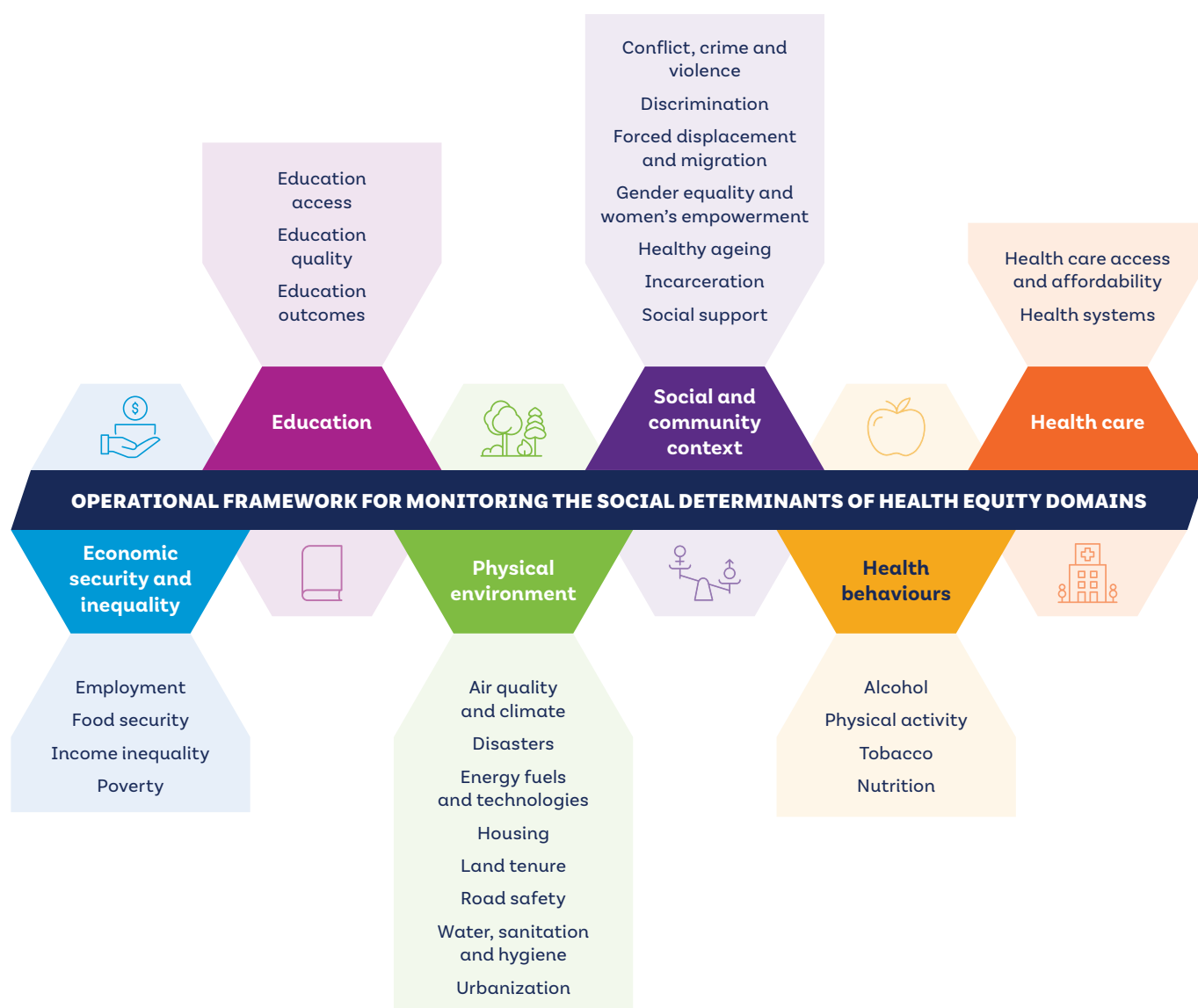
Key trends affecting the social determinants of health equity

1.3.1 Using the SDG indicators to measure progress in key social determinants of health equity

The Sustainable Development Goals (SDGs) were adopted by all United Nations Member States in 2015. The comprehensive targets for progress towards the SDGs encompass the areas considered by the Commission, while adding other dimensions of sustainable development with the overarching principle of “leaving no one behind”. The *WHO Operational Framework for Monitoring Social Determinants of Health Equity* (Fig. 4) provides an ongoing data resource for countries to assess progress by using a selection of the UN SDG indicators. Analysis of this data can

provide a picture of key progress, as well as where there is a need for greater focus on upstream action to tackle the highly interlinked SDG and social determinants agendas (69). Note that the Framework includes domains centered on health behaviours and access to health care which, while they contribute to inequities in health, are addressed extensively in other WHO technical reports, and so are not considered in detail in this report.

FIG. 4: Key SDG indicator areas relevant to monitoring the Social Determinants of Health Equity



1: Economic security and equality

Those living in extreme poverty (households where an individual earns or consumes less than the International Poverty Line of \$2.15 per day at 2017 prices, adjusted for inflation) experience some of the worst health outcomes globally. Since the 2008 Commission, the share of the global population living in extreme poverty has declined, from 15.7% in 2010 to 8.9% in 2019 (70, 71). However, the COVID-19 pandemic and subsequent shocks reversed these gains, increasing extreme poverty by nearly 90 million people (see Fig. 5). Recovery from this setback has delayed progress, so that on current trends, an estimated 6.9% of the global population will still be living in extreme poverty by 2030 (70).

Groups experiencing marginalization and discrimination, such as rural populations and women, are more likely to be living in extreme poverty. Children (those aged below 18) comprise a third of the global population and yet in 2023 accounted for more than half of those in extreme poverty, with poverty remaining one of the greatest risk factors for child morbidity and mortality. The COVID-19 pandemic slowed progress, leaving 30 million fewer children being lifted out of extreme poverty than projected (72). Countries facing chronic fragility (that is, where effective institutions may be inadequate or defunct) and conflict have seen extreme poverty rates stagnate above 40% in the past decade, while countries that have escaped this have cut their extreme poverty rates by more than half, with subsequent improvements in key health indicators such as child and maternal mortality (53, 54, 73).

Globally, economic inequalities between countries have declined since around 1980, as low- and middle-income economies have made gains, contributing to reductions in health inequities between countries. This trend went into reverse following the COVID-19 pandemic: 2022 saw an increase in inequality between countries for the first time in three decades (74). However, within many countries over the past two decades, economic inequality has increased markedly and is now responsible for a greater share of global economic inequality than between-country inequalities. Between 2000 and 2020, the gap between the average incomes of the top 10% and the bottom 50% of individuals within countries increased from approximately 12 times to 15 times, for 201 countries included in the World Income Inequality Database (75).

“Based on current trends, 6.9% of the global population will still be living in extreme poverty by 2030”

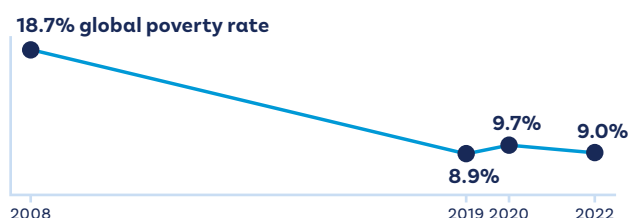
Social protection is crucial for improving health equity, yet coverage remains insufficient globally. Currently, more than 50% of the global population is covered by at least one social protection benefit, an increase from 42.8% in 2015. Most of the 3.8 billion people unprotected globally are in low-income countries (76).

FIG. 5: Progress against economic insecurity and inequality**PROGRESS IN ADDRESSING SOCIAL DETERMINANTS OF HEALTH AND ACHIEVEMENT OF SUSTAINABLE DEVELOPMENT GOALS****OPERATIONAL FRAMEWORK DOMAIN 1: ECONOMIC SECURITY AND INEQUALITY**

Since 2008, fewer people are living in poverty, but the COVID-19 pandemic and other crises caused a slight increase



Countries with **persistent unrest and conflict** have rates of extreme poverty of over **40%**



If current trends continue, by 2030:

around 575 million

people will live in **extreme poverty**

only one third

of countries will **halve** their **national poverty levels**

Between 1990–2019, **income inequalities between countries** decreased by **37%**, but COVID-19 reversed gains

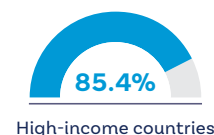
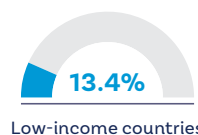


Income inequalities within countries are rising



Large gaps in social protection coverage between countries

Over **4 billion people** globally are **missing** at least one **social protection** benefit



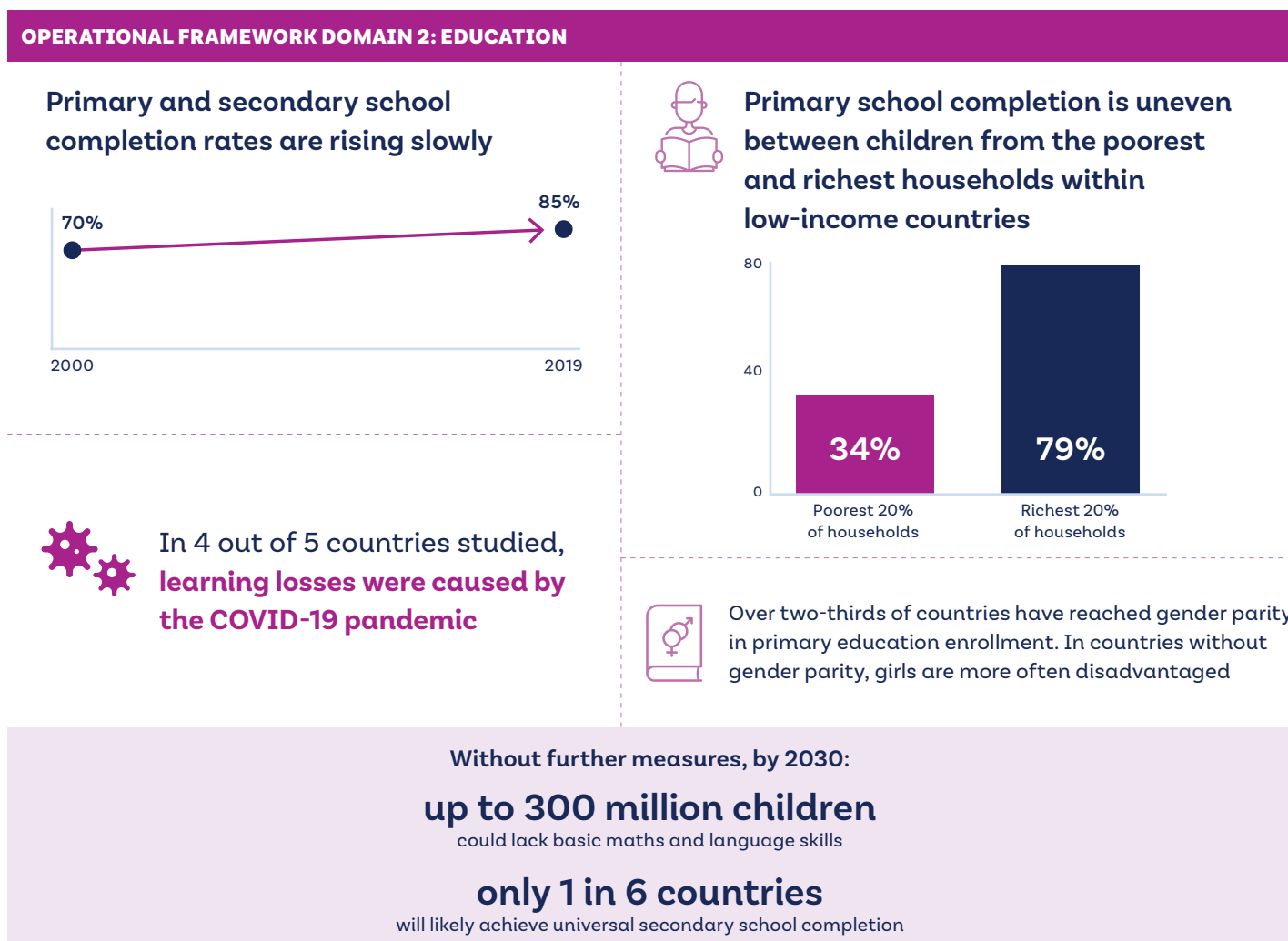
2: Education

Quality education for all, envisaged by SDG 4, was identified as fundamental by the Commission. Unfortunately, progress has been slow (see Fig. 6). A generation of children experienced significant learning losses due to the COVID-19 pandemic, not only affecting their ability to fulfil their educational potential, but also having potentially lifelong health impacts (70).

Educational inequalities persist both between and within countries in terms of school completion and quality of education. In high-income countries, 70–90% of children met minimum reading proficiency by the end of lower secondary school, compared to less than 60% in low- and middle-income countries for the years 2015 to 2019 (77). Countries in

sub-Saharan Africa have the lowest primary school completion rates, with less than two-thirds of children completing primary education (78). Groups experiencing marginalization – particularly girls, and children from poor households and rural areas – face significant barriers to education, leading to reduced skills and limited labour-market opportunities. Within low-income countries, these educational inequalities are stark: between 2014 and 2018, the primary school completion rate was 34% for children from the poorest 20% of households and 79% for children from the richest 20% (71). Similarly, within low-income countries, while educational inequalities between girls and boys have narrowed, they still persist. In 2018, the primary school completion rate was 62% for girls and 67% for boys. To meet national SDG 4 benchmarks, 79 low- and lower-middle-income countries still face an average annual financing gap of US\$ 97 billion each to expand access to high-quality education (78).

FIG. 6: Progress towards quality education for all



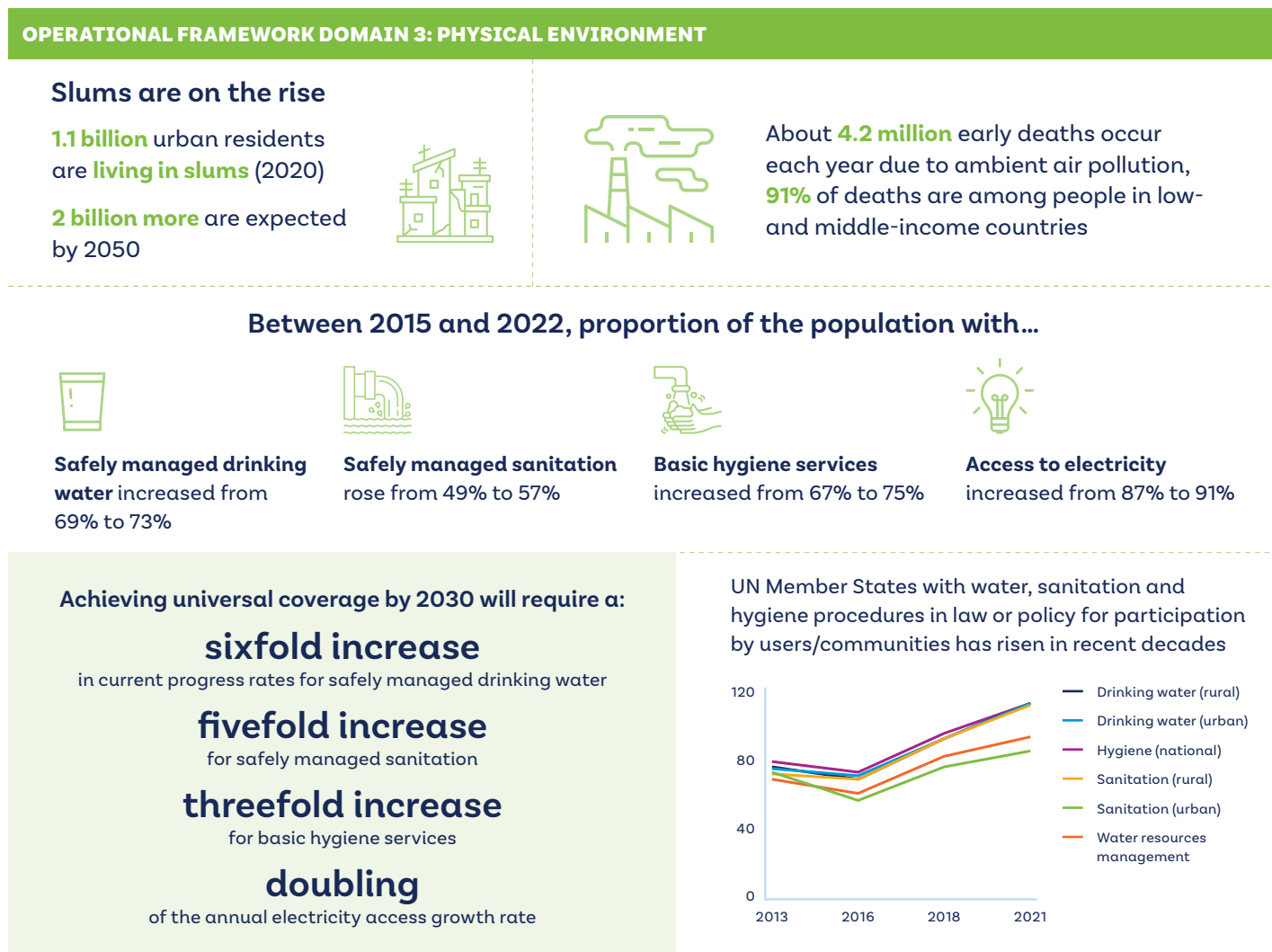
3: Physical environment

While there have been some improvements, there is still a long way to go to meet many of the SDGs related to physical environment. Access to safe and sufficient water, sanitation and hygiene, envisaged under SDG 6, has steadily increased, yet achieving universal coverage by 2030 will require faster rates of improvement. Stakeholder participation in water, sanitation and hygiene initiatives has been shown an effective approach to increasing access, and such participation has risen in many countries (70).

Access to electricity (SDG 7) has steadily increased in recent decades, rising from 87% to 91% of the global population between 2015 and 2022. However, the number of people lacking access to electricity grew in 2022 for the first time in a decade, with 685 million people lacking electricity, representing 10 million more than in 2021, due to disruptions to energy access from the pandemic and conflict. In 2022, international public financing for clean energy for developing countries was nearly half the level it was in 2016 (79).

The physical environment continues to be a key determinant of health inequities (see Fig.7). The number of people living in slums is on the rise. More than half the world's population currently resides in cities, and approximately a quarter of the global urban population lives in slums, with 2 billion more slum residents expected by 2050 (78). Air quality is also an important determinant of health, with the combined effects of ambient air pollution and household air pollution associated with almost 7 million premature deaths annually. Globally, population-weighted exposure to fine particulate matter declined by 9% from 2010 to 2019, but the level is significantly higher than the recommended air quality guidelines, with 99% of the global population breathing air that exceeds WHO air quality limits (80). In 2021, 31% of countries were yet to adopt legally mandated ambient air quality standards and 37% do not legally require national air quality monitoring, critical to understanding how air quality affects people (81).

“Poor air quality is associated with 7 million premature deaths annually”

FIG. 7: Progress towards a healthy physical environment

4: Social and community context

Social and community context encompasses many determinants, as per Fig.8. In this illustrative section, we focus on gender equality and forced displacement. Gender equality is critical to improving health equity. This includes direct impacts on maternal and child mortality rates, and there is emerging evidence that improving gender equality in countries is associated with increased life expectancy for both men and women (82, 83). Despite progress in some countries, the world is not on track to achieve full gender equality by 2030, with some key metrics likely to take centuries to achieve at the current rate of progress (70).

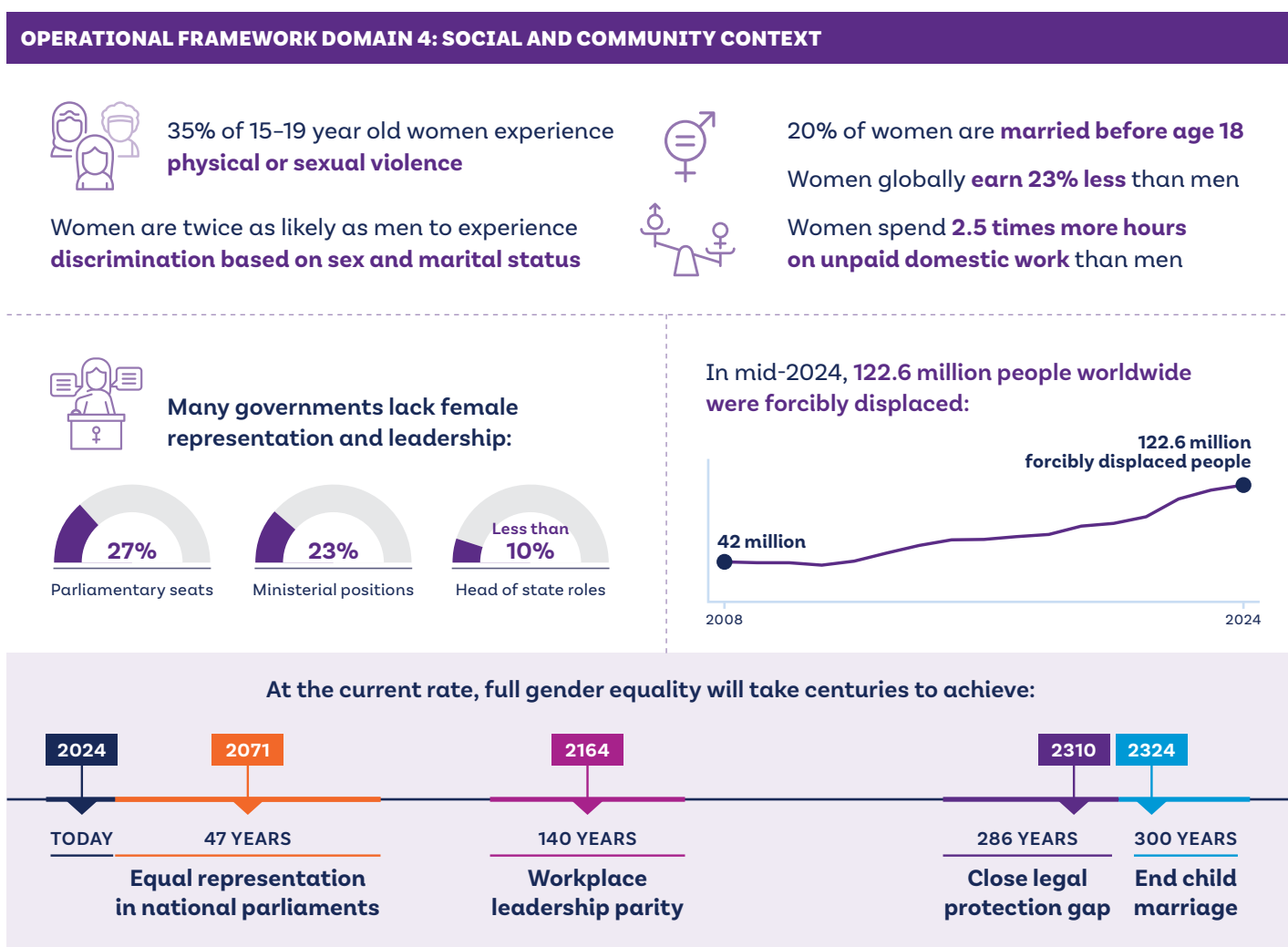
While international commitments have led to some improvements – such as declines in child marriage and female genital mutilation, and increased political representation for women – significant gender inequalities persist. Approximately 2.4 billion women of working age

“Women globally earn 23% less than men and spend 2.5 times more hours on unpaid domestic work”

lack equal economic opportunities (84). Women globally earn 23% less than men and spend 2.5 times more hours on unpaid domestic work (70). Women are also less likely to own or control land critical to their economic empowerment. Despite making up 43% of the global agricultural workforce, women comprise only 15% of all agricultural landholders (85). Gender-based violence remains widespread. In 2019, globally, 20% of women aged 20-24 were married before the age of 18, nearly half of married women lacked decision-making power over their sexual and reproductive health, and 35% of women aged 15-49 had experienced physical or sexual violence (78, 86, 87). Female genital mutilation affects one in three girls aged 15-19 in 30 countries in Africa and the Middle East, leading to severe health risks (88).

In recent years, the world has faced multiple crises that have resulted in the forced displacement and migration of millions of refugees, asylum-seekers, and internally displaced and stateless people. In mid-2024, nearly three times more people are displaced compared to 2008.

FIG. 8: Progress towards equitable social and community contexts



1.3.2 The COVID-19 pandemic and the social determinants of health equity

The COVID-19 pandemic underscored the relationships between the economy, livelihoods, health, and social vulnerability, and uncovered and exacerbated social, economic and health inequities (75, 89). The pandemic amplified a feedback loop: social inequalities pre-pandemic made the pandemic worse, and in turn, the pandemic made health inequities worse, including through some of the control measures.

An 84-country study showed higher national income inequality was associated with higher mortality 30 days into the epidemic (90). Another study of 141 countries found that more unequal countries (measured by the Gini coefficient) had higher mortality in the first wave of the pandemic (91), with similar findings in a study of all countries in the Organisation for Economic Co-operation and Development (OECD) (92). At subnational level – for example, in the United States (93), India (94) and Brazil (95) – the same has been found: the greater the degree of economic inequality, the greater the incidence of COVID-19 infection and mortality.

Groups that experienced increased rates of COVID-19 morbidity and mortality include poorer people, marginalized ethnic minorities and Indigenous Peoples, low-paid essential workers, migrants, populations affected by emergencies and conflicts, incarcerated populations, homeless people and people with disabilities (96, 97) (see Box 5). The prevalence of deaths in older age groups has been dismissed as a biological inevitability, masking the reality of avoidable deaths among older people and the neglect of care systems (98).

Inequities also hampered the success of response actions against the COVID-19-pandemic. To reduce the risk and scale of COVID-19 transmission, individuals, communities and governments implemented public health and social measures at an unprecedented scale from the onset of the pandemic. These measures refer to non-pharmaceutical interventions, such as hand washing, mask-wearing, physical distancing, school and business measures, and international travel and trade measures. While such measures were found to reduce transmission of COVID-19, especially when implemented early on and in bundles, their effectiveness also hinges on their uptake and people's adherence to them (110, 111, 112). The ability to adhere to public health and social measures was affected by existing social and economic inequalities, leaving many populations with less protection. For example, workers in the informal or precarious workforce were often unable to work from home, comply with quarantine and isolation recommendations, or maintain physical distancing due to the nature of their work or the risk of income loss. People living unhoused or

BOX 5. Racial, economic and disability-related inequalities during the COVID-19 pandemic in sample countries

Research into COVID-19 in diverse settings across regions has revealed cross-cutting patterns of inequalities that were prevalent during the pandemic and exacerbated its impacts. For example:

Racial inequalities

In the **United States** and the **United Kingdom**, there were racial/ethnic inequities in COVID-19 mortality in every age group (99, 100).

In **Brazil**, *Pardo* (mixed ethnic ancestry) and Black Brazilians admitted to hospital with COVID-19 had significantly higher risk of mortality than White Brazilians (hazard ratio 1.45 for *Pardo* Brazilians; 1.32 for Black Brazilians) (101).

Economic inequalities

In **Colombia**, data from March to October 2020 showed that those reliant on government-subsidized health insurance and those living in areas classified in the lower socioeconomic strata had a higher risk of dying from COVID-19 (102).

In **Japan**, a higher burden of COVID-19 cases and deaths was observed in prefectures with lower household incomes, a higher proportion of the population receiving public assistance, a higher unemployment rate, more household crowding, and higher smoking and obesity rates, compared with prefectures with the most social advantage (103). COVID-19 mortality rates were higher in areas with the greatest socioeconomic disadvantage for the first year of the pandemic (up to February 2021) (103).

In **Switzerland**, people living in more deprived neighbourhoods were less likely to be tested for COVID-19, but more likely to test positive, be admitted to hospital or die, compared with those in less deprived neighbourhoods (104).

In **South Africa**, a linear relationship was seen between subdistricts with high poverty and high COVID-19 death rates in Cape Town, with the highest levels of COVID-19 mortality in the poorest districts (105).

In **Hungary**, research shows that those living in more deprived municipalities, and districts with higher proportions of Roma people, had a lower risk of being identified as a confirmed COVID-19 case, but a higher risk of death from the virus (106).

In the Emilia-Romagna region of northern **Italy**, people living in the most disadvantaged census blocks in terms of index of deprivation, household crowding and foreign resident population showed an increased risk of death overall and from COVID-19 during the first outbreak peak of the pandemic (March and April 2020) (107).

Disability-related inequalities

In the **United Kingdom** and the **Republic of Korea**, more than half of the COVID-19 related deaths in 2020 occurred among people with disabilities (108, 109).

in overcrowded housing were often unable to access personal protective equipment like masks, clean water and soap, or to isolate when sick to prevent onward transmission. These conditions increased the chances of infection, and together with other socially determined risk factors, such as pre-existing co-morbidities and limited access to health care, created increased risk for severe illness and mortality (96).

In addition, the prolonged implementation of societally disruptive public health and social measures, such as travel and trade restrictions or school and business closures, led to a widening of health and socioeconomic inequities. This was particularly the case in the absence or unequal

provision of mitigation measures such as social protection, including paid sick leave, unemployment benefits and in-cash support for food and living costs (113, 114, 115).

There was evidence of the social gradient in access to non-COVID-19 health care during the pandemic in many countries, including Bangladesh, Kenya, Nigeria and Pakistan; and reduced diagnoses for common health problems in disadvantaged communities – for example, in the United Kingdom (116). Maternal mortality rates have increased since the pandemic in several settings (117, 118), and sexual and reproductive health outcomes have deteriorated (119, 120, 121). Health services in lower-income countries were more heavily affected than those in wealthier ones, and their time to recovery has been longer (122).

Spillover impacts of the pandemic to other social determinants have included increased poverty, food insecurity, increased violence, disruption to education, and mental health impacts, particularly among young people (96, 123). In the European Union (EU), an additional 40 out of every 100 young people reported mental health problems in 2022 compared with 2016 (124). In 2021, largely driven by COVID-19, the gender gap in food insecurity was one percentage point higher than in 2020, with 32% of women worldwide moderately or severely food insecure, compared with 27.6% of men (125). An additional malnutrition effect of the pandemic was growth in children being overweight or obese, in particular among socially vulnerable population groups, caused by increased intake of nutritionally poor, energy-dense foods, lower physical activity and inequality of opportunities for outdoor activities (126, 127, 128).

The availability of medical countermeasures, including vaccines and treatments to fight the COVID-19 pandemic, was delayed in lower-income nations, increasing health inequities between countries (129). Debt burdens were already a challenge, and many low- and middle-income countries emerged from the pandemic with larger amounts of sovereign debt, reducing their fiscal space to invest in recovery and necessary infrastructure to improve health (130).

1.3.3 The cost-of-living crisis

The legacy of COVID-19, combined with the impacts of recent conflicts, disasters and disease outbreaks, has driven high inflation, causing the livelihoods and health of billions of people to deteriorate. Inflation has broken multidecade records in high-income economies, while low- and middle-income countries – in particular, the poorest – have experienced even higher levels of inflation. The effect is to drive up the price of essential goods and services necessary for health, including housing, water, energy, food and transport. Such essentials make up a higher proportion of household expenditure for the poor (131, 132).

Increases in inflation have had damaging impacts on health equity (133). Estimates for 2023 show that since 2019, global hunger has risen by 152 million, to 733 million people, and 2.33 billion people were facing moderate or severe food insecurity, remaining well above pre-pandemic levels (134). The cost-of-living crisis has further widened socioeconomic inequalities in childhood obesity by rising the prices of health-promoting foods, which disproportionately affects disadvantaged families and communities already at risk of obesity (135).

In the first three months of the Ukraine war in 2022, 71 million people were pushed into poverty, with hotspots in the Balkans, the Caspian Sea region and sub-Saharan Africa, particularly the Sahel (136, 137). The increased costs of fuel and food are also impacting high-income countries, with many of the poorest struggling to heat their homes and feed themselves and their families. Commodity prices increased globally in 2023–2024, particularly for energy and food, with inflation affecting nations of all income levels.

Global economic growth and inflation have started to stabilize after bottoming out in 2023, but the cost of living remains high in many countries, with 38% of developing countries eligible for World Bank assistance, having a lower GDP than before the COVID-19 pandemic (138, 139). Sub-Saharan Africa is home to approximately 60% of those in extreme poverty. The World Bank estimates that economic growth in the region will rise from 2.6% in 2023 to 3% in 2024 and 4% in 2025–2026, with conflict and climate impacts dampening growth, and recommends that “policy should be geared toward sharing the growth benefits more equally across the population” (140).



1.4

Evidence and action on the social determinants of health equity

1.4.1 Growing evidence for, and action on, the social determinants of health equity

In the 15 years since the Commission reported, knowledge about the social determinants and how they shape health has grown (141). The increasing evidence base on the social determinants of health equity has served to validate the analysis of the Commission and reinforce its recommendations (142). Emerging evidence from practical implementation of approaches based on the social determinants of health is leading to refinements and improved practices, as well as adaptation in other settings (143, 144, 145).

In relation to mental health conditions, for example, there is an increasing appreciation of the social determinants as fundamental factors in determining population and individual mental health (146, 147). The relative neglect and exclusion of these factors from mental health discourse and practice can result in over-reliance on pharmacological treatments, to which there are inherent inequities in access. The use of broader psychological, social and economic interventions implemented collaboratively with communities and non-health sectors can help address the root causes of mental health conditions.

“In order to address the social determinants of depression and anxiety, States should put the fight against poverty and inequality at the heart of national strategies to improve mental health, and they should combat the rise of income and wealth inequalities, the informalization and destandardization of work, the contractualization of welfare, and housing segregation in urban areas”

– UN Special Rapporteur on extreme poverty and human rights, 2024

The measurement and analysis of the extent of health inequities has largely focused on high-income settings, and has not been accompanied by commensurate research and guidance on the most effective ways to take action (148, 149, 150, 151). There has been insufficient data to monitor whether health inequities change in response to policies, and practical tools to translate intentions into action have been lacking, undermining the appetite and capacity to take action (152, 153).

Many stakeholders are unsure of where to start to address the social determinants of health equity, or which approaches are most relevant and practical in their context. While the comprehensiveness of the “social determinants of health equity” approach is one of its strengths, the current evidence base does not easily support prioritization, and it is not always straightforward to make the short-term financial case for action (154, 155). How different actions are strategically combined remains an area for innovation and learning.

1.4.2 National and local strategies for action on the social determinants of health equity

National health equity action plans were among the implementation approaches that brought together governance and management of implementation in the Commission’s country action workstream (6). Since then, there have been a number of reviews by national governments to inform their strategic approaches to health and the social determinants of health equity, including in Australia, Brazil, Denmark, Ethiopia, Finland, Iceland, the Islamic Republic of Iran, Lao People’s Democratic Republic, New Zealand, Norway, Pakistan, the Philippines and the United Kingdom (England and Scotland) (12).

Health in All Policies (HiAP) approaches to address the social determinants of health have been used both at national and local levels in different development contexts. The SDGs have lent support to seeking synergies from HiAP approaches with co-benefits from environmental policies.

More recently, well-being economy approaches that incorporate the social determinants of health equity are being considered in many countries and regions (156). The term “well-being society” was used in the 2021 Geneva Charter for Well-being with the following meaning: “creating sustainable well-being societies, committed to achieving equitable health now and for future generations without breaching ecological limits” (157).

Implementation of policies and programmes on the social determinants of health continues to increase, with positive impacts in many local areas and a proliferation of approaches based on the social determinants of health equity led by local governments and mayors (158, 159). The SDH

Special Initiative is an action-learning initiative that has led to joined-up local-to-national action and the implementation of approaches at local demonstration and pioneer sites in eight countries and territories in three WHO Regions (15, 160).

While more implementation research is needed, examples of action on the social determinants of health equity implemented in towns, municipalities, cities, regions, islands and countries have helped build evidence for effective ways to support better and more equitable health, and provide a type of social learning that encourages uptake and social innovation. These examples have also strengthened understanding of the practical application of comprehensive strategies required to address the range of different social determinants of health (144). Some of this learning is captured in this report, but much is ongoing, as momentum builds and an increasing number of places adopt the social determinants of health equity agenda. These efforts must be evaluated for impact and lessons learnt to address barriers to implementation.

1.4.3 Intersectoral and Primary Health Care approaches

Efforts to reduce health inequities have often focused more narrowly on the fairer delivery of health services through ministries of health, expecting these services to lead to greater health equity. However, even with strong leadership from the health sector, the challenges of intersectoral work and the political accountabilities towards service provision can dissuade upstream action on key social determinants.

Global calls for the reorientation of health systems towards a primary health care (PHC) approach give renewed focus to intersectoral approaches like HiAP. They highlight the role of health sectors in considering barriers to health equity arising from other sectors. They also draw attention to intersectoral policy and community engagement as two of the three components of the PHC approach (161, 162, 163). But despite the broad vision of the 1978 Declaration of Alma-Ata, as well as the more recent articulation through the 2018 Declaration of Astana, many PHC initiatives have been limited in their ability to drive upstream action on the social determinants of health equity (164, 165).

“COVID-19, HIV and Mpox are reminders that inequities in health care cannot be reduced through action in the health sector alone”

Unsurprisingly, the mismatch in focus and lack of engagement of whole-of-government or whole-of society approaches has often failed to propel action on social determinants. COVID-19, HIV and Mpox are more recent reminders on a global scale that inequities in health care cannot be reduced through action in the health sector alone (166, 167). Social contract perspectives and transformative policies to address inequalities in the context of contemporary crises are critical, given that inequalities and crises reinforce and compound each other (168).

Recent documented experiences, such as the intersectoral collaborations that promoted civic participation and people-driven health care in the United Republic of Tanzania, as well as in Thailand, explicitly moved beyond the medical model to include innovative governance models and intersectoral approaches to the social determinants of health (169, 170, 171). Experiences in Lao People's Democratic Republic and Chile highlight the possibilities of using PHC-oriented health reforms to develop a comprehensive family and community health care model that incorporates a strong component of intersectoral work on social determinants (see Boxes 15 and 29) (163, 172).

1.4.4 Reflection on progress

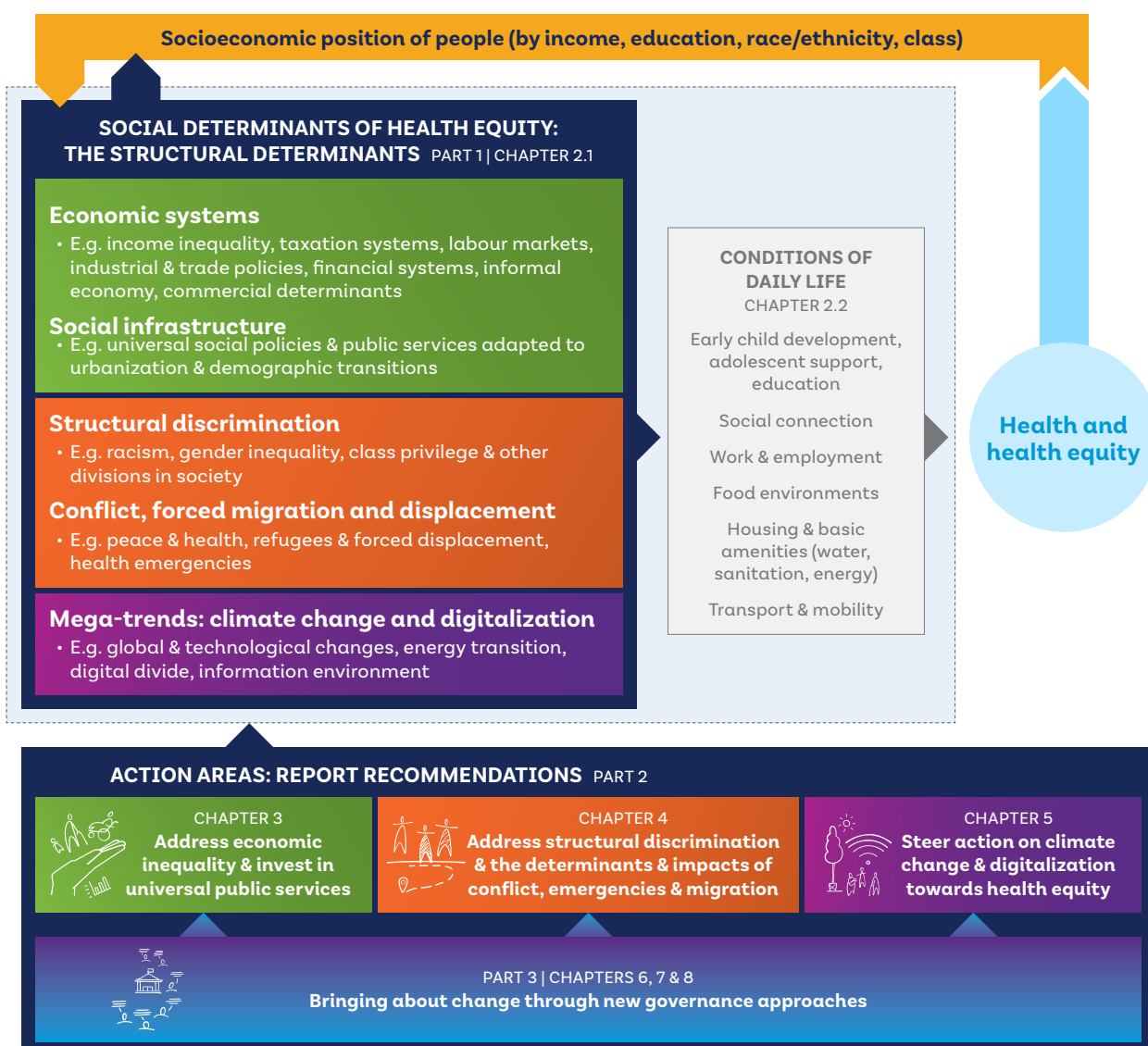
Overall, while the Commission's targets for greater health equity have not been met, there has been some progress reducing inequities in life expectancy, and in maternal, child and adult mortality rates between countries. However, wide health inequities persist – particularly within countries – related to social position and among those experiencing discrimination, and progress has stalled in many countries more recently. The fundamental diagnosis of the Commission still applies: “poor social policies and programmes, unfair economic arrangements, and bad politics” are a “toxic combination” that drives health inequities (6). Today's health inequities are a result of a political failure to address these structural determinants. Recent global crises – including conflict, climate change, inflation and the COVID-19 pandemic – have exposed even more starkly the consequences of unequal societies and the poor conditions experienced by a large proportion of the world's population, amplifying existing health inequities and creating new ones.

CHAPTER 2:

The social determinants of health equity

The social determinants of health equity are the structural political, economic, social and cultural systems that stratify societies and differentially shape living conditions across the life course. Section 2.1 describes the pathways and mechanisms through which these determinants impact health equity. Each of the social determinants of health equity in Fig. 9 is introduced in this section and is in turn the subject of corresponding recommendations in Chapters 3–5 of this report. Section 2.2 provides illustrative examples of how structures shape the conditions of daily life and impact health and health equity, focusing on prominent areas with strong evidence and for which action responses are possible.

FIG. 9: This report’s approach to outlining the Social Determinants of Health Equity



2.1

The social determinants of health equity

2.1.1 Economic systems

Economic systems are inextricably linked with health and health equity. At all scales and levels, from global to local, economic systems determine the flow of capital, the allocation of economic resources, the production and distribution of products, goods and services, and the conceptualization and valuing of economic development and growth. By determining the resources available to countries, economic systems affect inequities in income, wealth and power within and between nations, and abilities to invest in the social determinants of health equity. As the 2024 Nobel laureates in economics argue, the prosperity of nations is driven by strong societal institutions, with those countries which have invested in inclusive institutions able to chart a pathway to prosperity even from a low resource base, while countries that are trapped in extractive and exploitative patterns remain poor (173).

“The way markets operate, the role of the public sector, and economic inequalities are structural drivers of inequities in the conditions of daily life, mostly produced or modified by political choices”

Report of the Commission of the Pan American Health Organization on Equity and Health Inequalities in the Americas (31)

Macroeconomic volatility, economic shocks like the COVID-19 pandemic, and the cost-of-living crisis have severe negative impacts, increasing poverty and health inequities for all age groups and countries at all income levels (174, 175, 176). Correspondingly, programmatic interventions designed to protect populations against economic shocks and build broad-based resilience are critical to the conservation and extension of equity in health, as demonstrated by numerous countries on all continents (see boxes below for specific country examples) (177, 178).

Economic and political priorities shape resources available between and within countries, with direct impacts on health and health equity (179). They determine what is funded and to whom that funding is distributed. The availability of resources and how they are distributed can significantly constrain governments’ ability to finance actions that will directly

improve health equity, such as investments in universal public services and social infrastructure. Constraints on governments arising from economic systems can both harm and promote health and equity – for example, the challenges investing in public services due to a lack of fiscal space, or the potential for regulation and revenue-raising from health-harming commercial products or services (180, 181). The Council on the Economics of Health for All has advocated for new economic thinking that goes beyond fixing market failures, and supports policy-makers to actively create and shape an economy that delivers on goals critical to health equity. This thinking includes good education systems, decent working conditions and safe environments. It also requires organizing and financing health systems in a way that ensures access to effective services for all, at all ages. Essential to this mission is ensuring that the economy is understood as being in the service of people and the planet, rather than economic growth being all powerful and the end goal (182).

Some insight into the predominance of economic factors in people's health comes from analysis of the European Quality of Life Survey, which suggests that for respondents in Europe, income security is the largest single contributing factor to self-reported health, accounting for 35% of the difference between the most and least affluent quintiles. This is followed by housing deprivation and educational outcomes, both also important social determinants of health equity (Table 2) (13).

TABLE 2. European Health Equity Status Report decomposition analysis – leading factors driving inequalities in self-reported health between the most and least deprived 20%

Source: Analysis of the European Quality of Life Survey in: World Health Organization Regional Office for Europe. *Healthy, prosperous lives for all: the European Health Equity Status Report* (13).

Factor	% relative contribution
Making ends meet (income security)	35.0
Housing deprivation	15.7
Educational outcomes	13.3
Poor-quality health services	7.9
Fuel deprivation	5.5
Lack of trust	5.3
Not employed	5.0
Lack of green space	2.6
Unsafe neighbourhood	2.3
Working excessive hours	2.0

Poverty is detrimental to health

Poverty influences health and longevity through multiple mechanisms, including (but not limited to) barriers to accessing quality health care, safe and stable housing, and healthy and affordable foods, along with higher levels of exposure to environmental pollutants, and greater social exclusion. Living in extreme poverty on less than US\$ 2.15 a day denies people the ability to access vital services, including safe water, food, shelter and health care, which leads to significantly worse health outcomes (183). Under-five mortality is closely correlated with the proportion of the population living in extreme poverty (53, 184). Globally, four out of five people experiencing extreme poverty live in rural areas (185).

There is a strong correlation between poverty, life expectancy and GDP per capita below a threshold of approximately US\$ 15 000, after which life expectancy is more influenced by the policies and programmes available and their distributional impacts (139, 186, 187). More comprehensive measures of well-being and living conditions are needed alongside GDP, that integrate considerations of economic inequality and health outcomes.

Economic inequality and health

Absolute deprivation of resources for health, as characterized by poverty, worsens aggregate levels of population health and health inequities, but so do relative poverty and economic inequality – that is, the unequal distribution of income across the population. Economic inequality in a society is associated with worse overall population health outcomes, including life expectancy, infant mortality, obesity and mental health conditions, and with higher health inequities. In countries where data is available, there is a remarkably close association between an area's deprivation levels and health outcomes in social gradients. One mechanism through which inequality works relates to decreased social cohesion and trust in institutions, resulting in fewer investments in resources needed for health across the population, regardless of education, employment, income or wealth (188, 189).

The potential health gains from reducing economic inequality are significant. For example, WHO analysis predicts that eliminating wealth-related inequality in under-five mortality (that is, decreasing national mortality to the level of the richest 20% in each country) could help save the lives of 1.8 million children in low- and middle-income countries (190). Health gains from reducing economic inequality are also evident in high-income countries (191). An analysis of the United States across counties

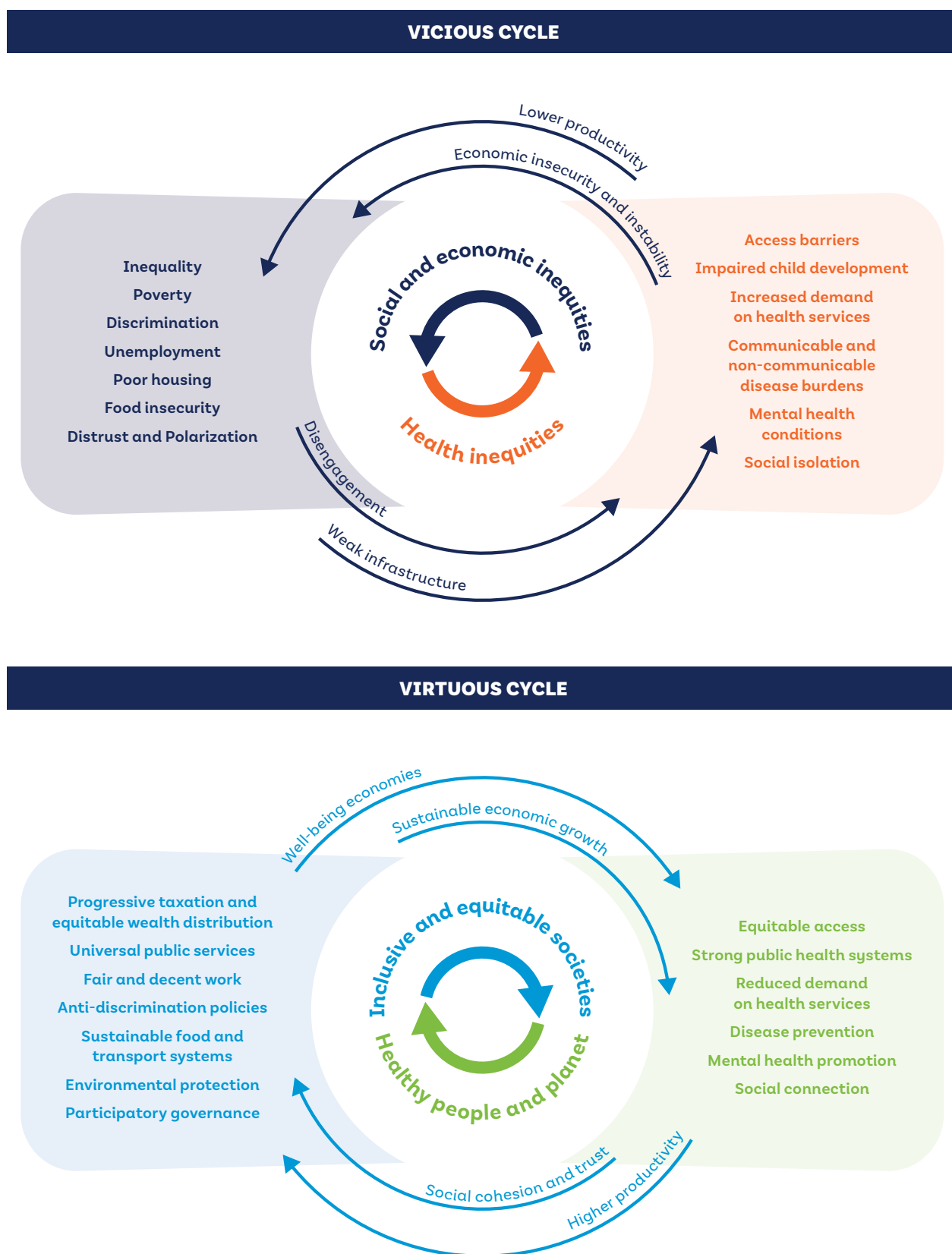
“The potential health gains from reducing economic inequality are significant”

showed that a one-percentage point increase in the Gini index (a measure of income inequality) was associated with a 4% increase in child maltreatment (192). Higher levels of income inequality are also significantly associated with increases in new HIV infections in the following year. Estimates show a 25% reduction in the Gini coefficient (a reduction in income inequality) would reduce the HIV incidence by 0.14 infections per 1000 people globally, and by 0.29 infections in the sample, excluding Africa. For African countries, this same reduction in income inequality would be associated with a reduction of 2.11 infections per 1000 people in the following year (193). A similar pattern was seen for AIDS-related deaths – increases in income inequality resulted in a significant association with increases in such deaths the following year. The effect is more pronounced for non-African countries: a 25% reduction in income inequality would be associated with a reduction in AIDS-related deaths of 17.39 per 100 000 deaths in the sample (193).

Poor health and inequality cost economies and society

As well as poverty and economic inequality leading to poor health, these health inequities and low levels of population health overall damage economies and societies more broadly. An unhealthy workforce undermines economic productivity and leads to increased and avoidable social protection and health care costs. Globally, it has been estimated that improved population health would add US\$ 12 trillion or 8% to global GDP by 2040, and 0.3% to global employment growth (194). Eighty per cent of these gains would come in low- and middle-income countries. For example, in Africa alone, healthy life years lost in 2015 cost the region US\$ 2.5 trillion (195).

While the relationship is complex, improving population health and reducing health inequities would tend to benefit economies, including in high-income countries. Fig. 10 outlines how the vicious cycle of inequity can be transformed to a virtuous cycle through the actions recommended in this report. A study examining the relationships between health and economic growth found a robust causal effect between health improvement and per capita growth rates in a sample of 26 high-income countries over the period 1960–2000 (196). One estimate suggested a 10% reduction in cardiovascular mortality was associated with a one percentage point increase in growth of per capita income (197). In 2020, during the COVID-19 pandemic, Latin America and the Caribbean suffered the largest economic contraction in the last 120 years, with a drop of 7% in GDP (198). In relation to inequity specifically, it was estimated that in the United Kingdom, health inequities cost an estimated £70.5 billion per year in reduced productivity, tax losses, and health and welfare costs, while improved health equity could save Australia 6.5 billion Australian dollars annually, as well as generate 8 billion dollars from the 170 000 people enabled to enter the workforce (199).

FIG. 10: Social determinants and health equity: from a vicious to a virtuous cycle

The commercial determinants of health

The commercial determinants of health are the commercial dimension of the social determinants of health. These are the activities undertaken by commercial actors that affect people's health, directly and indirectly, positively and negatively. Commercial actors not only shape health and health equity through the development, sale and marketing of their products, goods or services, but also in their impacts on the health of their workforce, on supply chains, and in shaping the norms and the regulatory and legal ecosystems that influence health (200, 201, 202). Commercial actors that produce goods and services that improve health can reduce health inequities when they support good employment conditions and have broader positive impacts on communities (203). However, health-harming sectors, such as the tobacco and alcohol industries, pose risks to health under all circumstances.

Just four industries – health-harming food and drinks, fossil fuels, alcohol and tobacco – account for at least a third of global preventable deaths, collectively in 2021 causing 19 million deaths annually (200, 204). Experience has shown that these industries can and will prevent and undermine public-sector action to limit health-harming products, services and practices, including by seeking to shape public discourse, and to bias or undermine research. They have also established opaque front groups to pursue their interests. The health-harming segment of the food industry has been particularly effective in influencing national governments to reduce or not implement regulations, and in marketing products which misinform about their effects on health, the environment and other social determinants (205, 206).



2.1.2 Social infrastructure adaptive to major transitions

Social infrastructure refers to the policies, services, resources and related spaces that people have access to, which enable them to fully participate in social, civic and economic life without barriers (25, 26). Investing in social infrastructure means building strong public institutions and strengthening social cohesion (207). Historically, governments that have invested in social infrastructure have ensured higher standards of living and health (208). Evidence suggests that the larger the social budget, the greater the benefits in terms of poverty and inequality reduction (209). Countries with broader, more generous welfare states have higher levels of health and lower levels of health inequality (6, 210).

Social infrastructure must be responsive to important transitions that are shaping our societies, our environment and our economies, including urbanization and demographic shifts. The exponential growth of cities, and demographic changes in age distribution due to declining birth and death rates, have major implications for health equity.

Urbanization

Urbanization has an impact on a number of important social determinants of health equity, including urban built environments, housing conditions, access to safe water and sanitation, food security, transport systems and air quality. Over 55% of the world's population lives in urban areas – a proportion expected to increase to 68% by 2050, driven by many of the structural determinants mentioned in this chapter, including climate change, conflict and economic opportunity. This growth is particularly rapid in Africa and Asia, regions which account for over 80% of the 1 billion people living in slums or informal settlements (211). Informal settlements are an evident manifestation of the impacts of a lack of investment in social infrastructure. When there are insufficient public institutions collectively managing, planning and steering investment in affordable housing, vast proportions of disadvantaged populations will be subject to poor-quality housing, undermining their health. Spatial inequality analysis is an emerging and increasingly important tool to understand how the determinants of health equity cluster, compound and interact.

Health inequities are frequently starkest in urban areas, sometimes varying from street to street. The social and demographic characteristics of urban areas impact health and levels of health inequities, including via

“Spatial inequality analysis is an increasingly important tool to understand how the determinants of health equity cluster, compound and interact”

population age and gender balance; the extent to which there are large proportions of migrants, unemployed residents, slum-dwellers or other socially vulnerable groups; and cultural factors. Groups made socially vulnerable are more likely to live in neighbourhoods near manmade or natural hazards, and with poor water supplies, sanitation, energy and transport services, or housing conditions (212). Urban services often fail to keep up with rapid and unplanned urbanization, and the needs of underprivileged urban areas are frequently not met. This is particularly the case for people already subject to exclusion due to their gender, race, caste or religion.

Exposure to levels of air and noise pollution are higher in cities. Sedentary lifestyles resulting from poor and inequitable transport systems, and the lack of space for safe walking, cycling and active living also make cities epicentres of the noncommunicable diseases epidemic – which is reinforced by increasingly poor diets with high amounts of processed food. Rapid and unplanned urbanization combined with climate and environmental change, increased global travel and trade, and other societal challenges has stimulated the emergence or re-emergence of vector-borne diseases (213, 214, 215, 216).

Demographic shifts

Demographic transitions carry with them the prospect of equity dividends at each stage of the life course, but also the risk of entrenching and worsening inequities if social infrastructure, policies and programmes do not reflect changing age structures (217, 218). In 2020, the number of people globally aged 65 years or over outnumbered that of children under five for the first time. By 2050, people aged 60 years or over will also outnumber adolescents and young people aged 15–24 years. Projections indicate that nearly 80% of the world's older population will live in low- and middle-income countries in 2050 (219).

A social determinants approach to health recognizes that older people are not a homogeneous group, but instead show the cumulative health impact of social and economic disadvantage or privilege that people have experienced from birth, and the opportunities and vulnerabilities they experience as they age. Evidence from longitudinal cohort studies documents the positive impacts of higher levels of education and wealth on health trajectories, leading to better health at older ages (220, 221). A comparative analysis of 42 countries with cross-sectional data on almost 152 000 older people documented a strong social gradient: older people with higher education levels have a higher cognitive capacity, at every age (49). WHO estimates that at least 142 million older people worldwide are unable to meet their basic needs, noting that women are more likely to live alone and in poverty than men, and after 80 years of age, women

are less able to meet their basic needs than men, with the gap widening with age (49). A 23-country longitudinal study offers an explanation for these differences, finding that living in a country where gender inequality is more pronounced increases women's risk of difficulty in five basic activities – bathing, dressing, feeding, toileting and getting in and out of bed – while men's risk does not increase (222). Additionally, 20–34% of older people experience loneliness or social isolation globally, with associated impacts on health and well-being (223, 224).

Labour policies and broader welfare and economic policies also shape opportunities for older people. A 12-country longitudinal study investigated older people's labour market experiences five years before and after country- and gender-specific retirement ages, noting that reaching “full pension age” did not signify a one-time labour market exit, as many older people work part-time, partially retire, or constantly move in and out of the labour market (225). Importantly, older adults with limited financial resources work longer, regardless of their health status. In particular, women have greater income insecurity in older age and have less access to contributory pension benefits (226).

At the other end of the life-course, evidence is beginning to accumulate that there are new vulnerabilities emerging among adolescents and young adults. Alarming increases in mental ill-health and a lack of well-being have been observed over the past decade in adolescents and young adults (aged 12 to 25). The evidence is remarkably consistent, albeit from a small group of Anglophone countries, and shows by far the highest rises in mental distress among young women (227). These studies show aggregate well-being steadily rising with age. Potential explanations for rapidly decreasing well-being among this age cohort include the impact of social media exposure and rising income and wealth inequality, which has increased economic precarity. Further investigation is needed into whether these changes are being seen in countries at all income levels, their potential causes and, crucially, effective remedies.

2.1.3 Structural discrimination

Structural discrimination refers to the rules, norms, routines, and patterns of attitudes and behaviours in institutions and other societal structures that represent obstacles to groups or individuals in achieving the same rights and opportunities available to most of the population (27). Structural discrimination occurs systematically across society and is embedded in prevailing cultures, legal mechanisms and economic structures. Often reflecting historical power relations and attitudes, it creates, perpetuates and reinforces actively harmful and unfair practices and outcomes that become normalized within society. Discrimination

is prohibited in international human rights law, and in most national constitutions and legal frameworks. However, to varying extents, structural discrimination exists in all societies and may be based on race, ethnicity, religion, migration status, Indigeneity, gender identity, sexual orientation, economic status and disability, among other characteristics.

The direct impacts of discrimination on health include poorer mental health (147, 218, 228), loss of self-esteem, increased stress (229), symptoms of trauma (230), reduced feelings of power, loss of liberty, and impacts from violence (231). Discrimination creates barriers to accessing health care, and results in people receiving worse-quality health care. Structural discrimination also has multiple, widespread negative impacts on key social determinants of health, including increased poverty and reduced access to education, employment, health care and other services, and safe, good-quality living conditions and environments.

“Experiencing discrimination undermines trust and confidence... which in turn affects individual and collective health”

Forms of discrimination often intersect, with many people experiencing discrimination across multiple compounding domains (232). Experiencing discrimination undermines trust and confidence in societies’ social, economic, political and cultural systems, which in turn affects individual and collective health. During the COVID-19 pandemic, it was notable that in many countries, vaccine hesitancy was higher among communities which were marginalized, excluded and discriminated against (233, 234, 235), and was related to a lack of trust, which had been undermined by experiences of discrimination and associated trauma.

Sexism and gender inequality

Women are disproportionately affected by poverty, discrimination, exploitation and sexual violence. The resulting health inequities are further compounded by heteronormative and patriarchal gender norms, and discrimination and disadvantages based on sexual orientation, gender identity, gender expression and sex characteristics (236). Studies of women and LGBTQI+ populations show that exposure to violence is associated with significant health consequences, including HIV acquisition and suicide (237, 238, 239).

While women tend to live longer than men on average, they often suffer from poorer health and quality of life (240). This is due to accumulated gender-based discrimination and disadvantages over their lifespan, including poorer economic conditions and gender pay gaps, multiple comorbidities, and inadequate social and health protection. Structural

barriers to gender equality exist and persist, to varying extent, in every country. The World Bank reports that only 14 countries globally have legal parity between men and women, with progress towards closing this gap being too slow (241).

In the world of work, these barriers manifest as gender differences in labour-force participation and pay, occupational segmentation, unequal working conditions, and women's disproportionate burden of unpaid domestic and care work (242). The care economy is a key arena in which gender inequities and their impact on women's health play out, with women carrying out three times the amount of unpaid care work than men (243). The total value of unpaid care and domestic work is estimated to be between 10% and 39% of GDP across countries (242), increasing the burden on carer's mental health and physical health and well-being. The ways in which discrimination intersects and accumulates can be seen in relation to migrant and ethnic minority female workers, who are often over-represented in jobs at the low end of the care workforce, and frequently deprived of workplace protection and rights (244, 245).

Racism

Racism is rooted in colonialism, slavery, ethnicity, caste and other systems that work to maintain historical hierarchical structures, and the unequal distribution of power and resources that sustains them (246). Racial discrimination is institutionalized in economic and legal systems, education, employment, housing, transport and food systems, all of which have major impacts on health and health equity (228). Racism intersects with other systems that reproduce inequity, including in the policing, justice and penal systems, where the disproportionate incarceration of minority groups has a direct impact on health outcomes (247, 248). The committee that monitors the implementation of the Convention on the Elimination of Racial Discrimination recently issued an important advance in the systematic consideration of racism and its impacts on health, through its General Recommendation related to the realization of the right to health (249).

Specific pathways from structural racism to negative health outcomes have been found across many health conditions – for example, in evidence from the United States in relation to mental health, cardiovascular outcomes, physical function, cognition, diabetes, risks of preterm birth, stillbirth, obesity and late cancer diagnosis (250, 251, 252).

Even where current laws are non-discriminatory, historical structural discrimination has left some population groups with fewer resources, which continues to impact on intergenerational inequity. For example, in the United States, people of colour were discriminated against by mortgage lenders until the early 1970s – a process called “redlining”,

preventing them from building equity to hand to the next generation (253). In the United Kingdom, substantial wealth transfer from parents to young adults was three times more likely for White people than young adults who are Pakistani or Bangladeshi (254).

There is a particular need to focus on the health legacies of colonialism on Indigenous Peoples. Although Indigenous Peoples make up just 6% of the global population, they account for about 19% of the extreme poor, and have a life expectancy up to 19 years lower than that of non-Indigenous Peoples (255, 256). Globally, Indigenous Peoples are subjected to discrimination, violence, impoverishment, cultural damage and, increasingly, the harmful impacts of climate change. In turn, they experience higher rates of physical illness and disease, food insecurity, poor living standards and mental health conditions, that have been associated with the negative impacts of colonialism (147, 255, 257, 258, 259). A key conclusion of the Pan American Health Organization's Commission on Equity and Health Inequalities in the Americas was that the legacy of colonialism continued to harm Indigenous and minority ethnic communities, and is evident in inequitable economic, social and legal systems (31).

Disability discrimination

People with disabilities face additional challenges in their social determinants of health and in accessing services, experiencing shorter and unhealthier lives (50, 260, 261). It has been estimated that in 2021, 16% of the world's population (some 1.3 billion people), had a disability, with nearly 80% living in low- and middle-income countries (262). As well as barriers to accessing appropriate housing, employment and essential services – including education and health care – many people with disability are subject to discrimination and exclusion.

Ageism

Ageism refers to the stereotypes, prejudice and discrimination directed towards others or oneself based on age. Ageism affects people of all ages, and deepens inequalities related to the social determinants of health, limiting access to essential services and opportunities. It is ubiquitous, with approximately one in two people globally estimated to hold some ageist beliefs. Discrimination based on age affects areas such as employment, health care, housing, social protection and community participation – crucial factors in determining an individual's health and well-being. Ageism intersects with other forms of discrimination, including those based on gender, race, disability and socioeconomic status, amplifying vulnerability and health inequities (218). Actions that dismantle discrimination and reduce inequality in socioeconomic conditions will raise the trajectory of healthy ageing for everyone (263).

2.1.4 Conflict, climate change, migration and displacement

Dynamic forces that are driving increased vulnerability in living conditions and health inequity include conflict, climate change, migration and displacement.

Conflict is a critical determinant of health. It disrupts essential services, destroys infrastructure and exacerbates vulnerabilities (264). The year 2024 saw more conflicts globally than any since 1946, many with huge death tolls and widespread consequences on health (265). In Gaza and across the occupied Palestinian territory, including east Jerusalem, and in Sudan, there are ongoing conflicts which have devastated health care systems. The health status of populations which were already very vulnerable has become critical.

“2024 saw more conflicts globally than any year since 1946”

In Gaza, where more than 40 000 people are estimated to have died in the conflict since late 2023, airstrikes and a lack of access to basic resources, including food, water and medical supplies, have left hospitals overwhelmed, compromising care for pregnant women, children and those with chronic conditions. The military destruction of a large proportion of buildings and infrastructure – including attacks on health care facilities and health workers – and the displacement of over 90% of the population have caused widespread suffering that will have long-lasting negative health consequences.

Similarly, in Sudan, conflict has led to attacks on health care facilities, leaving millions without access to medical care. With millions of displaced people living in dire conditions, the risks of malnutrition, disease outbreaks and gender-based violence are heightened. The collapse of health systems in both regions exemplifies the direct and indirect health consequences of conflict. When humanitarian organizations are prevented from accessing affected populations to provide essential services, the crisis is further deepened. The effects on these vulnerable populations will be long-lasting and not easily repaired.

The health impacts of conflict include mortality and injury as a direct result of fighting, but conflicts are also associated with higher levels of malnutrition, disability, injury, communicable diseases, poor mental health and post-traumatic stress disorder. Conflict also destroys physical and social infrastructure, with resulting population displacement, destruction of homes, food insecurity, disruption of livelihoods, and increasing poverty

and destitution (266, 267). The legacy of conflict and violence leaves deep scars in individuals and societies across the life course, through exposure to trauma. An estimated 70% of the global population has experienced at least one major traumatic event in their lifetime (268).

Approximately 80% of WHO's humanitarian caseload and 70% of disease outbreaks that the Organization responds to take place in fragile and conflict-affected settings, as illustrated by the 2024 outbreak of polio in Gaza. Countries that have avoided conflict have halved their poverty rates over the past decade, while in fragile and conflict affected states, poverty rates have been stuck at 40% or more (73). Morbidity and mortality rates for children under 5 years of age among crisis-affected populations may be as much as 20 times higher than those of children in more secure contexts (269). There is evidence to support the relationship between so-called horizontal inequality between social groups (by ethnicity, language, disability status and religion) and increased risk of violence and conflict (270). Conversely, health equity and robust health systems can also foster social cohesion and trust. As such, sustaining peace and achieving equitable health care are mutually dependent and reinforce objectives critical to long-term violence prevention (271).

The inequities of climate change

The 2023 report by the Intergovernmental Panel on Climate Change documents approximately 3.3–3.6 billion people living in contexts that are highly vulnerable to climate change (272). The climate crisis and global warming are already having myriad damaging impacts on physical and mental health, including increasingly frequent extreme weather events, such as heatwaves, storms and floods, as well as the disruption of food and water systems, and increases in zoonoses (diseases transmittable from animals to humans) and food-, water- and vector-borne diseases (273). As well as having direct health impacts, climate change damages the social determinants of health, particularly among those already experiencing poverty, exclusion and poor health. By 2030, an additional 68–135 million people could be pushed into extreme poverty because of climate change (274). The poorest 50% of the world's population contributed 12% of global greenhouse gas emissions in 2019, while the top 10% accounted for 48%. In terms of emissions growth between 1990 and 2019, the bottom 50% were responsible for 16% of the growth, while 23% of the growth came from the top 1% (275).

“The harms and health burdens of climate change are not shared equally”

The harms and health burdens of climate change are not shared equally. Disadvantaged groups and those with greater vulnerabilities suffer disproportionately from the adverse effects of climate change, creating a spiral that further increases exposure and deepens inequality (276, 277, 278, 279, 280). The largest adverse impacts are experienced in low- and middle-income countries and small-island developing states, and by women, Indigenous Peoples, small-scale food producers, people with disabilities, and low-income households, which face the greatest burden and have the least resources with which to adapt (281, 282, 283, 284).

Forced migration and displacement

Conflict, economic inequality and climate change are fuelling rising amounts of forced migration and displacement. One in eight people on the planet is either a migrant, a refugee or displaced from their home (285). The number of forcibly displaced people worldwide doubled between 2012 and 2019 (286) and surged to historically high levels in 2023 and 2024 (287). The rise in overall forced displacement – to over 120 million people, at an estimate, by May 2024 – was the 12th consecutive annual increase, and reflects both new and changing conflicts and a failure to resolve long-standing crises (288).

One driver is climate change. The United Nations High Commissioner for Refugees estimates that an average of 20 million people are forcibly displaced to other areas in countries all over the world by weather-related events every year (289). Climate-related migration destabilizes vulnerable and marginalized communities, exacerbates resource scarcity, and can ignite political tension (290). There is also clear evidence that climate change is fuelling migration out of affected areas such as sub-Saharan Africa (291).

The relationship between social determinants of health and migration is complex. Many migrants, refugees and displaced people face poorer health outcomes than the host populations (7). Migration itself can be influenced by social determinants of health equity, as people may be pushed to leave their home countries due to conflict or socioeconomic, political or environmental factors. On arrival in the host country, migrants often face inequalities in key social determinants of health equity compared to the resident population, due to discrimination, legal restrictions, language barriers, cultural differences and competency among service providers (292). In addition, leaving one's home community often means leaving behind established social support systems like family, friends and community networks. The loss of these support systems can have a negative health impact and may affect an individual's ability to cope with stressors.

2.2

Daily living conditions: inequities in the fabric of daily life

This report and its recommendations for action focus primarily on the upstream social determinants of health equity, because these drive inequities in the conditions of daily life that are so visible. There is also a continuing need for simultaneous actions to improve and reduce inequities in daily living conditions, and there is a strong evidence base for how to act on these conditions across the life course in many contexts.

2.2.1 Quality early childhood support and education

There is strong and clear evidence that poor living conditions in early childhood contribute to worse child health, as well as increased likelihood of morbidity and mortality in adulthood (293, 294). The combination of nutrition, security and safety, responsive caregiving and good early development constitutes positive early childhood conditions that set the foundations for health throughout life (295). Interventions to improve early childhood development have their greatest positive impacts on the most disadvantaged, including in relation to child care, early childhood education, child protection, social welfare and community development (296, 297, 298). Programmes to help improve the development of young children and to reduce adverse experiences can have a number of long-term benefits, including better mental and physical health (299). While many countries have boosted equitable access to high-quality childhood education, at the global level, children from high-income families have seven times greater access than those from low-income families (297).

Adverse childhood experiences – that is, chronic exposure of infants and children to stressors – negatively affect brain development, disturb endocrine and immune systems, create biological changes and adversely influence behaviour that increases the risk of difficulties in school and the workplace, and leads to poor health. Violence against children affects an estimated one billion children per year and has harmful impacts on health and well-being throughout life (300). There is a relationship between the rates at which children experience violence and many of the social determinants of health – including education; income and social protection; employment and job security; food and housing security, and affordable access to quality health services. In the United Kingdom, cross-sectional analyses of emergency hospital admissions for assault-related injuries, including for children, showed that higher rates were associated with higher economic deprivation, with a sixfold increase in admission rates between the wealthiest and the poorest quintiles of residence (301).

In Denmark, a large population-based study found that the longer a child lived in poorer circumstances, the higher the subsequent risks of self-harm and interpersonal violence offences, and vice versa for time spent growing up in affluent conditions (302).

The importance of adolescent health and education

Nearly 35% of the global disease burden may have its origins in behaviours and habits established in adolescence (303). In many high- and middle-income countries, deaths are now higher in older male adolescents and young male adults than in 1 to 4-year-olds (304, 305). Up to half of all mental health conditions start by 14 years (306). Many of the underpinning problems at this phase are structural, related to discrimination and poverty, and these conditions need addressing through investments in social infrastructure, including education systems and social services (307, 308). Research also highlights the role that education and social and developmental services can play in ensuring social inclusion and reducing isolation. Between 10% and 15% of adolescents experience loneliness and/or social isolation globally (223, 224).

People with higher educational attainment have better health and lifespans than their less educated peers at a population level (309). Societies with higher education of women have lower child mortality rates. Evidence from Demographic Health Surveys in low- and middle-income countries showed the risk of death in pregnancy of those with six years or less of education was twice that of those with more than 12 years' education (61). Estimates show that of 8.2 million fewer deaths in children younger than 5 years between 1970 and 2009 in 175 countries, 4.2 million (51.2%) could be attributed to increased educational attainment in women of reproductive age (310).

2.2.2 Work and employment conditions

Employment and work were a major focus of the Commission, as they represent key pathways through which inequitable economic, social and cultural systems directly affect health. Good-quality work includes job security, sufficient pay, progression, and safe and rewarding working conditions. Conversely, poor-quality work is insecure, poorly paid, monotonous, repetitive and/or hazardous; involves long hours, poses significant job strain (311) and requires a high effort, but provides low rewards and low pay, with a lack of recognition or opportunities for progression (312). Unemployment – often more prevalent in groups experiencing marginalization, such as people with disabilities and women – is also associated with poorer physical and mental health outcomes and higher mortality rates (313, 314, 315).

“Only 30% of workers globally have access to social protection if they are unwell while working”

According to WHO and International Labour Organization (ILO) estimates, exposures to occupational hazards in 2016 were responsible for 1.88 million deaths and 89.7 million disability-adjusted life years. An estimated 744 924 deaths were attributable to long working hours and 450 381 to exposure to particulate matter (316). Only 25% of workers globally are covered by occupational health services (317). Additionally, in many countries, employees in numerous sectors, including the health sector, are not protected through specific legal and policy instruments. Where workers are denied a voice at work, social inequalities are exacerbated. Workers' rights and protections, collective organizing and bargaining, and stability often lead to better-quality employment practices and working conditions (318, 319). Enforceable employment rights and protections have been codified in the ILO fundamental rights at work, including the right to join a union, to organize collectively to secure better conditions of work, and to a safe and healthy working environment (320).

Increasing global efforts to systematically incorporate information on injury, illness and disease from the workplace into public health surveillance systems could provide significant insights into industries, occupations and worker groups most impacted by disease, crises and health inequalities, and into where better protections are needed (321). In many countries, there is a high and increasing prevalence of working poverty in both the formal and informal sectors, disproportionately affecting women and young people (322, 323). In high-income countries, the rise of the “gig economy”, which has accelerated since COVID-19, undermines job security and associated protective contractual conditions (324). Only 30% of workers globally have access to social protection if they are unwell while working (325). Being unable to take leave when ill leads to delayed treatment and lengthens recovery processes. Sick leave is associated with lower all-cause mortality, heart disease and unintentional injuries, and its positive effects are seen in relatively short periods of less than five years (326).

2.2.3 Food environments

The social determinants of health equity, including their commercial dimension, contribute to inequitable food systems that promote health-harming diets and increase rates of malnutrition, obesity and associated noncommunicable diseases. Inequalities in malnutrition and access to nutritious foods persist between different social groups, exacerbated by recent crises and transitions (125, 327). Those in poverty and on low

incomes are less likely to be able to afford nutritious food, and they therefore face food insecurity, which contributes to inequitable burdens of infectious diseases, malnutrition and obesity.

Low-income populations and racial and ethnic minority populations in neighbourhoods of the lowest income, high deprivation and high segregation are disproportionately more likely to live in “food deserts” – geographic areas with limited access to affordable and healthy food options – which contribute to poor health outcomes (328). These populations also disproportionately live in “food swamps” – communities with far greater exposure to marketing and establishments like fast food restaurants, corner stores and liquor stores selling unhealthy or highly processed foods and beverages, than to affordable, nutritious food and beverage options (329, 330, 331). Food swamps are associated with poor health, including obesity and obesity-related chronic diseases, cancers and mortality (332, 333). Women and girls are more likely to experience all forms of malnutrition, including obesity (334, 335). Those living in emergency and conflict settings, people with disabilities and Indigenous Peoples also experience food insecurity and lack of access to healthy diets (314, 336). Populations at dietary disadvantage are also more likely to access food through public institutions, including prisons, homeless shelters, public schools, hospitals and care facilities, underlining the important role of the public sector in shaping food environments.

“Low-income and ethnic minority populations are disproportionately more likely to live in “food deserts” – geographic areas with limited access to affordable and healthy food options”

2.2.4 Housing

Housing is an important determinant of health and is shaped by structural determinants (337). Economic systems that generate poverty, low social infrastructure investment and discrimination lead to housing shortages and rising housing costs, which undermine access to good quality, affordable, safe and stable housing that supports health and well-being (338). Urbanization further exacerbates these challenges, accelerating the demand for affordable housing.

Poor-quality, insecure, overcrowded or unaffordable housing leads to poor health and health inequities related to socioeconomic position, ethnicity, age, geography, disability and migration status. In high-income settings, the burden of disease from inadequate housing is substantial (339).

For example, in a study in 2011 across 11 European countries, low indoor temperatures are estimated to result in 38 200 excess winter deaths per year, representing 12.8 excess deaths per 100 000 of the population (339). However, in low- and middle-income countries, the environmental burden of disease related to poor-quality housing is likely to be much greater (340). For example, in Guatemala, 89% of the lowest-income quintile have dirt floors, compared with 4% of the highest-income quintile (341). In the United States, repeated hospitalizations for childhood asthma are correlated with living in areas with the highest proportion of crowded housing conditions, the largest number of racial minorities and the highest neighbourhood-level poverty (341).

Housing insecurity and lack of affordability are severe and growing problems worldwide, leading to homelessness, overcrowding and lack of privacy, all of which have negative impacts on health (342). High housing costs lead to increased poverty and mean people cut back on other goods essential for health, including food, energy, transport and health care. Difficulty paying rent and mortgage costs exposes people to risks of eviction and foreclosure, and increases the likelihood that people have to move often (342, 343). There are approximately 150 million people worldwide without a home, and evidence indicates that providing them with housing can improve aspects of their health (344).

Discrimination in access to housing undermines access to a good-quality home for many groups. In some countries, certain groups – including Indigenous Peoples, minority populations, single-parent families, older people, women, and people with disabilities – are more likely to live in unsuitable housing. Migrants and other disadvantaged groups tend to be clustered in the most deprived and environmentally degraded neighbourhoods, with the fewest mobility, work and educational opportunities, lower access to health services, and below-average health.

Improving housing quality can yield important social and economic benefits – for example, efficient and safe thermal insulation enhances health and health equity, while also lowering expenditure on energy and reducing carbon emissions. Slum upgrading can significantly reduce poor health and water-related expenditure (345). Combining provision of quality housing with infrastructure and good-quality health care and emergency services provides a better chance for intervention success (340).

2.2.5 Energy

Fuel and energy poverty is a problem in many countries, perpetuating social and health inequities. Some 750 million people lack access to electricity. In addition, energy poverty is widespread, manifesting itself through significant household income spent covering energy bills, and

inability to pay for electricity services – which drives poor health (346). A third of the world’s population – 2.4 billion people – lacks access to clean fuels and technologies for cooking, exposing them to dangerous levels of indoor air pollution. The negative health impacts of environmental degradation resulting from lack of access to clean energy are also concentrated in poor and disadvantaged households (78).

In particular, the health of women and children suffers because it is generally their task to oversee fuel collection. Efficient energy use and energy-efficient appliances are strongly linked to lower levels of exposure to indoor air pollution and, particularly among children, to burns. In addition, they improve household finances by providing savings on energy costs. Improved efficiency can be initiated through government regulation and supported by the provision of incentives.

There is a risk that energy transition efforts could have a negative impact on equity. For example, a 2022 review of the literature on energy transition and energy poverty in the United States found evidence of greater negative impacts on lower-income households and on Black and Indigenous households and those of people of colour, even controlling for income, unless mitigation measures are taken (347).

2.2.6 Transport systems and mobility

Good quality, safe and affordable public transport systems are indispensable for health, enabling access to essential services, employment and social lives (342). They require strong public governance, planning and investment. Poor transport systems lead to increased risk of road traffic injuries, physical inactivity and respiratory diseases (see Box 6). Economically disadvantaged and marginalized communities are frequently disproportionately exposed to these risks. More broadly, transport systems also have a decisive influence on other social determinants of health equity, by enabling or inhibiting access to services and opportunities. This includes access to employment and education, health services, clean water, leisure, healthy food and green spaces, which in turn impact a number of health outcomes (348, 349). Transport is also a significant contributor to poor air quality and greenhouse gas emissions.

“Poor transport systems lead to increased risk of road traffic injuries, physical inactivity and respiratory diseases”

BOX 6. Transport and the burden of disease

Transport has a multi-faceted impact on health inequity, including through traffic crashes, as a cause of air pollution and as a determinant of physical activity.

Every year, 1.2 million people lose their lives on the world's roads, and as many as 50 million are injured (350). More than one quarter of those killed and injured are pedestrians and cyclists. Road traffic injuries are now the leading cause of death for children and young adults aged 5–29 years. Low- and middle-income countries account for more than 90% of all road traffic deaths, despite having less than 60% of the world's motor vehicles (351, 352).

Outdoor air pollution contributes to over 4 million deaths a year. WHO's Fifth Air Quality Database

covering over 6000 cities demonstrates that more than half of the world's urban population lives in cities that exceed recommended levels of fine particulate matter by more than six times, with people in low- and middle-income countries suffering the highest exposures (353, 354).

Globally, 31% of adults and 81% of adolescents aged 11–17 do not meet WHO recommendations for physical activity (355). Physical inactivity drives approximately 830 000 deaths from noncommunicable diseases per year (356). Poor pedestrian environments, including non-existent sidewalks or lack of sidewalk connectivity, obstacles on sidewalks, and poor signalling or design of street crossings create severe barriers to walking, and pose even larger risks for groups such as children and older people (357).

Countries that have invested little in public transport systems and where mobility is dependent on the use of private individual vehicles see health inequities exacerbated. Rapid motorization and poor public transport services mean that those with lower socioeconomic status are more likely to be pedestrians, cyclists or use unsafe modes of transport. These “vulnerable road users” comprise the majority of road traffic deaths. The link between road traffic fatality rates and public transport use is especially strong in larger cities (358, 359). Transportation barriers have also been linked to poorer health outcomes for people with disabilities (360).

Many places have taken positive action to improve transport systems and safety, which has shown positive impacts on health and the social determinants of health (361). The barriers to action improving transport systems include lack of fiscal space; entrenched social and economic inequalities, and discrimination. These are the focus of this report's recommendations.

2.3

Part 1 summary: the need for more political and policy action on the social determinants of health equity

While there has been some progress in improving health and reducing inequities, it has not been sufficiently widespread to meet the Commission's targets. Inequities between social groups within many countries are growing. The failure to make further progress is largely due to insufficient structural-level action to address the upstream social determinants of health equity outlined in this chapter, with economic inequalities, lack of commitment to social infrastructure, and structural discrimination continuing to drive health inequities. The combined impact of the COVID-19 pandemic, climate change, increasing levels of conflict, and a cost-of-living crisis are denying billions of people essential goods needed to survive and thrive, deepening disadvantage. Cumulatively, these crises have damaged health, increased inequities and thwarted progress towards achieving the Sustainable Development Goals by 2030 and meeting the Commission's targets. Lessons can be learned from scaling up approaches to address social determinants, including through intersectoral approaches and local-level action, but a more sustained focus on action to address structural determinants is required in order to reduce health inequities.



PART 1:

The state of social
determinants of
health equity

PART 2:

Tackling the social
determinants of
health equity on
multiple fronts

PART 3:

Bringing about
change through
new governance
approaches

CHAPTER 3:

Reducing economic inequality and investing in universal public services as a route to health equity

Reducing economic inequality, investing in social infrastructure and universal public services, and improving income security are essential actions to support health and reduce health inequities within countries. To achieve these aims, governments need to generate sufficient fiscal space for investment. Efficient, extensive and progressive taxation can be an important means of generating revenue.



3.1

Public services accessible to all

**RECOMMENDATION 3.1****Address economic inequality and invest in universal public services**

- 3.1.1 Use progressive taxation to expand fiscal space for income transfers and universal and equitable public services.
- 3.1.2 Provide adequate public funding for good quality, equitable infrastructure and services.
- 3.1.3 Move towards well-being economies and consider well-being budgeting.

Tax for universal public services

Affordable, accessible, high-quality universal public services are pivotal aspects of social infrastructure, which reduce health inequities by mitigating the impacts of low income, exclusion and discrimination (362). Modelling by WHO found that it would take just 0.1% of GDP invested in health, social protection, community infrastructure and active labour-market policies to markedly improve the health of 250 000 people living in a country with a population of 60 million, within four years (191). The larger the social budget, the greater the benefits in terms of poverty and inequality reduction (209), with countries with the highest levels of income inequality standing to gain the most from increased social spending (363). A review of public spending on health care across 173 countries showed that a per capita 1% increase in health spending was associated with a 0.59% drop in infant mortality and a 0.62% reduction in under-5 mortality, and that the impacts are particularly strong in low- and middle-income countries (364).

However, the gap in investing in the social infrastructure and services that underpin equity is large. It has been estimated that to meet the SDG targets, around US\$ 4 trillion of annual additional investment in public infrastructure and services is needed – approximately 4% of global GDP (365). Without this, by 2030, 84 million children and young people will be out of school, 300 million students will lack basic numeracy and literacy skills, one in four people will be without safe drinking water and half the world's urban population will lack access to public transportation (78, 77, 366).

“To meet the SDG targets, around US\$ 4 trillion of annual additional investment in public infrastructure and services is needed – approximately 4% of global GDP”

Countries whose lack of fiscal space prevents them from better investing in universal public services find their development severely hampered. Progressive taxation that increases as income levels rise can be an effective tool to combat income inequality (367, 368) and thus health inequities (369). Countries that move from flat to progressive tax systems have been able to increase their impact on reducing inequity (370). Redistributive policies have been applied with success in both low- and middle-income countries, as in Namibia (see Box 7), and in high-income countries such as Norway, which has levied wealth and resource extraction taxes to redistribute wealth, raise revenue for investment in public services and tackle rising inequality (see Box 8). Health taxes on health-harming products like tobacco, alcohol and sugary beverages also can play a key role (see Section 3.3 and Box 13). Raising prices on these products by 50% would raise US\$ 3.7 trillion over the next five years for public spending: \$2.1 trillion in low- and middle-income countries and \$1.6 trillion in high-income countries (371).

BOX 7. **Namibia: Vision 2030 and its impact on inequality**

Namibia ranks as the second most unequal country globally. Over the past two decades, the Namibian government has committed to reducing inequality through Vision 2030 and associated National Development Plans, like NDP 5 (372). Redistributive measures include in-kind transfers such as school feeding programmes, free text books and free medications and health care services, and cash transfers like the various social grants for the public. In the 2023-2024 financial year, the government expanded its safety net for vulnerable groups by increasing the monthly Vulnerable Children's Grant for households with children aged up to 16, the Disability Grant and the Old Age Grant (373). Although the old-age pension is given to all Namibians when they turn 60 years old, it has enabled many households to provide food and shelter for young children, as these are in most cases in the care of their grandparents. Namibia has consistently maintained continent-leading social spending as a share of GDP, allocating more than half of government spending for education and health (374).

These measures target the poorest segments of society: 33% of the poorest quintile received at least one type of transfer, and direct transfers represented up to 66.1% of their income, compared to 13.1% among the richest

quintile (375). Vision 2030 has also provided the framework for integrated fiscal and social development: the Harambee Prosperity Plan II (376), covering 2021-2025, aims to enhance service delivery, contribute to economic recovery and engender inclusive growth (377).

Without these measures, inequality would be 13 Gini points higher (378). Income from in-kind transfers accounts for about 78.2% of the reduction in the Gini coefficient (375). Social spending has been associated with lower infant mortality and fertility rates, and increased adult and youth literacy rates (379). A strong social protection infrastructure helped engender resilience to the impacts of the COVID-19 pandemic, with a programme of emergency income grants, assisted by special grants made available to small and medium-sized enterprises, protecting vulnerable populations from severe economic shocks (380). Targeted social spending has helped to reduce inequality in Namibia, but given the projected increase in poverty due to the pandemic and high inequality, social protection needs to be better targeted to reach people working in the informal sector, female-headed households, subsistence farmers, people with disabilities, young people and marginalized groups such as Indigenous communities (381).

BOX 8. Norway: Wealth and resource taxes to reduce economic inequality

Wealth and resource taxes have been an instrument used by many countries around the world to raise revenue, redistribute wealth and tackle rising inequality (382). Norway has one of the highest GDPs per capita in the world and is in the leading ranks of the United Nations Development Programme Human Development Index (383). Its redistributive policies, social security and large public sector are financed by high taxes (384). Revenues raised from wealth taxes (385) and taxes on natural resource-based industries (386) support the long-term financing of the Norwegian welfare state, which protects and promotes the economic and social well-being of its citizens and improves health equity.

Since 1882, Norway has imposed a wealth tax (*formuesskatt*) levied by central government and municipalities with a maximum rate of 1.1%. By narrowing the income gap between children from wealthy and less wealthy families, the wealth tax promotes a fairer distribution of income across generations with limited efficiency losses (387, 388). This policy intervention lowered the Gini coefficient by about 1 point (388) and contributes to long-term socioeconomic stability.

Norway has a well-established history of resource taxes: in hydropower, through licencing schemes, since 1911; adding a resource rent tax in 1997, and introducing a special tax on petroleum in 1975. New resource rent taxes were introduced on aquaculture in 2023 (389) and wind power in 2024 (390), with increased tax revenues expected from 2024 (391). The new wind-power tax has been structured to meet a number of objectives. Its effective rate is 25%, with an immediate deduction of investment costs, but there are generous transitional arrangements for existing wind farms. At least half the revenues raised accrue to host municipalities when they facilitate wind power, and government grants are available for local purposes, such as nature preservation (390, 392).

Historically, government investment in universal education, social protection and health coverage has been essential to higher standards of living (208). If Norway, with its strong provision of public goods – especially in health and education – can reduce income gaps for children through wealth taxes, then the impact of such taxes may be even more pronounced in countries with much lower provision of public goods (386).

Conversely, austerity policies such as those that followed the 2008 financial crisis have profound impacts on the social determinants of health, and therefore health and health equity. Evidence from a systematic review in Europe showed a strong association between increases in the number of children with low birth weight and the level of austerity (393). Evidence from the United Kingdom following the introduction of austerity policies in 2010 showed a widening gap in life expectancy between wealthier and more deprived regions, with gains

“Austerity policies such as those that followed the 2008 financial crisis have profound impacts on the social determinants of health, and therefore health and health equity”

in life expectancy stalling and healthy life expectancy declining across the country (39). Globally, public finance for investment in social infrastructure is reducing: the public sector as a percentage of GDP in countries worldwide has tended to decrease compared with 20 years ago (394).

Efforts to generate sufficient tax revenue have an international dimension. Greater global coherence on tax is needed to ensure that national efforts are not thwarted by individuals and companies moving their official addresses for tax purposes to lower-tax countries (395), and to halt the race to the lowest tax rate, as countries seek to attract businesses. At a global level, there are major ongoing negotiations and developments around international taxation, including the recent agreements towards minimum corporation tax, and current negotiations around tax cooperation at the UN (see Box 9).

BOX 9. Processes towards international tax coordination

Global Forum on Transparency and Exchange of Information for Tax Purposes (396):

The G20 declared the end of banking secrecy in 2009, and since then, the international community has achieved great success in the fight against offshore tax evasion. Working through the Global Forum, countries and jurisdictions have implemented robust standards that have prompted an unprecedented level of transparency in tax matters. Voluntary disclosure programmes, offshore tax investigations and related measures have helped identify close to €126 billion in additional revenues so far.

OECD Global Minimum Tax Agreement:

In 2021, 136 countries agreed to a global tax agreement to ensure that companies pay a minimum tax rate of 15%, bringing an end to a “race to the bottom” on corporate tax rates. The OECD, which steered the negotiations, estimated that this would generate an additional US\$ 150 billion globally in additional tax revenues (397).

The UN Committee of Experts on International Co-operation in Tax Matters:

In 2023, a United Nations General Assembly resolution to establish a potential convention to rework international tax procedures, tabled by

Nigeria and championed by African countries, passed by 125 votes to 48. A 19-member ad hoc committee has drafted terms of reference for a potential framework convention on international tax co-operation, for consideration by the Assembly, following which the text of the convention must be negotiated and ratified (398).

OECD base erosion and profit-sharing initiative (399):

To end tax avoidance by multinational enterprises, 145 countries are collaborating to stop them avoiding taxation in their home countries by shifting activities abroad to low- or no-tax jurisdictions. Such activities are estimated to cost countries US\$ 100–240 billion annually, equating to 4–10% of global corporate tax revenue and hurting developing economies disproportionately (400).

The move towards well-being economies

Increasingly, countries are seeking to drive a transition towards fairer, greener and healthier societies through well-being economy approaches (401), which connect well-being, health equity and social goals with budgeting processes (402, 403, 404, 405). Some higher-income countries are creating “well-being budgets” that widen the scope of financial budgets and influence how policy choices are made. These include approaches which prioritize a high level of investment in well-being and health equity, as well as those which value social and environmental co-benefits as part of investment decision-making.

Priorities of the New Zealand well-being budget framework, for example, have included tackling mental health, childhood programmes and universal education, fostering a sustainable economy, and improving employment (406). The budgeting approach is supported by the New Zealand Treasury’s Living Standards Framework, a localized, holistic measurement framework which looks at the country’s wealth as comprising financial and physical capital, human capability, social cohesion and the natural environment (407). Iceland and Australia have adopted similar approaches (408, 409, 410, 411).

Similarly, gender budgeting has been implemented to varying degrees by most G20 countries. For example, in Austria, the constitution requires reporting of gender equality achievements and outcomes, and auditing of gender programme impacts at every level of government (412).

Participatory budgeting has also proven successful at promoting inclusion and focusing government action on community priorities, including health – as shown in Rosario, Argentina, where funded projects included improvements to city parks, markets and community gardens (413) (see Chapter 6). Countries such as France and Wales have introduced legislation that includes a statutory requirement to report to Parliament regularly on the state of national well-being to inform policy (414, 415). Lithuania has passed legislation to support intersectoral action, including the use of a new modelling tool to connect quality of life with investment decision-making (416).

3.2

Development financing and investment



RECOMMENDATION 3.2

Champion development financing and investment that promote health equity

- 3.2.1 Ensure that the need for funding (fiscal space) to address the social determinants of health is included in approaches to taxation, development financing and debt relief.
- 3.2.2 Deliver and monitor development financing to support public investment in policies, actions and infrastructure that address the social determinants of health equity.

As well as income and health taxes, sources of finance for lower-income countries to invest in social determinants include foreign direct investment and official development assistance, yet debt burdens are still a major challenge. In 2023, official development assistance to low- and middle-income countries amounted to US\$ 223.7 billion (417) and foreign direct investment to US\$ 841 billion (418). Recipient countries can shape the impact of foreign direct investment by insisting that companies meet minimum commitments on wage and employment conditions, and contribute adequately to the provision of education, health services and housing infrastructure for local workers. Yet countries often lack the bargaining power to establish the strong regulatory environment needed to ensure commercial actors support health and health equity, and are held accountable to provide safe and healthy working environments (see 3.3). The impact of official development assistance on advancing health equity must be assessed carefully and over the long term. Some have suggested its impacts may be perverse, with one systematic review finding that development assistance for health was associated with subsequently reduced domestic health spending, possibly indicating that official development assistance had crowded out domestic effort (419).

Record levels of debt

Servicing debt is a crippling barrier to government investment in the social determinants of health equity. The total value of interest payments made by the world's 75 poorest countries, more than half of which are in Africa, has quadrupled over the past decade. In 2024, these countries will have to spend more than US\$ 185 billion, which amounts to roughly 7.5% of their combined GDP, to service their debts (420).

External debt constrains countries' fiscal space – especially in those with low resources – to invest in the social infrastructure that creates healthier and more equitable communities. Many of the world's low- and middle-income countries have had to slash public spending in order to make debt repayments to creditors and interest payments on their debts. In 2023, a record 54 countries allocated 10% or more of government revenues to interest payments, and in 2024, it was estimated that almost half of low- and middle-income countries will reduce their public expenditure (421). This shrinks finance for social and health spending that supports better health and well-being, leading to the negative austerity spiral (422).

In 2021, lower- and middle-income countries spent 27.5% of their budgets on debt service – twice their education spending, four times health care spending and nearly 12 times social protection spending (423). Evidence from several Asian economies shows a correlation between external debt and infant mortality and reduced life expectancy (424). A total of 3.3 billion people now live in countries that spend more on interest payments than on either education or health (421).

During the COVID-19 pandemic, many countries took on debt and emerged with high degrees of debt distress (425). Countries more vulnerable to climate change (426) and conflict (427) are also at higher risk of unsupportable debt, but it is often challenging for governments to escape their debt burden. To date, many holders of debt, particularly private institutions, have been unwilling to restructure or write off government debts (422).

International financial institutions have also at times constrained governments' abilities to grow social spending, so that they focus instead on reducing their indebtedness. Research in 16 West African countries with International Monetary Fund (IMF) programmes in the period 1995–2014 found that fewer than half of health care expenditure targets were met, and in several of these countries, the IMF advised against increases in social spending, out of concern that these increases would not be financially sustainable. The greater the number of IMF structural adjustment conditions taken on by countries, the lower the health spending (428). In addition, while social spending targets were missed, budget balancing conditions were consistently met, and often far exceeded (429, 430).

In view of the size of the debt crisis, the United Nations Conference on Trade and Development called in 2023 for urgent debt restructuring (431). An evaluation of the Highly Indebted Poor Countries Initiative, which redirects funds meant to service external debt towards the provision of social services and poverty reduction in eligible countries,

found statistically significant decreases in the under-5 mortality rate in sub-Saharan Africa as a result of the initiative. Eligible countries had faster decreases in under-5 mortality than those ineligible (although the ineligible countries were starting from a lower base) (432). Debt swaps for health are a promising modality of debt relief, as demonstrated by Spain's agreement to use a debt swap for health investments in Cameroon (see Box 10).

BOX 10. Spain and Cameroon: Debt swap for improving health

Debt swaps are innovative fiscal instruments that allow countries to convert their debt into investments in critical areas such as health (433).

In 2017, together with the Global Fund to Fight AIDS, Tuberculosis and Malaria, Spain agreed to waive debts owed by Cameroon in exchange for investments in domestic programmes in those areas, as part of the Global Fund's Debt2Health programme (434). Spain's Ministry of Economy cancelled a total of €24.1 million in outstanding debt, contingent on Cameroon's Ministry of Finance paying approximately 40% of that amount into the Global Fund, which has been earmarked for the fight against AIDS (434).

The debt swap significantly boosted health investments in Cameroon, contributing to improved health, particularly around HIV/AIDS (435). Programme funds have enabled 30 000 people in total to receive antiretroviral therapy, and together with additional resources mobilized by the Global Fund, have helped raise the treatment coverage rate for people living with HIV from 50% to 70% (436). Such initiatives highlight the importance of international cooperation in addressing global health challenges and promoting health equity.



3.3

Commercial determinants of health

**RECOMMENDATION 3.3****Analyse and address the commercial determinants of health**

- 3.3.1 Regulate health-harming commercial activities and maximize health-promoting commercial activities.
- 3.3.2 Address and manage conflicts of interest, to prevent negative impacts on population health and health equity.
- 3.3.3 Use the public sector to provide incentives for commercial activities that positively affect health and health equity.
- 3.3.4 Strengthen health equity considerations in global and regional trade processes, including over intellectual property.

Most commercial actors have health-promoting potential, but their capacity to deliver on this often depends on the governance contexts in which they operate. Addressing commercial determinants requires a systematic approach to understanding the balance of power and information between public and commercial actors. Where commercial actors are able to externalize the costs of harming health, they undermine the public interest, so strong public-sector leadership is needed to bring commercial actors into alignment with the public interest in health (202, 437). As a follow-up to this world report, the forthcoming *WHO Global Report on the Commercial Determinants of Health* intends to detail systematic approaches to maximize health benefits. Public health sectors can advocate for legislative changes that restrict the power of private industries (438).

Effective instruments to curb health-harming products include tax measures in relation to tobacco, alcohol, salt, trans-fatty acids and other foods (see Box 11, for example, on taxing sugar-sweetened beverages) (439, 440, 441, 442). A systematic review of health taxes in Latin America found that they reduced consumption of harmful goods (81% of studies), and had positive effects on revenue generation (71%) and health outcomes (82%) (443). Other strategies to limit exposure to unhealthier options include giving them less prominence on supermarket shelves, banning price promotions, and restricting marketing.

BOX 11. Mexico: Taxing sugar-sweetened beverages to tackle the obesity epidemic

Mexico has one of the highest rates of excess weight and obesity – a significant public health concern (444, 445). Lower-income and other disadvantaged groups are more likely to experience obesity (446, 447, 448), contributing to health inequities. Mexico is the world's leading consumer of sugar-sweetened beverages (449), which contribute substantially to its obesity epidemic (450).

To address this, the country has implemented several policies, including a tax on such beverages and mandatory front-of-pack labelling for sugary drinks and high-calorie non-essential foods.

The tax of one Mexican peso per litre of sugar-sweetened beverage came into effect in January 2014 (451). These measures aim to reduce the consumption of unhealthy beverages and provide consumers with better nutritional information.

Studies have shown an overall reduction in purchases of sugary drinks, especially of mid-priced beverages, following the tax implementation (452). Lower-income groups, which are more sensitive to price changes, have shown significant decreases in consumption. Modelling predicts that over 10 years, this would prevent 240 000 cases of obesity (453).

Across all industries and types of commercial actor, there is a need to ensure processes for transparently identifying and responding to perceived or actual conflicts of interest. Frameworks to address this problem and monitor transparency have been developed in academia, policy-making, and scientific studies (454, 455). Sector-specific tools have been created to help manage conflicts of interest in the alcohol (456) and nutrition sectors (457), including by WHO (458).

The public sector can improve the practices of commercial actors to support health and the social determinants of health by providing anchor institutions in communities. Public-sector institutions often have the power and influence to lead by example, and incentivize and shape commercial practices (459, 460, 461). Public procurement accounts for around US\$ 8.5 trillion of annual global expenditure, and across all types of countries accounts for about a third of spending (462), making it a powerful tool to drive change. Health-sensitive financing and investment is another such tool. A review by the global Association of Chartered Certified Accountants (463) found that public procurement policies at national and local levels have, for example, increased paid labour-force participation among women in Kenya (464); reduced carbon emissions and created jobs in the Republic of Korea (465); and reduced poverty and hunger among farmers in Brazil (466). Similarly, in the United Kingdom, contractors are required to demonstrate social value in the procurement and delivery process (467). Public food procurement has been an effective change instrument in the Philippines (see Box 12).

BOX 12. The Philippines: Quezon City Healthy Public Food Procurement Policy

In the Philippines, 69% of deaths are due to noncommunicable diseases, with cardiovascular diseases as the primary cause, leading to over 300 000 fatalities in 2021 (468). Dietary risk factors, including high intake of saturated fats, trans fats and cholesterol, contribute to approximately 81 000 of these deaths (469).

Quezon City, one of the most densely populated in the Philippines, with around 3.2 million people (470), committed to addressing NCD risk factors and sustainable development by joining the C40 Cities Climate Leadership Group and Partnership for Healthy Cities (471). In 2021, the city developed the Quezon City Healthy Public Food Procurement Policy (472, 473) to positively affect the overall health and nutrition of its residents. The policy mandates adherence to nutritional standards for all food provided in city-operated hospitals, schools, daycare centres, offices and events. It also supports urban agriculture and local sourcing of healthy food and ingredients.

To support the policy, the Grow QC Taskforce was established (474). This facilitated mobile kitchens and feeding programmes, distributed 5.5 tons of squash from local farmers to affected families,

and set up 295 feeding stations supporting up to 100 families each. It also developed a community kitchen to serve nutrient-enriched meals to the most undernourished and needy residents, opened an urban farm and hosted a harvest festival promoting local agriculture.

The policy has created an enabling environment for healthier food choices in Quezon City. It succeeded in reducing risk factors for noncommunicable diseases such as cardiovascular disease by providing more accessible healthy options. Over 60 000 patients and beneficiaries of city hospitals and shelters, along with more than 18 000 city employees, have received healthy meals (475). The policy has been endorsed by the Department of the Interior and Local Government, and the Department of Health, with plans to scale it up nationally (469).

Public food procurement, as exemplified by Quezon City, has been hailed as a potential game changer, but comprehensive approaches are needed to meet its transformative potential, accompanied by systematic analysis of what works and what does not (476).

Trade and intellectual property

Trade and investment policies can contribute to reducing inequalities between countries, but may also have the potential to drive within-country inequality higher unless carefully steered to ensure benefits spread to all. Trade agreements can be conditional on the avoidance of commercial practices that misuse or overuse products with potential to cause harm to human, animal or planetary health through supply chains (477), but the use of such conditionalities that support health equity has often been limited in practice. On the contrary, trade policies have been associated with multiple pathways for inequities in population health, including the offshoring of health-harming products or supply chains to areas with poorer occupational or environmental health protections; the transfer of health-hazardous products and waste, and the displacement of productive economic sectors with export-dependent ones (478, 479, 480, 481, 482, 483, 484, 485, 486, 487, 488).

Health exceptions and provisions exist in the General Agreement on Tariffs and Trade, the Technical Barriers to Trade Agreement, and the Agreement on Trade-Related Aspects of Intellectual Property Rights (489, 490). However, effective use of these protections has been undermined by narrow interpretation and the influence of commercial actors, including through legal challenges to both climate change mitigation measures and health-promoting regulation of food, alcohol, tobacco, toxic chemicals and medical devices. Health-related labelling (for example, for alcoholic beverages) has also been contested under World Trade Organization (WTO) rules on technical barriers to trade (491, 492).

The need to balance intellectual property rights with affordable pricing and access to new medicines is a well-known policy and equity issue (493, 494). The gap in access to life-saving HIV treatments at the end of the 1990s inaugurated a search for a new consensus, but conflicting priorities remain. Countries at all income levels are struggling to pay for many classes of medicines, including newer cancer drugs (495). Furthermore, despite the use of insulin for over a century, prices have remained high and availability poor, leaving it unaffordable in many contexts (496). The COVID-19 pandemic provided another example of the impact of market pricing and supply deals on global inequities in vaccine access (497, 498).

As well as the core role of the WTO, other multilateral agencies are active in advancing health equity-sensitive approaches in trade processes. For example, the World Intellectual Property Organization has negotiated a treaty to address the interface between intellectual property, genetic resources and traditional knowledge, establishing a provision in international law for a new disclosure requirement for patent applicants whose inventions are based on genetic resources and/or associated traditional knowledge (499). The United Nations Human Rights Council has adopted a resolution to establish an open-ended intergovernmental working group on transnational corporations and other business enterprises with respect to human rights (500).

3.4

Universal social protection

**RECOMMENDATION 3.4****Expand universal social protection coverage throughout life**

- 3.4.1 Ensure adequate income guarantees and care throughout the life course, including for people with disabilities or chronic health conditions.
- 3.4.2 Establish and broaden paid leave benefits for all workers, including in informal and precarious employment.

Social protection through income support or service provision sustains people through life and health events, and enables them to accumulate human, productive and financial assets. The International Covenant on Economic, Social and Cultural Rights establishes the right to health and social protection or “social security”. Higher public spending on social protection is associated with better health at the population level (501). Where countries have diminished social protection, life expectancy has fallen (502) and health inequalities increased (503). The COVID-19 pandemic highlighted the dangers of weak social protection systems, which contributed to the pandemic’s spread and prolonged unequal effects (504).

To optimize the impact on health equity while reducing inequality across the population, social protection needs high coverage across a range of different benefits. It must also reach excluded, underresourced groups such as the poor, older people, workers in the informal sector, people with disabilities, children and women. Evidence shows that filling the gaps in social protection across the life course mitigates the impact of adverse life events, provides support during critical life phases, and is a key route to health equity. Early-life interventions focusing on children with disabilities support their meaningful participation in society later on, while access to early rehabilitation can reduce the prevalence of chronic conditions (262). Income support for people experiencing unemployment protects their mental health (147, 505), is associated with better self-rated health among the socioeconomically disadvantaged (506), and mitigates the association between unemployment and suicide (507). Old-age pensions reduce poverty and insecurity (508, 509), with evidence of the health impact of pension systems from many countries, including England, South Africa and China (510, 511, 512).

“Higher public spending on social protection is associated with better health at the population level”

Social protection – a key component of social infrastructure

Important types of benefits include child and family benefits (such as cash benefits or food vouchers), parental leave (compensation for time off work to care for children, with access to maternity and child health care and nutrition), unemployment benefits (income support in case of earning loss, which was expanded in many countries during the COVID-19 pandemic), employment injury benefits (income replacement in case of occupational disease and work accidents), sickness benefits (including income replacement during absence from work due to illness) and pensions.

A study from Brazil showed that every additional US dollar spent on social protection can increase GDP by US\$ 2.9 within 25 months (513). Countries at all income levels which provide more generous minimum income benefits (including child and housing benefits, social assistance and tax credits) have lower mortality rates (514, 515). These welfare states, which provide more comprehensive benefits and are gender inclusive, achieve better health outcomes, especially for women (516). There are several pathways by which old-age pensions can improve important determinants of health, crucially by reducing poverty and insecurity (508, 509).

Income support, often in the form of cash transfers, is a key social protection instrument to reduce poverty in low- and middle-income countries. Unconditional cash transfers have been associated with a 37% reduction in the likelihood of having any illness, as well as reducing food insecurity, increasing school participation (517), and improving physical and mental health (147, 517, 518, 519).

The importance of sick-leave benefits

Only a third of the world's working-age population has income security protected by law in case of sickness (520). Those with such protections are largely in formal employment, but 61% of the global workforce is in the informal economy, where such protections are almost always absent (323). Many informal workers pay indirect taxes and other forms of fees and levies, but often receive no benefits in return (521). Where quality services exist, sickness benefits facilitate prompt access to medical care, promote mental health, aid recuperation and reduce the spread of diseases to the workplace and the community (522, 523, 524, 525). They also decrease employee turnover and burnout, increasing productivity and morale (526, 527, 528) and reducing the economic burden on minorities and women (529, 530). For example, tuberculosis patients experience variable, but often severe, levels of direct non-medical costs (mostly transport and nutrition) and income loss, which makes paid sick leave a crucial part of the income support needed to avoid catastrophic financial consequences, and enable patients to complete the care pathway (531).

“Disability benefits can lift people out of poverty and support social inclusion”

Insightful evidence comes from the United States, as one of the few advanced economies that does not provide mandatory sickness benefits. Evidence from the country’s varied local practice indicates that where sickness coverage exists, populations have lower rates of all-cause mortality, heart disease and unintentional injuries, and that these positive effects are seen in relatively short periods of less than five years (326). Evidence from European countries, where coverage is more likely to be universal, indicates that far from being a burden to economic performance, paid sick leave can provide health and economic gains for employers, workers and the economy (532).

Securing minimum income protections for populations already at higher risk of harmful environmental exposures or in ill-health helps to reduce psychosocial stress, as well as securing material needs, aiding recovery from illness, and increasing access to health services and education (518, 533, 534, 535, 536, 537, 538, 539, 540). Disability benefits can lift people out of poverty and support social inclusion (541). Benefits should be calculated so as to allow equal participation in society, and on the basis of the specific requirements of people with disabilities, as endorsed by international organizations and stakeholders, including the Global Action on Disability Network, WHO, the ILO, the Joint United Nations Programme on HIV/AIDS, the United Nations Children’s Fund, the United Nations Development Programme, and the UN Special Rapporteur on the rights of persons with disabilities, among others (542).

Direct access to income support for those who are sick should be complemented by addressing the needs of carers and others providing support. These are often unpaid, less likely to engage in employment and more likely to suffer from mental health conditions, with women disproportionately affected (147). Providing care directly through affordable public childcare centres and long-term care for older people transfers these care responsibilities from women to the state (543). Different social protection measures are available to redress inequities in care work, including universal health insurance packages, cash-for-care benefits, and compensating for time spent caring through pension credits or universal parental leave coverage (543). In recent decades, Viet Nam expanded its social protection to become universal and across the life course, with positive impacts on poverty and other social determinants that improve health and health equity (see Box 13).

BOX 13. Viet Nam: Universal social protection

Viet Nam is transitioning from lower- to middle-income country status (544), and improving its universal social protection system has been identified as crucial to this transition to address poverty (545). While Viet Nam's social protection system was originally established in the 1940s, concerns in the 1990s that poverty affected almost 60% of the population led to its expansion (546). Today, the multidimensional poverty rate is closer to 2% (547).

The country has a universal "life-cycle system" which recognizes the need for additional support for priority populations, including children, people with disabilities, minority communities living in remote areas, and older people (548). Targeted policy actions, such as modifying

eligibility criteria for social and health protection through labour laws; amendments to health insurance schemes; taxation reform and pension programmes, aim to help close gaps in access to social and health protection coverage across population groups.

Viet Nam's coverage rates outperform other lower-middle-income countries, with 38.3% of the population covered by at least one social protection cash benefit and 92% covered with national health insurance (549). Together with the country's steady, high annual economic growth rates, the government's progressive universal social protection policies have had evident impact, in improved standards of living and a significant reduction in poverty levels.

Social protection in support of child development

Social protection for 1.4 billion children remains elusive (76). Poverty and socioeconomic disadvantage blight child health, with negative effects persisting into adulthood and old age, entrenching health inequities across generations (550, 551, 552). But only 26% of children worldwide have access to family or cash benefits, with enormous variability between nations: 87% of children are covered in high-income countries; 23% in upper-middle-income countries; 21% in lower-middle-income countries and only 9% in low-income countries (553). Evidence for the positive impact of child-centred interventions is extensive. Cash transfers reduce child poverty and improve children's material circumstances, as well as household resilience (554); improve nutritional outcomes in children (533), and decrease illness (555). They also improve children's subjective well-being (554). Social protection interventions increase school attendance and reduce gender inequalities in education (556). Reducing poverty in childhood to avoid negative impacts on health requires policies to increase employment and wages of families living in poverty (557, 558).

"Social protection interventions increase school attendance and reduce gender inequalities in education"



Child-sensitive social protection systems are a core part of integrated health and care systems, including access to early childhood education and childcare services; education and skills development programmes and cash benefits, as well as health care and health insurance throughout life (559). In 2006, Chile made early childhood development a priority and adopted a comprehensive policy for children which combines universal and targeted benefits for marginalized subgroups (see Box 14).

BOX 14. Chile: Ensuring all children can reach their full potential

Chile's 2005 national survey created alarm by showing that 30% of children under 5 were not reaching their expected developmental milestones (560). This led the government to make child development a priority, under then-President Michelle Bachelet, a paediatrician and the country's first female president (561). In 2006, Chile adopted a comprehensive policy, *Chile Crece Contigo* ("Chile Grows with You"), aimed at helping all children reach their full potential, regardless of socioeconomic status (562). The policy combines universal benefits with targeted benefits for the more vulnerable, starting from gestation to 4 years old (563). *Chile Crece Contigo* coordinates activities across nine ministries, including health, education and social development.

Evaluation of the programme's impact showed the proportion of children under 5 with developmental delays declined nationally from 14% to 10% between 2006 and 2016–2017 (562). Specific benefits of participation in the programme included improved cognitive and motor skills, more parental engagement in activities that promote early childhood development (such as reading to their children and providing a stimulating home environment), a reduction in socioeconomic disparities, and increased access to health services such as vaccination and regular health check-ups. However, the programme still faces challenges in becoming fully consolidated, and comprehensive ongoing impact evaluation is needed (564).

Paid parental leave aims to secure parental income and parents' return to work while promoting gender equality and mother and infant well-being (565). Several individual studies from high-income countries show that introducing paid maternal or paid parental leave improves maternal physical and mental health outcomes and health-related behaviours (566), and reduces the likelihood of postnatal depression, with the benefits greater for low-resource and first-time mothers (567). Longer paid maternity leave is also associated with better mental health and longer duration of breastfeeding (565), while generous paid parental leave is associated with better infant and child health, particularly in terms of lower mortality rates (568). Sweden provides its citizens with parental leave for both parents, and affordable childcare and schooling, with positive socioeconomic impacts for families and the country's economy that impact health equity (see Box 15).

BOX 15. Sweden: Paid parental leave and childcare promoting gender equality and economic growth

Sweden has a long history of policies aimed at achieving gender equality by encouraging equal numbers of men and women to join the workforce and promoting equal pay. In 1974, the country became the first in the world to replace gender-specific maternity leave with paid parental leave, allowing both parents to stay at home with their children (569). As of 2024, Sweden offers both parents access to 480 days of shared leave with partial pay (570, 571). Parental benefit is paid at approximately 80% of the employee's regular salary (up to a ceiling) and flexible use is possible, with many collective agreements supplementing that further. There is a minimum guaranteed benefit for parents on low incomes or without an income.

In addition to paid parental leave, Sweden provides affordable childcare and schooling to help achieve work-life balance for parents (572, 573, 574). From the age of 1, children in

Sweden have the right to a place in preschool at a modest fee. School for children and young people is fully tax-financed, from preschool (*förskoleklass*) to upper secondary school, often including lunches. Parents have a right to paid sick leave to care for an unwell child under 12 years of age (575, 576).

In Sweden, extensive parental leave goes hand-in-hand with high labour-force participation, with nearly 90% of 25- to 64-year-olds being employed (572). Affordable childcare results in most parents in Sweden choosing to enrol their children in preschool and to return to work after their parental leave (572). Paid parental leave and access to affordable childcare contribute to high female labour-force participation rates, and reduce gender disparities in the workplace (569). Disadvantaged children enrolled in early education programmes in Sweden perform better academically than those not enrolled (577).

CHAPTER 4:

Tackle structural discrimination and the determinants and impacts of conflict, emergencies and migration

Health inequity is an important entry point to both recognize, and galvanize action on, inequities due to discrimination based on gender, race and other intersecting social factors.



BOX 16. Health inequities as a motivation for action

In 2023, Riana Brown and colleagues reported research in *Science* that should be a catalyst for action. Their goal was to identify what kind of information might trigger public support for tackling racial disparities. The answer was health. Knowledge about health inequities adversely

affecting Black Americans elicited greater support for political action than, for example, knowledge about economic disparities. Why? Because health was seen to have a “morally sacred” place in a person’s vision for what is or is not a fair society (367).

4.1

Structural discrimination



RECOMMENDATION 4.1

Take action against structural discrimination

- 4.1.1 Recognize and repair discrimination embedded in policies, laws, institutions and social norms.
- 4.1.2 Redress the negative impacts of colonization by developing standards for reparative justice that measure impacts on health.
- 4.1.3 Acknowledge Indigeneity as a determinant of health and health equity (shaped by discrimination and historical injustices, as well as protective factors), while upholding the provisions of the United Nations Declaration on the Rights of Indigenous Peoples.

International human rights frameworks are explicit in prohibiting many forms of discrimination, and most countries have constitutional provisions and national legislation that prohibit discrimination (578). However, while good policies and frameworks may exist on paper, adherence to them can be weak, and adequate enforcement is needed (4). A systematic review of workers and refugees in high-income countries found that structural discrimination in the forms of racism, poor housing conditions, immigration policies and exploitative labour practices were major determinants impacting health and access to health care services (579).

Concerted action involving national and local governments, civil society and legislators is still needed to remove structural barriers to health equity. This requires transparent, inclusive and participatory approaches

“Overcoming structural discrimination requires human rights principles and social justice to be embedded within institutions”

to tracking accountability for human rights-based laws and policies that improve the health of all, including groups experiencing discrimination (580). Groups which have historically faced stigma and discrimination, such as LGBTQI+ populations, have poorer health outcomes as a direct result, which are then compounded by the invisibility of marginalized populations when they are unwilling to identify themselves for fear of further stigma.

Overcoming structural barriers resulting from discriminatory legal and policy frameworks requires human rights principles and social justice to be embedded within institutions, supported by the active engagement of historically marginalized groups in the creation of equitable structures and opportunities (581, 582, 583). Action to address discrimination effectively must tackle the intersection across different dimensions of inequity and through diverse channels of support. For example, support for women’s participation in the labour force and decreasing their burden of care work is associated with lower gender inequalities in health (584), but according to the World Bank, in 155 out of 173 economies, at least one gender-based legal restriction exists on women’s employment and commerce capabilities, and in 18 countries, husbands have the legal right to prevent their wives from working (585).

In relation to disability, cross-sectoral initiatives include the “Housing First” approaches implemented in North America, which have been promising in supporting people with psychosocial disabilities, and transport-focused interventions in Cambodia and the United Republic of Tanzania, but in all these cases, implementation is needed at scale (586). Practical guidance to support key sectoral interventions to improve well-being for people with mental health conditions and psychosocial disabilities has recently been developed (147). For ethnic communities experiencing marginalization, approaches demonstrated to be effective in securing better health outcomes include supplemental income programmes, minimum wage policies, nutrition safeguard programmes, fair immigration policies, and ensuring reproductive rights (587).

Redressing racism through reparative justice

Reparative justice focuses on repairing past harms, stopping present harm and preventing the reproduction of harm. Reparations are being examined as a public health strategy to help eliminate racial inequalities in health (588). Modelling studies in the United States have found that reparation payments that would close the average racial wealth gap would be associated with reductions in the longevity gap by between 65% to 102.5% (589). Similarly, in examining the impacts of the COVID-19 pandemic, it has been suggested that had reparation payments to redress disparities been available, inequities in incidence would have diminished, with reductions in overcrowded housing, less exposure in frontline jobs and increased access to preventative measures (590).

One of the key pillars to reparative justice is “the right to the truth”. The Inter-American Commission on Human Rights has noted that history is rich with violations of international humanitarian law, yet with the consistent absence of complete, objective and truthful information about what transpired. States and civil society have developed approaches and initiatives to flush out the truth using a wide range of methods (591). As part of treaty processes with First Nations peoples, the state of Victoria in Australia has launched a truth-telling process (592). The Netherlands has apologized for its role in the transatlantic slave trade and planned a €200 million fund to address the consequences (593). In South Africa, reparation payments have been made within a truth and reconciliation framework (see Box 17). Canada has made long-running efforts to provide compensation to those harmed by discriminatory underfunding of the First Nations Child and Family Service, as well as to give effect to “Jordan’s principle” of “ensuring that First Nations children can access the products, services and supports they need, when they need them” (594, 595).

“Discrimination and dispossession from Indigenous land and territories, alongside loss of protective factors such as cultural identity, traditional medicine and unique Indigenous conceptualizations of health, drive inequitable health outcomes”

BOX 17. South Africa: Truth and reconciliation

Prior to democratic elections in 1994, South Africa was internationally ostracized for its racist apartheid policies, premised on the separate development of “White”, “Black”, “Coloured” and “Indian/Asian” populations. Beginning in the 1940s, a legal scheme of racial discrimination systematically dispossessed and disenfranchised non-white South Africans, as well as supporting separate development between white Afrikaans-speaking and white English-speaking subpopulations.

In 1995, the new government led by Nelson Mandela established the Truth and Reconciliation Commission to address South Africa’s long history of institutionalized racism (596). The initiative was a first step in the country’s process of healing and rebuilding relationships. The Commission’s goal was to investigate human rights violations carried out during apartheid and to provide a platform for survivors and perpetrators to testify (597).

The Committee on Reparations and Rehabilitation made comprehensive recommendations in 1998 (598). In 2003, President Thabo Mbeki announced a one-time payment of approximately 30 000 rand each to 18 000 victims who testified before the Truth and Reconciliation Commission (599).

Additionally, reparations programmes aimed at uplifting Black communities as a whole were designed as part of broader development programmes for all South Africans. To date, the reparation fund has disbursed 98 million rand to victims of apartheid. The fund’s allocation towards health care has increased access to essential medical services, and it has supported educational scholarships and business grants to empower previously disadvantaged communities. Funds have also been used to build infrastructure in Black communities, including schools, clinics and community centres, which have long-term benefits for social cohesion and economic development (600).

Indigenous Peoples’ determinants of health are distinct due to historical injustice, therefore the UN Permanent Forum on Indigenous Issues recognizes the importance of acknowledging Indigeneity as a social determinant of health (601, 602). Discrimination and dispossession from Indigenous land and territories, alongside loss of protective factors such as cultural identity, traditional medicine and unique Indigenous conceptualizations of health, drive inequitable health outcomes. A legacy of inequality and exclusion has made Indigenous Peoples more vulnerable to the impacts of climate change and natural hazards, including disease outbreaks such as COVID-19 (255). Indigenous Peoples often face discriminatory practices and differential treatment when accessing health care facilities, fuelling mistrust and discouraging the use of services (601). The revitalization of Indigenous Peoples’ cultural identity is essential to tackle discrimination and promote health and well-being.

The UN Declaration on the Rights of Indigenous Peoples provides a framework to address the underlying determinants of Indigenous Peoples' health, such as access to lands, territories and resources; the manifestation, practice, development and teaching of Indigenous spiritual and religious traditions; and the preservation and maintenance of traditional medicines and health practices (603).

The active involvement and leadership of Indigenous Peoples in developing and determining health programmes affecting them is central to efforts to overcome health inequities. Culturally safe interventions improve service acceptability and outcomes (604), with key elements including:

- ▶ two-way communication, based on listening and validation of Indigenous worldviews (605)
- ▶ a robust cohort of Indigenous health workers who can catalyse the formation of a trustful rapport and increase the likelihood of service use (605)
- ▶ the incorporation of Elders and kinship networks, complemented by linguistic interpretation services within primary care frameworks (606)
- ▶ respect for self-determination and incorporation of traditional knowledge into health decisions (606).



4.2

Emergencies, migration and conflict

**RECOMMENDATION 4.2****Promote health equity during emergencies, migration and conflict**

- 4.2.1 Recognize peace as a fundamental determinant of health equity, and health as a bridge to peace
- 4.2.2 Ensure emergency preparedness and response efforts incorporate the social determinants of health equity, including additional social protection measures.
- 4.2.3 Ensure displaced people and migrants have access to health and social services.

Health emergencies and conflict exacerbate existing health and social inequities by creating and concentrating vulnerability. Whether this vulnerability arises from conflict, disaster or a health emergency, the keys to effective preparedness and response are similar: adaptability of responses to rapidly changing contexts; programming efforts that recognize the differential needs of the most vulnerable, and adjustment of levels of support accordingly. In conflict situations, ensuring health and social services are available for migrants, displaced populations and refugees is paramount. Humanitarian aid can improve access to health care and services, but increasing attacks on the wounded and sick, medical personnel and medical facilities are universally regarded as serious violations of global norms, and the UN Security Council has demanded an end to impunity for those responsible (607, 608). Diplomacy can play a vital role in ensuring health-related concerns in emergencies and conflict are kept at the fore, and the social determinants are prioritized in wider diplomatic discussions, so they can directly shape global health norms at national, regional and global levels (609, 610). Deepening the connections between health and foreign policy can contribute to a safer, more peaceful and healthier world.

“Deepening the connections between health and foreign policy can contribute to a safer, more peaceful and healthier world”

Adequate social protection, including universal health coverage, is key to building epidemic and pandemic preparedness and ensuring health emergency responses do not deepen inequities, but protect both lives and livelihoods (611, 612, 613). Adaptive social protection systems are vital to mitigate the unintended negative consequences of emergency, public health and social measures in the face of rapidly unfolding crises (614), as shown in the COVID-19 pandemic (96). For example, cash transfers during the pandemic were associated with a decrease in food insecurity in countries that have different levels of income (615, 616, 617). They were also positively linked to self-reported health status (618); a higher likelihood of participating in volunteering (important for social cohesion); higher investment in children's education (619), and lower risk of depression and anxiety (620, 621). As Boxes 18–20 show, these direct support mechanisms, providing cash where and when it is needed most, are tools which can be deployed at scale, even in emergency settings where service delivery has been affected (622).

Targeting must be approached with an equity lens – for example, recognizing that those in the informal sector may be missed by support programmes. When support has reached informal workers, they have experienced better psychological well-being and increased food security (623). In conflict-affected areas, community health worker interventions may be effective and efficient in providing access to care (624). WHO has developed guidelines for providing health care, education and social support to communities adversely affected by health emergencies, and which have existing structures. Faith-based organizations provide many of these critical services and can be key partners in health emergencies (625, 626).



BOX 18. Togo: Emergency cash transfers to poor households during the COVID-19 pandemic

During the COVID-19 pandemic, the government of Togo devised a fully digital emergency cash transfer programme via mobile phones, targeting informal workers and poor households, called *Novissi*, meaning “solidarity” in the Éwé language (628, 629). The pandemic exacerbated economic vulnerabilities (630), necessitating urgent interventions to support the most disadvantaged populations.

Togo built the *Novissi* platform – a digital and contactless unconditional cash transfer programme, serving over 12% of the population – from scratch in just 10 days in response to COVID-19. The platform paid out more to women than men and was able to reach vulnerable populations (629, 631). A national voter registry containing information on individuals, their occupation and home location helped establish eligibility. To expand platform coverage, the poorest villages and neighbourhoods were selected through high-resolution satellite imagery and nationally representative household consumption data. Within those villages and neighbourhoods, the poorest individuals were then prioritized through machine learning algorithms using mobile phone metadata

and phone surveys generating predicted consumption patterns for 5.7 million individuals (70% of the population).

Individuals registered for the scheme by calling from a mobile phone with a valid and unique voter identification card. During the first phase of the programme, more than 500 000 beneficiaries received US\$ 22 million over two months, with 65% of beneficiaries being women (632). In October 2020, the programme was expanded to the 200 poorest cantons of Togo through a partnership with GiveDirectly (633). The new data-driven technologies deployed enabled rapid disbursement in a time of crisis and ensured that aid was directed towards the most vulnerable populations. The use of mobile money facilitated greater access to financial services for many who were previously unbanked, promoting financial inclusion and economic resilience among the poor (634).

The success of the *Novissi* programme during the pandemic led to its continued deployment, showcasing a scalable model that could be adapted for future emergency response efforts in other regions facing similar challenges.

Conflict situations and health emergencies are crises, but should not be regarded as a state of exception divorced from the ongoing socioeconomic realities which shape health outcomes. For example, the recently developed global architecture for health emergency prevention, preparedness response and resilience recognizes that protection of health cannot be disentangled from the social determinants of health equity. It advocates for intersectoral action to strengthen social welfare and protection, livelihoods, continuity of business and education, and food security (627).



BOX 19. India: Cash transfers to support people with low incomes during the COVID-19 pandemic

In March 2020, in response to the emerging economic repercussions of the COVID-19 pandemic, the national Government of India launched the *Pradhan Mantri Garib Kalyan* package, a comprehensive relief package of 1.7 trillion rupees (US\$ 20 billion) to support low-income individuals and families with cash benefits (635).

As of September 2020, key financial allocations and beneficiaries of the package included (635):

- ▶ Cash transfers: Some 309.5 billion rupees disbursed to women beneficiaries of the National Mission for Financial Inclusion.
- ▶ Support for vulnerable groups: Support provided to 28.1 million senior citizens,

widows and people with disabilities, in two instalments.

- ▶ Agricultural support: 178.9 billion rupees, benefiting 84.9 million farmers.
- ▶ Support for construction workers: 49.9 billion rupees to aid 18.2 million building and other construction workers through State Welfare Boards.

Under the initiative, over 420 million poor people benefited from the scheme and a total of 68.8 billion rupees in financial assistance was disbursed to ensure economic relief and support during the COVID-19 pandemic (635).

BOX 20. Democratic Republic of the Congo: Cash-for-work programme in response to Ebola

The large-scale outbreak of Ebola virus in the Democratic Republic of the Congo in 2018 occurred in a region where armed conflict between dozens of groups had been ongoing for more than 20 years. This situation presented a significant challenge for public health responses, partly due to the local population's deep distrust of external actors. Some people referred to the disease as manufactured and claimed that the millions of dollars spent on the response were part of a money-making scheme labelled the "Ebola business" (636, 637, 638).

From April 2019, to address these challenges, the government of the Democratic Republic of the Congo rolled out an emergency cash-for-work programme with support from the World Bank, to provide much-needed income support, and to help communities recover from the Ebola crisis and build resilience. The programme fostered trust by emphasizing active and meaningful

community engagement, and transparency in the use of resources (639). All adults were eligible, with participants selected by public lottery.

Immediate financial relief was provided to thousands of households, and employment opportunities created in a region where conflict had severely limited economic activities. By 30 September 2020, more than 35 000 direct beneficiaries (half of them women) had worked more than a million days and received about US\$ 3 million in payment (640). The construction of social infrastructure, such as health facilities, schools and roads, improved access to essential services and contributed to long-term community resilience (641). The programme's inclusive approach, particularly the adaptation of work requirements for vulnerable groups, ensured that its benefits reached those most in need (640).

The right of displaced people to health

Whether people move due to conflict, an emergency, displacement or migration, health outcomes can be at risk. During the migration process, access to water, sanitation and food are crucial, while other factors become more relevant during and after resettlement (7). Living conditions, including housing (642, 643) and employment (644, 645, 646, 647), are important determinants of the health of migrants, refugees and asylum-seekers (648, 649, 650). While evidence on structural approaches to migrant health needs is scarce, providing access to health and social insurance has been effective in some contexts (292), and community PHC models have been shown as better equipped to address the underlying health needs of migrants than those that focus only on providing medical care (651, 652). The provision of free or low-cost sexual and reproductive health services to women, including migrant and refugee women, promotes service use, contraceptive use and access to prenatal care (653). A variety of PHC interventions can be beneficial for the health of internally displaced people in conflict settings, including mental health and psychological support in schools (654); community-based baby feeding programmes with psychosocial intervention, malaria prevention or WASH interventions (655).

There is some evidence that intersectoral approaches for the prevention and control of malaria among migrant and mobile populations are effective, but most interventions are still only vertically organized and are therefore not fully integrated (656). There is more evidence suggesting positive outcomes from health interventions targeting migrant and refugee populations delivered at the individual (657, 658, 659) or community level. These include psychological interventions for post-traumatic stress disorder and depression in refugees (660); obesity prevention through behavioural interventions (661); health care workers acting as “community navigators”, providing culturally sensitive guidance to help people overcome barriers to accessing health care (662); uptake of vaccines (663), health literacy (664); and culturally safe maternal health care (665). However, the impact of these interventions on health equity is rarely evaluated (666).

CHAPTER 5:

Steer mega-trends towards health equity: climate change and digitalization

While economic inequity and structural discrimination have been persistent forces shaping health inequity, they do not exist in an unchanging world. At the outset of this millennium, two of the broad forces of change with the most profound impact on health equity are climate change and digitalization (667, 668). The green and digital economies are key pillars of the economic plans of many countries (669).



The climate crisis is reshaping the relationship between people and planet, and climate mitigation and adaptation will be increasingly dominant in establishing both opportunities to advance health equity, and constraints on doing so (273). The digital transformation has not only reshaped the social and economic world, but has opened new horizons for health action, as well as created new concerns with the potential to deepen health inequity (670). Harnessing the forces of both climate action and digitalization to promote health equity goals holds enormous promise to improve health and reduce health inequities, if attention to equity is prioritized.

5.1

The potential of climate action to accelerate health equity



RECOMMENDATION 5.1

Articulate and accelerate the health equity benefits of climate action and the preservation of biodiversity

- 5.1.1 Support the development and implementation of climate change mitigation and adaptation policies that maximize health equity benefits.
- 5.1.2 Ensure energy and food transitions are able to reduce energy poverty and food insecurity.
- 5.1.3 Ensure that climate action is a core objective of health systems.
- 5.1.4 Strengthen support for Indigenous communities in their stewardship of land and natural resources.

Aligning climate and health objectives has the potential to deliver substantial co-benefits, but this requires intersectoral action (671). WHO has estimated that taking action on climate change could save 7.5 million lives, and that the health gains from improved air quality through a rapid global transition to clean energy would repay transition costs twice over (672). The direct health and mortality impacts of extreme weather events can be reduced. Other benefits for health from action addressing climate change through a social determinants approach include promoting physical activity through increases in active transport; reducing ill health and mortality from cold housing through insulation, and supporting healthy dietary choices with a low environmental footprint.

As well as suffering the greatest impact of climate change, those living in precarious conditions have the least capacity to adapt. Subsistence farmers and low-income workers are often unable to avoid excessive heat (673). People with the lowest incomes are the most likely to depend for their survival on resources provided by nature, through sectors such as agriculture, forestry and fisheries. In low-income settings, climate impacts affect women and men differently: women and girls – responsible in many societies for collecting water and fuel – frequently have to walk further to do so, and are often the last to eat amid increasing food insecurity (674). A systematic review found that during and after extreme weather events, women, girls, and sexual and gender minorities are at increased risk of violence and exploitation, related to economic instability, food insecurity, mental stress, disrupted infrastructure, tradition and exacerbated gender inequality (675). Poor people and populations experiencing marginalization are less likely to have access to insurance and social protection, healthy housing or primary health care, making them particularly vulnerable to any shock affecting their lives and livelihoods.

Achieving the goals of the 2015 Paris Agreement (676) requires an inclusive, just energy transition, which recognizes the needs of different communities and gives priority to access to clean energy for those currently experiencing energy insecurity (677). Decarbonization of the energy supply can address energy poverty, create decent jobs and reduce air pollution, all of which support better health and health equity. A lack of electricity is a particular issue for sub-Saharan Africa, with three in five people without access (678). The transition to zero-carbon economies can help those who are most energy disadvantaged to leapfrog up the “energy ladder” directly to clean electricity, and reap the associated benefits in daily living conditions and health. The International Energy Agency estimates the number of people with access to electricity through a solar home system in sub-Saharan Africa has increased by about 25 million since 2019, topping 45 million in 2022 following sustained government programmes to reduce their costs in countries such as Kenya, Nigeria and Rwanda (678).

BOX 21. South Africa: Kuyasa Clean Development Mechanism to reduce emissions

The Kuyasa Clean Development Mechanism (CDM) project aims to improve the thermal performance and reduce carbon dioxide (CO₂) emissions of both existing and future housing units in Kuyasa, a low-income neighbourhood of Khayelitsha Township in south-eastern Cape Town, South Africa (679). The project, launched in 2008, involves retrofitting existing housing units with solar water heaters, ceiling insulation and low-energy, long-life compact fluorescent light bulbs.

The programme has upgraded about 2300 units and is planned for a period of 21 years, with a three-fold goal of enhancing energy efficiency, reducing greenhouse gas emissions and lowering energy costs for residents (680). It is carried out by the City of Cape Town in collaboration with the SouthSouthNorth Project, a network of institutions and experts that helps public projects through funding from the United Nations Framework Convention on Climate Change (UNFCCC) Clean Development Mechanism (681).

The project is expected to save about 2.8 tons of CO₂ per household, per year (682). Households benefit from an estimated annual saving of US\$ 110 on energy costs, a major boost to low-income families (683). The installation of solar water heaters and ceiling insulation improves the thermal performance of homes, making them more comfortable and healthier by maintaining stable indoor temperatures and reducing exposure to cold and damp conditions.

The programme also has positive health and safety effects, resulting from better insulated ceilings, greater warmth, and less dust and sand in houses (683). Among surveyed households, 81% indicated a decrease in frequency of respiratory illness, and the proportion of households experiencing illnesses twice or more a year dropped from 79% to 26% (684).

Community engagement in project implementation has fostered a sense of ownership and responsibility among residents, boding well for the sustainability of the project and its wider acceptance locally.

When designed and implemented with equity as a priority, climate mitigation strategies can have significant health equity co-benefits. For example, green building design to reduce flooding (685), and insulation and ventilation of housing (686), could both help to mitigate against extreme weather and higher energy costs for those living in low-lying areas or cold homes. The Canadian state of British Columbia introduced a carbon tax on consumption, the proceeds of which were used to reduce other taxes or to provide direct cash transfers to low-income consumers, with net progressive impact (687). Support to low-income households to improve energy efficiency and reduce bills can be highly effective – for example, as in the United Kingdom’s Warm Front programme (688). Also in the United Kingdom, research on environmental inequalities informed targeted funding decisions that benefit populations living in deprived areas who are most vulnerable to environmental risks.

“When designed and implemented with equity as a priority, climate mitigation strategies can have significant health equity co-benefits”

Bangladesh is on the front line of climate change. Sea temperatures in the Bay of Bengal have significantly increased, and Bangladesh is experiencing some of the fastest rises in sea level on record worldwide (689, 690). Given the low-lying land, floods have caused large-scale health crises and destruction of infrastructure and housing. However, green building design is underway to help protect the poorest. The World Bank's Multipurpose Disaster Shelter Project aims to give 14 million people in nine districts access to temporary safe havens, by building 552 new shelters, rehabilitating 450 existing shelters and constructing over 550 kilometres of evacuation roads. The shelters are used as primary schools during regular weather (691). On a smaller scale, 20 plinth shelters supported by the charity Friendship have been built with the involvement of local communities to raise shelters off the ground (692). Small loans have also been provided to help energy users install solar home systems, resulting in an increase in access to energy through clean sources (see Box 22).

BOX 22. Bangladesh: Grameen Shakti programme for renewable energy and health equity

In Bangladesh, access to electricity and clean energy technologies has historically been limited, particularly in rural areas (693). This lack of access has significant implications for health, economic development and gender equity (694).

In response, the Grameen Shakti programme began disseminating solar home systems in 1996, enabling energy users to purchase renewable energy equipment (695) through an innovative financing mechanism (696). So far, the programme has installed nearly 1.8 million solar-home-systems (697), 36 000 biogas plants (698), and nearly a million improved cooking stoves serving around 4 million beneficiaries (699). The initiative focuses on giving communities and women direct access to credit and training, facilitating gender empowerment and the deployment of clean energy technologies (700).

With assistance from the programme, electricity access in Bangladesh rose from 20% of the population in 2000 to 85% in 2022 (701). The installation of biogas plants and improved cooking stoves has reduced indoor air pollution,

which is a major health hazard in rural areas (702). A study by the International Centre for Diarrhoeal Disease Research, Bangladesh found that households using improved cooking stoves reported a significant reduction in respiratory illnesses, particularly among women and children (703). Access to credit and technical training has empowered women to become entrepreneurs and leaders in their communities. Research by the World Bank indicates that women's involvement in renewable energy projects has led to greater economic independence and improved social status (704).

According to a report by the Asian Development Bank, the renewable energy sector in Bangladesh has generated thousands of jobs, contributing to economic growth and poverty reduction (705). The environmental impact has been considerable: data from the United Nations Environment Programme suggests that Bangladesh's solar home system programme has helped the country cut down its emissions, with Grameen Shakti alone helping it avoid approximately 1.5 million tons of CO₂ emissions annually (706).

Food security is a critical element of climate transition. Innovative use of technology can help to conserve water through “precision” watering; diversifying crops and animals increases total systemic resistance to climate change and market volatility; climate-resistant seeds and climate-sensitive animal husbandry can improve production efficiency; better storage and mindful purchasing patterns can reduce food waste; and improved early warning systems to farmers can help to mitigate against disasters (707). In contexts where meat is overconsumed, diets that are less reliant on meat and more reliant on wholegrains, fruits, vegetables and pulses can also help to meet climate change and nutrition targets (708, 709).

Preserving biodiversity underpins global and food security, and ensures the continued availability of many medicines based on natural compounds found in plants and animals. Natural environments and green spaces provide an ecosystem that enhances human health and well-being (710). Healthy ecosystems also play a role in regulating diseases, preserving soil quality and climate stability (711).

Community involvement is key to ensuring that climate action is contextually relevant and responsive. Community-driven approaches have proved critical to sustainability. For instance, in drylands, farmer-managed natural regeneration of trees in arable areas has enabled



the restoration of over 5 million hectares of land in the Sahel (712). In Ethiopia between 2012 and 2015, the Managing Environmental Resources to Enable Transitions programme, which entails community-based watershed rehabilitation in rural landscapes, supported around 648 000 people, resulted in the rehabilitation of 25.4 million hectares of land in 72 severely food-insecure districts. This was achieved through social protection schemes that provided payment in exchange for work to build public assets (713).

By prioritizing and delivering on local climate action, health systems can build resilience, enhance preparedness, and reduce the impact of climate-related health risks. Decarbonizing health systems brings a host of benefits, as seen in the Solar for Health initiative in Zimbabwe (see Box 23). Local programmes, based on insight into effective preventative and mitigation strategies that work for specific communities, are needed (714, 715). Local knowledge is critical to effective health-response actions, brokered by trusted interlocutors over considerable periods of time (716). When systems collaborate with local residents, they gain valuable insights, enhance trust and create more effective interventions.

BOX 23. Zimbabwe: The Solar for Health initiative

Zimbabwe's health system, especially in rural areas, has been severely impacted by persistent energy shortages, exacerbated by climate change and erratic weather conditions.

To help meet these challenges, the United Nations Development Programme has been spearheading the Solar for Health initiative in 15 countries, including Zimbabwe, since 2017, as a means of connecting the energy and health sectors to promote health, while protecting the environment (717). With support from the United Nations Development Programme and the Global Fund to Fight AIDS, Tuberculosis and Malaria, the initiative in Zimbabwe aims to harness previously untapped solar energy sources to power critical social infrastructure (718). Zimbabwe has installed solar photovoltaic systems in over 400 health facilities, so that as of 2018, over 6.5 million people across the country were benefiting from reliable access to electricity for health

facilities. This enabled longer operating hours, better staff retention, and safer maternal care and life-saving procedures (718).

Solar energy also reduced electricity costs by up to 60%, allowing clinics to reinvest savings in sustaining and improving facilities and services (719). The Solar for Health initiative is estimated to fully recoup its cost within two to four years (719).

By providing reliable electricity in poor, remote and rural areas, the initiative has reduced health inequalities, particularly addressing the higher maternal mortality in rural areas and among more impoverished communities (720). It has contributed to the country's progress across at least three of the Sustainable Development Goals: SDG 13 (Climate action), SDG 7 (Affordable and Clean Energy), and SDG 3 (Good Health and Well-being).

“Restoring land tenure and compensating for land confiscation has improved the health of Indigenous communities”

Indigenous Peoples can play a crucial role as agents of change through ecosystem- and community-based practices that combine adaptation, mitigation and sustainable development (721). A key issue for Indigenous Peoples is land tenure. There is evidence that restoring land tenure and compensating for land confiscation has improved the health of Indigenous communities. For example, in New Zealand, this process has allowed Māori to create their own infrastructure, to begin to develop their own health care initiatives, and to advise health care authorities and governments on interventions to reduce health inequities between Māori and non-Māori communities (722). Restoring Indigenous land rights in Australia has also improved health (see Box 24).

BOX 24. Australia: Caring for land, country and Indigenous health

Indigenous Peoples around the world experience worse health than their non-Indigenous counterparts (723). In Australia, forced removal from ancestral lands, which Indigenous Australians had occupied for around 65 000 years (724), has been particularly damaging. The connection to the land holds profound cultural significance for Indigenous Peoples, and continued association with ancestral lands, and caring for them, is seen as vital to health (725, 726, 727, 728, 729).

A range of public policies since the 1970s has focused on restoring land to Indigenous Peoples, enabling them to care for country in ways that fulfil traditional requirements. Federal and State governments have acknowledged land rights through legislation. A landmark case in the Australian High Court, brought by Indigenous campaigner Eddie Mabo, successfully challenged the colonial assumption that the Australian continent was *terra nullius* (“nobody’s land”) when the English arrived (730).

The restoration of Indigenous land ownership and the ability to care for country is recognized as a health benefit and an important contributor to closing the gap between Australian Indigenous life expectancy and that of other Australians (731).

Evidence shows that Indigenous people involved in managing and caring for their land have lower rates of diseases such as diabetes, renal disease, hypertension and obesity (732, 733). They also report better self-esteem, self-identity and connection to culture compared to those not involved in land care (728). This holistic approach to health underscores the importance of cultural practices and land management in promoting physical and mental well-being.

Initiatives to restore Indigenous land rights have contributed to narrowing the life expectancy gap between Indigenous and non-Indigenous Australians (734).

5.2

The digital transformation and health equity



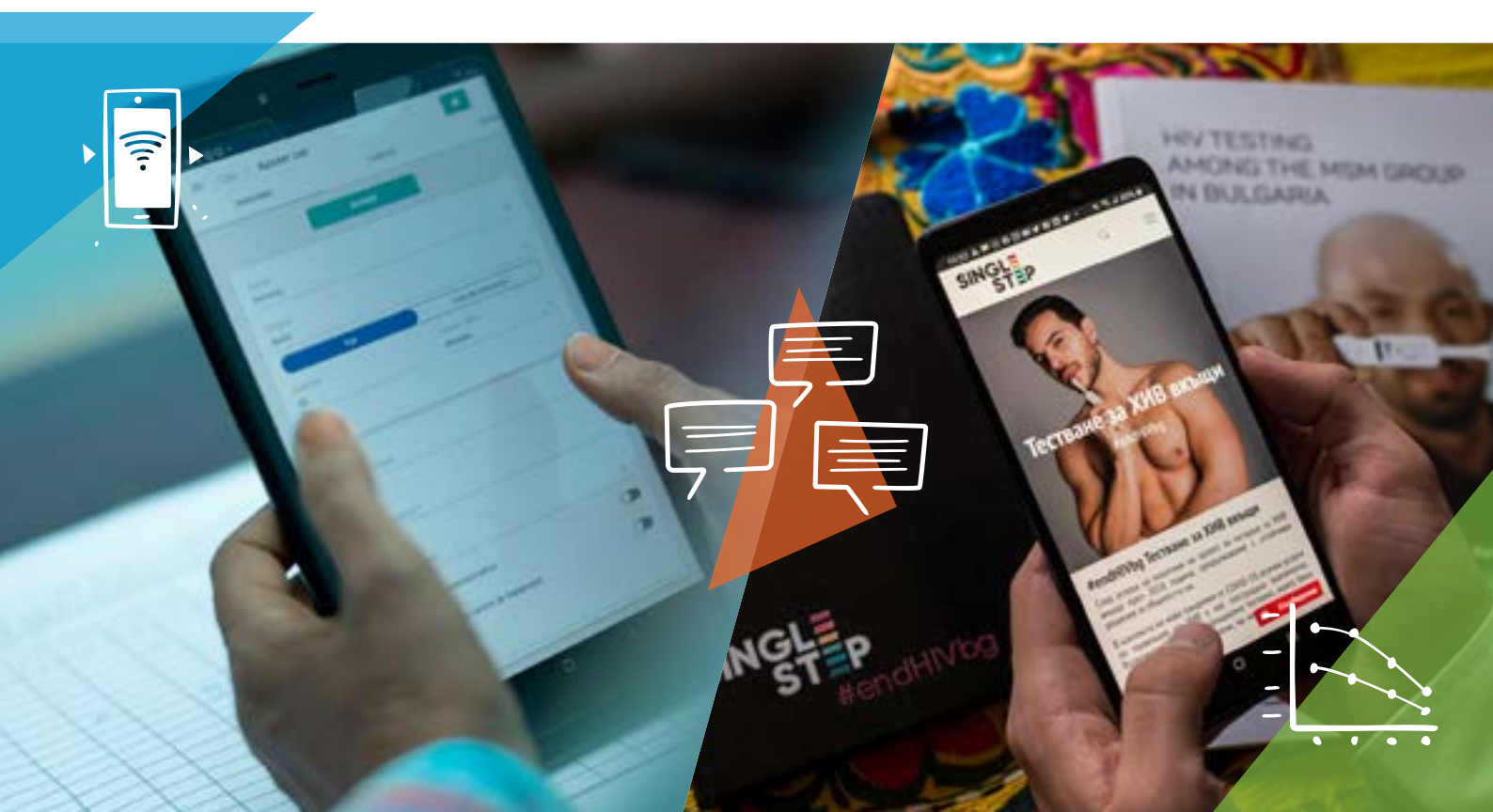
RECOMMENDATION 5.2

Steer the digital transformation in favour of health equity and the public good

5.2.1 Address the digital divide.

5.2.2 Ensure digital and technological transformations and artificial intelligence promote health for all.

Digital technologies, including the emergence of Artificial Intelligence (AI), are already rapidly transforming the world economy and how societies function, shaping health outcomes. Digitalization holds the potential to improve population and individual health – for example, through the promise of revolutionizing access to health and social services for those going without, which can help overcome physical barriers and workforce limitations, as demonstrated in the Islamic Republic of Iran (see Box 25). Yet digitalization and the opportunities associated with AI and other digital transformations are not equally available either between or within countries, and through disruption and possible existential risks, they could also cause significant harm and widen existing health inequities.



BOX 25. The Islamic Republic of Iran: Improving maternal and neonatal health care through telemedicine

Amid challenges posed by economic constraints and sanctions, the Islamic Republic of Iran has struggled to ensure health equity – a situation exacerbated by the significant influx of Afghan refugees, totalling 2.2 million by the end of November 2022 (735). To improve health care accessibility, the Ministry of Health and Medical Education embarked on a transformative telemedicine initiative, with WHO support, in Sistan and Baluchistan province, to enhance maternal and neonatal health care within neonatal intensive care units and primary health care systems (736).

The multipronged initiative established telemedicine infrastructure, launching mobile clinics, capacitating laboratory services and developing standardized procedures. Tele-consultancy and tele-visit services offered neonatal intensive care, including access to eye screening for premature babies, mobile

sonography and cardiography for newborns, and the products needed to support free-of-charge delivery services for vulnerable populations.

The WHO country office reported that significant strides have been made in enhancing access to essential services, with the procurement of delivery facilities, medical monitoring equipment, telecommunication devices and vehicles for providing mobile clinics and service monitoring. The telemedicine system in Sistan and Baluchistan facilitates timely access to health care services for 221 000 people, including 10 000 pregnant women. WHO personnel also report that the project outcomes have extended beyond the country's borders, with two paediatricians from Afghanistan and Yemen attending neonatal fellowship training in the Islamic Republic of Iran and benefiting from telemedicine devices for eye examination of premature babies.

The need for digital access and to mitigate related health risks

With increasing information and services available online, digital access is now seen as a human right, and is vital for access to health care and other essential services that influence the social determinants of health (737, 738). For example, alongside enabling people to access health and banking services, digital platforms can help connect rural communities to wider markets, and improve their economic outlook and health. Digital technologies also have the potential to reduce social isolation and loneliness (739). In education, virtual learning environments and distance learning have opened up programmes to low-income students and those in remote areas who would otherwise be excluded.

In 2022, a third of the world's population lacked digital access, largely comprising those on the lowest incomes. Older people, ethnic minorities and Indigenous Peoples are less likely to use the internet, and low-income groups and people living in rural areas have lower internet access rates and speed (740, 741, 742, 743). In general, women have lower adoption levels for digital technologies than men (744). People with disabilities can also face additional barriers to accessing digital technologies equitably (745).

“Digital technologies, in particular social media, can contribute to feelings of isolation and worsened mental health”

The increasing body of evidence for the harmful impact of the digital transformation on child and adolescent health has led to growing government action. Negative impacts include the significant amount of potentially harmful online content available, particularly to children and young people, such as inappropriate marketing of health-harming or dangerous products, encouragement of violence, and explicit sexual content. Digital technologies, in particular social media, can contribute to feelings of isolation and worsened mental health (746), and can place people – especially children and young people – at risk of exploitation (747). They can also be highly addictive, exacerbating previous addictions such as online gambling, or generating new addictive behaviours such as gaming (748, 749).

Evidence from high-income Anglophone countries shows alarming increases in mental health conditions among adolescents and young adults over the past decade, and corresponding decreases in well-being in this age cohort (227). Young women in particular are experiencing mental distress at much higher levels than previously. The longstanding “U-shaped” distribution of well-being, whereby adolescents and older people had the poorest well-being, is disappearing, and instead there is a linear gradient as well-being steadily increases with age. The timing of this shift suggests the cause may be a generation of digital natives more exposed to social media use, but a lot more investigation is needed into the causal pathways, the extent to which the same patterns are seen in lower-income countries, and to what extent there is differential impact across different socioeconomic groups.

A number of population health actions have been identified to mitigate these harmful impacts, including delaying uptake through age restrictions and school bans; warning labels and health promotion efforts, and reducing children’s use of digital media and devices by deploying technical features. Other approaches centre on establishing financial deterrents, creating digital-free spaces, providing digital education and increasing digital literacy, changing norms and creating alternatives, and directly addressing harms through appropriate governance.

Digitalization, equity and the world of work

A clear way in which digitalization will affect health and health equity is through its impacts on the quality and availability of employment. Yet many of these impacts are still unknown (750, 751, 752, 753, 754). New ways of working provide solutions for classic occupational health problems – for example, by using robots for hazardous work, or AI for monotonous

“An unregulated rollout of digitalization is likely to create new groups of winners and losers, with major consequences for social equity, including health outcomes”

tasks. But there are urgent, unanswered questions about how such a transformation will be managed. An unregulated rollout is likely to create new groups of winners and losers, with major consequences for social equity, including health outcomes (755). While digital platforms could contribute to wider formalization of informal work, platform deployment tends to be based on business models which lead to upward redistribution, and avoidance of existing social and health protection requirements (756).

The global spread of automation also risks reversing decades of poverty reduction by eliminating formal-sector jobs in regions where employment, not labour, is scarce (757). Given this risk, carefully considered mitigation measures are needed to prevent such scenarios from negatively impacting outcomes for workers, and to avoid large increases in the need for social protection and other services, and the potential costs to public finances. Such strategies may include AI and technological impact assessments in relation to employment and to health equity (757); tax and policy environments that place human well-being at the centre of economic policy; and education and retraining specifically to help workers enter sectors with growth, such as health care, the green economy and digital solutions. Income support or universal income solutions can also help address technological unemployment, while job placement services can match skills with available opportunities. For example, with the rise of the internet and related workforce, multiple countries have enacted legislation to protect those engaged in work through online services (see Box 26).



BOX 26. Global efforts to protect gig workers' rights

There are an estimated 154 million to 435 million digital platform workers (or “gig workers”) globally, employed as taxi drivers, domestic workers or food delivery drivers, and in other industries that use online technology as an exchange medium (758). This form of employment can lead to compromised health and well-being, due to occupational health and safety risks, precarity (for example, psychological distress, lack of health and social insurance) and platform-based vulnerabilities (for example, work-related stress from surveillance; social isolation and financial risks) (759).

Several jurisdictions have begun taking steps to recognize gig workers, improve their working conditions and enhance their access to social protection. In 2023, **Australia** adopted legislation to enable the independent workplace tribunal to set minimum standards for “employee-like”

workers that work for or via a digital labour platform (760). **Argentina and Chile** have enacted laws covering working hours, rest periods, remuneration and leave entitlements for gig workers (761, 762). In **Seattle, USA**, the local Office of Labour Standards addresses offences against gig workers related to pay and leave entitlements and working conditions, and has established minimum pay rates for transportation company drivers, along with mechanisms to advocate for drivers' rights (763, 764).

Legislative efforts to address unfair employment conditions for gig workers have not kept pace with the growth of the platform economy. The effectiveness of these policy initiatives is still in the early stages. However, the introduction of bold industrial relations policies is essential to protect those in this growing employment sector.

Big data and artificial intelligence

Digitalization and AI are changing the information and political decision-making landscape, including for health, with the potential to undermine trust in institutions. The dissemination of misinformation poses unique threats to health, as evidenced during the COVID-19 pandemic (765). Algorithmic governance is an emerging area that requires particular attention, with specific efforts needed to combat the algorithmic bias that has contributed to health inequalities (766).

Big data can support more responsive and accurate policies and programmes. For example: Georgia has implemented blockchain technology to create a transparent, secure and corruption-free system for tracking land ownership and transactions (767); AI systems in many locations help prioritize medical treatments and manage health care resources more efficiently (768), and the Scottish government uses geospatial big data to identify strategic outcomes and inform community funding decisions (769). However, governance of the sector is needed so that tools and analyses are developed and shared with those who will benefit from them, and are built with diversity in mind. Tools built on big data and AI algorithms must be monitored to ensure that they do not reinforce historical discrimination.

In “Our Common Agenda”, the UN Secretary-General proposes that “every adult should have affordable access to digital networks, as well as digitally-enabled financial and health services built on emerging principles and best practices” (770). Drawing on this, the Global Digital Compact adopted at the UN’s Summit of the Future in September 2024 set a new benchmark for inclusive digital transformation, which aims to bring together governments, private-sector entities and civil society organizations in a joint commitment to increase connectivity for all; prevent the fragmentation of the internet to maintain a unified global digital space; protect individual privacy and data, and promote the ethical development and use of AI. The compact recognizes digital technologies as a global public good and encourages their development and use for the benefit of all, while applying human rights principles in the digital sphere, including freedom of expression, privacy, and protection from discrimination and misleading content (771).

An increasingly complex health information environment

Health information has always been an important determinant of health. Lack of access to timely and accurate information has deepened health inequities and been a marker of persistently poor health outcomes. However, the proliferation of health information channels and the explosion in the volume of health information has created new challenges, as well as opportunities, in addressing the determinants of health. The emergency context of disease outbreaks and pandemics shows clearly how crucial good health information is for rapid and effective responses, but also how these responses can be undermined by the malign influence of misinformation and disinformation. These health information issues cross all manner of health conditions and their determinants, including nutrition, physical activity, injury prevention and mental health (772).

Much of the effort to manage health information, including misinformation and disinformation, has fallen to health institutions, which are often not equipped to deal with the contemporary information environment. Efforts focused only on better communication of accurate health information, including techniques such as fact checking, are not adequate to deal with the volume and diversity of misinformation and disinformation (773). Instead, efforts need to be able to tackle the drivers of health information production and consumption. The fact that misinformation is 70% more likely to be spread than verifiable information, and that falsehoods spread six times faster than truths, illustrates the scale of the challenge (774). New partnerships are needed between health authorities, the private sector (including platform owners) and civil society, to establish the governance frameworks that can systematically promote good information and call out bad information, in particular regarding its use to exacerbate health inequities.



PART 1:

The state of social
determinants of
health equity

PART 2:

Tackling the social
determinants of
health equity on
multiple fronts

PART 3:

Bringing about
change through
new governance
approaches

CHAPTER 6:

Local government, communities and civil society taking action



6.1

Releasing the power of local government


RECOMMENDATION 6.1
Equip local government to reduce health inequities

- 6.1.1 Strengthen the role of local government to implement community-centred actions for health equity.
- 6.1.2 Ensure healthy housing and built environments, including through universal design principles.
- 6.1.3 Equip local government to support age-friendly communities and combat social isolation and loneliness.

Local government is recognized as an essential institution in the design and implementation of policies to improve health and well-being through the social determinants of health. The movement towards the “localization” of the SDGs – which has significant crossovers with the social determinants approach – reflects this (775). In many countries, local governments manage services that have a direct impact on people’s lives, including urban planning, housing, transport, social services, and early childhood and skills development programmes.

Given their proximity and accountability to communities, local governments are more inclined to understand the interconnected nature of the social determinants of health equity, and the barriers that hinder progress in achieving it. By creating alliances and motivating stakeholders, local governments can strengthen the design and implementation of programmes (776). This improves accountability in service provision and facilitates democratic participation (777).

Cities offer significant opportunities to transform a nation’s health and ensure greater health equity. In several countries, public health responsibilities have been devolved to local governments, which are responsible by law for acting to promote the health and well-being of citizens in their area. In some places, such as in Norway, local governments are specifically mandated to reduce social inequalities in health (778, 779). Local-level action can improve key social determinants and health outcomes (780, 781), as shown by cases in the British city of Coventry (782), the Colombian municipality of Paipa (783) or in Chile (see Box 27).

BOX 27. Chile: Connecting social and health services at the local level

In Chile, health services and social support have historically been fragmented, creating significant barriers for vulnerable populations in accessing essential benefits (784). Many people, particularly those facing economic hardship or health challenges, struggle to navigate the complex systems of social services and health care. This lack of coordination often leads to missed opportunities for timely support, exacerbating health disparities and limiting access to critical resources.

In response, including as part of the SDH Special Initiative, municipalities across the country have implemented a local intersectoral management model, *Gestión Social Local*, to integrate health and social services using three main tools (785):

- ▶ An integrated platform that securely connects individual and family social and health information with public benefits data, establishing referral procedures among various municipal and primary health service providers.

- ▶ A comprehensive catalogue of benefits, services and social support, integrated into the platform, that facilitates referrals and ensures families have access to all social and health benefits they are entitled to.
- ▶ A locally agreed referral system for connecting individuals with necessary services.

The WHO country office in Chile reports that as of August 2024, 324 municipalities had adopted the *Gestión Social Local* model, with 209 receiving ongoing support and training. Approximately 6000 municipal staff members were trained as *Gestión Social Local* operators, significantly increasing the efficiency of service delivery. WHO staff also report that data from a pilot programme implemented in 13 municipalities showed a successful referral rate between social and health units, with specific categories identified for social assistance, such as economic hardship and health-related financial support.

To fulfil their role in supporting the social determinants of health and health equity, local governments need sufficient funds – via taxation or transfers from the national government – and a supportive relationship with other levels of government (48, 777). However, wide cross-sectoral responsibilities are often not accompanied by sufficient funding (48, 786). Higher per capita municipal spending on services including education, housing and libraries has been associated with lower odds of severe maternal morbidity in the United States (787). Conversely, in England, cuts to local government are correlated with deteriorating health and exacerbating inequity (788).

The concentration of ongoing poverty and poor social determinants of health which characterizes rural living conditions for much of the world's population also needs to be addressed, and there are promising examples of action. In South Africa, the Rural Health Advocacy Project produced a guide to meeting the needs of rural contexts in health policy, strategic planning and resourcing (789). Similarly, in the United States, the Rural Health Information Hub's "SDH in Rural Communities Toolkit" provides resources and programme models to address social determinants in rural communities (790).

BOX 28. Lao People's Democratic Republic: The CONNECT trust-building initiative

In Lao People's Democratic Republic, essential health care access is challenged by social determinants of health such as poverty, education, remote locations, transport costs, distance to health services, lack of emergency health funding, lack of accessible health information, and low levels of trust in health providers. In a culturally diverse society, language barriers, early marriage, adolescent pregnancy and existing gender norms among some communities also impact use of health care, especially among pregnant women.

Service uptake varies widely by socioeconomic status, ethnicity and geographical location – for example, the proportion of Lao ethnic children fully vaccinated is more than double that of some other ethnic groups (791, 792). During COVID-19 there was further disruption of essential health care uptake, such as routine immunization, antenatal care and delivery with skilled birth attendants (793), and increased inequality due to the economic impacts of restrictions and inflation (794). Previous interventions aimed at promoting community health have mostly failed to result in sustainable change, partly because narrow technical solutions were deployed to tackle health governance issues, resulting in limited systematic impact (795).

The Community Network Engagement for Essential Healthcare and COVID-19 Responses through Trust (CONNECT) was developed by the Lao People's Democratic Republic Ministries of Health and Home Affairs, with WHO support (796). Development started during COVID-19 and became part of the SDH Special Initiative in 2021. The network aims to enhance existing governance and community structures and capacities to address underlying social determinants of health, while also strengthening trust between villagers, health services and local government. Through a series of participatory workshops, it supports community representatives, health workers and local government staff to build trusting relationships and identify context-specific social determinants of health, to co-design solutions.

Evaluations of CONNECT (797, 798) show that the initiative successfully addressed language and cultural barriers through an adaptable, non-literate approach, using translation, accessible health information provided through audience-friendly two-way communication, and working with local leaders to build relationships and respectful communication.

The impact of the approach was seen in an outreach campaign in northern Lao People's Democratic Republic, where COVID-19 vaccination uptake increased seven-fold compared with previous rounds among remote ethnically and religiously diverse populations (799, 800). Home visits and face-to-face communication helped deliver accessible health information to people with disabilities, as well as identify vulnerable families and their support needs and opportunities.

In southern Lao People's Democratic Republic, the WHO country office reports that community-led transport support schemes were developed in collaboration with health centres to enable pregnant women in remote communities to identify due dates and access health centres for delivery. Village funds and joint saving schemes were also established for families requiring support to access health care. In an area where women had preferred to give birth at home rather than in the health centre, due to lack of trust in health providers, the team connected the local birth attendant with the centre and worked on a new approach towards pregnant women, focusing on feelings and respectful care. WHO staff also report that in areas where early marriage and adolescent pregnancy rates were high, village authorities provided respectful non-stigmatizing counselling jointly with the families of adolescents, offering support especially where the young couples were poor or had low education levels, leading to increased use of antenatal care.

At a national level, the Ministries of Health and Home Affairs are collaborating to provide family counselling services to young couples, to boost respectful family planning and registration.

Housing and transport

Local government responsibilities for the built environment, including housing and transport, are critical to health equity (801). Affordable housing, such as public housing, can be promoted through providing public housing directly; zoning for and funding a supply of appropriate dwellings; providing subsidies, such as housing vouchers, or using tax mechanisms, such as low-income housing tax credits (342).

As well as influencing rates of mortality and risk of noncommunicable diseases, housing interventions are key to combating infectious and vector-borne diseases. For example, in Zambia, window and door screening, combined with water and environmental management, reduced the incidence of malaria in children under 5 by 50–75% in a period prior to use of dichloro-diphenyl-trichloroethane (DDT) (802). Equipping housing with safety devices such as smoke and carbon monoxide alarms, stair gates and window guards reduces hazards that lead to accidental injuries (342). Interventions directed at improving warmth can lead to health improvement, while housing that is affordable to heat is linked to improved health, and may improve social relationships (803). Programmes to address homelessness are also associated with improvements in health equity, as demonstrated in New Zealand (Box 29).



BOX 29. New Zealand: Health and housing interventions

In Aotearoa, New Zealand, poor housing conditions such as mould, dampness and overcrowding contribute to health inequities, particularly among Māori, Pacific peoples and other communities experiencing marginalization (804).

Launched in 2017, the Housing First programme facilitates secure housing and wraparound support for people with complex health and mental health needs. It addresses homelessness by offering stable housing first, followed by tailored health and social services (805). Initially piloted in Auckland and funded by the central government, Housing First is now rolled out nationally. Five years after implementation, Housing First has reduced hospitalizations among study participants by 44%, mental health inpatient stays by 63% and criminal charges by 43% (806). It has also increased employment among previously homeless people (806).

Established in 2013 and expanded to the remainder of the country in early 2023, the Healthy Homes Initiative aims to increase the number of children living in warm, dry and healthy homes by addressing insulation, heating

and ventilation issues, enhancing their health and reducing housing-related hospitalisations (807). The programme focuses on pregnant people, low-income *whānau* (Indigenous extended family groups) with children aged between 0 and 19 years old who have been hospitalised with a specified housing-related condition, and individuals with, or at risk of, Rheumatic Fever. Contracted providers work with eligible *whānau* to provide education and access to interventions to create warm, dry and healthy homes, such as insulation, beds and bedding, curtains, and support finding alternative accommodation if needed. The programme operates through a partnership between providers, government agencies, and regional and local stakeholders. A five-year outcomes evaluation published in November 2024 shows that the initiative reduced hospitalizations by 18.6%, improved housing quality, and led to better school attendance and employment outcomes among families (808).

By improving equitable access to safe, stable housing, these interventions have contributed to improving equity, particularly for Māori, Pacific peoples and other marginalized groups.

More liveable and healthier places can be promoted through better transport planning (809). For example, modelling suggests that land-use changes towards more compact cities, accompanied by policies promoting walking, cycling and the use of public transport, would result in improvements in diabetes, cardiovascular disease and respiratory disease among citizens of Boston, Copenhagen, Delhi, London, Melbourne and São Paulo, with an overall gain of 420–826 disability-adjusted life-years per 100 000 people. Such policies should be paired with the provision of safe walking and cycling infrastructure, to avoid an increase in road trauma for pedestrians and cyclists, and increase physical activity levels (810). Exposure to trees in cities has also been associated with health and climate change mitigation benefits, but a health equity lens in planning new green areas is indispensable to avoid widening inequalities, since higher socioeconomic groups tend to live in areas with higher green density (811). WHO's Urban Health Initiative is modelling sustainable transport scenarios to reduce air pollution and improve health equity in cities across the world, including Accra, Ghana (see Box 30).

BOX 30. Ghana: Sustainable transport to reduce air pollution and improve health equity

Accra, Ghana, faces significant challenges with air pollution, primarily from vehicle emissions (812). It was a pilot city in WHO's Urban Health Initiative, which models sustainable transport scenarios to reduce air pollution and improve health equity using the Integrated Sustainable Transport Health Assessment Tool (49, 813).

The initiative proposed a shift from passenger cars to electrified public transport, along with increased walking and cycling. This approach aims to reduce emissions and promote physical activity among residents. Modelled impacts suggest these policies could result in up to

5500 averted deaths by reducing air pollution over 35 years, and avert an additional 33 000 deaths through increased physical activity due to enhanced walking and cycling infrastructure between 2015 and 2050 (813, 814). The economic value of these health gains is calculated to be about US\$ 15 billion over 35 years (815).

This substantial economic benefit underscores the value of investing in sustainable transport infrastructure, which not only enhances public health but also reduces health care costs and improves economic productivity.

Where communities experiencing marginalization are better able to participate in policy processes, transport systems are better able to improve equity and justice (816). Universal design principles can be mandated for urban planning and public transport investments, not only to ensure accessibility for people with disability, but also to promote the general inclusiveness of neighbourhoods and transport networks.

Healthy places across the life course

A successful approach to protect health and prevent diseases in cities requires action at all stages of life. For example, public spaces designed for children are crucial for their well-being and development, fulfilling their needs and rights to play, interact socially and engage in physical activities. By doing so, they greatly enhance the quality of children's everyday lives and support their personal growth and health. Likewise, ensuring that public spaces are breastfeeding-friendly will encourage this important practice, helping breastfeeding become the social norm. This is not just a matter of creating dedicated spaces, but also of ensuring urban settings – including workplaces, parks, shopping centres and community spaces – are accessible, comfortable and safe for breastfeeding.

Interventions to promote public spaces for breastfeeding mothers and for children can be initiated through mayoral visions; formal policies, plans and programmes; catalytic inputs from global and civil society organizations, and grassroots demand. Through the Child Friendly Cities Initiative, the realization of child rights is promoted at the local level via a unique network of municipal governments, civil society organizations, the private sector, academia, the media and children (817).

BOX 31. Brazil: Child-friendly public spaces in the city of Recife

Neighbourhood early childhood learning and development facilities can benefit early childhood development (818, 819, 820).

In the city of Recife in Brazil, the mayor initiated the *Primeiro A Infância* (“Early Childhood”) programme in 2017 to make public spaces safer and friendlier for young children and their caregivers. Child-friendly public spaces were piloted in two marginalized neighbourhoods, with interventions to claim public spaces for children in vulnerable hillside communities deprived of open space. Tactical interventions included improving stairways, sloping alleys and vacant land, and adding child-friendly elements in public spaces to make them engaging for children, mainly through painting bright murals on walls and floors.

Pre- and post-implementation surveys showed increased usage of newly constructed public spaces by children and their caregivers, as families found leisure areas on their doorsteps, and an increase in perceived safety in at least one neighbourhood. A survey conducted as part of the government’s *Mais Vidas Nos Morros* (“More Life in the Hills”) programme revealed the percentage of people feeling “happy” increased by 41.4%. People who spent time in public spaces rose by 65.7%, and 64.3% more children went outside the house to play (821, 822).

Rather than “reinventing the wheel”, the mayor of Recife built on existing initiatives, using the “child priority zone” model promoted by the Bernard van Leer Foundation’s Urban95 programme, and leveraging the existing “Mother Owl” programme for *Primeiro A Infância*.

“Age-friendly environments and communities... enable older people to age safely, while retaining autonomy and health”

Most older people want to “age in place” – to remain at home in their familiar surroundings and maintain the relationships that are important to them. Age-friendly environments and communities that promote inclusion of all older people, and remove barriers to accommodate the loss of capacity, enable older people to age safely, continue to develop personally and contribute to their communities, while retaining autonomy and health. Policies that include integrated services for older people within national universal health coverage (UHC) schemes, and make lifelong learning opportunities available, promote healthy ageing (823).

By acting on physical and social environments, cities can play a major role in contributing to healthy ageing, improving access to services and allowing people to age well (824). Age-friendly cities and communities can help address ageism, better integrate health and social care, enable people to choose where they want to live as they age, and create more resilience in the face of humanitarian crises and emergencies such as the COVID-19 pandemic (824). It is essential that age-friendly communities are attentive to equity and that low-income and excluded groups also benefit from them, including those marginalized due to ethnicity, race, sexual orientation, gender identity, expression or sex characteristics.

6.2

Community empowerment and civil society



RECOMMENDATION 6.2

Support community engagement and civil society

6.2.1 Incorporate representative community engagement and social participation in policy processes and delivery.

6.2.2 Create the enabling conditions that maximize the capabilities of independent and inclusive civil society to promote health equity.

Good governance for action on the social determinants of health requires the meaningful engagement of people, communities and civil society in decision-making – from priority setting to policy formulation, through to implementation and review. Civil society and community-based organizations have made improvements in the social determinants of health possible through advocacy, monitoring, mobilization of communities, technical support and training, and– crucially – by giving voice to those who are most disadvantaged (825). At the Seventy-seventh World Health Assembly, Member States committed to strengthen and sustain regular and meaningful social participation processes for health (826). Governance systems must have clear and transparent accountability to local communities to ensure health and well-being, and action on the social determinants of health equity (777, 827).

Besides public and community engagement initiatives, civil society organizations are well positioned to represent and support the engagement of excluded communities, and can often deliver services at a lower cost, with greater efficacy, than other sectors (828, 829). Civil society has long been recognized by WHO as a key actor in the design and implementation of policies for health and well-being (830). WHO has set up a Civil Society Commission to support it in its engagement with civil society to achieve health for all (831). Civil society organizations can also be mediators in complex and contested policies, bringing a strong advocacy role and diverse views and expertise to policy-making (828). Grassroots and community-based organizations are vital for understanding and building awareness of the determinants of health inequalities, and pushing for inclusivity and change at the local and national levels (832, 833). For example, in the United Kingdom in 2024, the Make Health Equal campaign, run by a coalition of academia and civil society organizations, was effective in using mixed media to

raise awareness and concern among the general public about key social determinants of health, helping shift views away from individual responsibility for health outcomes (834).

The Make Health Equal campaign to raise awareness of health inequalities and differences in life expectancy in London and Manchester, United Kingdom



Health Equals' #MakeHealthEqual campaign billboard in Manchester, United Kingdom, 2024
Infinity Outdoor Ltd



Health Equals' #MakeHealthEqual campaign launch in London, United Kingdom, 2024
Health Equals / PA Media/ David Parry Media Assignments

“Without social participation, ‘there can be no true health’”

As well as highlighting the vital role of local government, the COVID-19 pandemic brought to the forefront the crucial role of civil society organizations globally, supporting the engagement of excluded communities, providing social assistance, responding to health care needs, and coordinating and collaborating with national and local governments and businesses (835). Many civil society groups expanded their missions to respond to the challenges of the pandemic, important elements including the use of their pre-existing relationships and participatory platforms and spaces with regional, national and local governments, and strong integration within the communities they serve (836). In many cases, civil society moved quickly to fill the gaps left by slow-moving authorities (837, 838, 839). For example, the emergence of collective kitchens led mostly by women to fight food insecurity shows how civil society self-organized to mitigate the impact of the pandemic on the conditions of daily living (840). Community-based organizations are especially well positioned because of the trusting relationships they build. As key contributors to social infrastructure, they can offer lessons for governments on building trust (841).

Excluding diverse and representative communities and their trusted leaders or representatives in defining problems and designing interventions can result in increasing health inequities (842). Power and power relations influence the availability and nature of participatory spaces for health, determining who is included and who is not, often aggravating social inequalities (843). For example, governments can establish legal barriers that prevent people and organizations from fully participating, damaging social networks and impeding opportunities for community groups to express their views and be heard. Other barriers can include a lack of resources and failure to provide sufficient resources to community organizations. Social participation, defined by WHO as “empowering people, communities and civil society through inclusive participation in decision-making processes that affect health, across the policy cycle and at all levels of the system”, is necessary to redress power imbalances, address conflicts of interest and create safe spaces for participants to meaningfully engage and influence decisions. As noted by WHO, without social participation, “there can be no true health” (844), but it requires political commitment, government capacities, time and material resources. Health for all is not primarily a technical endeavour, but one that requires political navigation and influencing those with power, as has been advocated by longstanding global networks and movements, such as the People’s Health Movement (845). In contexts with weak governance systems, community-led initiatives have achieved gains in health accountability by confronting power (846, 847).

BOX 32. Tunisia: Societal dialogues for health

Following the 2011 revolution, Tunisia faced widening socio-economic disparities, inequities in health care access, and increasing noncommunicable disease rates (848). These issues, coupled with deteriorating public health care and exclusion of marginalized groups, became central to public debate.

The Societal Dialogue, a participatory governance process, was launched in 2012 to reform health care, ensure coverage for

vulnerable populations, and foster citizen participation (849). It involved three phases: a health system diagnosis, development of the National Health Plan 2030, and the plan's implementation. Over 3400 participants contributed to defining the strategic direction for Tunisia's health system (849). The focus on social determinants represented a shift in the country's approach to health care, emphasizing a people-centered system, accountable to citizens and aiming for equitable access and financing.

The degree to which people and communities are facilitated and supported to influence a country's governance processes and the decisions that affect their lives is an important aspect of governance for health equity (850), as in Tunisia (see Box 32). Examples of good practice show that investment in social participation, civil society networks and capacity building, and sustained participatory platforms are the vital underpinnings to build trust (851).



The United Nations High Commissioner for Human Rights identified five essential ingredients to optimize civil society's transformative potential: a robust legal framework, compliant with international standards, that safeguards public freedoms and effective access to justice; a political environment conducive to civil society work; access to information; avenues for civil society participation in decision-making processes, and long-term support and resources for civil society (852).

Evidence shows that community engagement, when rooted in the continuous and intentional process of relationship building between stakeholders, promotes good understanding of locally specific needs and aspirations. It enables different types of actors to collaborate, share resources and work together to achieve joint health goals, and helps to address social determinants of health equity (853, 854, 855), including in specific areas such as:

- ▶ child health (856), child immunization and vaccine equity (857, 858, 859)
- ▶ the prevention and control of COVID-19 (860, 861)
- ▶ mosquito-control interventions (862)
- ▶ sexual and reproductive health interventions in conflict settings (863)
- ▶ the design, implementation and evaluation of nutrition interventions to reduce chronic diseases in Indigenous populations in the United States (864)
- ▶ the reduction of population weight gain (865).

Effective community engagement requires systemic implementation. In Brazil, for example, formal social participation mechanisms create a space for community engagement that is integral to health system governance nationwide (827). It is complemented by participatory budgeting, which has been in place since 1989, with evidence that involved municipalities were highly responsive to popular preferences when allocating expenditures, and spent more on sanitation and health services, seeing a reduction in infant mortality (866).

In relation to other health determinants and outcomes, robust evidence from the United Kingdom indicates that community engagement can improve housing management, has the potential to increase the quality of local services, and improves community well-being (867). Community-based participatory research and community-engaged research are widely used approaches, with numerous systematic reviews documenting positive outcomes (868). The systematic use of behavioural insights can also help inform community actions. The development of effective

“A lack of social connection is one of the key mechanisms that create and perpetuate health inequities”

solutions will necessarily be place-based and often complex, requiring policy-makers to overcome the bias towards well-tried but not necessarily effective solutions (869).

A lack of social connection is one of the key mechanisms that create and perpetuate health inequities, but a range of actions can be taken by communities and government, especially at the local level, to build social connection, as outlined in Box 33.

BOX 33. Social connection and health equity

Social isolation and loneliness are pervasive in most places in the world, negatively impacting physical and mental health and longevity (223). In OECD countries, there is consistent evidence that interventions providing social support (such as befriending activities or social skills development) and those directed at expanding someone's social network and connections decrease loneliness in the short term (870, 871).

Effective interventions can be implemented across the life course and in different groups, including people with mental illnesses, disabilities or chronic conditions; parents and caregivers, and immigrants and refugees (147, 872). Community-level strategies can include action on transport, digital inclusion and the built environment, as well as promotion of volunteering and age-friendly communities (223, 873). For example, local community facilities and active engagement in green and “blue” (waterside) spaces can be used to promote the social connectedness and mental health of specific groups (874), while intergenerational mixing spaces can support stronger social cohesion (875). This is particularly important for population groups at greater risk of social disconnection, such as those with disabilities or mobility issues, older people and those with chronic illness. Evidence shows that among older individuals, green communal spaces and allotment gardens contribute to social inclusion

and a sense of community, while living closer to local resources and perceiving local areas to have good facilities are associated with greater levels of social participation (876, 877, 878).

Incorporating design elements that both encourage social interaction – such as seating areas, good acoustics and features such as effective lighting that promote feelings of safety – and address barriers to access and use among marginalized populations helps strengthen social infrastructure. Improving local transport systems – for example, through free or subsidized use of travel for certain population groups – can encourage use of local facilities and connect people. In England, age-friendly transport policies that provide free bus passes on local services to those aged 60 years or over have not only increased transport use, but have improved contact with friends and family, and reduced feelings of loneliness in this age group (879).

While much of the evidence-gathering in relation to interventions to address loneliness is from high-income countries, countries at all income levels face similar underlying issues. The “Friendship Bench” initiative in Zimbabwe used grandmothers as lay counsellors and had a significant impact on mental health symptoms among participants. The model has since spread to Malawi, Tanzania and New York City (880, 881).

BOX 34. Germany: Federal Government's Strategy to Counter Loneliness

In Germany in 2023, the Federal Government adopted its Strategy to Counter Loneliness (882), through cross-sectoral social action and integrative planning at local district level. Beforehand, the Federal Government went through a large-scale consultative process. The strategy pursues 111 measures within five objectives to strengthen social connectedness and social interaction, in order to prevent and alleviate loneliness. Among other factors, the strategy describes the potential of integrative social planning at the municipal level, with urban development funding supporting social infrastructure. It emphasizes a holistic approach,

integrating various sectors to address loneliness, including health care, social services, urban planning and community organizations. It aims to equip service providers with the necessary skills to identify and support individuals experiencing loneliness, promoting integrated social planning at the local district level to create environments that support social interaction and connectedness. This includes designing public spaces, community centres and transportation systems that facilitate social engagement among residents, particularly the young and older people.

IMMUNISER C'EST PROTEGER VOS ENFANTS



CHAPTER 7:

Build a health and care sector that ensures action on the social determinants, equitable access and genuine participation



7.1

Universal health coverage, financing and primary health care

**RECOMMENDATION 7.1**
Achieve universal health coverage through progressive health financing and primary health care approaches

- 7.1.1 Improve equitable access to a continuum of quality primary health services addressing both physical and mental health.
- 7.1.2 Use pooled government resources to minimize out-of-pocket costs and finance health services.
- 7.1.3 Increase the share of health and care sector funding to meet essential needs, to avoid marginalization and discrimination.
- 7.1.4 Facilitate equitable access to health technologies, from research and development through to manufacturing and delivery.

Ensuring UHC through the primary health care (PHC) approach improves health and health equity. There is a clear association between life expectancy at birth and UHC indicators in OECD countries and others, such as Brazil, China and India (883). In recent years, the extension of coverage has stalled, and financial protection – ensuring that people do not avoid necessary health care due to financial barriers – has worsened (884). In 2021, about 4.5 billion people – half the global population – were not fully covered by essential health services (884). The most cited barriers to achieving UHC include: financing mechanisms and their effects, such as patients' out-of-pocket expenditure and impoverishment; governance issues, including policy inconsistency; inequity in income or service coverage; weak regulation and supervision mechanisms, and poverty (885).

Out-of-pocket expenditure and co-payments remain a significant burden on populations, pushing many people into greater poverty and worse health. They are a notable barrier in accessing health care across the globe, acting as both a determinant of health inequities, and an obstacle to delivering universal health services to the whole population. The number of people globally living in households which experience catastrophic out-of-pocket health expenditure was estimated to be 2 billion in 2019, with

poorest households more likely to incur this financial hardship (884). Even small expenditures in absolute terms can be devastating for poor families; around 1.3 billion people were pushed into poverty, or were already poor and saw their situation worsen, due to such payments, including 300 million people already living in extreme poverty (884).

Some analysis has suggested that the commitment to equity within UHC policy documentation is largely rhetorical, and, as a consequence, impacts are not substantive enough (886). WHO has developed guidance on equity and UHC which encourages countries to: categorize services into priority classes and expand coverage for high-priority services to everyone; eliminate out-of-pocket payments, while increasing mandatory, progressive prepayment with pooling of funds; and take measures to ensure that disadvantaged groups are not left behind (887).

The effective realization of UHC requires a fair financing system, which distributes the burden according to ability to pay, and the benefits of spending according to need (888). Progressivity in health financing means that households with higher incomes contribute a higher share of their income than those with lower incomes (888). The Lancet Global Health Commission on financing PHC asserted that in most low- and middle-income countries, out-of-pocket payments must be reduced to a level that can only be achieved through increasing allocations to PHC from general taxes, as set out in Chapter 3.



High levels of out-of-pocket expenditure indicate a poorly functioning and inequitable health system. Conversely, those health systems that have introduced universal coverage schemes to cap out-of-pocket costs have seen rates of impoverishment fall, and improvements in equitable access to services. An independent assessment concluded that after implementing its Universal Coverage Scheme, Thailand was able to reduce catastrophic health expenditure from 6.8% to 2.8% among those members in the poorest quintile, as well as impoverishment as a result of payment for medicines and health services (889).

In low-income countries, both more domestic financing and more development assistance are needed to expand resources for PHC (890). Several health reforms, including new forms of financing, have been implemented with positive effects in the Islamic Republic of Iran (891), and in some African countries, national health insurance schemes have generally reduced catastrophic health expenditure. However, some feature important gaps, leaving behind disadvantaged groups, such as informal workers, rural residents or those with lower educational levels (892, 893).

Medicines constitute a high proportion of out-of-pocket health expenditure – as much as 70% in some cases. Medication expenditures for patients suffering from diabetes, one of the most important noncommunicable diseases driving out-of-pocket treatment costs in low- and middle-income countries, can push households to reduce expenditure on other basic needs, such as food (894). Not only are households impoverished by drug costs, but spending is often wasted on ineffective remedies, with the most vulnerable populations at risk of exploitation by unscrupulous operators, and prey to the unregulated sale of counterfeit medications (895).

Primary Health Care approaches

PHC is situated at the interface between health services and the social determinants of health equity. The PHC approach is a whole-of-society approach to health that aims to maximize the level and distribution of health and well-being through three components: (a) primary care and essential public health functions as the core of integrated health services (b) intersectoral policy and action on the social determinants of health equity, and (c) empowered people and communities (162). PHC thus extends beyond service delivery to outline an approach for the whole system, incorporating intersectoral action, as well as empowering people and communities as co-creators of their health. The service delivery component itself is also broader than primary care services alone, and includes public health services that have a population perspective and natural synergies, and require a coordinated approach for integrated delivery (163).

“Countries that have implemented primary health care, especially at the first level of contact with the health system, have healthier populations, fewer health-related inequalities, and lower health care costs”

Countries that have implemented PHC, especially at the first level of contact with the health system, have healthier populations, fewer health-related inequalities, and lower health care costs (896, 897, 898). PHC-focused initiatives in low- and middle-income countries are linked to improvements in access to health care among the poor, reductions in child mortality and, in some cases, reductions in wealth-based inequalities in mortality (896). The adoption of strong PHC models has been associated with equity, both in low- and middle-income countries (899) and in Europe (900). Estimates indicate that closing the funding gap in PHC would avoid over 60 million deaths worldwide and increase global life expectancy by 3.7 years by 2030 compared to a 2015 baseline (901). Integrating mental health services in primary care has several advantages, including reducing stigma for patients and their families, improving access, improving the treatment of patients with comorbidities and, crucially, strengthened prevention, as PHC workers are frontline professionals providing the first level of contact for individuals, their families and the broader community (902). PHC organizations can also gather data on local social determinants of health to design and deliver new interventions aimed at reducing health inequalities in noncommunicable diseases and improving population health outcomes (903). Boxes 35 and 36 showcase examples of PHC approaches improving health equity.

BOX 35. Chile: Establishing the social determinants of health equity in PHC reform

Action on the social determinants of health equity has been identified as a key axis in the Chilean health reforms currently underway. Chile is a pathfinder country in the SDH Special Initiative (15).

Seven municipalities in three regions of the country are building up a theory of change to address the social determinants of health equity. They have established multisectoral teams, supported by regional equity committees, to bring together community leaders, civil society, academics and policy-makers to conduct a

situation analysis of the main health inequities that affect their districts and design a response action plan.

A diverse set of problems has been prioritized through these workshops, including access to quality water; caregiver health and support, and the retention of children in vulnerable situations in the preschool education system. Community asset mapping and social determinants monitoring systems anchored at the local level are being developed to guide the deployment of local action plans over the next two years.

BOX 36. Thailand: Achieving Universal Health Coverage through strong Primary Health Care

Thailand has prioritized long-term investment in resilient health systems and achieving Universal Health Coverage (UHC) and social protection for all citizens (904). By 2002, the entire population was covered by three public health insurance schemes: civil servants and their dependents by the Civil Servant Medical Benefit Scheme; private-sector employees by the Social Health Insurance Scheme, and the rest of the population by the Universal Coverage Scheme – including underprivileged and vulnerable populations. These schemes aim to ensure health care access for all citizens, offering benefit packages and operating mechanisms that allow UHC beneficiaries to access health services without facing bankruptcy, and emphasizing health promotion, disease prevention, treatment and rehabilitation (905).

These efforts contributed to an increase in the UHC service coverage index from 68% in 2010 to 82% in 2021, against a global average of 68% in

2021 (906), alongside a decrease in incidence of catastrophic out-of-pocket payments from 2.8% of total households in 2010 to 2.1% in 2021 (vs. a global average of 13.5% in 2021) (907).

Access to quality primary health care services, especially in remote and rural areas, has been a key priority for Thailand. In 2019, the country introduced legislation known as the Primary Health Care System Act, which advocates for a family doctor care team, to positively impact access to quality primary care for all (908). An extensive network of well-functioning primary health care facilities that meet people's health needs, regardless of age, income, gender or ethnicity, is crucial for achieving UHC and health equity.

Health systems that truly deliver on equity and discrimination

Discrimination based on gender, caste, ethnicity, disability, Indigeneity, migratory status, race, age, religion or skin colour affects health systems worldwide (909), negatively impacting their availability, acceptability, accessibility and quality of care (910, 911, 912, 913, 914, 915, 916, 917, 918, 919). Health systems should redress health inequities and avoid reinforcing them through sexism, racism, ableism and other manifestations of discrimination. Gender discrimination is a strong determinant of health care access and outcomes (920). Although the elimination of user fees for maternal and child health services has reduced barriers, women in many low-income settings who are poor, adolescent, migrant or from ethnic minorities still avoid health facilities for reasons including mistreatment and abuse (920). Poor women and those from rural areas are more likely to work in informal employment and be excluded from health protection schemes. Women are less likely than men to be diagnosed with cardiovascular disease, although as many women as men are affected. Gender inequality and discriminatory laws and policies continue to impede access to sexual health and HIV services in many parts of the world (920).

As successful responses to the HIV epidemic have demonstrated, approaches to disease control are more effective and sustainable when they affirm the needs and perspectives of key populations (921). In the context of HIV, explicit responses to the needs of men who have sex with men, people who inject drugs, and sex workers were pivotal on the pathway to eliminating HIV as a public health problem (922).

Efforts are needed to recruit health and care workers more equitably, including from diverse racial and social class backgrounds (923). Evidence from the United States shows that Black, Hispanic/Latinx and Native American people are more likely to work in underserved communities; a more diverse workforce can be equipped with a diversity of perspectives to address structural biases in health care systems (924). In addition, providing culturally safe, trauma- and violence- informed, and contextually tailored care are key dimensions of equity-oriented PHC for Indigenous populations (925). In New Zealand, training in cultural safety is a requirement for the registration of nurses (926). In Guatemala, Community Defenders for the Right to Health, leaders in their own communities, address systematic exclusion and discrimination against Indigenous communities (847). They gather complaints from service users, such as not receiving required medicines or being disrespected by providers, identify recurrent patterns and, together with advisors from a local civil society organization, bring them to public officials for resolution (847).

There is increasing acknowledgement that community health workers can effectively deliver a range of prevention and treatment services (927). Evidence shows that community health workers are effective in reaching marginalized groups and extending health care access to those living in rural areas, with lower educational or socioeconomic status – although their impact on health equity within these groups can still be improved (928). In Brazil, where community health workers cover 67% of the population, their introduction produced significant health improvements, reduced inequity in health care use, and allowed for the collection of more accurate mortality data (929). WHO has produced guidelines to optimize community health worker programmes, with recommendations including the engagement of communities in defining their health needs and in selecting community health workers and mobilizing local resources (927). Local autonomy over resource allocation has also been linked with lower rates of people needing ambulatory care among First Nations communities in Canada (930).

Achieving greater health equity requires going beyond interventions to improve individual health outcomes. It involves meaningful engagement in the political process to generate the structural changes set out in this report, and to shape the way health care systems are financed, and the distribution of resources within them (931). Community control of health services can be effective with sufficient public funding (see Box 37).

BOX 37. Australia: Aboriginal Community Controlled Health Services

In Australia, Aboriginal Community Controlled Health Services emerged in the early 1970s to address racism, poor access to mainstream health services and low-quality care, and the impacts of ongoing colonization (932, 933, 934). The first such service, the Aboriginal Medical Service Co-operative, opened in Redfern, New South Wales, in 1971 (935) as part of a social movement emphasizing community control and self-determination for Aboriginal and Torres Strait Islander peoples. This was influenced by the Redfern Black Power movement, following Charles Perkins's 1965 "Freedom Ride" civil rights protest rally in New South Wales (934, 935, 936).

Aboriginal Community Controlled Health Services quickly spread, and today, there are over 145 across Australia, representing the most successful

and comprehensive model of primary health care in the country (937). Primarily funded by different levels of government, these services operate under principles of self-determination and community control, providing culturally safe health services that strengthen connections to culture and land (934, 938, 939). Alongside treatment, they focus on disease prevention and health promotion, and address local social determinants of health equity, such as employment, housing, alcohol supply and the justice system. They are significant employers of Aboriginal and Torres Strait Islander peoples (933, 939). The governance structure of Aboriginal Community Controlled Health Services includes community members through boards and other engagement methods, such as forums and feedback mechanisms, ensuring services are aligned with community needs (939, 940).

Equitable access to health technologies

Inequities in access to vaccines, therapeutics, diagnostics and other health technologies during the COVID-19 and mpox pandemics exacerbated health inequities between and within countries (941, 942). A key issue is whether access problems in low- and middle-income countries can be solved within the flexibilities of the current intellectual property system, or whether substantially new approaches are required (943).

A more fundamental resolution of inequitable access to medical technologies requires change in the balance of power, capacity and knowledge which makes it inevitable that medical innovation is unequally distributed. An argument has been made that for vaccine production systems to be truly equitable and provide health security to all countries, it is not enough to produce vaccines locally; rather, the production process needs also to be owned and run locally, and considered part of a national policy (944).

One proposal to build equity into global research and development in health suggests that all actors – public- and private-sector researchers; funders, policy-makers and regulators – should address a checklist for research and development in the global public interest, including that investments in capacity are made for all countries, and that all contributions are recognized fairly, and all benefits shared equitably (945).

Affordability can be built into product development pathways at the outset. For example, the Drugs for Neglected Diseases initiative has two fundamental principles in the drug development work it supports: the need to ensure that drugs are affordable and accessible in an equitable manner to patients who need them, and the desire to develop drugs as global public goods whenever possible (946). The vaccine development organization CEPI, which was one of the earliest movers to support vaccine development when news of SARS-CoV-2 first emerged, requires access considerations to be included in all its partnership agreements (947). However, these globally supported partnerships are the exception, rather than the rule, in prioritizing affordability in research and development aimed at biomedical innovation.

The COVID-19 pandemic also demonstrated the inequities in global knowledge distribution. As soon as successful vaccines were on the horizon, there were calls to decentralize production, addressing some of the most acute supply-chain issues. However, lead times to establish manufacturing capacities in low- and middle-income countries where these were lacking proved too lengthy. In a context where Africa produces less than 2% of the drugs used on the continent, the African Union had already recognized the need for an African Medicines Agency, which was established by treaty in 2019 and received sufficient national ratifications to come into force in November 2021 (948). The mRNA vaccine technology transfer hub is promoting equitable access (see Box 38).

BOX 38. The mRNA vaccine technology transfer hub

The COVID-19 pandemic highlighted global inequities in access to vaccines and health care technology (949). Developing countries often face challenges in acquiring advanced medical technologies and manufacturing capabilities, leading to disparities in vaccine availability and public health outcomes (950).

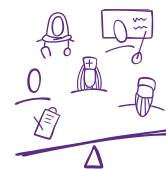
To address these inequities, WHO established the mRNA technology transfer programme, a global initiative that aims to improve health and health security by establishing sustainable, locally owned mRNA manufacturing capabilities in and for low- and middle-income countries (951). The programme is based around a technology transfer “hub”, Afrigen, located in South Africa. This will provide the technology development,

training and transfer, while 15-20 “spokes” in low- and middle-income countries across the world will receive training and technology from the hub, then produce and sell products commercially.

The hub has developed a new prototype COVID-19 vaccine that shows good in vitro results. Animal safety and immunogenicity studies are underway. Training and technology transfer has started for different manufacturers, and research and development work has started at the South African consortium on a tuberculosis mRNA vaccine. The hub has enhanced the technical skills of local scientists and health care professionals – capacity building that extends beyond vaccine production to other areas of biotechnology and public health.

7.2

The health and care workforce

**RECOMMENDATION 7.2****Build and retain a health and care workforce capable of delivering equity**

- 7.2.1 Recognize and reward work in the care economy, including informal care, and demonstrate the role of the health and care sector as a healthy and equitable employer.
- 7.2.2 Develop human capacity in health, social protection, education, labour, local government and service organizations, to enhance intersectoral efforts to address the social determinants of health equity.

While there is growing evidence for the importance of addressing social determinants to improve health and reduce health inequity, and there have been some successes, significant capacity building is required to improve social determinants across all sectors and levels of government. Health workforce education and training often fails to integrate the social determinants approaches, underscoring the need for curricular content and effective teaching approaches on social determinants, including in ongoing professional development across multiple roles, such as policy-makers, health system administrators, clinicians and social care workers (952, 953). A PHC-oriented health and care workforce is well placed to take intersectoral actions on health equity. It is important to engage with networks and initiatives for interdisciplinary and implementation research, to identify the multiple social determinants that influence health and health inequities, and use evidence to inform interventions. The health workforce must also build capacity to ensure civil society and public participation in the design and delivery of health care, and to enable effective intersectoral collaborations.

Medical professionals can be effective advocates of the social determinants of health approach, as those working in more deprived communities directly see in their daily practice the effects of poverty and inequalities on health. They are usually well-respected and listened to by authorities, hence the importance of different professional bodies, including the World Medical Association, stating their commitment to reducing health inequities by acting on social determinants (954, 955, 956). However, for this action to take place effectively in everyday practice, it is necessary to develop a health workforce with an adequate mix of disciplines, to train practitioners in the social determinants of health equity and to add related activities to job descriptions (957). At the local level, the health workforce can advocate for healthier communities by making the case for action on issues such as housing, access to green

“The health workforce can advocate for healthier communities by making the case for action on issues such as housing, access to green spaces, and air pollution”

spaces, or air pollution. To make such advocacy possible, they need to be supported in their roles and provided with adequate training and resourcing (see Box 39).

Approaches to addressing the social determinants of health equity in curricula and training should include sharing best practices and experiences, and engaging stakeholders – including patients, families and professionals from other sectors – and ensuring the prioritization and participation of groups experiencing marginalization (958). There is a need to train more people from within the communities that they serve. Evidence shows that embedding social determinants training in medical schools, and training graduates who are committed to work in underserved communities, improves health outcomes in such places. For example, in the Philippines, mothers from communities receiving care from medical graduates and final-year students who received socially accountable health professional education were more likely to receive perinatal examinations and to have a youngest child with normal birthweight, who has received all immunizations and is still breastfed at 6 months of age, than those cared for by professionals not receiving this education. These communities were also more likely to have more equitable access to essential maternal health care services (959). The graduates trained in social accountability were also more likely to work in rural areas, came from lower socioeconomic backgrounds, and had more positive attitudes towards community service (960).

BOX 39. The SDH Special Initiative actions on training

The Pan American Health Organization has undertaken specific capacity building on intersectoral action with staff in over 16 pioneer municipalities that are leading redesign of systems under the SDH Special Initiative (961).

Similarly, in Morocco in 2023, the Ministry of Health, supported by WHO, trained a pool of 24 professionals from different ministerial departments, academia and non-governmental

organizations in intersectoral action to address the social determinants of health. Participants gained tools and knowledge to integrate considerations of health and health equity impacts into public policies. The course on Health in All Policies now forms the basis of training materials to be included in a platform at the National Public Health School to galvanize a network of experts around the social determinants of health.

In 2020, the global health workforce amounted to 65 million people (962). However, it is not equally distributed throughout the world or within countries, following an “inverse care” pattern, by which richer countries and the more affluent areas within all countries (for example, urban vs rural) with better population health have more physicians than those more disadvantaged areas where people have poorer health and need health professionals the most. In 2020, high-income countries had 6.5 times more health workers than low-income countries, accounting for population size (962). WHO estimates that by 2035, there will be a global shortfall of 12.9 million health care professionals, with low- and middle-income countries suffering the brunt of this shortage. These “inverse care” gaps need to be acted on to make the health sector more equity-oriented. Digital technologies, if deployed to support equity, ethics and integration with existing health teams, may contribute to improving equitable access to services in the context of workforce shortages (963).

**“Women face high levels of persistent discrimination
in the formal health and care economy”**

Gender equality in the health and care economy

Women face high levels of persistent discrimination in the formal health and care economy. While they represent almost 70% of the global health and social workforce (964), it is estimated that they hold only 25% of senior roles (965), while experiencing a 24 percentage point pay gap compared to men globally, accounting for age, education and other factors (964). This impacts women throughout their life courses, increasing the odds of their retiring into poverty (964). Ethnic minority employees are also discriminated against, with evidence widely available from countries such as the United Kingdom and the United States, where ethnic minority health care workers are less likely to be promoted and achieve leadership roles than their White colleagues, and are paid less for the same jobs (966).

Delivering UHC requires the creation of 18 million new jobs in health and social care, most of them in low- and middle-income countries, so barriers that keep women from entering and advancing in the system must be removed (967). Additionally, in the same way that having more women in political leadership positions improves educational outcomes and income prospects for girls, it is likely that having more women in health leadership will support greater health equity policies and practice (968).

Women's input into health systems has an annual financial value of over US\$ 3 trillion, half of which is through unpaid work (967, 969). Women perform around 75% of all unpaid care and domestic work worldwide (970). Their role in providing informal care is often not recognized, despite strong evidence indicating that unpaid domestic care work has a highly significant economic impact, estimated to account for between 17% and 32% of total GDP in Europe (964). However, unpaid care work is not included in GDP calculations, making it invisible in economic data used in policy-making. Unpaid work creates gender gaps in employment, work and low access to pensions (970), which are determinants of health. Informal carers suffer from higher loneliness levels (971), higher cardiovascular disease incidence (972), worse mental health (973), more emotional exhaustion (974) and Burnout Syndrome that negatively affects their quality of life (975). Negative health effects are especially seen in female and married caregivers, and those providing intensive care (976). Informal care and social costs should be incorporated into national accounts to adequately measure the economic impact of diseases (977, 978, 979, 980, 981, 982, 983, 984, 985, 986).

BOX 40. Uruguay: The National Care System to support unpaid caregivers and care workers

Time-use surveys conducted in Uruguay in 2007 and 2013 revealed that women spent two thirds of their week on unpaid work and only one third on paid work, while men experienced the opposite. In response, in November 2015, the country's Parliament adopted the Care Act (987) entitling all children, people with disabilities and older people to access care services. The law established the National Integrated Care System (*Sistema Nacional Integrado de Cuidados*), which was launched in 2016 (988) with the overarching goal of expanding existing care services for the dependent population and creating new ones. The system's objectives include to change the division of labour between women and men within households, to support unpaid caregivers, and promote the formalization of care work and workers' continuing education and training (989).

Uruguay's care system has an innovative governance model, which includes the Secretary of Care who executes the decisions made by the

National Care Board (*Junta Nacional de Cuidados*), an intersectoral body that integrates several government institutions (987). This board is advised by the Care Advisory Council, composed of universities, civil society organizations, trade unions and business chambers (990).

The most recent time-use survey in 2022 indicates improvements in co-responsibility between the state and families, and between men and women within households (991). The burden of unpaid work decreased for women and increased for men between 2013 and 2022, suggesting progress towards gender equality in care responsibilities. The formalization of paid caregivers has provided professional recognition and improved working conditions for care workers, contributing to better-quality care services. These services have also been expanded to ensure that all dependent individuals, regardless of age or disability, have access to essential care.

CHAPTER 8:

Governance and data for action

Recommendations to tackle the structural conditions that have led to the perpetuation of health inequities across the world are set out in this report. In order to deliver those recommendations and reduce inequities in health, strong governance for health equity is needed, informed by up-to-date, disaggregated health and social determinants data.



8.1

Governments in the driving seat

**RECOMMENDATION 8.1****Strengthen focus on social determinants in health systems and policy platforms**

- 8.1.1 Integrate addressing the social determinants of health equity in all health strategies, policies, emergency preparedness and response plans, and public health laws.
- 8.1.2 Establish coordination and accountability mechanisms for intersectoral collaboration and community engagement for health at all levels of government.

The social determinants of health equity are not a discrete “programme” that can be rolled out by a single agency: they can only be tackled through approaches that are prioritized, coordinated and coherent across government and other sectors. As set out in Chapter 3, an empowered and resourced public sector is crucial for addressing social determinants of health equity. National governments set economic and social policies, develop legislation and establish regulatory frameworks, and shape cultural discourses. As the relationship between health and income, level of education, or access to services is graded, the greater people’s social disadvantage, the worse their health. In order to raise and flatten the gradients, there is a need to use the principle of “proportionate universalism” for achieving greater health equity, so that programmes are universal, but delivered with greater intensity where there is greater need, and adapted to address specific vulnerabilities.

Health officials can help to convene, lead and encourage these intersectoral approaches, but impact will be limited without an overarching whole-of-government and whole-of-society government strategy that prioritizes social development, community well-being and equity for health. Governments must champion the formation of networks across departments and services working on equity and well-being.

“The social determinants of health equity are not a discrete “programme” that can be rolled out by a single agency: they can only be tackled through approaches that are prioritized, coordinated and coherent across government and other sectors”

A collaborative health sector

Health care organizations can shift their traditional focus on medical interventions to integrate the social determinants of health into their practice. This goes beyond intervening to redress inequity in access or treatment, although that is also essential. Partnerships between the health sector and other sectors – including local government, community organizations, workplaces and schools – can improve daily living conditions within communities, influencing people's health (992).

Embracing the PHC approach is crucial to this strengthened focus on social determinants in health systems and policy platforms. Doctors and health care professionals can incorporate the social determinants of health equity into their daily practice – for example, by making a detailed social history of the patient, in addition to their medical history (992). Recent examples of this include in Spain, where the Ministry of Health has produced recommendations for regions (which are mostly responsible for health) to collect data on social determinants as part of the patient digital health record, including income, migratory status, social participation, social and family support, and area deprivation (993). But individual doctors' work with their communities must be supported by the wider system's commitment to address determinants (992), with the health sector also playing a leading role in advocating for action on the social determinants of health equity at national and international policy levels, as well as in creating awareness among the public (992, 994).

The health sector is also responsible for a sizeable proportion of countries' budgets worldwide. As such, it can have a positive impact on health through procurement, requiring contractors to be good employers, and influencing the economies and conditions of local communities. The health sector is also an employer, a provider of services, the owner of buildings and other assets, and a leading or anchor organization in the community. Through all these different roles, it can promote positive change (995).

Prioritizing social determinants in policies, strategies and laws

Some countries and regions have integrated the structural determinants of health into their legislative frameworks – among them Norway (996), Quebec (997) and South Australia (998, 999) – using Health in all Policies (HiAP) as a key mechanism for cross-sectoral action (see recommendation below) (1000). The Norwegian example illustrates that embedding the social determinants in law is a necessary first step that raises awareness (1001) and helps keep the approach on the agenda, but in order to have impact, sufficient funds and the involvement of all administrative levels are needed (1002, 1003).

Colombia has integrated the social determinants of health as pillars in successive public health plans and strategies. The Colombian *Plan Decenal de Salud Pública* (Decennial Public Health Plan) 2012–2021 aimed at developing a model to manage the social determinants of health and

reduce health inequalities, while the subsequent 10-year plan for the period 2022–2031 explicitly aims to “strengthen intersectoral articulation to manage the social determinants of health” (1004). A narrative review of policies identified subsequent important advances at the national level, including the creation of an Observatory for Monitoring Health Inequalities, and the Health Equity Strategy (*PASE a la Equidad en Salud*) (1005).

Political differences in framing social determinants are important, and can help embed social determinants approaches in differing political contexts. Policy is substantially influenced by the way it is framed (1006). For example, in Scotland in 2012, legislation for minimum unit-pricing of alcohol gained enough support for approval when it was framed as an environmental and social issue. However, legislative changes to address obesity faced greater challenges in Australia, France and Switzerland when they were framed as addressing issues affecting individuals and amenable to health promotion and education strategies (1007, 1008). Human rights framing was used in Nepal to achieve support for legislation on gender-based violence (1009). Similarly, countries adopting a well-being economy approach have framed trade issues as being wider than merely concerning economic prosperity (1010). Integrating the social determinants of health in national public health laws can help protect the focus on health inequalities and the social determinants of health equity from changes in government, according to analysis in Scandinavian countries (1011).

Emergency preparedness and response plans have in the past overlooked the social determinants, prioritizing clinical care over promoting healthy populations (1012). The pandemic showed that if the social determinants of health for groups are not addressed, including income security, there will be lower compliance with public health measures, and health inequities will worsen during and following a pandemic (1013).

BOX 41. **Brazil: Integrating social determinants of health in infectious disease control**

In 2023, Brazil established an Interministerial Committee for the Elimination of Tuberculosis and Other Socially Determined Diseases (1014). This committee aims to promote intersectoral actions that contribute to the elimination of diseases that pose a public health problem in Brazil, with a strong focus on social determinants of health.

Brazil’s approach to controlling neglected infectious diseases by addressing social

determinants of health leverages initiatives such as the country’s *Programa Bolsa Família* conditional cash transfer initiative, established in 2004 (1015). A 2020 study published in *The Lancet* demonstrated that, among other impacts, the cash-transfer beneficiaries showed greater adherence to leprosy multidrug therapy and had higher cure rates (1016). The study also showed wider positive health and social impacts of the programme (1017).

Intersectoral collaboration is recognized as an essential mechanism to address global health challenges, including health inequity and achieving the SDGs (161, 1018, 1019). HiAP approaches advocate for formal governance and accountability structures, and use the Commission's comprehensive causation framework of the social determinants of health and health equity. The HiAP approach has been endorsed by global commitments under WHO health promotion conferences in the 2013 Helsinki Statement and subsequent World Health Assembly 67.12 resolution (1020). A review from Eastern Mediterranean countries noted that those with stronger multisectoral engagement during the COVID-19 pandemic achieved better COVID-19-related outcomes (1021). After the pandemic, many of the formal governance and accountability mechanisms put in place to manage social, economic and health interests were dissolved or transmuted into "One Health" mechanisms that integrate human, animal and environmental health, and maintain a biomedical focus on the human-animal interface, without taking a comprehensive social determinants approach, as proposed by HiAP.

Successful intersectoral mechanisms require: 1) governance and accountability 2) opportunities for leadership at all levels of government 3) ways of working across silos that encourage collaboration, not competition, and 4) financing, resources and human capabilities for working across sectors (170). The health sector can play an important role in developing HiAP and intersectoral approaches, including influencing the agendas, policies and laws of other sectors. This requires understanding how different sectoral policies impact on health, as well as policy priorities in other sectors (1022).



BOX 42. United States: California's Health in All Policies Task Force

California is the largest and most diverse state in the United States, with nearly 40 million people and no single ethnic group forming a majority of the population. The state faces significant health equity and racial equity challenges, including climate disasters such as extreme droughts and wildfires, high rates of chronic disease, and large inequities along racial, gender and socioeconomic lines. For instance, during the COVID-19 pandemic, death rates for Black Californians were 19% higher, Latinx Californians 8% higher and Pacific Islander Californians 82% higher than White Californians (1023).

The California Health in All Policies (HiAP) Task Force (170) was established in 2010 to take a comprehensive approach to addressing equity-related challenges (1024). The Task Force is

supported through a collaborative relationship between the cabinet-level Strategic Growth Council, the California Department of Public Health, and the non-governmental Public Health Institute. Over nearly 12 years of operation, the Task Force has evolved from focusing on the social determinants of health and equity, to explicitly addressing structural and institutional racism as key determinants of health inequities.

Among the promising results from California's focus on intersectoral action to address structural determinants of health inequities is the work of the Office of Neighbourhood Safety to achieve a 79% reduction in firearm assaults and a 62% fall in homicides in Richmond between 2007 and 2022 (1025).

Several key governing strategies for intersectoral action have been recommended for low- and middle-income countries, but also apply to high-income countries (1018). These include framing the issues strategically; clearly defining the roles of each sector; using existing structures when possible; understanding the key actors and political environment; paying attention to the role of non-state actors; addressing conflicts of interest; distributing leadership across sectors and levels; developing financing and monitoring systems to encourage collaboration; strengthening implementation processes and organizational capacity; and supporting mutual learning and implementation research.

A research agenda

While a body of indicative evidence and country good practice has been developed on progress in tackling the social determinants of health equity, a systematic research agenda – in particular, a focus on implementation research – is still lacking. For example, while a number of jurisdictions have championed HiAP approaches, there is little robust evidence of their impact on health outcomes, nor of the conditions under which such approaches become politically salient or lose their political appeal.

“The knowledge frontiers of the social determinants of health equity are largely unmapped”

The timing and combination of successful efforts to address inequity is complex, but critical for a meaningful impact. The sequencing of different measures may make the difference between effective and ineffective strategies, but there is very little systematic evidence to guide decision-making on these issues. The knowledge frontiers of the social determinants of health equity are largely unmapped, and represent a key opportunity to accelerate and sustain national and global efforts. Innovative solutions need to be catalogued, tested and replicated, and knowledge shared across jurisdictions to identify ways in which systems can change their behaviours to deliver greater equity in health. Key elements of a research agenda on social determinants of health equity are outlined in Box 43.

BOX 43. Key elements of a research agenda on the social determinants of health equity

Emphasis on intersectionality and structural determinants:

Future research must adopt intersectional frameworks that account for intersecting social determinants – including structural discrimination and economic inequality – to better inform targeted interventions and policies.

Integration of community-centred research:

Research on the social determinants of health equity must prioritize participatory and community-centred methodologies, ensuring that the voices and lived experiences of marginalized populations shape the knowledge-production process and its policy applications.

Focus on actionable evidence: To address persistent knowledge-to-action gaps, research should emphasize actionable outputs, co-created

with policy-makers and practitioners to ensure feasibility and relevance in diverse contexts.

Inclusion of non-traditional knowledge systems:

Recognizing the value of non-traditional knowledge systems, including Indigenous and local knowledge, can provide novel insights and support more inclusive approaches to addressing health inequities.

Strengthening governance research: Research must also focus on governance processes, identifying barriers and enablers for policy adoption and implementation across different contexts, including the roles of transnational actors and multisectoral partnerships

8.2

Data and monitoring

**RECOMMENDATION 8.2****Monitor social determinants of health equity**

8.2.1 Strengthen statistical infrastructure and build capacity for the use of disaggregated data to measure progress on health equity.

8.2.2 Utilize new technologies and novel data sources to fill data gaps and inform action on social determinants.

Governments, international organizations and communities need to be able to measure and track the progress of both the social determinants underlying health inequities, and the impacts of policy and programmatic actions on health equity (1026). Monitoring the social determinants of health equity at national and subnational levels is essential, in conjunction with insights directly from communities, to identify and understand issues and needs, and to determine the best course of action for national and local governments to improve health equity. Data helps identify and prioritize particular issues related to the social determinants of health equity, and track policies and investments addressing them, thus promoting accountability. Good quality, disaggregated data is needed to advance intersectoral processes, and drives accountability, transparency and service improvement.

Health and health equity impact assessments are decision-support tools that help evaluate the potential health impacts and other impacts of an intervention or policy. These assessments can be used to support monitoring and evaluation of health equity at the national level, and to inform the development of interventions and policies that reduce health inequalities (1027, 1028, 1029, 1030, 1031). This can be placed at the heart of PHC-oriented health system reforms currently underway in different countries (1032). Health and health equity impact assessments are also a useful tool to address living conditions across the life course, as well as the social determinants of health equity, because they can influence decision-makers in non-health sectors. The process can allow for informed debates on policies and interventions, and provide essential information to decision-makers about the benefits of interventions and impacts on inequalities (1033, 1034). Strong health equity impact assessments rely on clear theories of change, the availability of good-quality disaggregated data, and the collection of new data through mixed methods when existing data is unavailable.

Unfortunately, most countries lack systems for timely monitoring of health inequities or of the social determinants of health equity, and do not evaluate the impact of policies, despite this being one of the Commission's key recommendations (1035). The WHO package of tools and resources for health inequality and social determinants monitoring sets a standard for action and capacity building in this area (see Box 44).

However, data on social determinants is disproportionately concentrated in high-income countries. Improving data collection and availability in low- and middle-income countries is key to culturally appropriate prioritization and the identification of areas where action has not been taken (1038). To enable more effective monitoring and evaluation, countries may need to augment their statistical infrastructure; improve human and financial resources and the capacity for monitoring; enable intersectoral collaboration and data-sharing across sectors, and develop governance systems sufficient to ensure that evidence informs policy-making and action (151).

BOX 44. WHO resources for health inequality and social determinants monitoring

Health inequality monitoring

Health Equity Assessment Toolkit (HEAT and HEAT Plus): A software application that facilitates the exploration, analysis and reporting of health inequality data (1027).

Health inequality monitoring: harnessing data to advance health equity: A comprehensive resource for health inequality monitoring, which covers a range of topics, from the importance and general applications of such monitoring, to ways to align monitoring activities to generate impact. Includes step-by-step manuals and accompanying workbooks that provide practical guidance on the application of a five-step cycle of inequality monitoring (1036).

Health Inequality Monitoring eLearning courses:

Free, self-directed online courses covering the foundations of health inequality monitoring, applications to specific health topics, and skills-building.

Social determinants monitoring: conditions and actions

Operational Framework for Monitoring Social Determinants of Health Equity: A universal set of indicators and guidance, aimed at governments and their partners, for monitoring social determinants of health and related actions, adaptable to diverse national contexts. The Framework supports countries in using data to drive policy action across sectors to improve health equity (69).

Global databases

Health Inequality Data Repository:

The largest global collection of disaggregated data about health and the determinants of health, facilitating national monitoring of both health inequalities and social determinants (1037).

BOX 45. Thailand: Implementation of the WHO Operational Framework for Monitoring Social Determinants of Health Equity

In 2023, Thailand explored how to apply the draft WHO *Operational Framework for Monitoring Social Determinants of Health Equity* to strengthen equity monitoring (1044). Of the 63 recommended indicators in the Operational Framework, 46 were identified as feasible for the national context. These will be prioritized in close consultation with stakeholders, policy users and the National Statistics Office.

The 46 indicators identified for monitoring align closely with Thailand's equity principles and reflect stakeholder priorities from both

government and community levels, ensuring the indicators are relevant and actionable in the Thai context. The Operational Framework aligns with and supports the equity principles of the 3rd National Health System Charter 2022 (1045, 1046), whose 12 chapters provide guidance for strengthening health and other sectoral policies and strategies, to foster an equitable Thai health system. The Operational Framework and health equity dashboard, once fully developed, will support the monitoring of progress towards the health equity goal enshrined in the Health Systems Charter.

There is a strong case for the importance of intersectional analysis to better understand the “causes of causes” in social and health inequalities (1039, 1040), and countries must develop systems capable of identifying issues and measuring progress towards them. However, experiences from individual studies in countries such as Finland and Mexico indicate that data availability does not, on its own, ensure its use by decision-makers or health care professionals (1041), and that legal mandates and a regulatory evaluation agency can be essential in institutionalizing monitoring and evaluation (1042). Having sufficient data enables countries to prioritize action to improve daily living conditions, guided by monitoring and data. In Colombia, for example, data and monitoring have helped the government prioritize policies focused on early childhood development; opportunities for education and first employment; improved housing conditions, and social protection for families and populations living in vulnerable circumstances (1043). Box 45 demonstrates the process launched by Thailand to achieve improvements in monitoring.

Innovative tools, methods and data sources – including those from communities – can be used to enhance our knowledge and understanding of health inequalities and the social determinants. For example, a data dashboard for urban American Indian/Alaska Native people created by the Urban Indian Health Institute in the United States shows how new data sources can include groups traditionally marginalized in data collection (1047).

“Emerging data sources provide new opportunities to capture information on social determinants... together with AI-enabled analysis tools”

Additionally, emerging data sources beyond traditional government-supported ones, including big data, provide new opportunities to capture information on social determinants (1048), together with AI-enabled analysis tools. For example, readily available data on health determinants from clinical electronic health records can be extracted, subject to suitable permissions, using natural language processing tools (1049). In addition, community-level data can be attainable from public sources and can be linked to health data at a local level to assess the effect of social and community factors on individual health outcomes (1050), or to monitor the impacts of local interventions. Data protection and privacy requirements must be fully adhered to in such developments.

BOX 46. Canada: The Pan-Canadian Health Inequalities Reporting Initiative

Launched in 2016, the Pan-Canadian Health Inequalities Reporting Initiative (1051) aims to strengthen health inequalities measurement, monitoring and reporting capacity in Canada (1052), supporting surveillance and research activities, informing policy and programme decision-making, and enabling the monitoring of progress in reducing health inequalities over time. The initiative is led by the Public Health Agency of Canada, in collaboration with the Pan-Canadian Public Health Network, Statistics Canada, the First Nations Information Governance Centre, the Canadian Institute for Health Information and other partners. It integrates measures for more than 200 indicators of health outcomes, risk factors and social determinants of health, disaggregated across a range of socioeconomic and sociodemographic variables such as sex/gender, age, income, education, employment, occupation, immigrant status, Indigenous

identity, race/ethnicity, urban/rural residence, material and social deprivation, functional health/participation, and sexual orientation (1053).

To date, the initiative has helped provide a deeper understanding of health inequalities in Canada. The collaboration between federal, provincial and territorial partners, as well as Indigenous representatives and other stakeholders, has enabled policy-makers and health practitioners to access detailed, disaggregated data on health statuses and social determinants of health across different demographics and regions.

As a result, policies and programmes can be better tailored to the specific needs of vulnerable populations, with the aim of reducing health inequality across the country.



CHAPTER 9:

**A renewed vision of
health informed by
actions to address the
social determinants
of health equity**

This report has shown that there has been some progress on the original targets for reducing health inequalities, set by the WHO Commission on Social Determinants of Health in 2008, but much more action is needed. As new frameworks for development unfold post-2030, it is important to accelerate progress on the Sustainable Development Goals now, through the policy actions recommended to address the social determinants of health equity, as well as to influence future development frameworks.

The social, rather than solely biomedical or behavioural, roots of population health need to be recognized. Leaders should not confuse health care services with population health outcomes, nor make the assumption that health care can be delivered effectively and equitably without attention to its social context.

Global, national and local leaders in public and private spheres and in social movements need to seize opportunities to reinvigorate and purposefully pursue an agenda for change that renews the vision of health and prioritizes equity and people-centred development. This will support the necessary broader coalition of actors, bridging scientific and social disciplines, development contexts and different levels of action. Using social innovation and equity principles in the face of increasing complexity and interconnected systems is as important as technical innovation. COVID-19 demonstrated that without adhering to equity principles, new technologies guided by market forces alone do not reach people in diverse socioeconomic and geographic settings. Without taking up social interventions to build trust in government and social cohesion, protection from COVID-19 and its negative socioeconomic impacts would have remained the preserve of those with privilege and wealth, while sowing the seeds of increased contagion and economic destruction.

Replacing negative reinforcing cycles in economic, environmental and social systems with positive reinforcing cycles is core to addressing the social determinants of health equity (see Fig. 10, page 39). Only through comprehensive and systemic action will it be possible to achieve long, healthy lives for all, as proposed in the Sustainable Development Agenda, while healing the fractures in society that inequality in all its dimensions represents.

Our shared vision of humanity, its future, and that of our planet, is encapsulated in the 2030 Sustainable Development Agenda. This agenda and related frameworks provide a comprehensive view of what is important for prosperity and well-being. The Social Summit in 2025 will update the 1995 Copenhagen Declaration on Social Development, and address many of the social determinants of health equity outlined in this report, such as investing in social infrastructure through building universal comprehensive social protection systems, housing and education for all, and removing structural discrimination.

Ways to take action

This report has outlined actions, backed by good evidence, that will reduce health inequities. A comprehensive strategic framework for actions on the social determinants of health equity will deliver towards a vision for social development. Working within such a framework can help establish links between population health, political priorities, and national and local economies, and bring understanding of which sets of policy elements will work together more harmoniously. As this world report outlines, the four interlocking strategies needed to make lasting contributions to addressing the social determinants of health equity cover:



1. Addressing economic inequality and investing in social infrastructure and universal public services.
2. Overcoming structural discrimination.
3. Managing the challenges and opportunities posed by the mega-trends of climate change and the digital transformation in a way that systematically promotes health equity.
4. Putting in place the governance arrangements that will maximize the coherence and impact of action.

There are many policy entry points to address the social determinants of health equity, including efforts around “Health Equity in All Policies”, support for well-being economies and societies, and PHC approaches. These strategies have a common aim to bring health, social and environmental impacts more centrally into economic, social and development thinking. The four strategic action areas identified in this report, and the subsequent recommendations, reinforce one another and will produce compounding benefits.

“Multidisciplinary thinking, community engagement and intersectoral public policy dialogue are essential to bridge policy silos and action areas”

The report’s recommendations are signposts, and their assembly and sequencing is highly context-dependent. There is no silver bullet or any one sequence of recommendations that can be applied to address all problems. Comprehensive strategies should pay attention to:

- ▶ accountability and transparency mechanisms to address the social determinants of health equity in all policies, including purposeful whole-of-government and whole-of-society strategies
- ▶ recognition of health inequities as a summary indicator of social problems and challenges
- ▶ pathways to dealing with the challenges of social, generational and technological transitions
- ▶ information and data sharing, and related analytic tools and trend analysis to drive action
- ▶ policy systems that identify the theory of change and strategic relevance of combinations of policy actions.

The economy is an essential component of a comprehensive strategy to rectify health inequities. Labour markets and financing for public and social infrastructure, social protection and education are all essential to the social determinants of health equity. The economic actors in a well-being economy value outcomes related to belonging, voice, trust, tolerance and solidarity, and seek to ensure that the profitability of commercial activities is not maximized at the expense of these outcomes.

The learning and innovation policy laboratories fostered by the SDH Special Initiative demonstrate the value of multilevel action strategies. Strategies conceived solely at the national level, and that do not work bidirectionally from local to national levels, are unlikely to be sustainable. However, many of the structural determinants covered in this report’s action areas require national commitment. This means that comprehensive strategies will need to identify processes of mutual reinforcement between local and national levels.

Multidisciplinary thinking, community engagement and intersectoral public policy dialogue are essential to bridge policy silos and action areas. These characteristics are also common to integrative approaches to human-centred development and to a renewal of a PHC approach designed to integrate the social determinants of health.

For systemic action on the social determinants of health equity, collections of actors are required. While comprehensive strategies can be useful for those directing larger policy processes, every stakeholder, every person, can make a difference within their scope of action – whether orchestrated or not. This is a similar message to that for actions addressing the climate crisis, which is so intertwined with health equity.

National and local government – central leadership

National and local governments need to place health equity responsibilities at the highest governmental level and as a central purpose of government. To achieve this, they need to commit to addressing the main action areas from this report, and to ensure cohesive policy implementation across diverse departments, agencies and legal frameworks.

Governments must integrate health equity into financial strategies, domestically and internationally. This involves collaborating with international development finance, while enhancing domestic revenue through progressive taxation to fund social policies that are transformative, integrative, coordinated and responsive to people. Governments can foster shared commitment by supporting collaborative partnerships and planning, measuring well-being, carrying out joint monitoring and evaluation, and institutionalizing training and capacity development for collaboration across governmental agencies, and across not-for-profit civil society organizations.



Health leaders in national and local government

If health leaders are to be good advocates of addressing the social determinants of health equity, they need to be good partners for development in all its facets. This requires that they participate in coalitions and partnerships with other sectors. Within these shared contexts, the health community can build the evidence and carry out accompanying advocacy for action on the social determinants of health equity and corresponding PHC approaches, preventive health, health promotion, well-being societies, and the right to health as part of sustainable development.

Civil society

Civil society organizations play a critical role in advancing action on the social determinants of health equity, informed by the voices and experiences of communities themselves. Civil society is crucial in monitoring and advocating for improvements to various social determinants, such as employment conditions, gender equality and environmental protection. Its involvement is also essential for holding national and international multilateral organizations accountable, and ensuring that social determinants of health equity are prioritized in public and policy discourse. Non-profit organizations can contribute to health equity by supporting access to educational and vocational training, as a condition of decent employment, and investments in social infrastructure.

The private sector

Private-sector actors can become key partners for achieving health equity goals when they share the health objectives of the public sector, are held accountable for delivering on their commitments, and are subject to safeguards against conflicts of interest. Leaders in the private sector must commit to and enhance each other's accountability for best-in-class standards and agreements – particularly in employment and environmental practices across their supply chains, ensuring healthy and equitable working conditions for all. Commercial leaders from across industries and sectors must also join and support those in the business community innovating and striving for health equity and the well-being economy.

Multilateral action across the United Nations and international financing institutions

While WHO has a core role to play, a much broader partnership across the UN and multilateral system is required for deep and lasting progress in the multiple social domains that shape health equity. Multilateral institutions can help enable national efforts, both by codifying and disseminating international norms and standards, and by providing

resources and support to countries and other actors in making progress towards equity. Global agencies can identify health equity as a shared goal, and leverage a unified framework of indicators on the social determinants of health equity as a key feature to monitor human development progress. Interagency collaboration in thematic working groups will support a focus on the social determinants of health equity. International financial institutions also have an important role, with a responsibility to ensure that aid and debt relief effectively support policies and actions to address these determinants, with associated health equity performance indicators.

WHO

The World Health Organization will continue to foster global and national policy norms and standards, including mechanisms for policy coherence for health equity, and to support evidence gathering, the investment case for action, convening diverse partners, and capacity building. WHO will also continue to strengthen global technical capacities to advocate for action on the social determinants of health equity in multilateral forums, and support Member States to share technical knowledge and experiences of intersectoral policy action and strategies for health equity. Additionally, WHO will support the evaluation of the social determinants of health equity through monitoring frameworks, global and national health surveillance systems, and health equity policy assessment tools.

Research institutions

Research institutions need to be central to generating partnerships between practitioners at all levels, and through local and international academic networks. Assessment of complex social strategies remains an important part of the implementation research agenda. Researchers must disseminate evidence about actions and strategies on the social determinants of health equity. They must allocate funding specifically for this area, support the development of indicators and evaluation of interventions, and establish open-access virtual networks to enhance global and South-South knowledge sharing. These institutions are also tasked with addressing the brain drain from low- and middle-income countries, and eliminating racial and gender biases in research teams, proposals, designs, practices and reports, as well as counterbalancing bias and supporting collaborative national health and social policy research. Their efforts are crucial in building a robust evidence base that informs policy and practice globally regarding action on the social determinants of health equity.

TABLE 1. Recommendations for action on social determinants of health equity

Recommendation	Report section	Key actions
Action area 1: Address economic inequality and invest in universal public services (Part 2, Chapter 3)		
Address economic inequality and invest in universal public services. 	3.1	Use progressive taxation to expand fiscal space for income transfers and universal, equitable public services. Provide adequate public funding for good-quality, equitable infrastructure and services, and ensure that effective mechanisms are in place to safeguard quality and secure equity. Move towards well-being economies and consider well-being budgeting.
Promote the social determinants of health equity in development financing and investment. 	3.2	Ensure that the need for funding (fiscal space) to address the social determinants of health equity is included in approaches to development financing, debt relief and taxation. Deliver and monitor development financing to support public investment in policies, actions and infrastructure that address the social determinants of health equity.
Analyse and address the commercial determinants of health. 	3.3	Analyse, legislate on and regulate commercial activities that negatively affect health and health equity, while maximizing the health-promoting capacity of the private sector. Address and manage conflicts of interest in policy environments, to prevent negative impacts on population health and health equity. Use the scale of the public sector to incentivize commercial activities that positively affect health and health equity – for example, through mandating public procurement that requires sustainable, safe and healthy products, and safe and fair labour standards throughout supply chains. Strengthen health equity considerations in global and regional trade processes, including in relation to intellectual property.
Increase comprehensiveness and expand coverage of universal social protection systems across the life course. 	3.4	Ensure adequate income guarantees and care throughout the life course, including for people with disabilities or chronic health conditions. Build and expand paid leave benefits for sickness and parental leave, including for the precariously employed and informal workers.

Recommendation	Report section	Key actions
Action area 2: Tackle structural discrimination and the determinants and impacts of conflict, emergencies and migration (Part 2, Chapter 4)		
Address structural discrimination. 	4.1	Recognize and repair discrimination embedded in policies, laws, institutions and social norms. Redress the negative impacts of colonization by developing standards for reparative justice that measure impacts on health. Acknowledge Indigeneity as a determinant of health and health equity (shaped by discrimination and historical injustices, as well as protective factors), while upholding the provisions of the United Nations Declaration on the Rights of Indigenous Peoples.
Address and protect the social determinants of health equity in emergencies, migration and conflict. 	4.2	Recognize peace as a fundamental determinant of health equity, and health as a bridge to peace. Ensure preparedness and response efforts for health emergencies reflect social determinants of health equity considerations at their core, including the equitable and rapid rollout of social protection. Ensure the rights of, and support, displaced people and migrants to access health and social services without discrimination.
Action area 3: Steer action on climate change and digitalization towards health equity (Part 2, Chapter 5)		
Articulate and accelerate the health equity benefits of climate action and the preservation of biodiversity. 	5.1	Support the development and implementation of climate change mitigation and adaptation policies that maximize health equity and co-benefits. Enable the energy and food transitions to prioritize addressing energy poverty and food insecurity. Integrate health systems more centrally in climate and environment strategies, and ensure that local and equitable climate action is a core objective of health systems. Strengthen support for Indigenous communities in their stewardship of land and natural resources, and recognize and address the impacts of environmental degradation, land confiscation and climate-related migration.
Steer the digital transformation in favour of health equity and the public good. 	5.2	Address the digital divide. Prevent harmful impacts of digital and technological transformations and artificial intelligence on health and health equity, including in relation to labour markets and social cohesion.

Recommendation	Report section	Key actions
Action area 4: Bringing about change through new governance approaches (Part 3, Chapters 6–8)		
Empower local governments to address health inequities. 	6.1	Strengthen the role of local government with the functions and resources to implement community-centred actions for health equity. Ensure that urban, rural and territorial planning, transport and housing investments are underpinned by approaches that deliver healthy housing and built environments, and adopt universal design principles. Equip local government to address health equity across the life course, including in supporting age-friendly communities, and combating social isolation and loneliness.
Support community engagement and civil society. 	6.2	Incorporate representative community engagement and social participation in local policy processes, and in health and social service decision-making and delivery. Create the enabling conditions that maximize the capabilities of independent and inclusive civil society to address the social determinants of health equity.
Achieve universal health coverage through progressive health financing and primary health care approaches. 	7.1	Improve equitable access to a continuum of quality health services, addressing both physical and mental health across the life course, by strengthening primary health care-orientation within health systems. Minimize patients' out-of-pocket expenditure, and finance health services from pooled government resources. Increase the share of health and care sector funding in proportion to need, with special intensity for populations experiencing marginalization, and recognize and address discrimination within health systems. Facilitate equitable access to health technologies, from research and development, through to manufacturing and equitable delivery.
Build and retain a health and care workforce capable of delivering equity. 	7.2	Recognize and reward work in the care economy, including informal care, and demonstrate the role of the health and care sector as a driver of health equity and social inclusion, by ensuring decent and safe working conditions and addressing pay and representation gaps. Develop human capacity in health, social protection, education, labour, local government and service organizations to enhance intersectoral efforts to address the social determinants of health equity.
Strengthen focus on social determinants in health systems and policy platforms. 	8.1	Integrate addressing the social determinants of health equity in all health strategies, policies, emergency preparedness and response plans, and public health laws. Establish coordination and accountability mechanisms for intersectoral collaboration and community engagement for health (including Health in All Policies functions) at all levels of government.
Monitor social determinants of health equity. 	8.2	Strengthen statistical infrastructure, institutionalize and build capacity for the use of disaggregated data to measure progress on the social determinants of health equity, including the distribution of socially produced health risks across social groups and geographic areas; the mechanisms that drive inequities, and actions to address them. Utilize new technologies and novel data sources to fill data gaps and inform action on the social determinants of health equity.

CONCLUSION:

Equity is a political choice

Despite a comprehensive diagnosis and set of actions recommended by the Commission on the Social Determinants of Health in 2008, progress on reducing inequalities in health has been slow, and inequity is killing millions. To enable improvements in living conditions and address underlying structural determinants, change is required, from policy-setting and legislation, through to implementation. Governments and politicians are in the driving seat.

This report recognizes that improvements in health equity have been too slow and, alarmingly, inequities between social groups within countries are often deepening. In the pursuit of prosperity, countries have too often failed to integrate equity and fairness into policies. Globally, we have also failed to integrate an appreciation of intergenerational equity, further undermined by the existential threats of climate change, war and conflict.

Health equity stands at the core of sustainable development. The social determinants of health equity provide a lens on what “the good life” consists of, and what stands in the way of its realization, fairly and for all. The agenda to address the social determinants of health equity is necessarily broad and can feel like a call to achieve “everything, everywhere, all at once”. But this complexity must not be an excuse for inaction. Practical actions as outlined in this report will deliver improvements in health equity, and communities, governments, thinkers and leaders are already showing the way forward. Seizing on their example, and knitting together disparate efforts into a comprehensive strategy of action, will be of immense benefit to all of humanity for generations to come.



References

1. World Health Organization. GHE: Life expectancy and healthy life expectancy [website]. 2024 (<https://www.who.int/data/gho/data/themes/mortality-and-global-health-estimates/ghe-life-expectancy-and-healthy-life-expectancy>, accessed 2 May 2024).
2. UN DESA. UN DESA Policy Brief No. 145: On the importance of monitoring inequality in life expectancy [website]. 2022 (<https://www.un.org/development/desa/dpad/publication/un-desa-policy-brief-no-145-on-the-importance-of-monitoring-inequality-in-life-expectancy/>, accessed 25 September 2024).
3. Norheim OF, Chang AY, Bolongaita S, Barraza-Lloréns M, Fawole A, Gebremedhin LT et al. Halving premature death and improving quality of life at all ages: cross-country analyses of past trends and future directions. *Lancet* 2024. ([https://doi.org/10.1016/S0140-6736\(24\)02417-6](https://doi.org/10.1016/S0140-6736(24)02417-6)).
4. PAHO Commission, Institute of Health Equity. The Final Report of the Commission of the Pan American Health Organization on Equity and Health Inequalities in the Americas. 2019 (<https://www.instituteofhealthequity.org/resources-reports/the-final-report-of-the-commission-of-the-pan-american-health-organisation-on-equity-and-health-inequalities-in-the-americas>, accessed 7 December 2023).
5. Build Back Fairer: Achieving Health Equity in the Eastern Mediterranean Region: Report of the Commission on Social Determinants of Health in the Eastern Mediterranean Region. World Health Organization. Regional Office for the Eastern Mediterranean, 2021 (<https://iris.who.int/handle/10665/348185>, accessed 22 July 2024). License: CC BY-NC-SA 3.0 IGO.
6. Commission on Social Determinants of Health. Closing the gap in a generation: health equity through action on the social determinants of health: final report of the Commission on Social Determinants of Health. World Health Organization, 2008 (<https://iris.who.int/handle/10665/43943>, accessed 2 May 2024).
7. World Health Organization. World Report on the Health of Refugees and Migrants, 1st ed. Geneva, World Health Organization. 2022 (<https://iris.who.int/handle/10665/360404>). License: CC BY-NC-SA 3.0 IGO.
8. van den Berg M, Flavel J, Schram A, Friel S, Gesesew HA, Baum F. Social, cultural and political conditions for advancing health equity: examples from eight country case studies (2011–2021). *BMJ Glob Health* 2024;9:e015694.
9. Marmot MG, Wilkinson RG, editors. Social determinants of health, 2. ed., reprinted. Oxford, Oxford Univ. Press, 2011.
10. Braveman P, Egerter S, Williams DR. The social determinants of health: coming of age. *Annu Rev Public Health* 2011;32:381–398.
11. Dahlgren G, Whitehead M. Policies and strategies to promote social equity in health. Background document to WHO – Strategy paper for Europe. Arbetsrapport 1991. Arbetsrapport; (https://ideas.repec.org/p/hhs/ifswps/2007_014.html, accessed 21 May 2024).
12. Donkin A, Goldblatt P, Allen J, Nathanson V, Marmot M. Global action on the social determinants of health. *BMJ Glob Health* 2018;3:e000603.
13. World Health Organization. Healthy, prosperous lives for all: the European Health Equity Status Report. World Health Organization Regional Office for Europe, 2019 (<https://iris.who.int/handle/10665/326879>, accessed 9 April 2024). License: CC BY-NC-SA 3.0 IGO.
14. World Health Organization. A Conceptual Framework for Action on the Social Determinants of Health. Geneva, World Health Organization, 2020 (<https://www.who.int/publications-detail-redirect/9789241500852>, accessed 2 May 2024).
15. Solar O, Valentine N, Castedo A, Brandt GS, Sathyandran J, Ahmed Z et al. Action on the social determinants for advancing health equity in the time of COVID-19: perspectives of actors engaged in a WHO Special Initiative. *Int J Equity Health* 2023;21:193.
16. Committee on Community-Based Solutions to Promote Health Equity in the United States, Board on Population Health and Public Health Practice, Health and Medicine Division, National Academies of Sciences, Engineering, and Medicine. Communities in Action: Pathways to Health Equity (Weinstein JN, Geller A, Negussie Y, Baciu A, editors). Washington (DC). National Academies Press, 2017:24624. (<https://doi.org/10.17226/24624>).
17. McGinnis JM, Williams-Russo P, Knickman JR. The case for more active policy attention to health promotion. *Health Aff Proj Hope* 2002;21:78–93.
18. Hood CM, Gennuso KP, Swain GR, Catlin BB. County Health Rankings. *Am J Prev Med* 2016;50:129–135.

19. Spiegel JM, Breilh J, Yassi A. Why language matters: insights and challenges in applying a social determination of health approach in a North-South collaborative research program. *Glob Health* 2015;11:9.
20. Evans T, Brown H. Road traffic crashes: operationalizing equity in the context of health sector reform. *Inj Control Saf Promot* 2003;10:11-12.
21. World Health Organization. Handbook on health inequality monitoring with a special focus on low- and middle-income countries. Geneva, World Health Organization, 2013 (<https://iris.who.int/handle/10665/85345>, accessed 2 May 2024).
22. O'Neill J, Tabish H, Welch V, Petticrew M, Pottie K, Clarke M et al. Applying an equity lens to interventions: using PROGRESS ensures consideration of socially stratifying factors to illuminate inequities in health. *J Clin Epidemiol* 2014;67:56-64.
23. Finland. Ministry of Social Affairs and Health. Health in all policies: Helsinki statement. Framework for country action. Geneva, World Health Organization, 2014 (<https://iris.who.int/handle/10665/112636>, accessed 8 January 2025).
24. World Health Organization. Commercial determinants of health [website]. 2023 (<https://www.who.int/news-room/fact-sheets/detail/commercial-determinants-of-health>, accessed 16 November 2024).
25. Joshi A, Aldrich DP. Corraling a Chimera: A Critical Review of the Term Social Infrastructure. *SSRN Electron J* 2022. (<https://doi.org/10.2139/ssrn.4310160>).
26. Renner A-T, Plank L, Getzner M. Introduction to the Handbook of Social Infrastructure. Cheltenham: Edward Elgar Publishing; 2024: 1-17 (<https://doi.org/10.4337/9781800883130>).
27. Najcevska M. Structural Discrimination-Definitions, approaches and trends. Office of the United Nations High Commissioner for Human Rights, 2015 (<https://www.ohchr.org/Documents/Issues/Racism/IWG/Session8/MirjanaNajcevska.doc>).
28. Seventy-fourth World Health Assembly WHA74.16 Agenda item 22.1 Social Determinants of Health. World Health Organization, 2021 (https://apps.who.int/gb/ebwha/pdf_files/WHA74/A74_R16-en.pdf).
29. World Health Organization. Closing the health equity gap: policy options and opportunities for action. Geneva, World Health Organization, 2013 (<https://iris.who.int/handle/10665/78335>, accessed 22 July 2024).
30. Yerramilli P, Chopra M, Rasanathan K. The cost of inaction on health equity and its social determinants. *BMJ Glob Health* 2024;9:e012690.
31. Commission of the Pan American Health Organization on Equity and Health Inequalities in the Americas. Just Societies: Health Equity and Dignified Lives. Report of the Commission of the Pan American Health Organization on Equity and Health Inequalities in the Americas. Washington (DC), Pan American Health Organization, 2019 (<https://iris.paho.org/handle/10665.2/51571>).
32. Institute of Health Equity. Review of social determinants and the health divide in the WHO European Region: final report. Institute of Health Equity, 2013 (<https://www.instituteofhealthequity.org/resources-reports/review-of-social-determinants-and-the-health-divide-in-the-who-european-region-final-report>, accessed 23 April 2025).
33. World Health Organization. Rio Political Declaration on Social Determinants of Health [website]. 2011 (<https://www.who.int/publications/m/item/rio-political-declaration-on-social-determinants-of-health>, accessed 15 April 2024).
34. Anderson I, Robson B, Connolly M, Al-Yaman F, Bjertness E, King A et al. Indigenous and tribal peoples' health (The Lancet-Lowitja Institute Global Collaboration): a population study. *Lancet* 2016;388:131-157.
35. Kataoka A, Fukui K, Sato T, Kikuchi H, Inoue S, Kondo N et al. Geographical socioeconomic inequalities in healthy life expectancy in Japan, 2010-2014: An ecological study. *Lancet Reg Health - West Pac* 2021;14:100204.
36. Dwyer-Lindgren L, Kendrick P, Kelly YO, Sylte DO, Schmidt C, Blacker BF et al. Life expectancy by county, race, and ethnicity in the USA, 2000-19: a systematic analysis of health disparities. *Lancet* 2022;400:25-38.
37. Chetty R, Stepner M, Abraham S, Lin S, Scuderi B, Turner N et al. The Association Between Income and Life Expectancy in the United States, 2001-2014. *JAMA* 2016;315:1750-1766.
38. Dwyer-Lindgren L, Baumann MM, Li Z, Kelly YO, Schmidt C, Searchinger C et al. Ten Americas: a systematic analysis of life expectancy disparities in the USA. *Lancet* 2024;404:2299-2313.
39. Marmot M, Allen J, Boyce T, Goldblatt P, Morrison J. Marmot Review 10 Years On. Institute of Health Equity, 2020 (<https://www.instituteofhealthequity.org/resources-reports/marmot-review-10-years-on>, accessed 16 October 2023).
40. Kinge JM, Modalsli JH, Øverland S, Gjessing HK, Tollånes MC, Knudsen AK et al. Association of Household Income with Life Expectancy and Cause-Specific Mortality in Norway, 2005-2015. *JAMA* 2019;321:1916-1925.

41. Organisation for Economic Co-operation and Development. Health at a Glance 2021: OECD Indicators. Paris: OECD Publishing; 2021 Health at a Glance; (<https://doi.org/10.1787/ae3016b9-en>).
42. European Public Health Association. Closing the life expectancy gap of Roma in Europe. Brussels, EPHA, 2018 (<https://epha.org/closing-the-life-expectancy-gap-of-roma-in-europe/>, accessed 22 October 2024).
43. Mokhayeri Y, Mahmoudi M, Haghdoust AA, Amini H, Asadi-Lari M, Naieni KH. How within-city socioeconomic disparities affect life expectancy? Results of Urban HEART in Tehran, Iran. *Med J Islam Repub Iran* 2014;28:80.
44. World Health Organization. Adult mortality rate (probability of dying between 15 and 60 years per 1000 population) [website]. ([https://www.who.int/data/gho/data/indicators/indicator-details/GHO/adult-mortality-rate-\(probability-of-dying-between-15-and-60-years-per-1000-population\)](https://www.who.int/data/gho/data/indicators/indicator-details/GHO/adult-mortality-rate-(probability-of-dying-between-15-and-60-years-per-1000-population))), accessed 23 April 2025).
45. Bahk J, Lynch JW, Khang Y-H. Forty years of economic growth and plummeting mortality: the mortality experience of the poorly educated in South Korea. *J Epidemiol Community Health* 2017;71:282–288.
46. Korda RJ, Biddle N, Lynch J, Eynstone-Hinkins J, Soga K, Banks E et al. Education inequalities in adult all-cause mortality: first national data for Australia using linked census and mortality data. *Int J Epidemiol* 2020;49:511–518.
47. Long D, Mackenbach JP, Klokgieters S, Kalédiené R, Deboosere P, Martikainen P et al. Widening educational inequalities in mortality in more recent birth cohorts: a study of 14 European countries. *J Epidemiol Community Health* 2023;77:400–408.
48. Goldblatt P, Castedo A, Allen J, Lionello L, Bell R, Marmot M et al. Rapid review of inequalities in health and wellbeing in Norway since 2014. Institute of Health Equity, 2023 (<https://www.instituteofhealthequity.org/resources-reports/rapid-review-of-inequalities-in-health-and-wellbeing-in-norway-since-2014>, accessed 6 June 2023).
49. World Health Organization. (2020). Decade of healthy ageing: baseline report. World Health Organization. (<https://iris.who.int/handle/10665/338677>) License: CC BY-NC-SA 3.0 IGO.
50. Smythe T, Kuper H. The association between disability and all-cause mortality in low-income and middle-income countries: a systematic review and meta-analysis. *Lancet Glob Health* 2024;12:e756–e770.
51. Fiorillo A, Luciano M, Pompili M, Sartorius N. Editorial: Reducing the Mortality Gap in People with Severe Mental Disorders: The Role of Lifestyle Psychosocial Interventions. *Front Psychiatry* 2019;10:434.
52. Thornicroft G. Physical health disparities and mental illness: the scandal of premature mortality. *Br J Psychiatry J Ment Sci* 2011;199:441–442.
53. UNICEF. Levels and trends in child mortality 2024 [website]. United Nations Children's Fund, 2025 (<https://data.unicef.org/resources/levels-and-trends-in-child-mortality-2024/>, accessed 15 April 2025).
54. United Nations Maternal Mortality Estimation Inter-agency Group. Trends in Maternal Mortality 2000 to 2023: Estimates by WHO, UNICEF, UNFPA, World Bank Group and UNDESA/Population Division. Geneva, World Health Organization, 2025 (<https://iris.who.int/handle/10665/381012>). License: CC BY-NC-SA 3.0 IGO.
55. Calvert C, John J, Nzvere FP, Cresswell JA, Fawcus S, Fottrell E et al. Maternal mortality in the covid-19 pandemic: findings from a rapid systematic review. *Glob Health Action* 2021;14:1974677. (<https://doi.org/10.1080/16549716.2021.1974677>)
56. The Fund for Peace. Fragile States Index Annual Report 2023. Washington (DC), The Fund for Peace, 2023 (<https://fragilestatesindex.org/2023/06/14/fragile-states-index-2023-annual-report/>, accessed 15 April 2025).
57. Scheil-Adlung X. Global evidence on inequities in rural health protection. New data on rural deficits in health coverage for 174 countries. International Labour Organization, Social Protection Department, 2015 ESS Paper Series (SECSOC) – ESS 47; (<https://www.ilo.org/publications/global-evidence-inequities-rural-health-protection-new-data-rural-deficits>, accessed 23 April 2025.)
58. Anyamele OD. Urban and rural differences across countries in child mortality in sub-Saharan Africa. *J Health Care Poor Underserved* 2009;20:90–98.
59. Menashe-Oren A, Stecklov G. Urban-rural disparities in adult mortality in sub-Saharan Africa. *Demogr Res* 2018;39:136–176.
60. Alkema L, Chou D, Hogan D, Zhang S, Moller A-B, Gemmill A et al. National, regional, and global levels and trends in maternal mortality between 1990 and 2015 with scenario-based projections to 2030: a systematic analysis by the United Nations Maternal Mortality Estimation Inter-Agency Group. *Lancet* 2016;387:462–474.
61. Karlsen S, Say L, Souza J-P, Hogue CJ, Calles DL, Gülmezoglu AM et al. The relationship between maternal education and mortality among women giving birth in health care institutions: Analysis of the cross sectional WHO Global Survey on Maternal and Perinatal Health. *BMC Public Health* 2011;11:606.

62. Australian Institute of Health and Welfare. Australia's mothers and babies: Maternal deaths [website]. 2023 (<https://www.aihw.gov.au/reports/mothers-babies/maternal-deaths-australia>, accessed 26 September 2024).
63. Hoyert D. Maternal Mortality Rates in the United States, 2020. National Center for Health Statistics (U.S.), 2022 (<https://dx.doi.org/10.15620/cdc:113967>).
64. Guimarães RM, Reis LGC, De Souza Mendes Gomes MA, Magluta C, De Freitas CM, Portela MC. Tracking excess of maternal deaths associated with COVID-19 in Brazil: a nationwide analysis. *BMC Pregnancy Childbirth* 2023;23:22.
65. MBRRACE-UK. Maternal mortality 2020-2022 [website]. 2024 (<https://www.npeu.ox.ac.uk/mbrrace-uk/data-brief/maternal-mortality-2020-2022>, accessed 31 May 2024).
66. Samuel O, Zewotir T, North D. Decomposing the urban-rural inequalities in the utilisation of maternal health care services: evidence from 27 selected countries in Sub-Saharan Africa. *Reprod Health* 2021;18:216.
67. Rossen LM, Ahrens KA, Womack LS, Uddin SFG, Branum AM. Rural-Urban Differences in Maternal Mortality Trends in the US, 1999-2017: Accounting for the Impact of the Pregnancy Status Checkbox. *Am J Epidemiol* 2022;191:1030-1039.
68. Brown HK, Ray JG, Chen S, Guttmann A, Havercamp SM, Parish S et al. Association of Preexisting Disability with Severe Maternal Morbidity or Mortality in Ontario, Canada. *JAMA Netw Open* 2021;4:e2034993.
69. World Health Organization. Operational framework for monitoring social determinants of health equity. Geneva, World Health Organization, 2024 (<https://iris.who.int/handle/10665/375732>). License: CC BY-NC-SA 3.0 IGO.
70. United Nations Department of Economic and Social Affairs. The Sustainable Development Goals Report 2024. New York, United Nations, 2024 (<https://unstats.un.org/sdgs/report/2024/>, accessed 25 September 2024).
71. United Nations Department of Economic and Social Affairs. The Sustainable Development Goals Report 2020. New York, United Nations, 2020 (<https://unstats.un.org/sdgs/report/2020/>, accessed 23 October 2024).
72. Salmeron Gomez D, Engilbertsdottir S, Cuesta Leiva JA, Newhouse D, Stewart D. Global Trends in Child Monetary Poverty According to International Poverty Lines. Washington (DC): World Bank; 2023 (<http://doi.org/10.1596/1813-9450-10525>).
73. World Bank. Fragility and Conflict: On the Front Lines of the Fight against Poverty [website]. 2020 (<https://www.worldbank.org/en/topic/poverty/publication/fragility-conflict-on-the-front-lines-fight-against-poverty>, accessed 25 March 2024).
74. Mahler D, Lakner C. The Impact of COVID-19 on Global Inequality and Poverty. Policy Research Working Papers. 2022. (<https://openknowledge.worldbank.org/entities/publication/54fae299-8800-585f-9f18-a42514f8d83b>, accessed 20 October 2024).
75. Chancel L, Piketty T, Saez E, Zucman G, Duflo E, Banerjee AV, editors. *World Inequality Report 2022*. Cambridge, Massachusetts; London, England, The Belknap Press of Harvard University Press, 2022.
76. International Labour Office. *World Social Protection Report 2024-26: Universal social protection for climate action and a just transition*, 1st ed. Geneva, International Labour Office, 2024.
77. United Nations Department of Economic and Social Affairs. *The Sustainable Development Goals Report 2022*. New York, United Nations, 2022 (<https://unstats.un.org/sdgs/report/2022/>, accessed 23 October 2024).
78. United Nations Department of Economic and Social Affairs. *The Sustainable Development Goals Report 2023: Special Edition*. New York, United Nations, 2023 (<https://desapublications.un.org/publications/sustainable-development-goals-report-2023-special-edition>, accessed 4 April 2025).
79. IEA, IRENA, UNSD, World Bank, WHO. *Tracking SDG 7: The Energy Progress Report 2024*. Washington (DC), 2024 (<https://trackingsdg7.esmap.org/downloads>, accessed 2 October 2024).
80. Li C, van Donkelaar A, Hammer MS, McDuffie EE, Burnett RT, Spadaro JV et al. Reversal of trends in global fine particulate matter air pollution. *Nat Commun* 2023;14:5349.
81. UNEP. *Regulating Air Quality: the First Global Assessment of Air Pollution Legislation* | UNEP - UN Environment Programme. UNEP, 2021 (<https://www.unep.org/resources/report/regulating-air-quality-first-global-assessment-air-pollution-legislation>, accessed 26 September 2024).
82. Brinda EM, Rajkumar AP, Enemark U. Association between gender inequality index and child mortality rates: a cross-national study of 138 countries. *BMC Public Health* 2015;15:97.
83. Pinho-Gomes A-C, Peters SAE, Woodward M. Gender equality related to gender differences in life expectancy across the globe gender equality and life expectancy. *PLOS Glob Public Health* 2023;3:e0001214.

84. Nearly 2.4 Billion Women Globally Don't Have Same Economic Rights as Men [website]. (<https://www.worldbank.org/en/news/press-release/2022/03/01/nearly-2-4-billion-women-globally-don-t-have-same-economic-rights-as-men>, accessed 26 September 2024).
85. International Fund for Agricultural Development. These numbers prove that rural women are crucial for a better future. But they're not getting what they need to succeed [website]. 2022 (<https://www.ifad.org/en/w/explainers/these-numbers-prove-that-rural-women-are-crucial-for-a-better-future>, accessed 20 October 2024).
86. World Bank. Gender-Based Violence (Violence Against Women and Girls) [website]. 2019 (<https://www.worldbank.org/en/topic/socialsustainability/brief/violence-against-women-and-girls>, accessed 23 October 2024).
87. United Nations Population Fund (UNFPA). Child marriage [website]. (<https://www.unfpa.org/child-marriage>, accessed 23 October 2024).
88. UNFPA. FGM Annual Report 2018. New York, UNFPA (<https://www.unfpa.org/fgm-annual-report>, accessed 23 October 2024).
89. Development Initiatives. Inequality: Global trends [website]. 2023 (<https://devinit.org/resources/inequality-global-trends/>, accessed 2 May 2024).
90. Elgar FJ, Stefaniak A, Wohl MJA. The trouble with trust: Time-series analysis of social capital, income inequality, and COVID-19 deaths in 84 countries. *Soc Sci Med* 2020;263:113365.
91. Davies J. Economic Inequality and COVID-19 Death Rates in the First Wave, a Cross-Country Analysis. *SSRN Electron J* 2021. (<https://doi.org/10.2139/ssrn.3812074>).
92. Wildman J. COVID-19 and income inequality in OECD countries. *Eur J Health Econ* 2021;22:455–462.
93. Oronce CIA, Scannell CA, Kawachi I, Tsugawa Y. Association Between State-Level Income Inequality and COVID-19 Cases and Mortality in the USA. *J Gen Intern Med* 2020;35:2791–2793.
94. Israelsen S, Malji A. COVID-19 in India: A Comparative Analysis of the Kerala and Gujarat Development Models' Initial Responses. *Prog Dev Stud* 2021;21:397–418.
95. Demenech LM, Dumith SDC, Vieira MECD, Neiva-Silva L. Desigualdade econômica e risco de infecção e morte por COVID-19 no Brasil. *Rev Bras Epidemiol* 2020;23:e200095.
96. World Health Organization. COVID-19 and the social determinants of health and health equity: evidence brief. Geneva, World Health Organization, 2021 (<https://iris.who.int/handle/10665/348333>). License: CC BY-NC-SA 3.0 IGO.
97. Williamson EJ, McDonald HI, Bhaskaran K, Walker AJ, Bacon S, Davy S et al. Risks of covid-19 hospital admission and death for people with learning disability: population-based cohort study using the OpenSAFELY platform. *BMJ* 2021;n1592.
98. Help Age International. COVID: millions of older people have died without being counted [website]. 2022 (<https://www.helpage.org/news/covid-millions-of-older-people-have-been-dying-without-being-counted/>, accessed 2 May 2024).
99. Marmot M, Allen J, Goldblatt P, Herd E, Morrison J. Build Back Fairer: The COVID-19 Marmot Review. Institute of Health Equity, 2020 (<https://www.instituteofhealthequity.org/resources-reports/build-back-fairer-the-covid-19-marmot-review>, accessed 31 May 2024).
100. Variation in racial/ethnic disparities in COVID-19 mortality by age in the United States: A cross-sectional study | PLOS Medicine [website]. (<https://journals.plos.org/plosmedicine/article?id=10.1371/journal.pmed.1003402>, accessed 2 May 2024).
101. Baqui P, Bica I, Marra V, Ercole A, Schaar M van der. Ethnic and regional variations in hospital mortality from COVID-19 in Brazil: a cross-sectional observational study. *Lancet Glob Health* 2020;8:e1018–e1026.
102. Cifuentes MP, Rodriguez-Villamizar LA, Rojas-Botero ML, Alvarez-Moreno CA, Fernández-Niño JA. Socioeconomic inequalities associated with mortality for COVID-19 in Colombia: a cohort nationwide study. *J Epidemiol Community Health* 2021;75:610–615.
103. Yoshikawa Y, Kawachi I. Association of Socioeconomic Characteristics with Disparities in COVID-19 Outcomes in Japan. *JAMA Netw Open* 2021;4:e2117060.
104. Riou J, Panczak R, Althaus CL, Junker C, Perisa D, Schneider K et al. Socioeconomic position and the COVID-19 care cascade from testing to mortality in Switzerland: a population-based analysis. *Lancet Public Health* 2021;6:e683–e691.
105. Higher COVID-19 mortality in low-income communities in the City of Cape Town – a descriptive ecological study - PMC [website]. (<https://www.ncbi.nlm.nih.gov/pmc/articles/PMC8264385/>, accessed 2 May 2024).
106. Oroszi B, Juhász A, Nagy C, Horváth JK, McKee M, Ádány R. Unequal burden of COVID-19 in Hungary: a geographical and socioeconomic analysis of the second wave of the pandemic. *BMJ Glob Health* 2021;6:e006427.
107. Di Girolamo C, Bartolini L, Caranci N, Moro ML. Socioeconomic inequalities in overall and COVID-19 mortality during the first outbreak peak in Emilia-Romagna Region (Northern Italy). *Epidemiol Prev* 2020;44:288–296.

108. Bosworth ML, Ayoubkhani D, Nafilyan V, Foubert J, Glickman M, Davey C et al. Deaths involving COVID-19 by self-reported disability status during the first two waves of the COVID-19 pandemic in England: a retrospective, population-based cohort study. *Lancet Public Health* 2021;6:e817–e825.
109. An C, Lim H, Kim D-W, Chang JH, Choi YJ, Kim SW. Machine learning prediction for mortality of patients diagnosed with COVID-19: a nationwide Korean cohort study. *Sci Rep* 2020;10:18716.
110. World Health Organization. WHO Public Health and Social Measures Initiative [website]. (<https://www.who.int/initiatives/who-public-health-and-social-measures-initiative>, accessed 24 July 2024).
111. Fadlallah R, El-Jardali F, Karroum LB, Kalach N, Hoteit R, Aoun A et al. The effects of public health and social measures (PHSM) implemented during the COVID-19 pandemic: An overview of systematic reviews. *Cochrane Evid Synth Methods* 2024;2:e12055.
112. Walport PSM. Executive Summary to the Royal Society report “COVID-19: examining the effectiveness of non-pharmaceutical interventions”. *Philos Trans R Soc Math Phys Eng Sci* 2023;381:20230211.
113. Osterrieder A, Cuman G, Pan-Ngum W, Cheah PK, Cheah P-K, Peerawaranun P et al. Economic and social impacts of COVID-19 and public health measures: results from an anonymous online survey in Thailand, Malaysia, the UK, Italy and Slovenia. *BMJ Open* 2021;11:e046863.
114. Shared Responsibility, Global Solidarity: Responding to the socio-economic impacts of COVID-19. United Nations, 2020 (<https://unsdg.un.org/sites/default/files/2020-03/SG-Report-Socio-Economic-Impact-of-Covid19.pdf>).
115. Li L, Taeihagh A, Tan SY. A scoping review of the impacts of COVID-19 physical distancing measures on vulnerable population groups. *Nat Commun* 2023;14:599.
116. Maddock J, Parsons S, Di Gessa G, Green MJ, Thompson EJ, Stevenson AJ et al. Inequalities in healthcare disruptions during the COVID-19 pandemic: evidence from 12 UK population-based longitudinal studies. *BMJ Open* 2022;12:e064981.
117. Emmanue F, Ahmad A, Reza T, Shahzad K, Rehman FU, Malik M et al. Indirect effects of COVID-19 pandemic on reproductive, maternal, newborn and child health services in Pakistan. *East Mediterr Health J* 2022;28:258–265.
118. NewsWise. COVID-19 in Pregnancy: Studying racial disparities and adverse birth outcomes [website]. 2022 (https://www.newswise.com/coronavirus/share-share-share-email-rounds-beata-mostafavi-february-14-2022-5-00-am-covid-19-in-pregnancy-studying-racial-disparities-and-adverse-birth-outcomes/?article_id=766111&ta=home, accessed 2 May 2024).
119. Meherali S, Adewale B, Ali S, Kennedy M, Salami B (Oladunni), Richter S et al. Impact of the COVID-19 Pandemic on Adolescents’ Sexual and Reproductive Health in Low- and Middle-Income Countries. *Int J Environ Res Public Health* 2021;18:13221.
120. Singh L, Abbas SM, Roberts B, Thompson N, Singh NS. A systematic review of the indirect impacts of COVID-19 on sexual and reproductive health services and outcomes in humanitarian settings. *BMJ Glob Health* 2023;8:e013477.
121. COVID-19 and sexual and reproductive health in low- and middle-income countries. Countdown 2030 Europe, 2022 (https://countdown2030europe.org/storage/app/media/uploaded-files/COVID-19%20and%20SRH%20in%20LMICs_C2030E_Feb2022.pdf).
122. Kurowski C, Kumar A, Mieses R, Julio C, Schmidt M, Silfverberg DV. Health Financing in a Time of Global Shocks: Strong Advance, Early Retreat. Washington (DC), The International Bank for Reconstruction and Development / The World Bank, 2023 “Double Shock, Double Recovery” Paper Series; (<https://openknowledge.worldbank.org/entities/publication/846332e1-406e-4958-ab76-cdbbb968678a>, accessed 2 May 2024).
123. United Nations Children’s Fund. COVID-19 and children [website]. 2022 (<https://data.unicef.org/covid-19-and-children/>, accessed 2 May 2024).
124. World Health Organization. Transforming the health and social equity landscape: promoting socially just and inclusive growth to improve resilience, solidarity and peace. Copenhagen, World Health Organization Regional Office for Europe, 2023 (<https://iris.who.int/handle/10665/370945>). License: CC BY-NC-SA 3.0 IGO.
125. FAO, IFAD, UNICEF, WFP, WHO. The State of Food Security and Nutrition in the World 2022. Rome, FAO, 2022 (<https://doi.org/10.4060/cc0639en>).
126. Ferentinou E, Koutelekos I, Pappa D, Manthou P, Dafogianni C. The Impact of the COVID-19 Pandemic on Childhood Obesity: A Review. *Cureus* 2023. (<https://doi.org/10.7759/cureus.45470>).
127. Ochoa-Moreno I, Taheem R, Woods-Townsend K, Chase D, Godfrey KM, Modi N et al. Projected health and economic effects of the increase in childhood obesity during the COVID-19 pandemic in England: The potential cost of inaction. *Baggaley RF, editor. PLOS ONE* 2024;19:e0296013.

128. Nour TY, Altıntaş KH. Effect of the COVID-19 pandemic on obesity and its risk factors: a systematic review. *BMC Public Health* 2023;23:1018.
129. Orpo P, Adhanom Ghebreyesus T. Building an economy for health and well-being. *Bull World Health Organ* 2024;102:298-298A.
130. World Bank. Poverty and Shared Prosperity 2022: Correcting Course. Washington (DC): World Bank; 2022 (<https://doi.org/10.1596/978-1-4648-1893-6>).
131. Ag and Food Statistics: Charting the Essentials – Food Prices and Spending | Economic Research Service [website]. ([https://www.ers.usda.gov/data-products/ag-and-food-statistics-charting-the-essentials/food-prices-and-spending?](https://www.ers.usda.gov/data-products/ag-and-food-statistics-charting-the-essentials/food-prices-and-spending?utm_source=chatgpt.com)utm_source=chatgpt.com, accessed 28 February 2025).
132. Chai A, Moneta A. Retrospectives: Engel Curves. *J Econ Perspect* 2010;24:225-240.
133. Movsisyan A, Wendel F, Bethel A, Coenen M, Krajewska J, Littlecott H et al. Inflation and health: a global scoping review. *Lancet Glob Health* 2024;12:e1038-e1048.
134. Food and Agriculture Organization (FAO). The state of food security and nutrition in the world 2024: financing to end hunger, food insecurity and malnutrition in all its forms. Rome, FAO, 2024.
135. Robinson E. Obesity and the cost of living crisis. *Int J Obes* 2023;47:93-94.
136. Global Network Against Food Crises. Global Report on Food Crises (GRFC) 2024 [website]. 2024 (<https://www.fsinplatform.org/report/global-report-food-crises-2024>, accessed 2 May 2024).
137. World Food Programme. WFP Global Operational Response Plan: Update #10 – February 2024 [website]. 2024 (<https://www.wfp.org/publications/wfp-global-operational-response-plan>, accessed 2 May 2024).
138. World Bank. Global Economic Prospects, June 2024. Washington (DC), World Bank, 2024 (<https://doi.org/10.1596/978-1-4648-2058-8>).
139. Chrimes T, Gootjes B, Ayhan Kose M, Wheeler C. The Great Reversal: Prospects, Risks, and Policies in International Development Association Countries. World Bank (<https://www.worldbank.org/en/research/publication/prospects-risks-and-policies-in-IDA-countries>, accessed 8 November 2024).
140. World Bank. The World Bank in Africa [website]. (<https://www.worldbank.org/en/region/afr/overview>, accessed 17 November 2024).
141. Yao Q, Li X, Luo F, Yang L, Liu C, Sun J. The historical roots and seminal research on health equity: a referenced publication year spectroscopy (RPYS) analysis. *Int J Equity Health* 2019;18:152.
142. Plamondon KM, Bottorff JL, Caxaj CS, Graham ID. The integration of evidence from the Commission on Social Determinants of Health in the field of health equity: a scoping review. *Crit Public Health* 2020;30:415-428.
143. Dean E. Marmot Places: the areas taking a proactive local approach to health inequalities. *BMJ* 2024;384:q654.
144. Institute of Health Equity. Marmot Places [website]. (<https://www.instituteofhealthequity.org/taking-action/taking-action/what-is-a-marmot-place.pdf>, accessed 31 May 2024).
145. Marmot M, Allen J. A programme for greater health equity for the next UK Government. *Lancet* 2024;403:2675-2677. ([https://doi.org/10.1016/S0140-6736\(24\)01243-1](https://doi.org/10.1016/S0140-6736(24)01243-1)).
146. De Schutter O. Report of the Special Rapporteur on extreme poverty and human rights. The burnout economy: poverty and mental health. UN Office of High Commissioner for Human Rights, 2024 (<https://undocs.org/Home/Mobile?FinalSymbol=A%2F79%2F162&Language=E&DeviceType=Desktop&LangRequested=False>, accessed 31 October 2024).
147. Guidance on mental health policy and strategic action plans. Module 1. Introduction, purpose and use of the guidance. Geneva, World Health Organization, 2025 (<https://iris.who.int/handle/10665/380465>, accessed 6 March 2025).
148. Östlin P, Schrecker T, Sadana R, Bonnefoy J, Gilson L, Hertzman C et al. Priorities for Research on Equity and Health: Towards an Equity-Focused Health Research Agenda. *PLoS Med* 2011;8:e1001115.
149. Silva M, Loureiro A, Cardoso G. Social determinants of mental health: A review of the evidence. *Eur J Psychiatry* 2016;30:259-292.
150. Rasanathan K, Diaz T. Research on health equity in the SDG era: the urgent need for greater focus on implementation. *Int J Equity Health* 2016;15:202.
151. Sadana R, Harper S. Data Systems Linking Social Determinants of Health with Health Outcomes: Advancing Public Goods to Support Research and Evidence-Based Policy and Programs. *Public Health Rep* 2011;126:6-13.
152. Freudenberg N, Lee K, Buse K, Collin J, Crosbie E, Friel S et al. Defining Priorities for Action and Research on the Commercial Determinants of Health: A Conceptual Review. *Am J Public Health* 2021;111:2202-2211.

153. Setting Global Research Priorities for Urban Health, 1st ed. Geneva, World Health Organization. 2022 (<https://iris.who.int/handle/10665/363443>). License: CC BY-NC-SA 3.0 IGO.
154. Miller G, Beaudin-Seiler B, Franco N. Do Investments in the Social Determinants of Health Reduce Health Care Costs? Research Consortium for Healthcare Value Assessment, 2021 (https://www.hcvalueassessment.org/application/files/7416/1981/6045/Research_Consortium_Research_Brief_No_4_-_Social_Determinants_of_Health.pdf).
155. Holt-Lunstad J. Social Connection as a Public Health Issue: The Evidence and a Systemic Framework for Prioritizing the "Social" in Social Determinants of Health. *Annu Rev Public Health* 2022;43:193-213.
156. Wellbeing Economy Alliance. Wellbeing Economy Alliance [website]. 2022 (<http://weall.org/>, accessed 15 March 2024).
157. The Geneva Charter for Well-being. 2021. (<https://www.who.int/publications/m/item/the-geneva-charter-for-well-being>, accessed 25 September 2024).
158. World Health Organization. Action on Social Determinants of Health Equity [website]. 2024 (<https://www.who.int/initiatives/action-on-the-social-determinants-of-health-for-advancing-equity>, accessed 28 April 2024).
159. Sadana R, Blas E. What Can Public Health Programs Do to Improve Health Equity? *Public Health Rep* 2013;128:12-20.
160. Goldblatt P, Castedo A, Allen J, Lionello L, Bell R, Marmot M et al. Rapid review of inequalities in health and wellbeing in Norway since 2014. Institute of Health Equity, 2023 (<https://www.instituteofhealthequity.org/resources-reports/rapid-review-of-inequalities-in-health-and-wellbeing-in-norway-since-2014>, accessed 31 May 2024).
161. Health in All Policies as part of the primary health care agenda on multisectoral action. World Health Organization, 2018 (<https://iris.who.int/bitstream/handle/10665/326463>, accessed 8 April 2024). License: CC BY-NC-SA 3.0 IGO.
162. Operational framework for primary health care: transforming vision into action. Geneva, World Health Organization, 2020 (<https://iris.who.int/handle/10665/337641>, accessed 22 July 2024). License: CC BY-NC-SA 3.0 IGO.
163. Rajan D, Rouleau K, Winkelmann J, Kringos D, Jakab M, Khalid F, editors. Implementing the primary health care approach: a primer. Geneva, World Health Organization, 2024 (<https://iris.who.int/handle/10665/376777>). License: CC BY-NC-SA 3.0 IGO.
164. International Conference on Primary Health Care. Declaration of Alma-Ata [website]. 1978 (<https://www.who.int/teams/social-determinants-of-health/declaration-of-alma-ata>, accessed 4 June 2024).
165. Global Conference on Primary Health Care: Astana, Kazakhstan. Declaration of Astana. (<https://www.who.int/publications/item/WHO-HIS-SDS-2018.61>, accessed 25 September 2024).
166. Kota KK, Hong J, Zelaya C, Riser AP, Rodriguez A, Weller DL et al. Racial and Ethnic Disparities in Mpox Cases and Vaccination Among Adult Males – United States, May–December 2022. *MMWR Morb Mortal Wkly Rep* 2023;72:398-403.
167. Abrams EM, Szefer SJ. COVID-19 and the impact of social determinants of health. *Lancet Respir Med* 2020;8:659-661.
168. United Nations Research Institute for Social Development (UNRISD). Crises of Inequality: Shifting Power for a New Eco-Social Contract. UNRISD (<https://www.unrisd.org>, accessed 26 September 2024).
169. World Health Organization Regional Office for Africa. A journey towards Health in All Policies in Tanzania. 2019 (<https://www.afro.who.int/publications/journey-towards-health-all-policies-tanzania>, accessed 26 September 2024).
170. Working Together for Equity and Healthier Populations: Sustainable Multisectoral Collaboration Based on Health in All Policies Approaches, 1st ed. Geneva, World Health Organization. 2023 (<https://iris.who.int/handle/10665/372714>). License: CC BY-NC-SA 3.0 IGO.
171. National Health Security Office – Thailand. The way forward of Thai UHC: Address quality and safe care, social determinants of health, primary care, and multi-sectoral collaboration. [website]. 2022 (<http://eng.nhso.go.th/view/1/DescriptionNews/The-way-forward-of-Thai-UHC-Address-quality-and-safe-care-social-determinants-of-health-primary-care-and-multi-sectoral-collaboration-/487/EN-US>, accessed 23 October 2024).
172. Valentine N, Solar O, Mendez F, Elfeky S. Learnings from the Special Initiative for Action on the Social Determinants of Health are important for strengthening health equity strategies [website]. 2023 (<https://www.medicusmundi.ch/en/advocacy/publications/mms-bulletin/health-determinants/kapitel-1/text-von-who-nadine-valentine>, accessed 26 September 2024).
173. Royal Swedish Academy of Sciences. The Prize in Economic Sciences 2024 [website]. 2024 (<https://www.nobelprize.org/prizes/economic-sciences/2024/press-release/>).

174. Broadbent P, Thomson R, Kopasker D, McCartney G, Meier P, Richiardi M et al. The public health implications of the cost-of-living crisis: outlining mechanisms and modelling consequences. *Lancet Reg Health – Eur* 2023;27:100585.
175. Alam K, Mahal A. Economic impacts of health shocks on households in low and middle income countries: a review of the literature. *Glob Health* 2014;10:21.
176. Headey DD, Ruel MT. Economic shocks predict increases in child wasting prevalence. *Nat Commun* 2022;13:2157.
177. Neves JA, Vasconcelos FDAGD, Machado ML, Recine E, Garcia GS, Medeiros MATD. The Brazilian cash transfer program (Bolsa Família): A tool for reducing inequalities and achieving social rights in Brazil. *Glob Public Health* 2022;17:26–42.
178. Valombola AT, Omomowo KE. The Impact of Child Social Grant in the Livelihood of Households in a Rural Constituency in Namibia. *J Asian Afr Stud* 2024;00219096241300443.
179. WHO Commission on Macroeconomics and Health. Macroeconomics and health: investing in health for economic development / Report of the Commission on Macroeconomics and Health. *Macroéconomie Santé Invest Dans Santé Pour Dév Économique Rapp Comm Macroéconomie Santé* 2001. (<https://iris.who.int/handle/10665/42435>).
180. Naik Y, Baker P, Ismail SA, Tillmann T, Bash K, Quantz D et al. Going upstream – an umbrella review of the macroeconomic determinants of health and health inequalities. *BMC Public Health* 2019;19:1678.
181. Korpi W, Palme J. The Paradox of Redistribution and Strategies of Equality: Welfare State Institutions, Inequality, and Poverty in the Western Countries. *Am Sociol Rev* 1998;63:661.
182. WHO Council on the Economics of Health for All. Health for all: transforming economies to deliver what matters: final report of the WHO Council on the Economics of Health for All. Geneva, World Health Organization, 2023 (<https://iris.who.int/handle/10665/373122>). License: CC BY-NC-SA 3.0 IGO.
183. Coates MM, Ezzati M, Robles Aguilar G, Kwan GF, Vigo D, Mocumbi AO et al. Burden of disease among the world's poorest billion people: An expert-informed secondary analysis of Global Burden of Disease estimates. *PLoS ONE* 2021;16:e0253073.
184. Our World in Data. Child mortality vs. share in extreme poverty [website]. 2023 (<https://ourworldindata.org/grapher/extreme-poverty-vs-child-mortality>, accessed 2 October 2024).
185. United Nations. Extreme Poverty in Developing Countries Inextricably Linked to Global Food Insecurity Crisis, Senior Officials Tell Second Committee [website]. 2023 (<https://press.un.org/en/2023/gaef3590.doc.htm>, accessed 2 October 2024).
186. Chenery HB, Schultz TP, Srinivasan TN, Strauss J, Behrman JR. *Handbook of Development Economics*. Elsevier, 1988.
187. Galea S. *Macrosocial Determinants of Population Health*. New York, NY, Springer, 2007 (<https://doi.org/10.1007/978-0-387-70812-6>).
188. Rothstein B, Uslaner EM. All for All: Equality, Corruption, and Social Trust. *World Polit* 2005;58:41–72.
189. Rözer JJ, Volker B. Does income inequality have lasting effects on health and trust? *Soc Sci Med* 2016;149:37–45.
190. World Health Organization. WHO releases the largest global collection of health inequality data [website]. 2023 (<https://www.who.int/news/item/20-04-2023-who-releases-the-largest-global-collection-of-health-inequality-data>).
191. WHO Regional Office for Europe. Healthy, prosperous lives for all: the European health equity status report. 2019. (<https://apps.who.int/iris/handle/10665/326879>). License: CC BY-NC-SA 3.0 IGO.
192. Zhang L, Simmel C, Nepomnyaschy L. Income inequality and child maltreatment rates in US counties, 2009–2018. *Child Abuse Negl* 2022;130:105328.
193. Ataguba JE-O, Birungi C, Cunial S, Kavanagh M. Income inequality and pandemics: insights from HIV/AIDS and COVID-19 – a multicountry observational study. *BMJ Glob Health* 2023;8:e013703.
194. McKinsey Global Institute. Prioritizing health: A prescription for prosperity [website]. 2020 (<https://www.mckinsey.com/industries/healthcare/our-insights/prioritizing-health-a-prescription-for-prosperity>, accessed 2 May 2024).
195. World Health Organization. A Heavy Burden: The Productivity Cost of Illness in Africa. Brazzaville, World Health Organization Regional Office for Africa, 2019 (<https://iris.who.int/handle/10665/312173>). License: CC BY-NC-SA 3.0 IGO.
196. Aghion P, Howitt P, Murtin F. The Relationship Between Health and Growth: When Lucas Meets Nelson-Phelps. *Rev Econ Inst* 2010;2. (<https://doi.org/10.3386/w15813>).
197. Sassi F. *Obesity and the economics of prevention: fit not fat*. Paris, OECD, 2010.

198. Economic Commission for Latin America and the Caribbean. Economic Survey of Latin America and the Caribbean, 2021. Santiago, Economic Commission for Latin America and the Caribbean, 2021 (<https://repositorio.cepal.org/server/api/core/bitstreams/b888a9e3-9c0b-4e22-8b52-6e159f40829e/content>).
199. Brown L, Thurecht L, Nepal B. The cost of inaction on the social determinants of health. Canberra, Catholic Health Australia, 2012 CHA-NATSEM Second Report on Health Inequalities; (<https://apo.org.au/node/29676>, accessed 2 May 2024).
200. The Lancet. Series from the Lancet Journals: Commercial Determinants of Health. Lancet. 2023. (<https://www.thelancet.com/series/commercial-determinants-health>).
201. Hiscock R, Bauld L, Amos A, Fidler JA, Munafò M. Socioeconomic status and smoking: a review. *Ann N Y Acad Sci* 2012;1248:107-123.
202. Gilmore AB, Fabbri A, Baum F, Bertscher A, Bondy K, Chang H-J et al. Defining and conceptualising the commercial determinants of health. *Lancet* 2023;401:1194-1213.
203. Institute of Health Equity, Legal & General. The Business of Health Equity: The Marmot Review for Industry. (<https://www.instituteoftheequity.org/resources-reports/the-business-of-health-equity-the-marmot-review-for-industry>, accessed 20 May 2024).
204. World Health Organization. Commercial Determinants of Noncommunicable Diseases in the WHO European Region. Copenhagen, World Health Organization. Regional Office for Europe, 2024 (<https://iris.who.int/handle/10665/376957>, accessed 22 July 2024). License: CC BY-NC-SA 3.0 IGO.
205. Mialon M, Vandevijvere S, Carriedo-Lutzenkirchen A, Bero L, Gomes F, Petticrew M et al. Mechanisms for addressing and managing the influence of corporations on public health policy, research and practice: a scoping review. *BMJ Open* 2020;10:e034082.
206. The Lancet. Unveiling the predatory tactics of the formula milk industry. *Lancet* 2023;401:409.
207. Klinenberg E. Palaces for the People. Penguin Random House, 2018 (<https://www.penguinrandomhouse.com/books/557044/palaces-for-the-people-by-eric-klinenberg/>, accessed 5 June 2024).
208. Lindert PH. Growing Public: Social Spending and Economic Growth since the Eighteenth Century: Volume 1: The Story. Cambridge, Cambridge University Press, 2004 (<https://doi.org/10.1017/CBO9780511510717>).
209. Ocampo JA, Gómez-Arteaga N. Social protection systems in Latin America: An assessment. International Labour Organization, 2016 Extension of Social Security – Working Paper – ESS 52; (<https://www.ilo.org/publications/social-protection-systems-latin-america-assessment>, accessed 2 May 2024).
210. Rubin J, Taylor J, Krapels J, Sutherland A, Felician MF, Liu JL et al. Are better health outcomes related to social expenditure? A cross-national empirical analysis of social expenditure and population health measures. Santa Monica, Calif., and Cambridge, UK, RAND Corporation, 2016 (https://www.rand.org/pubs/research_reports/RR1252.html, accessed 13 March 2024).
211. Population Division – United Nations. 2018 Revision of World Urbanization Prospects [website]. 2018 (https://population.un.org/wup/?_gl=1*6hfjx5*_ga*MjA0NDA3MzI5OS4xNzA5ODA3Mz*_ga_TK9BQL5X7Z*MTcxNzU5NDgwOC43LjAuMTcxNzU5NDgwOC4wLjAuMA, accessed 5 June 2024).
212. Kawachi I, Berkman LF. Neighborhoods and Health. Oxford University Press, 2003 (<https://books.google.com.br/books?id=KtXnCwAAQBAJ>).
213. Hosking J, Braubach M, Buss D, Khayesi M, Filho VP, de Sá TH. Towards a global framework for transport, health and health equity. *Environ Int* 2022;169:107472.
214. Chala B, Hamde F. Emerging and Re-emerging Vector-Borne Infectious Diseases and the Challenges for Control: A Review. *Front Public Health* 2021;9. (<https://doi.org/10.3389/fpubh.2021.715759>).
215. Kolimenakis A, Heinz S, Wilson ML, Winkler V, Yakob L, Michaelakis A et al. The role of urbanisation in the spread of Aedes mosquitoes and the diseases they transmit—A systematic review. *PLoS Negl Trop Dis* 2021;15:e0009631.
216. Fournet F, Jourdain F, Bonnet E, Degroote S, Ridde V. Effective surveillance systems for vector-borne diseases in urban settings and translation of the data into action: a scoping review. *Infect Dis Poverty* 2018;7:99.
217. Sadana R, Soucat A, Beard J. Universal health coverage must include older people. *Bull World Health Organ* 2018;96:2-2A.
218. Global Report on Ageism. 1st ed. Geneva, World Health Organization. 2021 (<https://iris.who.int/handle/10665/340208>). License: CC BY-NC-SA 3.0 IGO.
219. World Health Organization. Ageing and health [website]. (<https://www.who.int/news-room/fact-sheets/detail/ageing-and-health>, accessed 26 September 2024).

220. Wu Y-T, Daskalopoulou C, Muniz Terrera G, Sanchez Niubo A, Rodríguez-Artalejo F, Ayuso-Mateos JL et al. Education and wealth inequalities in healthy ageing in eight harmonised cohorts in the ATHLOS consortium: a population-based study. *Lancet Public Health* 2020;5:e386–e394.
221. Moreno-Agostino D, Daskalopoulou C, Wu Y-T, Koukounari A, Haro JM, Tyrovolas S et al. The impact of physical activity on healthy ageing trajectories: evidence from eight cohort studies. *Int J Behav Nutr Phys Act* 2020;17:92.
222. Lee J, Meijer E, Phillips D, Hu P. Disability Incidence Rates for Men and Women in 23 Countries: Evidence on Health Effects of Gender Inequality. Newman AB, editor. *J Gerontol Ser A* 2021;76:328–338.
223. World Health Organization. Social isolation and loneliness among older people: advocacy brief [website]. 2023 (<https://www.who.int/publications-detail-redirect/9789240030749>, accessed 12 March 2024).
224. Barreto M, Victor C, Hammond C, Eccles A, Richins MT, Qualter P. Loneliness around the world: Age, gender, and cultural differences in loneliness. *Personal Individ Differ* 2021;169:110066.
225. Madero-Cabib I, Corna L, Baumann I. Aging in Different Welfare Contexts: A Comparative Perspective on Later-Life Employment and Health. Raymo JM, editor. *J Gerontol Ser B* 2020;75:1515–1526.
226. Abels M, Arribas-Banos L, Demarco G. The gender pension gap: What does it tell us and what should be done about it? [website]. 2023 (<https://blogs.worldbank.org/en/jobs/gender-pension-gap-what-does-it-tell-us-and-what-should-be-done-about-it>, accessed 6 June 2024).
227. Twenge J, Blanchflower D. Declining Life Satisfaction and Happiness Among Young Adults in Six English-speaking Countries. Cambridge, MA, National Bureau of Economic Research, 2025 (<https://doi.org/10.3386/w33490>).
228. Selvarajah S, Corona Maioli S, Deivanayagam TA, De Morais Sato P, Devakumar D, Kim S-S et al. Racism, xenophobia, and discrimination: mapping pathways to health outcomes. *Lancet* 2022;400:2109–2124.
229. Scholars Strategy Network. How Discrimination Hurts Health and Personal Wellbeing [website]. 2014 (<https://scholars.org/contribution/how-discrimination-hurts-health-and-personal-wellbeing>, accessed 21 May 2024).
230. Healthy People 2030. Discrimination [website]. 2024 (<https://survey.alchemer.com/s3/7793088/2024-Phase-1-HP2030-Microsite-Survey>, accessed 21 May 2024).
231. Mental Health Foundation. Stigma and discrimination [website]. 2023 (<https://www.mentalhealth.org.uk/explore-mental-health/a-z-topics/stigma-and-discrimination>, accessed 21 May 2024).
232. Sen G, Iyer A, Mukherjee C. A Methodology to Analyse the Intersections of Social Inequalities in Health. *J Hum Dev Capab* 2009;10:397–415.
233. Kadambari S, Vanderslott S. Lessons about COVID-19 vaccine hesitancy among minority ethnic people in the UK. *Lancet Infect Dis* 2021;21:1204–1206.
234. Morgan KM, Maglalang DD, Monnig MA, Ahluwalia JS, Avila JC, Sokolovsky AW. Medical Mistrust, Perceived Discrimination, and Race: a Longitudinal Analysis of Predictors of COVID-19 Vaccine Hesitancy in US Adults. *J Racial Ethn Health Disparities* 2023;10:1846–1855.
235. Office for National Statistics. Coronavirus and vaccine hesitancy, Great Britain [website]. 2021 (<https://www.ons.gov.uk/peoplepopulationandcommunity/healthandsocialcare/healthandwellbeing/bulletins/coronavirusandvaccinehesitancygreatbritain/9august2021#coronavirus-and-vaccine-hesitancy-data>, accessed 4 June 2024).
236. Springer KW, Hankivsky O, Bates LM. Gender and health: relational, intersectional, and biosocial approaches. *Soc Sci Med* 1982 2012;74:1661–1666.
237. El-Bassel N, Mukherjee TI, Stoicescu C, Starbird LE, Stockman JK, Frye V et al. Intertwined epidemics: progress, gaps, and opportunities to address intimate partner violence and HIV among key populations of women. *Lancet HIV* 2022;9:e202–e213.
238. Hottes TS, Ferlatte O, Gesink D. Suicide and HIV as leading causes of death among gay and bisexual men: a comparison of estimated mortality and published research. *Crit Public Health* 2015;25:513–526.
239. World Health Organization. Violence against women [website]. 2024 (<https://www.who.int/news-room/fact-sheets/detail/violence-against-women>, accessed 5 June 2024).
240. World Economic Forum. Closing the Women's Health Gap to Improve Lives and Economies: A \$1 Trillion Opportunity to Improve Lives and Economies. 2024 (<https://www.weforum.org/publications/closing-the-women-s-health-gap-a-1-trillion-opportunity-to-improve-lives-and-economies/>, accessed 2 October 2024).
241. World Bank. Women, Business and the Law 2023. Washington (DC), World Bank, 2023 (<https://hdl.handle.net/10986/39462>, accessed 2 May 2024).

242. UN Secretary General. Women's economic empowerment in the changing world of work: report of the Secretary-General. New York, United Nations, 2016 (<https://digitallibrary.un.org/record/856760>, accessed 2 May 2024).
243. The Gender Pay Gap in the Health and Care Sector: A Global Analysis in the Time of COVID-19, 1st ed. Geneva, World Health Organization. 2022 (<https://iris.who.int/handle/10665/358057>). License: CC BY-NC-SA 3.0 IGO.
244. Women on the move: migration, care work and health. Geneva, World Health Organization, 2017 (<https://iris.who.int/handle/10665/259463>, accessed 9 January 2025). License: CC BY-NC-SA 3.0 IGO.
245. Seedat S, Rondon M. Women's wellbeing and the burden of unpaid work. *BMJ* 2021;374:n1972.
246. Devakumar D, Selvarajah S, Abubakar I, Kim S-S, McKee M, Sabharwal NS et al. Racism, xenophobia, discrimination, and the determination of health. *Lancet* 2022;400:2097-2108.
247. LeMasters K, Brinkley-Rubinstein L, Maner M, Peterson M, Nowotny K, Bailey Z. Carceral epidemiology: mass incarceration and structural racism during the COVID-19 pandemic. *Lancet Public Health* 2022;7:e287-e290.
248. Bhugra D, Wijesuriya R, Gnanapragasam S, Persaud A. Black and minority mental health in the UK: Challenges and solutions. *Forensic Sci Int Mind Law* 2020;1:100036.
249. Committee on the Elimination of Racial Discrimination. General recommendation No. 37 (2024) on racial discrimination in the enjoyment of the right to health. (<https://www.ohchr.org/en/documents/general-comments-and-recommendations/general-recommendation-no-37-2024-racial>, accessed 25 September 2024).
250. LaFave SE, Suen JJ, Seau Q, Bergman A, Fisher MC, Thorpe RJ et al. Racism and Older Black Americans' Health: a Systematic Review. *J Urban Health Bull N Y Acad Med* 2022;99:28-54.
251. Anim SB, Spurlark R, Turkson-Ocran R-A, Bohr N, Soco C, Simonovich SD. A Systematic Review of the Relationship Between Discrimination, Racism, and Type 2 Diabetes Healthcare Outcomes for Black Americans. *J Racial Ethn Health Disparities* 2023.
252. Williams DR, Lawrence JA, Davis BA. Racism and Health: Evidence and Needed Research. *Annu Rev Public Health* 2019;40:105-125.
253. Egede LE, Walker RJ, Campbell JA, Linde S, Hawks LC, Burgess KM. Modern Day Consequences of Historic Redlining: Finding a Path Forward. *J Gen Intern Med* 2023;38:1534-1537.
254. Boileau B, Sturrock D. Bank of mum and dad drives increasing economic inequalities in early adulthood [website]. 2023 (<https://ifs.org.uk/news/bank-mum-and-dad-drives-increasing-economic-inequalities-early-adulthood>, accessed 9 April 2024).
255. World Bank. Indigenous Peoples [website]. 2023 (<https://www.worldbank.org/en/topic/indigenouspeoples>, accessed 21 March 2024).
256. Alberta Health [online database]. Health indicator: life expectancy at birth by First Nations status. Edmonton: Government of Alberta. 2024. (<https://open.alberta.ca/opendata/health-indicator-life-expectancy-at-birth-by-first-nations-status>, accessed 27 February 2025).
257. Rodríguez-Díaz CE, Ramos JGP, Adriana Garriga-López. How Is Colonialism a Sociostructural Determinant of Health in Puerto Rico? *AMA J Ethics* 2022;24:E305-312. (<https://doi.org/10.1001/amajethics.2022.305>).
258. Axelsson P, Kukutai T, Kippen R. The field of Indigenous health and the role of colonisation and history. *J Popul Res* 2016;33:1-7. (<https://doi.org/10.1007/s12546-016-9163-2>).
259. Pan American Health Organization. Just Societies: Health Equity and Dignified Lives. Executive Summary of the Report of the Commission of the Pan American Health Organization on Equity and Health Inequalities in the Americas. Revised Edition. PAHO, 2019 (<https://iris.paho.org/handle/10665.2/51571>).
260. Krahn GL, Walker DK, Correa-De-Araujo R. Persons with Disabilities as an Unrecognized Health Disparity Population. *Am J Public Health* 2015;105:S198-S206.
261. O'Leary L, Hughes-McCormack L, Dunn K, Cooper S. Early death and causes of death of people with Down syndrome: A systematic review. *J Appl Res Intellect Disabil* 2018;31:687-708.
262. Global Report on Health Equity for Persons with Disabilities, 1st ed. Geneva, World Health Organization. 2022 (<https://iris.who.int/handle/10665/364834>). License: CC BY-NC-SA 3.0 IGO.
263. Sadana R, Blas E, Budhwani S, Koller T, Paraje G. Healthy Ageing: Raising Awareness of Inequalities, Determinants, and What Could Be Done to Improve Health Equity. *Gerontologist* 2016;56:S178-S193.
264. World Health Organization. First International Conference on Health Promotion, Ottawa, 21 November 1986 [website]. 2024 (<https://www.who.int/teams/health-promotion/enhanced-wellbeing/first-global-conference>, accessed 25 March 2024).
265. Uppsalla University - Department of Peace and Conflict. UCDP - Uppsalla Conflict Data Program [website]. (<https://ucdp.uu.se/>, accessed 25 September 2024).

266. Garry S, Checchi F. Armed conflict and public health: into the 21st century. *J Public Health* 2020;42:e287–e298.
267. Asi YM. How war kills: the overlooked threats to our health. JHU Press, 2024.
268. Benjet C, Bromet E, Karam EG, Kessler RC, McLaughlin KA, Ruscio AM et al. The epidemiology of traumatic event exposure worldwide: results from the World Mental Health Survey Consortium. *Psychol Med* 2016;46:327–343.
269. Nurturing Care for Early Childhood Development. Nurturing care for children living in humanitarian settings. 2020. (<https://nurturing-care.org/nurturing-care-in-humanitarian-settings/>, accessed 25 March 2024).
270. UNESCO, editor. Challenging inequalities: pathways to a just world. Paris, UNESCO Publishing, 2016. World Social Science Report, No. 2016.
271. World Health Organization. WHO Global Health and Peace Initiative (GHPI) [website]. (<https://www.who.int/initiatives/who-health-and-peace-initiative>, accessed 20 October 2024).
272. Core Writing Team, Lee H and Romero J (eds). Summary for Policymakers. In *Climate Change 2023: Synthesis Report. Contribution of Working Groups I, II and III to the Sixth Assessment Report of the Intergovernmental Panel on Climate Change*. Geneva, IPCC, 2023 (https://www.ipcc.ch/report/ar6/syr/downloads/report/IPCC_AR6_SYR_SPM.pdf).
273. Romanello M, Walawender M, Hsu S-C, Moskeland A, Palmeiro-Silva Y, Scamman D et al. The 2024 report of the Lancet Countdown on health and climate change: facing record-breaking threats from delayed action. *Lancet* 2024;404:1847–1896.
274. Jafino BA, Walsh BJ, Rozenberg J, Hallegatte S. Revised Estimates of the Impact of Climate Change on Extreme Poverty by 2030. Washington (DC). World Bank Group, 2020 Policy Research Working Paper no. WPS 9417, COVID-19 (Coronavirus). (<https://documents.worldbank.org/en/publication/documents-reports/documentdetail/706751601388457990/Revised-Estimates-of-the-Impact-of-Climate-Change-on-Extreme-Poverty-by-2030>, accessed 24 May 2024).
275. Chancel L. Global carbon inequality over 1990–2019. *Nat Sustain* 2022;5:931–938.
276. Nazrul IS, Winkel, John, Department of Economic and Social Affairs. Climate change and social inequality. New York, UN, Department of Economic and Social Affairs, 2017 DESA working paper. (<https://digitallibrary.un.org/record/3859027?ln=en&v=pdf>, accessed 2 May 2024).
277. Prina M, Khan N, Akhter Khan S, Caicedo JC, Peycheva A, Seo V et al. Climate change and healthy ageing: An assessment of the impact of climate hazards on older people. *J Glob Health* 2024;14:04101.
278. Proulx K, Daelmans B, Baltag V, Banati P. Climate change impacts on child and adolescent health and well-being: A narrative review. *J Glob Health* 2024;14:04061.
279. Conway F, Portela A, Filippi V, Chou D, Kovats S. Climate change, air pollution and maternal and newborn health: An overview of reviews of health outcomes. *J Glob Health* 2024;14:04128.
280. Etzel RA, Weimann E, Homer C, Arora NK, Maimela G, Prats EV et al. Climate change impacts on health across the life course. *J Glob Health* 2024;14:03018.
281. Hallegatte S, Rozenberg J. Climate change through a poverty lens. *Nat Clim Change* 2017;7:250–256. (<https://doi.org/10.1038/nclimate3253>).
282. World Health Organization, CBM, International Federation of Red Cross and Red Crescent Societies, International Organization for Migration, United Nations Children's Fund (UNICEF). Guidance note on disability and emergency risk management for health. Geneva, World Health Organization, 2013 (<https://iris.who.int/handle/10665/90369>, accessed 9 January 2025).
283. Pacific Disability Forum. Disability and Climate Change in the Pacific: Findings from Kiribati, Solomon Islands, and Tuvalu. Pacific Disability Forum, 2022 (<https://pacificdisability.org/wp-content/uploads/2022/08/PDF-Final-Report-on-Climate-Change-and-Persons-with-Disabilities.pdf>).
284. The Economic and Social Commission for Asia and the Pacific (ESCAP). Building disability-inclusive societies in Asia and the Pacific: assessing progress of the Incheon strategy. United Nations, 2017 (<https://repository.unescap.org/handle/20.500.12870/3040>, accessed 25 July 2024).
285. World Health Organization. Rabat Declaration adopted to improve refugee and migrant health [website]. 2023 (<https://www.who.int/news/item/16-06-2023-rabat-declaration-adopted-to-improve-refugee-and-migrant-health>, accessed 29 March 2024).
286. Global Alliance on War, Conflict and Health. Global Alliance on War, Conflict and Health Launch Event [website]. 2022 (<https://www.warconflicthealth.org/launch2022>, accessed 25 March 2024).
287. UNHCR. Global Trends Report [website]. 2024 (<https://www.unhcr.org/uk/global-trends>, accessed 18 June 2024).

288. UNHCR. Figures at a glance [website]. (<https://www.unhcr.org/about-unhcr/who-we-are/figures-glance>, accessed 18 June 2024).
289. International Refugee Assistance Project. Advocacy Memo on Climate Displacement. 2022 (<https://refugeerights.org/wp-content/uploads/2022/11/Climate-Displacement-Advocacy-Memo-Nov-2022.pdf>).
290. The White House. Report on the Impact of Climate Change on Migration. 2021 (<https://www.whitehouse.gov/wp-content/uploads/2021/10/Report-on-the-Impact-of-Climate-Change-on-Migration.pdf>).
291. Negev M, Teschner N, Rosenthal A, Levine H, Lew-Levy C, Davidovitch N. Adaptation of health systems to climate-related migration in Sub-Saharan Africa: Closing the gap. *Int J Hyg Environ Health* 2019;222:311–314.
292. Evagora-Campbell M, Zahidie A, Buse K, Rabbani F, Hawkes S. Promoting labour migrant health equity through action on the structural determinants: A systematic review. *J Migr Health* 2022;5:100082.
293. Smith GD, Kuh D. Commentary: William Ogilvy Kermack and the childhood origins of adult health and disease. *Int J Epidemiol* 2001;30:696–703.
294. Forsdahl A. Observations throwing light on the high mortality in the county of Finnmark: Is the high mortality today a late effect of very poor living conditions in childhood and adolescence? *Int J Epidemiol* 2002;31:302–308.
295. Black MM, Walker SP, Fernald LCH, Andersen CT, DiGirolamo AM, Lu C et al. Early childhood development coming of age: science through the life course. *Lancet Lond Engl* 2017;389:77–90.
296. Bruun Jensen B, Currie C, Dyson A, Eisenstadt N, Melhuish E. Early years, family and education task group: Review of social determinants of health and the health divide in the WHO European Region. World Health Organization. Regional Office for Europe, 2013 (<https://iris.who.int/handle/10665/344815>, accessed 2 May 2024).
297. Hahn RA, Barnett WS, Knopf JA, Truman BI, Johnson RL, Fielding JE et al. Early Childhood Education to Promote Health Equity: A Community Guide Systematic Review. *J Public Health Manag Pract* 2016;22:E1–E8.
298. Human Early Learning Partnership and the Commission on Social Determinants of Health. Early child development: a powerful equalizer: final report for the World Health Organization's Commission on the Social Determinants of Health. 2007. (<https://iris.who.int/handle/10665/69729>, accessed 25 July 2024).
299. Roberts, J, Donkin A, Marmot M. Opportunities for reducing socioeconomic inequalities in the mental health of children and young people—Reducing adversity and increasing resilience. *J Public Ment Health* 2016;15:4–18.
300. Hillis S, Mercy J, Amobi A, Kress H. Global Prevalence of Past-year Violence Against Children: A Systematic Review and Minimum Estimates. *Pediatrics* 2016;137:e20154079.
301. Bellis MA, Hughes K, Anderson Z, Tocque K, Hughes S. Contribution of violence to health inequalities in England: demographics and trends in emergency hospital admissions for assault. *J Epidemiol Community Health* 2008;62:1064–1071.
302. Mok PLH, Antonsen S, Pedersen CB, Carr MJ, Kapur N, Nazroo J et al. Family income inequalities and trajectories through childhood and self-harm and violence in young adults: a population-based, nested case-control study. *Lancet Public Health* 2018;3:e498–e507.
303. Akseer N, Mehta S, Wigle J, Chera R, Brickman ZJ, Al-Gashm S et al. Non-communicable diseases among adolescents: current status, determinants, interventions and policies. *BMC Public Health* 2020;20:1908.
304. Viner RM, Coffey C, Mathers C, Bloem P, Costello A, Santelli J et al. 50-year mortality trends in children and young people: a study of 50 low-income, middle-income, and high-income countries. *Lancet Lond Engl* 2011;377:1162–1174.
305. UN Inter-agency Group for Child Mortality Estimation (UN IGME). Levels and trends in child mortality: report 2021. UN IGME. 2021 (<https://www.who.int/publications/m/item/levels-and-trends-in-child-mortality-report-2021>, accessed 2 May 2024).
306. Orben A, Tomova L, Blakemore S-J. The effects of social deprivation on adolescent development and mental health. *Lancet Child Adolesc Health* 2020;4:634–640.
307. Colizzi M, Lasalvia A, Ruggeri M. Prevention and early intervention in youth mental health: is it time for a multidisciplinary and trans-diagnostic model for care? *Int J Ment Health Syst* 2020;14:23.
308. Heckman JJ. Investing in Disadvantaged Young Children is an Economically Efficient Policy. New York, 2006 (https://jenni.uchicago.edu/investing/invest-disadv_2005-12-22_247pm_awb.pdf).
309. Raghupathi V, Raghupathi W. The influence of education on health: an empirical assessment of OECD countries for the period 1995–2015. *Arch Public Health* 2020;78:20.

310. Lim SS, Updike RL, Kaldjian AS, Barber RM, Cowling K, York H et al. Measuring human capital: a systematic analysis of 195 countries and territories, 1990–2016. *Lancet* 2018;392:1217–1234.
311. Karasek R (Robert A). *Healthy work: stress, productivity, and the reconstruction of working life*. New York: Basic Books, 1990 (<http://archive.org/details/healthyworkstres0000kara>, accessed 5 June 2024).
312. Siegrist J. Adverse health effects of high-effort/low-reward conditions. *J Occup Health Psychol* 1996;1:27–41.
313. Linn MW, Sandifer R, Stein S. Effects of unemployment on mental and physical health. *Am J Public Health* 1985;75:502–506.
314. United Nations Department of Economic and Social Affairs. *Disability and Development Report: Realizing the Sustainable Development Goals by, for and with Persons with Disabilities*. United Nations. 2019. (<https://www.un.org/en/desa/un-disability-and-development-report---realizing-sdgs-and-persons-disabilities>, accessed 20 March 2025).
315. Emerson E, Hatton C, Baines S, Robertson J. The association between employment status and health among British adults with and without intellectual impairments: cross-sectional analyses of a cohort study. *BMC Public Health* 2018;18:401.
316. WHO/ILO Joint Estimates of the Work-Related Burden of Disease and Injury, 2000–2016: *Global Monitoring Report*, 1st ed. Geneva, World Health Organization and International Labour Office. 2021 (<https://iris.who.int/handle/10665/345242>). License: CC BY-NC-SA 3.0 IGO.
317. Rantanen J, Lehtinen S, Valenti A, Iavicoli S. A global survey on occupational health services in selected International Commission on Occupational Health (ICOH) member countries. *BMC Public Health* 2017;17:787.
318. Fujishiro K, Ahonen EQ, Winkler M. Investigating Employment Quality for Population Health and Health Equity: A Perspective of Power. *Int J Environ Res Public Health* 2022;19:9991.
319. Benach J, Muntaner C, Solar O, Santana V, Quinlan M. *Employment Conditions Knowledge Network. Employment, Work and Health Inequalities: A Global Perspective*. Icaria Editorial, 2014 (<https://www.upf.edu/web/greds-emconet/emconet>, accessed 5 June 2024).
320. International Labour Organization. *ILO Fundamental Principles and Rights at Work* [website]. (<https://www.ilo.org/ilo-declaration-fundamental-principles-and-rights-work>).
321. Armenti K, Sweeney M, Lingwall C, Yang L. Work: A Social Determinant of Health Worth Capturing. *Int J Environ Res Public Health* 2023;20:1199.
322. United Nations Department of Economic and Social Affairs. *The Sustainable Development Goals Report 2021*. New York, United Nations, 2021 (<https://unstats.un.org/sdgs/report/2021/>, accessed 2 May 2024).
323. *Women and men in the informal economy: A statistical picture*. Third edition. Geneva, International Labour Office, 2018 (http://www.ilo.org/global/publications/books/WCMS_626831/lang--en/index.htm, accessed 15 March 2024).
324. Rani U, Gobel N. *Job instability, precarity, informality, and inequality: Labour in the gig economy*. The Routledge Handbook of the Gig Economy. Routledge; 2022.
325. UN General Assembly Economic and Social Council. *Progress towards the Sustainable Development Goals: towards a rescue plan for people and planet*. Report of the Secretary-General (special edition) (Advance Unedited Version) [website]. 2023 (<https://hlpf.un.org/2023/documentation>, accessed 2 May 2024).
326. Kim D. Paid Sick Leave and Risks of All-Cause and Cause-Specific Mortality among Adult Workers in the USA. *Int J Environ Res Public Health* 2017;14:1247.
327. United Nations Statistics Division. *Goal 11: Make Cities and Human Settlements Inclusive, Safe, Resilient, and Sustainable | SDG Report 2022* [website]. (<https://unstats.un.org/sdgs/report/2022/goal-11/>, accessed 24 April 2024).
328. Andretti B, Cardoso LO, Honório OS, de Castro Junior PCP, Tavares LF, da Costa Gaspar da Silva I et al. Ecological study of the association between socioeconomic inequality and food deserts and swamps around schools in Rio de Janeiro, Brazil. *BMC Public Health* 2023;23:120.
329. Bridle-Fitzpatrick S. Food deserts or food swamps? A mixed-methods study of local food environments in a Mexican city. *Soc Sci Med* 2015;142:202–213.
330. Sushil Z, Vandevijvere S, Exeter DJ, Swinburn B. Food swamps by area socioeconomic deprivation in New Zealand: a national study. *Int J Public Health* 2017;62:869–877.
331. *Food marketing exposure and power and their associations with food-related attitudes, beliefs and behaviours: a narrative review*. Geneva, World Health Organization, 2022 (<https://iris.who.int/handle/10665/351521>). License: CC BY-NC-SA 3.0 IGO.

332. Cooksey-Stowers K, Schwartz M, Brownell K. Food Swamps Predict Obesity Rates Better Than Food Deserts in the United States. *Int J Environ Res Public Health* 2017;14:1366.
333. Bevel MS, Tsai M-H, Parham A, Andrzejak SE, Jones S, Moore JX. Association of Food Deserts and Food Swamps with Obesity-Related Cancer Mortality in the US. *JAMA Oncol* 2023;9:909.
334. UN Human Rights Council Advisory Committee Drafting Group on the Right to Food. Study on discrimination in the context of the right to food. Geneva, 2010 (<https://digitallibrary.un.org/record/698166>, accessed 2 May 2024).
335. Office of the High Commissioner for Human Rights. Report of the Special Rapporteur on the right to food. Note by the Secretariat. (A/HRC/31/51). [website]. 2015 (https://ap.ohchr.org/documents/dpage_e.aspx?si=A/HRC/31/51, accessed 24 May 2024).
336. Schwartz N, Buliung R, Wilson K. Disability and food access and insecurity: A scoping review of the literature. *Health Place* 2019;57:107-121.
337. Richards G, Carpenter J, Okpalugo E, Howard D, Heneghan C. Assessing housing exposures and interventions that impact healthy cities: a systematic overview of reviews. *Perspect Public Health* 2023;17579139231180756.
338. The Lancet. Housing: an overlooked social determinant of health. *Lancet* 2024;403:1723.
339. Braubach M, Jacobs D, Ormandy D. Environmental burden of disease associated with inadequate housing: a method guide to the quantification of health effects of selected housing risks in the WHO European Region. Geneva, World Health Organization, 2011 (<https://www.who.int/publications-detail-redirect/9789289057899>, accessed 2 May 2024).
340. Haines A, Bruce N, Cairncross S, Davies M, Greenland K, Hiscox A et al. Promoting Health and Advancing Development through Improved Housing in Low-Income Settings. *J Urban Health* 2013;90:810-831.
341. Asthma and Allergy Foundation of America. Asthma Disparities in America: A Roadmap to Reducing Burden on Racial and Ethnic Minorities. 2011 (<https://aafa.org/wp-content/uploads/2022/08/asthma-disparities-in-america-burden-on-racial-ethnic-minorities.pdf>, accessed 2 May 2024).
342. WHO housing and health guidelines. Geneva, World Health Organization, 2018 (<https://iris.who.int/handle/10665/276001>, accessed 9 January 2025). License: CC BY-NC-SA 3.0 IGO.
343. Vázquez-Vera H, Palència L, Magna I, Mena C, Neira J, Borrell C. The threat of home eviction and its effects on health through the equity lens: A systematic review. *Soc Sci Med* 2017;175:199-208.
344. Onapa H, Sharpley CF, Bitsika V, McMillan ME, MacLure K, Smith L et al. The physical and mental health effects of housing homeless people: A systematic review. *Health Soc Care Community* 2022;30:448-468.
345. Turley R, Saith R, Bhan N, Rehfuess E, Carter B. Slum upgrading strategies involving physical environment and infrastructure interventions and their effects on health and socio-economic outcomes. *Cochrane Database Syst Rev* 2013;CD010067.
346. World Health Organization. Energy: shared interests in sustainable development and energy services. Geneva, World Health Organization, 2013 Social determinants of health sectoral briefing series; 5; (<https://iris.who.int/handle/10665/91591>, accessed 26 September 2024).
347. Benneer LS. Energy Justice, Decarbonization, and the Clean Energy Transformation. *Annu Rev Resour Econ* 2022;14:647-668. (<https://doi.org/10.1146/annurev-resource-111920-022328>).
348. van Kempen E, Babisch W. The quantitative relationship between road traffic noise and hypertension: a meta-analysis. *J Hypertens* 2012;30:1075-1086.
349. Badland H, Schofield G. Transport, urban design, and physical activity: an evidence-based update. *Transp Res Part Transp Environ* 2005;10:177-196.
350. World Bank. The High Toll of Traffic Injuries: Unacceptable and Preventable. Washington (DC), World Bank, 2017 (<https://openknowledge.worldbank.org/server/api/core/bitstreams/2cbffdb9-f726-5771-8e94-50f902624f56/content>, accessed 27 February 2025).
351. World Health Organization. Road traffic injuries [website]. 2023 (<https://www.who.int/news-room/fact-sheets/detail/road-traffic-injuries>, accessed 5 June 2024).
352. Global status report on road safety 2023. Geneva, Switzerland, World Health Organization, 2023 (<https://iris.who.int/handle/10665/375016>, accessed 27 February 2025). License: CC BY-NC-SA 3.0 IGO.
353. World Health Organization. Air pollution [website]. 2024 (<https://www.who.int/health-topics/air-pollution>, accessed 5 June 2024).
354. World Health Organization. Air quality database 2022 [website]. 2022 (<https://www.who.int/data/gho/data/themes/air-pollution/who-air-quality-database/2022>, accessed 5 June 2024).

355. World Health Organization. Physical activity [website]. 2024 ([https://www.who.int/news-room/fact-sheets/detail/physical-activity#:~:text=31%25%20of%20adults%20and%2080,2030%2C%20from%20the%202010%20baseline](https://www.who.int/news-room/fact-sheets/detail/physical-activity#:~:text=31%25%20of%20adults%20and%2080,2030%2C%20from%20the%202010%20baseline,), accessed 5 June 2024).
356. Global Burden of Disease Collaborative Network. Global Burden of Disease Study 2019 (GBD 2019) Results. Institute for Health Metrics and Evaluation, 2020 (<https://vizhub.healthdata.org/gbd-results/>).
357. World Health Organization. Pedestrian Safety: A road safety manual for decision-makers and practitioners, Second edition. Geneva: World Health Organization, 2023 (<https://iris.who.int/handle/10665/367419>, accessed 5 March 2025). License: CC BY-NC-SA 3.0 IGO.
358. Economic Commission for Latin America and the Caribbean. Public transport, well-being and inequality: coverage and affordability in the city of Montevideo. Economic Commission for Latin America and the Caribbean, 2017 (<https://www.cepal.org/en/publications/42665-public-transport-well-being-and-inequality-coverage-and-affordability-city>, accessed 2 May 2024).
359. Local Governments for Sustainability. Promoting safe and sustainable cities with public transport for the SDGs. 2020 (<https://www.uitp.org/publications/promoting-safe-and-sustainable-cities-with-public-transport-for-the-sdgs/>, accessed 2 May 2024).
360. Kett M, Cole E, Turner J. Disability, Mobility and Transport in Low- and Middle-Income Countries: A Thematic Review. Sustainability 2020;12. (<https://doi.org/10.3390/su12020589>).
361. Stankov I, Meisel JD, Sarmiento OL, Delclòs-Alió X, Hidalgo D, Guzman LA et al. Uncovering physical activity trade-offs in transportation policy: A spatial agent-based model of Bogotá, Colombia. Int J Behav Nutr Phys Act 2024;21:54.
362. McCartney G, Hearty W, Arnot J, Popham F, Cumbers A, McMaster R. Impact of Political Economy on Population Health: A Systematic Review of Reviews. Am J Public Health 2019;109:e1-12.
363. Rubin J, Taylor J, Krapels J, Sutherland A, Felician MF, Liu JL et al. Are better health outcomes related to social expenditure? A cross-national empirical analysis of social expenditure and population health measures. 2016. (https://www.rand.org/pubs/research_reports/RR1252.html, accessed 13 March 2024).
364. Sun D, Ahn H, Lievens T, Zeng W. Evaluation of the performance of national health systems in 2004-2011: An analysis of 173 countries. PLOS ONE 2017;12:e0173346.
365. United Nations. Developing countries face \$4 trillion investment gap in SDGs [website]. 2023 (<https://news.un.org/en/story/2023/07/1138352>, accessed 23 October 2024).
366. UN World Water Development Report 2023. UN Water (<https://www.unwater.org/publications/un-world-water-development-report-2023>, accessed 23 October 2024).
367. Carbonell-Nicolau O, Llavador H. Inequality reducing properties of progressive income tax schedules: The case of endogenous income: Inequality reducing properties. Theor Econ Internet 2018;13:39-60.
368. Joumard I, Pisu M, Bloch D. Tackling income inequality: The role of taxes and transfers. OECD J Econ Stud 2012;2012:37-70.
369. Abdallah C, Coady D, Gupta S, Hanedar E. The Quest for the Holy Grail: Efficient and Equitable Fiscal Consolidation in India. 2015.
370. Barrios S, Ivaškaitė-Tamošiūnė V, Maftai A, Narazani E, Varga J. Progressive Tax Reforms in Flat Tax Countries. East Eur Econ Internet 2020;58:83-107.
371. Clark H, Lofthus C, Marten R, Rasanathan K. Health taxes: missed opportunities for health and health-care financing. Lancet 2024;404:1905-1907.
372. Government of Namibia. National Development Plan 5 (NDP5). 2021 (<https://www.npc.gov.na/wp-content/uploads/2021/11/NDP5.pdf>).
373. Mbombo M. Press Release: Rising Cost of Living Prompts Increase in Social Grants for the Public. Republic of Namibia, Office of the President, Ministry of Gender Equality, Poverty Eradication and Social Welfare, 2023 (<https://mgpesw.gov.na/documents/792320/0/Press+release+SOCIAL+GRANTS+INCREASE+NOTICE.pdf/d047e7f6-0574-c479-0f34-c65438a825c8>, accessed 5 March 2025).
374. Núñez PS. Fiscal Commitment to Social Spending: Namibia [website]. 2023 (<https://www.sdg16.plus/policies/fiscal-commitment-to-social-spending-namibia/>, accessed 22 May 2024).
375. Sulla V, Zikhali P, Schuler PM, Jellema JR. Does fiscal policy benefit the poor and reduce inequality in Namibia? Namibia, World Bank. 2017 (<https://documents.worldbank.org/en/publication/documents-reports/documentdetail/991551497258273367/Does-fiscal-policy-benefit-the-poor-and-reduce-inequality-in-Namibia>, accessed 22 May 2024).
376. Government of Namibia. Harambee Prosperity Plan II (HPP II): 2021-2025. Windhoek, Namibia. Office of the President, Republic of Namibia. 2021 (<https://hpii.gov.na>, accessed 7 January 2025).

377. Government of Namibia. Executive Summary: Health Policy and Strategic Plan (HPP II) [website]. 2016 (<https://hpii.gov.na>, accessed 24 March 2025).
378. Sulla V, Zikhali P, Facundo P. Inequality in Southern Africa: An Assessment of the Southern African Customs Union. South Africa, World Bank, 2022 (<https://documents.worldbank.org/pt/publication/documents-reports/documentdetail/099125303072236903/p1649270c02a1f06b0a3ae02e57eadd7a82>, accessed 9 April 2024).
379. De UK. Fiscal Policy Changes in Namibia after Independence: Are Namibians Better Off under Independent Decision Making? 2020. (<https://papers.ssrn.com/abstract=3637214>, accessed 22 May 2024).
380. Orkoh E, Nghishikomesho Hasholo E, Gyimah Sackey F, Asravor RK. COVID-19 emergency income grant and food security in Namibia. *Dev South Afr* 2024;41:290–310.
381. United Nations. Economic Commission for Africa. Poverty, vulnerability and inequality in Namibia: an estimation of recent trends. Addis Ababa, United Nations. Economic Commission for Africa, 2024 (<https://hdl.handle.net/10855/50301>, accessed 5 March 2025).
382. International Monetary Fund. Finance & Development, March 2018. Washington (DC): International Monetary Fund; 2018 (<https://doi.org/10.5089/9781484343418.022>).
383. WHO European Health Information Gateway. UNDP Human Development Index (HDI) (Norway) [website]. (<https://gateway.euro.who.int/en/hfa-explorer/hdi/norway/#Dp51s2QN64>, accessed 17 December 2025).
384. Haddinga DB Sean Bray, Joost. Insights into the Tax Systems of Scandinavian Countries. 2023. (<https://taxfoundation.org/blog/scandinavian-social-programs-taxes-2023/>, accessed 18 December 2024).
385. Government of Norway. Amendment to the 2022 Fiscal Budget Proposal: A new and fairer course for the whole of Norway [website]. 2021 (<https://www.regjeringen.no/en/aktuelt/a-new-and-fairer-course-for-the-whole-of-norway/id2884947/>, accessed 18 December 2024).
386. Minister of Finance Trygve Slagsvold Vedum. Henry George Lecture at St. Johns University: Norway's Historical Experience with Taxing and Regulating the Exploitation of Its Natural Resources [website]. 2024 (<https://www.regjeringen.no/en/aktuelt/henry-george-lecture-at-st-johns-university-norways-historical-experience-with-taxing-regulating-the-exploitation-of-its-natural-resources/id3062492/>, accessed 18 December 2024).
387. Thoresen TO, Ring MAK, Nygård OE, Epland J. A Wealth Tax at Work. *CESifo Econ Stud* 2022;68:321–361.
388. Berg K, Hebous S. Does a Wealth Tax Improve Equality of Opportunity? Evidence from Norway. Washington (DC), International Monetary Fund, 2021 IMF Working Papers; (<https://doi.org/10.5089/9781513574417.001>).
389. Norwegian Directorate of Fisheries. Resource rent tax on aquaculture [website]. (<https://www.fiskeridir.no/English/Aquaculture/resource-rent-tax-on-aquaculture>, accessed 18 December 2024).
390. Government of Norway. The National Budget 2024: The Government will introduce a resource rent tax on onshore wind power from 2024 [website]. 2023 (<https://www.regjeringen.no/en/aktuelt/the-government-will-introduce-a-resource-rent-tax-on-onshore-wind-power-from-2024/id2997403/>, accessed 18 December 2024).
391. Ministry of Finance, Norway. Profits from natural resources will be better distributed [website]. 2022 (<https://www.regjeringen.no/en/aktuelt/profits-from-natural-resources-will-be-better-distributed/id2929123/>, accessed 5 December 2024).
392. Jonas Aartun, Ingvar Gjerdem, Ole Kristian Grime. Resource rent tax on onshore wind from 2024 [website]. 2023 (<https://norway.dlapiper.com/en/news/resource-rent-tax-onshore-wind-2024>, accessed 18 December 2024).
393. Rajmil L, Hjern A, Spencer N, Taylor-Robinson D, Gunnlaugsson G, Raat H. Austerity policy and child health in European countries: a systematic literature review. *BMC Public Health* 2020;20:564.
394. WHO Council on the Economics of Health for All. Strengthening public sector capacity, budgets, and dynamic capabilities towards Health for All. Geneva, World Health Organization. Council Brief No. 4. 2022 (https://cdn.who.int/media/docs/default-source/council-on-the-economics-of-health-for-all/who_council4a_councilbrief4.pdf, accessed 27 September 2024).
395. KLIPPE WVD. The Global South is fighting for a voice in global tax rules. CIP 2022. (<https://internationalpolicy.org/publications/the-global-south-is-fighting-for-a-voice-in-global-tax-rules/>, accessed 12 June 2024).
396. Organisation for Economic Co-operation and Development. Global Forum on Transparency and Exchange of Information for Tax Purposes [website]. 2024 (<https://www.oecd.org/tax/transparency/>, accessed 12 June 2024).
397. Organisation for Economic Co-operation and Development. International tax reform: OECD releases technical guidance for implementation of the global minimum tax [website]. 2023 (<https://www.oecd.org/tax/beps/international-tax-reform-oecd-releases-technical-guidance-for-implementation-of-the-global-minimum-tax.htm>, accessed 18 June 2024).

398. Seventy-ninth United Nations General Assembly. Item 16 (f) of the provisional agenda, 30 August 2024: Macroeconomic policy questions: promotion of inclusive and effective international cooperation on tax matters at the United Nations. New York: United Nations, 2024 (A/79/33: <https://financing.desa.un.org/sites/default/files/2024-09/2415701E.pdf>, accessed 5 March 2025).
399. Organization for Economic Cooperation and Development. Addressing Base Erosion and Profit Shifting [website]. 2013 (<https://www.oecd.org/tax/addressing-base-erosion-and-profit-shifting-9789264192744-en.htm>, accessed 12 June 2024).
400. Organization for Economic Cooperation and Development Inclusive Framework on BEPS. Base erosion and profit shifting [website]. 2024 (<https://www.oecd.org/tax/beps/>, accessed 12 June 2024).
401. World Health Organization. Health in the Well-being Economy: Background Paper: Working together to achieve healthy, fairer, prosperous societies across the WHO European Region. Copenhagen, World Health Organization Regional Office for Europe, 2023 (<https://iris.who.int/handle/10665/366228>, accessed 22 July 2024). License: CC BY-NC-SA 3.0 IGO.
402. Council of the EU. Economy of Wellbeing: the Council adopts conclusions [website]. 2019 (<https://www.consilium.europa.eu/en/press/press-releases/2019/10/24/economy-of-wellbeing-the-council-adopts-conclusions/#:~:text=The%20Council%20today%20adopted%20conclusions%20on%20the%20Economy,their%20wellbeing%20at%20the%20centre%20of%20policy%20design>).
403. Organisation for Economic Co-operation and Development. The economy of well-being: Creating opportunities for people's well-being and economic growth. Paris: OECD Publishing; 2019 OECD Statistics Working Papers (<https://doi.org/10.1787/498e9bc7-en>).
404. Wellbeing Economy Governments [website]. (<https://wellbeingeconomy.org/wego>, accessed 5 December 2024).
405. Wellbeing Economy Alliance [website]. (<https://weall.org/>, accessed 23 March 2025).
406. New Zealand Parliament. Public Finance (Wellbeing) Amendment Bill. 2019 (https://www.parliament.nz/en/pb/sc/make-a-submission/document/52SCFE_SCF_BILL_91229/public-finance-wellbeing-amendment-bill, accessed 5 December 2024).
407. The Treasury. The Living Standards Framework (LSF) 2021. Wellington, New Zealand, 2021 (<https://www.treasury.govt.nz/publications/tp/living-standards-framework-2021#purpose-and-background>, accessed 5 December 2024).
408. Narayan AK, Kommunuri J. New Zealand Government's budgetary response to the Covid-19 pandemic. *Pac Account Rev* 2021;33:596–602.
409. World Health Organization. Harnessing the benefits of well-being policies and investments for health. 2023. (<https://www.who.int/europe/publications/m/item/harnessing-the-benefits-of-well-being-policies-and-investments-for-health>, accessed 15 March 2024).
410. Wellbeing Economy Alliance. For an economy in service of life [website]. 2022. (<http://weall.org/>, accessed 15 March 2024).
411. Welsh Government. Business Wales. Foundational Economy. 2023. (<https://businesswales.gov.wales/foundational-economy>, accessed 15 March 2024).
412. Alonso-Albarran V, Currstine T, Preston G, Soler A, Tchelishvili N, Weerathunga S. Gender budgeting in G20 countries. Washington (DC), International Monetary Fund, 2021 IMF working paper, No. WP/21, 269; (<https://doi.org/10.5089/9781616354510.001>).
413. Schugurensky D, Mook L. Participatory budgeting and local development: Impacts, challenges, and prospects. *Local Dev Soc* 2024;1–13.
414. National Assembly for Wales. Well-being of Future Generations (Wales) Act. 2015. (<https://www.legislation.gov.uk/anaw/2015/2/contents/enacted>).
415. Journal officiel de la République française. LOI n° 2015-411 du 13 avril 2015 visant à la prise en compte des nouveaux indicateurs de richesse dans la définition des politiques publiques. 2015.
416. World Health Organization. Shifting towards fiscally resilient, healthy societies: Finding common ground between health, central banks and finance. World Health Organization Regional Office for Europe, 2023 (<https://iris.who.int/handle/10665/376415>). License: CC BY-NC-SA 3.0 IGO.
417. Organisation for Economic Co-operation and Development. April 2024 – preliminary figures [website]. (<https://public.flourish.studio/story/2315218/>, accessed 23 April 2024).
418. United Nations Conference on Trade and Development (last). Global Investment Trends Monitor, No. 46. 2024 (https://unctad.org/system/files/official-document/diaeiainf2024d1_en.pdf, accessed 20 May 2024).
419. Lu C, Schneider MT, Gubbins P, Leach-Kemon K, Jamison D, Murray CJL. Public financing of health in developing countries: a cross-national systematic analysis. *Lancet* 2010;375:1375–1387.

420. UNDP. United Nations Statement to the Development Committee: Joint Ministerial Committee of the Boards of Governors of the Bank and the Fund [website]. 2024 (<https://www.undp.org/speeches/united-nations-statement-development-committee-joint-ministerial-committee-boards-governors-bank-and-fund-0>, accessed 18 November 2024).
421. United Nations Trade and Development. A World of Debt 2024. Geneva, United Nations Trade and Development, 2024 (<https://unctad.org/publication/world-of-debt>, accessed 12 June 2024).
422. Stubbs T, Kentikelenis A, Gabor D, Ghosh J, McKee M. The return of austerity imperils global health. *BMJ Glob Health* 2023;8:e011620.
423. Walker J, Martin M, Seery E, Abdo N, Kamande A, Lawson M. The Commitment to Reducing Inequality Index 2022. Oxford: Development Finance International, Oxfam; 2022 (<https://doi.org/10.21201/2022.9325>).
424. Ma Y, Hu M, Zafar Q. Analysis of the Impact of External Debt on Health in an Emerging Asian Economy: Does FDI Matter? *Front Public Health* 2022;10:824073.
425. Kose MA, Nagle P, Ohnsorge F, Sugawara N. What Has Been the Impact of COVID-19 on Debt? Turning a Wave into a Tsunami. World Bank Group, Prospects Group, 2021 Policy Research Working Paper 9871; (<https://documents1.worldbank.org/curated/en/801991638297695658/pdf/What-Has-Been-the-Impact-of-COVID-19-on-Debt-Turning-a-Wave-into-a-Tsunami.pdf>).
426. International Institute for Environment and Development. World's least-developed countries spend twice as much servicing debts as they receive in climate finance: data tables. 2024 (https://www.iied.org/sites/default/files/uploads/2024/10/Climate_finance_vs_debt_payments_tables_update.pdf, accessed 18 December 2024).
427. Jennifer Keller, Daniel Nogueira-Budny. Debt sustainability in fragile and conflict-affected states: heightened risks and needs. 2022. (<https://ieg.worldbankgroup.org/blog/debt-sustainability-fragile-and-conflict-affected-states-heightened-risks-and-needs>, accessed 18 December 2024).
428. Stubbs T, Kentikelenis A, Stuckler D, McKee M, King L. The impact of IMF conditionality on government health expenditure: A cross-national analysis of 16 West African nations. *Soc Sci Med* 2017;174:220–227.
429. King L, Kentikelenis A, Stubbs T. IMF conditionality and development policy space, 1985–2014. *Rev Int Polit Econ* 2016;23:543–82.
430. Meurs M, Seidelmann L, Koutsoumpa M. How healthy is a “healthy economy”? Incompatibility between current pathways towards SDG3 and SDG8. *Glob Health* 2019;15:83.
431. United Nations Conference on Trade and Development. Trade and Development Report 2023. (https://unctad.org/system/files/official-document/tdr2023_en.pdf, accessed 5 December 2023).
432. Oryema JB, Gymish-Brempong K, Picone G. The impact of debt relief on under five mortality rate in Sub-Saharan Africa. *Glob Public Health* 2017;12:795–807.
433. World Bank. How are Countries Trading Debt for Development? A Closer Look at Debt Swaps. 2024 (<https://www.worldbank.org/en/topic/debt/brief/how-are-countries-trading-debt-for-development>, accessed 18 December 2024).
434. The Global Fund. Debt2Health Overview: Collaboration Through Financial Innovation. (https://www.theglobalfund.org/media/12284/publication_debt2health_overview_en.pdf, accessed 16 December 2024).
435. Hu Y, Wang Z, Zhou S, Yang J, Chen Y, Wang Y et al. Redefining Debt-to-Health, a triple-win health financing instrument in global health. *Glob Health* 2024;20:39.
436. The Global Fund. Spain's Debt Swap Programs Have Improved Health Indicators in Cameroon, Ethiopia and DRC [website]. 2023 (<https://www.theglobalfund.org/en/updates/2023/2023-11-06-spain-debt-swap-improved-health-indicators-cameroon-ethiopia-drc/>, accessed 4 March 2025).
437. Loewenson R, Godt S, Chanda-Kapata P. Asserting public health interest in acting on commercial determinants of health in sub-Saharan Africa: insights from a discourse analysis. *BMJ Glob Health* 2022;7:e009271.
438. Sharpe S, McIlhone K, Hawke S, Ameratunga S. A health sector response to the commercial determinants of health. *N Z Med J* 2022;135:69–84.
439. WHO Technical Manual on Tobacco Tax Policy and Administration, 1st ed. Geneva, World Health Organization, 2021 (<https://iris.who.int/handle/10665/340659>). License: CC BY-NC-SA 3.0 IGO.
440. WHO Technical Manual on Alcohol Tax Policy and Administration, 1st ed. Geneva, World Health Organization, 2023 (<https://iris.who.int/handle/10665/374284>). License: CC BY-NC-SA 3.0 IGO.
441. SHAKE the salt habit. Geneva, World Health Organization, 2016 (<https://iris.who.int/handle/10665/250134>, accessed 30 September 2024).

442. World Health Organization. REPLACE Trans fat-free [website]. (<https://www.who.int/teams/nutrition-and-food-safety/replace-trans-fat>, accessed 30 September 2024).
443. Miracolo A, Sophiea M, Mills M, Kanavos P. Sin taxes and their effect on consumption, revenue generation and health improvement: a systematic literature review in Latin America. *Health Policy Plan* 2021;36:790–810.
444. Barquera S, Hernández-Barrera L, Trejo B, Shamah T, Campos-Nonato I, Rivera-Dommarco J. Obesidad en México, prevalencia y tendencias en adultos. *Ensanut 2018-19. Salud Pública México* 2020;62:682–692.
445. Oropeza Abúndez C, Instituto Nacional de Salud Pública, Mexico, editors. Encuesta nacional de salud y nutrición 2012: resultados nacionales, Segunda edición. Cuernavaca, Morelos, México México, D.F, Instituto Nacional de Salud Pública, 2013.
446. World Health Organization. The Inequality Epidemic: Low-Income Teens Face Higher Risks of Obesity, Inactivity and Poor Diet. WHO/Europe, 2024 (<https://www.who.int/europe/news-room/23-05-2024-the-inequality-epidemic--low-income-teens-face-higher-risks-of-obesity--inactivity-and-poor-diet>, accessed 16 December 2024).
447. Żukiewicz-Sobczak W, Wróblewska P, Zwoliński J, Chmielewska-Badora J, Adamczuk P, Krasowska E et al. Obesity and poverty paradox in developed countries. *Ann Agric Environ Med* 2014;21:590–594.
448. Anekwe CV, Jarrell AR, Townsend MJ, Gaudier GI, Hiserodt JM, Stanford FC. Socioeconomics of Obesity. *Curr Obes Rep* 2020;9:272–279.
449. Lara-Castor L, Micha R, Cudhea F, Miller V, Shi P, Zhang J et al. Sugar-sweetened beverage intakes among adults between 1990 and 2018 in 185 countries. *Nat Commun* 2023;14:5957.
450. Bray GA, Nielsen SJ, Popkin BM. Consumption of high-fructose corn syrup in beverages may play a role in the epidemic of obesity. *Am J Clin Nutr* 2004;79:537–543.
451. Enrique Peña Nieto, President of the United Mexican States. Decree amending, adding, and repealing various provisions of the Value Added Tax Law; the Special Tax on Production and Services Law; the Federal Rights Law; enacting the Income Tax Law; and repealing the Business Tax Law and the Cash Deposit Tax Law (DECRETO por el que se reforman, adicionan y derogan diversas disposiciones de la Ley del Impuesto al Valor Agregado; de la Ley del Impuesto Especial sobre Producción y Servicios; de la Ley Federal de Derechos, se expide la Ley del Impuesto sobre la Renta, y se abroga la Ley del Impuesto Empresarial a Tasa Única, y la Ley del Impuesto a los Depósitos en Efectivo). 2013. (https://www.dof.gob.mx/nota_detalle.php?codigo=5325371&fecha=11/12/2013).
452. Salgado Hernández JC, Ng SW, Colchero MA. Changes in sugar-sweetened beverage purchases across the price distribution after the implementation of a tax in Mexico: a before-and-after analysis. *BMC Public Health* 2023;23:265.
453. Basto-Abreu A, Barrientos-Gutiérrez T, Vidaña-Pérez D, Colchero MA, Hernández-F. M, Hernández-Ávila M et al. Cost-Effectiveness of the Sugar-Sweetened Beverage Excise Tax in Mexico. *Health Aff (Millwood)* 2019;38:1824–1831.
454. Mialon M. An overview of the commercial determinants of health. *Glob Health* 2020;16:74.
455. Wood B, Williams O, Nagarajan V, Sacks G. Market strategies used by processed food manufacturers to increase and consolidate their power: a systematic review and document analysis. *Glob Health* 2021;17:17.
456. Addressing and managing conflicts of interest in alcohol control policies. Geneva, World Health Organization. 2022 (<https://iris.who.int/handle/10665/352517>, accessed 13 March 2024). License: CC BY-NC-SA 3.0 IGO.
457. Ralston R, Hil SE, Da Silva Gomes F, Collin J. Towards Preventing and Managing Conflict of Interest in Nutrition Policy? An Analysis of Submissions to a Consultation on a Draft WHO Tool. *Int J Health Policy Manag* 2020;1.
458. Supporting member states in reaching informed decision-making on engaging with private sector entities for the prevention and control of noncommunicable diseases: a practical tool. Geneva, World Health Organization. 2024 (<https://iris.who.int/handle/10665/378209>, accessed 15 October 2024). License: CC BY-NC-SA 3.0 IGO.
459. World Health Organization Regional Office for Europe. Public-private partnerships for health care infrastructure and services: policy considerations for middle-income countries in Europe. (<https://www.who.int/europe/publications/i/item/9789289058605>, accessed 30 September 2024).
460. Amann M, K. Roehrich J, EBig M, Harland C. Driving sustainable supply chain management in the public sector: The importance of public procurement in the European Union. *Supply Chain Manag Int J* 2014;19:351–366.
461. City of Boston. Mayor Wu signs ordinance to divest City funds from the fossil fuel industry [website]. (<https://www.boston.gov/news/mayor-wu-signs-ordinance-divest-city-funds-fossil-fuel-industry>, accessed 30 September 2024).

462. Martins A, Guerry L. Public Procurement: 12 Ways Governments Can Unlock Better Value. 2018 (<https://www.oliverwyman.com/our-expertise/insights/2018/oct/12-ways-governments-can-unlock-better-value.html>, accessed 25 October 2023).
463. ACCA Global. New models of public procurement: a tool for sustainable recovery [website]. 2020 (https://www.accaglobal.com/gb/en/professional-insights/pro-accountants-the-future/New_models_public_procurement.html?_gl=1*mwm154*_ga*MTg5MDA4NjUwMS4xNjk4MjM1NjEx*_ga_J7W3P5MX6E*MTY5ODIzNTYxMC4xLjAuMTY5ODIzNTYxM_C4wLjAuMA, accessed 25 October 2023).
464. Mohammed R. Progress on the Economic Empowerment of Female Entrepreneurs in Kenya's 30% Preferential Public Procurement Policy. Walden Diss Dr Stud 2019. (<https://scholarworks.waldenu.edu/dissertations/7562>).
465. OECD. Going Green: Best Practices for Public Procurement. 2015 (<https://www.oecd.org/gov/public-procurement/green/>, accessed 25 October 2023).
466. Cervantes-Zapana M, Yagüe JL, De Nicolás VL, Ramirez A. Benefits of public procurement from family farming in Latin-American countries: Identification and prioritization. *J Clean Prod* 2020;277:123466. (<https://doi.org/10.1016/j.jclepro.2020.123466>).
467. Cabinet Office (United Kingdom), Department for Culture, Media and Sport, and Department for Digital, Culture, Media & Sport. Procurement Policy Note 06/20 – taking account of social value in the award of central government contracts [website]. (<https://www.gov.uk/government/publications/procurement-policy-note-0620-taking-account-of-social-value-in-the-award-of-central-government-contracts>, accessed 25 October 2023).
468. World Health Organization. Global health estimates: Leading causes of death [website]. 2024 (<https://www.who.int/data/gho/data/themes/mortality-and-global-health-estimates/ghle-leading-causes-of-death>, accessed 27 February 2025).
469. Resolve to Save Lives. Policy in Practice: Healthy Public Food Procurement in Quezon City, Philippines. 2022 (https://resolvetosavelives.org/wp-content/uploads/2023/06/RTSL_Case-Study_Quezon-City-Healthy-Public-Food-Procurement_2022.pdf).
470. C40 Cities. Quezon City, The Philippines [website]. (<https://www.c40.org/cities/quezon-city/>, accessed 16 December 2024).
471. C40 Cities. C40 Cities [website]. (<https://www.c40.org/>, accessed 16 December 2024).
472. Quezon City Government. EO 16: Establishing the Quezon City Healthy Food Procurement Policy. Quezon City, 2021 (<https://quezoncity.gov.ph/wp-content/uploads/2021/08/EO-16-Establishing-The-Quezon-City-Healthy-Food-Procurement-Policy.pdf>, accessed 16 December 2024).
473. Quezon City Government. QC First LGU to Implement Healthy Food Procurement Policy [website]. 2021 (<https://quezoncity.gov.ph/qc-first-lgu-to-implement-healthy-food-procurement-policy/>, accessed 16 December 2024).
474. C40 Cities. C40 Bloomberg Awards: Quezon City [website]. 2022 (<https://www.c40.org/case-studies/c40-bloomberg-awards-quezon-city/>, accessed 16 December 2024).
475. Resolve to Save Lives. A healthy public food procurement policy for the Philippines' largest city: Quezon City's new policy is a model for the whole country [website]. (<https://resolvetosavelives.org/cardiovascular-health/success/quezon-city/>, accessed 16 December 2024).
476. Swensson LFJ, Hunter D, Schneider S, Tartanac F. Public food procurement as a game changer for food system transformation. *Lancet Planet Health* 2021;5:e495–e496.
477. Koivusalo M. Policy space for health and trade and investment agreements. *Health Promot Int* 2014;29:i29–i47.
478. McCartney G, Hearty W, Arnot J, Popham F, Cumbers A, McMaster R. Impact of Political Economy on Population Health: A Systematic Review of Reviews. *Am J Public Health* 2019;109:e1–e12.
479. Barlow P, McKee M, Basu S, Stuckler D. The health impact of trade and investment agreements: a quantitative systematic review and network co-citation analysis. *Glob Health* 2017;13:13.
480. O'Connor D, Hou D, Ye J, Zhang Y, Ok YS, Song Y et al. Lead-based paint remains a major public health concern: A critical review of global production, trade, use, exposure, health risk, and implications. *Environ Int* 2018;121:85–101.
481. Goodman D, Arisco N, Jaacks LM. Synthetic Chemical Trade as a Potential Driver of Global Health Disparities and Data Gaps on Synthetic Chemicals in Vulnerable Populations. *Curr Environ Health Rep* 2020;7:1–12.
482. Newman SJ, Schulte K, Morellini MM, Rahal C, Leasure DR. Offshoring emissions through used vehicle exports. *Nat Clim Change* 2024;14:238–241.

483. Thow AM. Trade liberalisation and the nutrition transition: mapping the pathways for public health nutritionists. *Public Health Nutr* 2009;12:2150–2158.
484. Kumm M, Kinnunen P, Lehtikainen E, Porkka M, Queiroz C, Röö E et al. Interplay of trade and food system resilience: Gains on supply diversity over time at the cost of trade independency. *Glob Food Secur* 2020;24:100360.
485. Adhikari J, Shrestha M, Paudel D. Nepal's growing dependency on food imports: A threat to national sovereignty and ways forward. *Nepal Public Policy Rev* 2021;68–86.
486. Zhang Q, Jiang X, Tong D, Davis SJ, Zhao H, Geng G et al. Transboundary health impacts of transported global air pollution and international trade. *Nature* 2017;543:705–709.
487. Li R, Zhang J, Krebs P. Global trade drives transboundary transfer of the health impacts of polycyclic aromatic hydrocarbon emissions. *Commun Earth Environ* 2022;3:1–13.
488. Schram A, Aisbett E, Townsend B, Labonté R, Baum F, Friel S. Toxic trade: the impact of preferential trade agreements on alcohol imports from Australia in partner countries. *Addict* 2020;115:1277–1284.
489. World Trade Organization. TRIPS and public health [website]. 2024 (https://www.wto.org/english/tratop_e/trips_e/pharmapatent_e.htm, accessed 6 June 2024).
490. WHO Framework Convention on Tobacco Control. Health exceptions and recognition of public health in GATT, TBT and TRIPS [website]. (<https://portal-uat.who.int/fctcapps/fctcapps/fctc/kh/legalchallenges/health-exceptions-and-recognition-public-health-gatt-tbt-and-trips>, accessed 30 September 2024).
491. Barlow P, Stuckler D. Globalization and health policy space: Introducing the WTO health dataset of trade challenges to national health regulations at World Trade Organization, 1995–2016. *Soc Sci Med* 1982 2021;275:113807.
492. Barlow P, Gleeson D, O'Brien P, Labonté R. Industry influence over global alcohol policies via the World Trade Organization: a qualitative analysis of discussions on alcohol health warning labelling, 2010–19. *Lancet Glob Health* 2022;10:e429–e437.
493. Public health, innovation and intellectual property rights – Report of the Commission on Intellectual Property Rights, Innovation and Public Health. Geneva, World Health Organization, 2006 (<https://iris.who.int/handle/10665/43460>, accessed 15 October 2024).
494. Durkin A, Anne Sta Maria P, Willmore B, Kapczynski A. Addressing the Risks That Trade Secret Protections Pose for Health and Rights. *Health Hum Rights J* 2021;23:129–143.
495. Technical report: pricing of cancer medicines and its impacts: a comprehensive technical report for the World Health Assembly Resolution 70.12: operative paragraph 2.9 on pricing approaches and their impacts on availability and affordability of medicines for the prevention and treatment of cancer. Geneva, World Health Organization. 2018 (<https://iris.who.int/handle/10665/277190>). License: CC BY-NC-SA 3.0 IGO.
496. Ewen M, Joosse H-J, Beran D, Laing R. Insulin prices, availability and affordability in 13 low-income and middle-income countries. *BMJ Glob Health* 2019;4:e001410.
497. Freeman T, Baum F, Musolino C, Flavel J, McKee M, Chi C et al. Illustrating the impact of commercial determinants of health on the global COVID-19 pandemic: Thematic analysis of 16 country case studies. *Health Policy* 2023;134:104860. (<https://doi.org/10.1016/j.healthpol.2023.104860>).
498. Díaz FE, Arruvito L, Geffner J. The future of vaccination in Latin America: learning from the COVID-19 pandemic. *Curr Opin Immunol* 2023;85:102390. (<https://doi.org/10.1016/j.coi.2023.102390>).
499. World Intellectual Property Organization (WIPO). WIPO Member States Adopt Historic New Treaty on Intellectual Property, Genetic Resources and Associated Traditional Knowledge [website]. 2024 (https://www.wipo.int/pressroom/en/articles/2024/article_0007.html, accessed 30 September 2024).
500. Office of the High Commissioner for Human Rights. Open-ended intergovernmental working group on transnational corporations and other business enterprises with respect to human rights [website]. (<https://www.ohchr.org/en/hr-bodies/hrc/wg-trans-corp/igwg-on-tnc>, accessed 30 September 2024).
501. McCartney G, Hearty W, Arnot J, Popham F, Cumbers A, McMaster R. Impact of Political Economy on Population Health: A Systematic Review of Reviews. *Am J Public Health* 2019;109:e1–e12.
502. Seaman R, Walsh D, Beatty C, McCartney G, Dundas R. Social security cuts and life expectancy: a longitudinal analysis of local authorities in England, Scotland and Wales. *J Epidemiol Community Health* 2023;78:82–87.
503. Simpson J, Albani V, Bell Z, Bambra C, Brown H. Effects of social security policy reforms on mental health and inequalities: A systematic review of observational studies in high-income countries. *Soc Sci Med* 1982 2021;272:113717.
504. Razavi S, Behrendt C, Bierbaum M, Orton I, Tessier L. Reinvigorating the social contract and strengthening social cohesion: Social protection responses to COVID-19. *Int Soc Secur Rev* 2020;73:55–80.

505. Hillier-Brown F, Thomson K, McGowan V, Cairns J, Eikemo TA, Gil-González D et al. The effects of social protection policies on health inequalities: Evidence from systematic reviews. *Scand J Public Health* 2019;47: 655–665.
506. Shahidi FV, Muntaner C, Shankardass K, Quiñonez C, Siddiqi A. The effect of unemployment benefits on health: A propensity score analysis. *Soc Sci Med* 2019;226:198–206.
507. Shand F, Duffy L, Torok M. Can Government Responses to Unemployment Reduce the Impact of Unemployment on Suicide? *Crisis* 2022;43:59–66.
508. Li Q, Dong L, Zhang L. Have pensions reduced the relative poverty? ----- empirical analysis from China CHARLS data. *Heliyon* 2023;9:e22711.
509. Lee K. Old-age poverty in a pension latecomer: The impact of basic pension expansions in South Korea. *Soc Policy Adm* 2022;56:1022–1040.
510. Albani V, Brown H, Vera-Toscano E, Kingston A, Eikemo TA, Bambra C. Investigating the impact on mental wellbeing of an increase in pensions: A longitudinal analysis by area-level deprivation in England, 1998–2002. *Soc Sci Med* 2022;311:115316.
511. Riumallo Herl C, Kabudula C, Kahn K, Tollman S, Canning D. Pension exposure and health: Evidence from a longitudinal study in South Africa. *J Econ Ageing* 2022;23:100411.
512. Zhang D, Wang Y, Jiao Y. The Impact of Social Pension Schemes on the Mental Health of the Chinese Elderly: A Mediating Effect Perspective of Two-Way Intergenerational Support. *Int J Environ Res Public Health* 2022;19:8721.
513. Sanches M da S, Carvalho LB de. Multiplier effects of social protection: a SVAR approach for Brazil. *Int Rev Appl Econ* 2023;37:93–112.
514. Nelson K, Fritzell J. Welfare states and population health: the role of minimum income benefits for mortality. *Soc Sci Med* 1982 2014;112:63–71.
515. Lundberg O, Yngwe MÅ, Bergqvist K, Sjöberg O. DRIVERS final scientific report: The role of income and social protection for inequalities in health, evidence and policy implications. Report produced as part of the “DRIVERS for Health Equity” project (<http://health-gradient.eu/>). Stockholm, Centre for Health Equity Studies, 2014 (https://eurohealthnet.eu/wp-content/uploads/documents/Before%202016/15_Projects_DRIVERS_RoleOfIncomeAndSocialProtectionForInequalitiesInHealth.pdf, accessed 18 March 2024).
516. McAllister A, Fritzell S, Almroth M, Harber-Aschan L, Larsson S, Burström B. How do macro-level structural determinants affect inequalities in mental health? A systematic review of the literature. *Int J Equity Health* 2018;17:180.
517. Yoshino CA, Sidney-Annerstedt K, Wingfield T, Kirubi B, Viney K, Boccia D et al. Experiences of conditional and unconditional cash transfers intended for improving health outcomes and health service use: a qualitative evidence synthesis. *Cochrane Database Syst Rev* 2023.
518. Owusu-Addo E, Renzaho AMN, Smith BJ. The impact of cash transfers on social determinants of health and health inequalities in sub-Saharan Africa: a systematic review. *Health Policy Plan* 2018;33:675–696.
519. Zimmerman A, Garman E, Avendano-Pabon M, Araya R, Evans-Lacko S, McDaid D et al. The impact of cash transfers on mental health in children and young people in low-income and middle-income countries: a systematic review and meta-analysis. *BMJ Glob Health* 2021;6.
520. International Labour Office. World Social Protection Report 2020–22: Social protection at the crossroads – in pursuit of a better future. Geneva, International Labour Office, 2021 World Social Protection Report, No. 2020/2022.
521. Rogan M. Tax Justice and the Informal Economy: A Review of the Debates. Women in Informal Employment: Globalizing and Organizing, 2019 (<https://www.wiego.org/publications/tax-justice-and-informal-economy-review-debates>, accessed 15 March 2024).
522. Sandner L, Scheil-Adlung X. Evidence on Paid Sick Leave: Observations in Times of Crisis. 2010;2010:313–321.
523. Amoranto G. Paid Sick Leave as a Tool for COVID-19 Control. Asian Development Bank, 2020 (<https://www.adb.org/publications/paid-sick-leave-covid-19-control>, accessed 17 October 2023).
524. Pichler S, Wen K, Ziebarth NR. COVID-19 Emergency Sick Leave Has Helped Flatten the Curve in the United States. *Health Aff Proj Hope* 2020;39:2197–2204.
525. Piper K, Youk A, Iii AEJ, Kumar S. Paid sick days and stay-at-home behavior for influenza. *PLOS ONE* 2017;12:e0170698. (<https://doi.org/10.1371/journal.pone.0170698>).
526. Davison HK, Blackburn AS. The Case for Offering Paid Leave: Benefits to the Employer, Employee, and Society. *Compens Benefits Rev* 2023;55:3–18.

527. Townsend B, Friel S, Freeman T, Schram A, Strazdins L, Labonte R et al. Advancing a health equity agenda across multiple policy domains: a qualitative policy analysis of social, trade and welfare policy. *BMJ Open* 2020;10:e040180.
528. Saint-Martin A, Inanc H, Prinz C. Job Quality, Health and Productivity: An evidence-based framework for analysis. Paris, OECD, 2018 (<https://doi.org/10.1787/a8c84d91-en>).
529. Piper K, Youk A, James AE, Kumar S. Paid sick days and stay-at-home behavior for influenza. Olson DR, editor. *PLOS ONE* 2017;12:e0170698.
530. O'Campo P, Molnar A, Ng E, Renahy E, Mitchell C, Shankardass K et al. Social welfare matters: A realist review of when, how, and why unemployment insurance impacts poverty and health. *Soc Sci Med* 2015;132:88–94. (<https://doi.org/10.1016/j.socscimed.2015.03.025>).
531. Lönnroth K, Tessier L, Hensing G, Behrendt C. Income security in times of ill health: the next frontier for the SDGs. *BMJ Glob Health* 2020;5:e002493.
532. Sandner L, Scheil-Adlung X. Evidence on Paid Sick Leave: Observations in Times of Crisis. *Intereconomics* 2010;2010:313–321. (<https://www.intereconomics.eu/contents/year/2010/number/5/article/evidence-on-paid-sick-leave-observations-in-times-of-crisis.html>, accessed 17 October 2023).
533. Manley J, Balarajan Y, Malm S, Harman L, Owens J, Murthy S et al. Cash transfers and child nutritional outcomes: A systematic review and meta-analysis. *BMJ Glob Health* 2020;5.
534. Owusu-Addo E, Cross R. The impact of conditional cash transfers on child health in low- and middle-income countries: a systematic review. *Int J Public Health* 2014;59:609–618.
535. Siddiqi A, Rajaram A, Miller SP. Do cash transfer programmes yield better health in the first year of life? A systematic review linking low-income/middle-income and high-income contexts. *Arch Dis Child* 2018;103:920–926.
536. Målvist M, Yuan B, Trygg N, Selling K, Thomsen S. Targeted Interventions for Improved Equity in Maternal and Child Health in Low- and Middle-Income Settings: A Systematic Review and Meta-Analysis. *PLOS ONE* 2013;8:e66453.
537. McGuire J, Kaiser C, Bach-Mortensen AM. A systematic review and meta-analysis of the impact of cash transfers on subjective well-being and mental health in low- and middle-income countries. *Nat Hum Behav* 2022;6:359–370.
538. The Impact of Monetary Poverty Alleviation Programs on Children's and Adolescents' Mental Health: A Systematic Review and Meta-Analysis Across Low-, Middle-, and High-Income Countries. ScienceDirect [website]. (<https://www.sciencedirect.com/science/article/pii/S1054139X22003329>, accessed 3 June 2024).
539. Aragão FBA, Arcêncio RA, Fuentealba-Torres M, Carneiro TSG, Souza LLL, Alves YM et al. Impact of social protection programs on adults diagnosed with Tuberculosis: systematic review. *Rev Bras Enferm* 2021;74:e20190906.
540. Richterman A, Steer-Massaró J, Jarolimova J, Luong Nguyen LB, Werdenberg J, Ivers LC. Cash interventions to improve clinical outcomes for pulmonary tuberculosis: systematic review and meta-analysis. *Bull World Health Organ* 2018;96:471–483.
541. UN Partnership to Promote the Rights of Persons with Disabilities. Towards inclusive social protection to advance the rights of persons with disabilities [website]. 2023 (https://unprpd.org/programme/towards_inclusive_social_protection_to_advance_the_rights_of_persons_with_disabilities/, accessed 18 March 2024).
542. ILO Social Protection. Joint Statement: Towards Inclusive Social Protection Systems Supporting Full and Effective Participation of Persons with Disabilities [website]. 2019 (<https://www.social-protection.org/gimi/gess/ShowProjectWiki.action?id=3209&pid=2840>, accessed 12 June 2024).
543. Organisation for Economic Co-operation and Development. How can social protection address women's unpaid care work? Enabling Women's Economic Empowerment. Paris: OECD Publishing; 2019 (<https://doi.org/10.1787/c8eb9246-en>).
544. USAID. Improving Governance for Economic Growth [website]. (<https://www.usaid.gov/vietnam/improving-governance-for-economic-growth>).
545. Ramla Khalidi. Strengthening Social Protection to Promote Sustainable Poverty Reduction [website]. 2022 (<https://www.undp.org/vietnam/news/strengthening-social-protection-promote-sustainable-poverty-reduction>).
546. Olivier M. Viet Nam Social Security Legal Reform: An integrated framework. International Labour Organization, 2022 (<https://www.social-protection.org/gimi/Media.action;jsessionid=DI5vcnoqx22SkmMl1YVPprEiSBxceGYgsdDtUjgjXwjbZ8dYMpcE!1393577045?id=18868>, accessed 18 December 2024).

547. Oxford Poverty and Human Development Initiative. Multidimensional Poverty Index 2023: Unstacking Global Poverty – Data for High Impact Action: Briefing note for countries on the 2023 Multidimensional Poverty Index, Viet Nam. 2023 (<https://hdr.undp.org/sites/default/files/Country-Profiles/MPI/VNM.pdf>, accessed 16 December 2024).
548. Do Thi Thu. Policy Paper on the Social Protection System in Vietnam (Version 4 – 2022/08/14). United Nations Development Programme (UNDP), 2022 (https://erc.undp.org/api/download?filePath=%2Fdocuments%2F13558%2Fmgmtresponse%2Fkeyaction%2Fdoc_3277302067405545622UNDP_T220404_Report_FINAL.doc).
549. International Labour Organization. Vietnam country profile [website]. (<https://www.social-protection.org/gimi/ShowCountryProfile.action?iso=VN>, accessed 12 December 2024).
550. Galobardes B, Lynch JW, Davey Smith G. Childhood socioeconomic circumstances and cause-specific mortality in adulthood: systematic review and interpretation. *Epidemiol Rev* 2004;26:7–21.
551. Cooper K, Stewart K. Does money affect children's outcomes? York, Joseph Rowtree Foundation, 2013 (<https://www.jrf.org.uk/care/does-money-affect-childrens-outcomes>, accessed 18 March 2024).
552. Wickham S, Anwar E, Barr B, Law C, Taylor-Robinson D. Poverty and child health in the UK: using evidence for action. *Arch Dis Child* 2016;101:759–766.
553. ILO, UNICEF. More than a billion reasons: The urgent need to build universal social protection for children. Geneva and New York, ILO and UNICEF, 2023 (http://www.ilo.org/global/publications/books/WCMS_869602/lang--en/index.htm, accessed 15 March 2024).
554. UNICEF Innocenti Research Centre (last), Tirivayi N, Waidler J, Otchere F. Cash Transfers – Past, Present and Future: Evidence and Lessons Learned from the Transfer Project. 2021. (<https://doi.org/10.18356/26642166-2021/07>).
555. Huang C, Singh K, Handa S, Halpern C, Pettifor A, Thirumurthy H. Investments in children's health and the Kenyan cash transfer for orphans and vulnerable children: Evidence from an unconditional cash transfer scheme. *Health Policy Plan* 2017;32:943–955.
556. Bastagli F, Hagen-Zanker J, Sturge G. Cash transfers: what does the evidence say? [website]. 2016 (<https://odi.org/en/publications/cash-transfers-what-does-the-evidence-say-a-rigorous-review-of-impacts-and-the-role-of-design-and-implementation-features/>, accessed 15 March 2024).
557. Boccia D, Maritano S, Pizzi C, Richiardi MG, Lioret S, Richiardi L. The impact of income-support interventions on life course risk factors and health outcomes during childhood: a systematic review in high income countries. *BMC Public Health* 2023;23:744.
558. Spencer N, Raman S, O'Hare B, Tamburlini G. Addressing inequities in child health and development: towards social justice. *BMJ Paediatr Open* 2019;3:e000503.
559. UNICEF and ILO. Towards Universal Social Protection for Children: Achieving SDG 1.3. ILO-UNICEF Joint Report on Social Protection for Children. UNICEF and ILO, 2019 (<https://www.unicef.org/reports/towards-universal-social-protection-children-achieving-sdg-13>, accessed 15 March 2024).
560. Ministry of Health, Chile. Chile National Quality of Life and Health Survey 2006 (Encuesta Nacional de Calidad de Vida y Salud (ENCAVI)). Santiago, Chile, 2006 (<https://ghdx.healthdata.org/record/chile-national-quality-life-and-health-survey-2006>).
561. Presidential Advisory Council for the Reform of Childhood Policies in Chile. The future of children is always today. Proposals of the Presidential Advisory Council for the reform of childhood policies. 2006 (<https://catalogo.academia.cl/cgi-bin/koha/opac-detail.pl?biblionumber=12707>, accessed 9 April 2025).
562. Milman HM, Castillo CA, Sansotta AT, Delpiano PV, Murray J. Scaling up an early childhood development programme through a national multisectoral approach to social protection: lessons from Chile Crece Contigo. *The BMJ* 2018;363:k4513.
563. Ministry of Social Development, Chile. What is Chile Grows with You? Chile Crece Contigo, Unit of the Division of Promotion and Social Protection, Ministry of Social Development, Chile. 2015 (<https://www.crececontigo.gob.cl/wp-content/uploads/2015/11/que-es-Chile-Crece-2015.pdf>).
564. World Bank. 10 Years of Chile Grows with You (Chile Crece Contigo): Key Components and Lessons Learned for the Setting Up of Comprehensive Child Development Support Systems. 2018 (<https://documents.worldbank.org/en/publication/documents-reports/documentdetail/992351537159031673/10-years-of-chile-grows-with-you-chile-crece-contigo-key-components-and-lessons-learned-for-the-setting-up-of-comprehensive-child-development-support-systems>, accessed 12 December 2024).
565. Borrell C, Palència L, Muntaner C, Urquía M, Malmusi D, O'Campo P. Influence of Macrosocial Policies on Women's Health and Gender Inequalities in Health. *Epidemiol Rev* 2014;36:31–48. (<https://doi.org/10.1093/epirev/mxt002>).

566. Bütikofer A, Riise J, Skira M. The Impact of Paid Maternity Leave on Maternal Health. *Am Econ J Econ Policy* 2021;13:67-105.
567. Bilgrami A, Sinha K, Cutler H. The impact of introducing a national scheme for paid parental leave on maternal mental health outcomes. *Health Econ* 2020;29:1657-1681.
568. Nandi A, Jahagirdar D, Dimitris MC, Labrecque JA, Strumpf EC, Kaufman JS et al. The Impact of Parental and Medical Leave Policies on Socioeconomic and Health Outcomes in OECD Countries: A Systematic Review of the Empirical Literature. *Milbank Q* 2018;96:434-471.
569. Duvander A-ZE. Swedish parental leave and gender equality: achievements and reform challenges in a European perspective. Stockholm, Institutet för framtidsstudier, 2005.
570. Swedish Social Insurance Agency (Försäkringskassan). Parental benefits [website]. (<https://www.forsakringskassan.se/english/parents/when-the-child-is-born/parental-benefit>).
571. The Swedish Association of University Teachers and Researchers. Parental leave: Parental allowance and supplementary parental allowance [website]. (<https://sulf.se/en/work-salary-and-benefits/parental-leave/parental-allowance-and-supplementary-parental-allowance/>).
572. Sweden.se. Work-life balance: Sweden has made it easier to combine career with family life. Here's how [website]. (<https://sweden.se/work-business/working-in-sweden/work-life-balance>, accessed 16 December 2024).
573. Dutta R. Sweden's Universal Subsidized Daycare for Children [website]. 2024 (<https://www.sdg16.plus/policies/swedens-universal-subsidized-daycare-for-children>).
574. Sweden.se. The Swedish School System: Education is key in Sweden. It is tax-financed, and compulsory from the age of 6. [website]. (<https://sweden.se/life/society/the-swedish-school-system>, accessed 16 December 2024).
575. Swedish Social Insurance Agency (Försäkringskassan). Care of a sick child [website]. (<https://www.forsakringskassan.se/english/parents/care-of-a-sick-child-vab>, accessed 16 December 2024).
576. Organisation for Economic Co-operation and Development. Labour force participation rate [website]. 2022 (<http://data.oecd.org/emp/labour-force-participation-rate.htm>, accessed 27 May 2024).
577. Hogrebe N, Strietholt R. Does non-participation in preschool affect children's reading achievement? International evidence from propensity score analyses. *Large-Scale Assess Educ* 2016;4:2.
578. Raub A, Latz I, Sprague A, Stein MA, Heymann J. Constitutional Rights of Persons with Disabilities: An Analysis of 193 National Constitutions. *Harv Hum Rights J* 2016;29:203-240.
579. Yang B, Kelly C, Shamputa IC, Barker K, Thi Kim Nguyen D. Structural Origins of Poor Health Outcomes in Documented Temporary Foreign Workers and Refugees in High-Income Countries: A Review. *Healthcare* 2023;11:1295. (<https://doi.org/10.3390/healthcare11091295>).
580. World Health Organization. Public health, innovation and intellectual property rights – Report of the Commission on Intellectual Property Rights, Innovation and Public Health. Copenhagen, World Health Organization Regional Office for Europe, 2023 (<https://iris.who.int/bitstream/handle/10665/368236/WHO-EURO-2023-7022-46788-68195-eng.pdf?sequence=1&isAllowed=y>).
581. Egede, L.E, Walker, R.J, Williams, J.S. Addressing Structural Inequalities, Structural Racism, and Social Determinants of Health: a Vision for the Future. *J GEN INTERN MED* 2024;39:487-497.
582. Anderson LM, Adeney KL, Shinn C, Safranek S, Buckner-Brown J, Krause LK. Community coalition-driven interventions to reduce health disparities among racial and ethnic minority populations. *Cochrane Public Health Group, editor. Cochrane Database Syst Rev* 2015;2015. (<https://doi.org/10.1002/14651858.CD009905.pub2>).
583. Ahmed S, Chase LE, Wagnild J, Akhter N, Sturridge S, Clarke A et al. Community health workers and health equity in low- and middle-income countries: systematic review and recommendations for policy and practice. *Int J Equity Health* 2022;21:49.
584. Palència L, De Moortel D, Artazcoz L, Salvador-Piedrafita M, Puig-Barrachina V, Hagqvist E et al. Gender Policies and Gender Inequalities in Health in Europe: Results of the SOPHIE Project. *Int J Health Serv Plan Adm Eval* 2017;47:61-82.
585. World Bank. Women, Business and the Law 2016: Getting to Equal. Washington, DC, World Bank, 2015.
586. Gréaux M, Moro M.F, Kamenov K. et al. Health equity for persons with disabilities: a global scoping review on barriers and interventions in healthcare services. *Int J Equity Health* 2023;22:236.

587. Clark E C, Cranston E, Polin T et al. Structural interventions that affect racial inequities and their impact on population health outcomes: a systematic review. *BMC Public Health* 2022;22:2162.
588. FXB Center for Health and Human Rights at Harvard. Making the Public Health Case for Reparations. 2022 (<https://fxb.harvard.edu/racial-justice/reparations/>, accessed 25 October 2023).
589. Himmelstein KEW, Lawrence JA, Jahn JL, Ceasar JN, Morse M, Bassett MT et al. Association Between Racial Wealth Inequities and Racial Disparities in Longevity Among US Adults and Role of Reparations Payments, 1992 to 2018. *JAMA Netw Open* 2022;5:e2240519. (<https://doi.org/10.1001/jamanetworkopen.2022.40519>).
590. Richardson ET, Malik MM, Darity WA, Mullen AK, Morse ME, Malik M et al. Reparations for Black American descendants of persons enslaved in the U.S. and their potential impact on SARS-CoV-2 transmission. *Soc Sci Med* 1982 2021;276:113741.
591. Inter-American Commission on Human Rights. The Right to Truth in the Americas [website]. 2014 (<https://www.oas.org/en/iachr/Default.asp>, accessed 12 June 2024).
592. Yoorrook Justice Commission. Home [website]. (<https://yoorrookjusticecommission.org.au/>, accessed 12 June 2024).
593. Government of the Netherlands. Government apologises for the Netherlands' role in the history of slavery [website]. 2022 (<https://www.government.nl/latest/news/2022/12/19/government-apologises-for-the-netherlands-role-in-the-history-of-slavery>).
594. Government of Canada. Final settlement agreement on Compensation and Agreement-in-Principle for long-term reform of First Nations Child and Family Services and Jordan's Principle [website]. 2024 (<https://sac-isc.gc.ca/eng/1646942622080/1646942693297#chp4-3>, accessed 12 December 2024).
595. Farge E. UN rights chief urges states to act on slavery reparations. *Reuters*, 19 April 2024, Section World. (<https://www.reuters.com/world/un-rights-chief-urges-states-act-slavery-reparations-2024-04-19/>, accessed 12 June 2024).
596. Niyitunga EB. Variables influencing truth reconciliation commissions to promote post-conflict peacebuilding and statebuilding in Africa: a South African experience. *Front Polit Sci* 2024;6:1355951.
597. Clark JN. Reconciliation via Truth? A Study of South Africa's TRC. *J Hum Rights* 2012;11:189–209.
598. Truth and Reconciliation Commission. Final Report of the Truth and Reconciliation Commission – Volume 5 Chapter 5. South African Government, 1998 (<https://www.justice.gov.za/trc/report/finalreport/Volume5.pdf>).
599. Government of South Africa. Government Notice No. R. 1660: Regulations Regarding Reparation to Victims. *Government Gazette*, 2003 (https://www.justice.gov.za/legislation/regulations/r2003/2003_r1660_gg25695-nat-unity.pdf).
600. Gibson JL. The Contributions of Truth to Reconciliation: Lessons From South Africa. *J Confl Resolut* 2006;50:409–432.
601. Permanent Forum on Indigenous Issues, twenty-third session (15 - 26 April 2024, New York). Improving the health and wellness of Indigenous Peoples globally: operationalization of Indigenous determinants of health: note / by the Secretariat [website]. 2024 (https://documents.un.org/symbol-explorer?s=E/C.19/2024/5&i=E/C.19/2024/5_1709247950656, accessed 12 June 2024).
602. Condo Riveras SF, Mcglade H, Roth G. Indigenous Determinants of Health in the 2030 Agenda. New York, UN Permanent Forum on Indigenous Issues (<https://docs.un.org/en/E/C.19/2023/5>, accessed 22 October 2024).
603. United Nations General Assembly. Declaration on the Rights of Indigenous People. 2007. (https://www.un.org/development/desa/indigenouspeoples/wp-content/uploads/sites/19/2018/11/UNDRIP_E_web.pdf).
604. Moloney A, Stuart L, Chen Y, Lin F. Healthcare professionals' cultural safety practices for indigenous peoples in the acute care setting – a scoping review. *Contemp Nurse* 2023;59:272–293.
605. De Zilva S, Walker T, Palermo C, Brimblecombe J. Culturally safe health care practice for Indigenous Peoples in Australia: A systematic meta-ethnographic review. *J Health Serv Res Policy* 2022;27:74–84.
606. Poitras M T, Vaillancourt V, Canapé A, et al. Culturally safe interventions in primary care for the management of chronic diseases of urban Indigenous People: a scoping review. *Fam Med Community Health* 2022;10:e001606.
607. Safeguarding Health in Conflict Coalition, Insecurity Insight. SHCC Report 2021: UNRELENTING VIOLENCE: Violence Against Health Care in Conflict. Safeguarding Health in Conflict Coalition, 2021 (<https://www.safeguardinghealth.org/sites/shcc/files/SHCC%25202021%2520Unrelenting%2520Violence%2520FINAL.pdf>, accessed 26 March 2024).

608. World Health Organization. WHO Surveillance System Attacks on Healthcare (SSA) [website]. 2024 (<https://extranet.who.int/ssa/LeftMenu/Index.aspx>, accessed 26 March 2024).
609. Katz R, Kornblet S, Arnold G, Lief E, Fischer JE. Defining Health Diplomacy: Changing Demands in the Era of Globalization. *Milbank Q* 2011;89:503–523.
610. Kickbusch I, Liu A. Global health diplomacy—reconstructing power and governance. *Lancet* 2022;399:2156–2166.
611. Barron GC, Laryea-Adjei G, Vike-Freiberga V, Abubakar I, Dakkak H, Devakumar D et al. Safeguarding people living in vulnerable conditions in the COVID-19 era through universal health coverage and social protection. *Lancet Public Health* 2022;7:e86–e92.
612. Tediosi F, Lönnroth K, Pablos-Méndez A, Raviglione M. Build back stronger universal health coverage systems after the COVID-19 pandemic: the need for better governance and linkage with universal social protection. *BMJ Glob Health* 2020;5:e004020.
613. World Health Organization. Role of social protection in reducing the burden of public health and social measures during the COVID-19 pandemic: evidence review. Geneva, World Health Organization, 2024 (<https://iris.who.int/handle/10665/379704>). License: CC BY-NC-SA 3.0 IGO.
614. Bowen T, Del Ninno C, Andrews C, Coll-Black S, Gentilini U, Johnson K et al. Adaptive Social Protection: Building Resilience to Shocks. Washington (DC): World Bank, 2020 (<https://doi.org/10.1596/978-1-4648-1575-1>).
615. Bryant A, Follett L. Hunger relief: A natural experiment from additional SNAP benefits during the COVID-19 pandemic. *Lancet Reg Health – Am* 2022;10:100224.
616. Makkar S, Manivannan JR, Swaminathan S, Travasso SM, John AT, Webb P et al. Role of cash transfers in mitigating food insecurity in India during the COVID-19 pandemic: a longitudinal study in the Bihar state. *BMJ Open* 2022;12:e060624.
617. Kumar A, Mishra AK, Saroj S, Rashid S. Government transfers, COVID-19 shock, and food insecurity: Evidence from rural households in India. *Agribus N Y N* 2022;38:636–659.
618. Ohrnberger J. Economic shocks, health, and social protection: The effect of COVID-19 income shocks on health and mitigation through cash transfers in South Africa. *Health Econ* 2022;31:2481–2498.
619. Londoño-Vélez J, Querubin P. The Impact of Emergency Cash Assistance in a Pandemic: Experimental Evidence from Colombia. *Rev Econ Stat* 2022;104:157–165.
620. Berkowitz SA, Basu S. Unmet Social Needs and Worse Mental Health after Expiration of COVID-19 Federal Pandemic Unemployment Compensation. *Health Aff (Millwood)* 2021;40:426–434.
621. Berkowitz SA, Basu S. Unemployment Insurance, Health-Related Social Needs, Health Care Access, and Mental Health During the COVID-19 Pandemic. *JAMA Intern Med* 2021;181:699–702.
622. Gentilini U. Cash Transfers in Pandemic Times: Evidence, Practices, and Implications from the Largest Scale Up in History. 2022 (<https://openknowledge.worldbank.org/entities/publication/7cdbc28a-22ce-5e86-a4bc-640442d46ef8>, accessed 8 October 2024).
623. Cañedo AP, Fabregas R, Gupta P. Emergency cash transfers for informal workers: Impact evidence from Mexico. *J Public Econ* 2023;219:104820.
624. Werner K, Kak M, Herbst CH, Lin TK. The role of community health worker-based care in post-conflict settings: a systematic review. *Health Policy Plan* 2023;38:261–274.
625. World Health Organization. World Health Organization strategy for engaging religious leaders, faith-based organizations and faith communities in health emergencies [website]. 2021 (<https://www.who.int/publications-detail-redirect/9789240037205>, accessed 18 October 2023).
626. Hess S, Smith S, Umachandran S. Faith as a complex system: engaging with the faith sector for strengthened health emergency preparedness and response. *Lancet Glob Health* 2024;12:e1750–e1751.
627. Strengthening the global architecture for health emergency prevention, preparedness, response and resilience. World Health Organization, 2023 (<https://www.who.int/publications/m/item/strengthening-the-global-architecture-for-health-emergency-prevention--preparedness--response-and-resilience>, accessed 8 October 2024).
628. World Bank. Prioritizing the poorest and most vulnerable in West Africa: Togo's Novissi platform for social protection uses machine learning, geospatial analytics, and mobile phone metadata for the pandemic response. [website]. 2021 (<https://www.worldbank.org/en/results/2021/04/13/prioritizing-the-poorest-and-most-vulnerable-in-west-africa-togo-s-novissi-platform-for-social-protection-uses-machine-l>, accessed 5 December 2024).
629. Debenedetti L. Togo's Novissi Cash Transfer: Designing and Implementing a Fully Digital Social Assistance Program during COVID-19. Togo, 2021 (<https://poverty-action.org/sites/default/files/publications/Togo-Novissi-Cash-Transfer-Brief-August%202021.pdf>, accessed 16 December 2024).

630. World Bank. Economic Update for Togo: The Coronavirus Is Exerting Pressure on the Economy. 2020 (<https://www.worldbank.org/en/country/togo/publication/economic-update-for-togo-the-coronavirus-is-exerting-pressure-on-the-economy>, accessed 18 December 2024).
631. Regional Innovation Centre UNDP Asia-Pacific. A COVID cash transfer programme in Togo that gives more money to women. The Innovation Dividend Podcast, EP 9. (<https://undp-ric.medium.com/cina-lawson-a-covid-cash-transfer-programme-that-gives-more-money-to-women-in-togo-2386c5dff49>).
632. Government of Togo. The Government is making changes to the "NOVISSI" program [website]. 2020 (<https://novissi.gouv.tg/wp-content/uploads/2020/06/20200617-CP-Revue-Programme-Novissi-FR-1.pdf>, accessed 18 December 2024).
633. GiveDirectly. FAQ | GiveDirectly [website]. (<https://www.givedirectly.org/faq/>).
634. Davidovic S, Nunhuck S, Prady D, Tourpe H. Beyond the COVID-19 crisis: a framework for sustainable government-to-person mobile money transfers. Washington (DC), International Monetary Fund, 2020 IMF working paper, No. WP/20, 198; (<https://doi.org/10.5089/9781513557670.001>).
635. Government of India. Pradhan Mantri Garib Kalyan Package (PMGKP) [website]. (<https://www.india.gov.in/spotlight/pradhan-mantri-garib-kalyan-package-pmgkp>, accessed 27 May 2024).
636. Vinck P, Pham PN, Bindu KK, Bedford J, Nilles EJ. Institutional trust and misinformation in the response to the 2018–19 Ebola outbreak in North Kivu, DR Congo: a population-based survey. *Lancet Infect Dis* 2019;19:529–536.
637. Park S-J, Brown H, Wema KM, Gobat N, Borchert M, Kalubi J et al. "Ebola is a business": an analysis of the atmosphere of mistrust in the tenth Ebola epidemic in the DRC. *Crit Public Health* 2023;33:297–307.
638. Gabe Joselow and Linda Givetash. Congo's Ebola response threatened by conspiracy theories, rumors. NBC News, 20 April 2019. (<https://www.nbcnews.com/news/world/congo-s-ebola-response-threatened-conspiracy-theories-rumors-n994156>).
639. Gallego-Ayala GK Jordi. The social response to Ebola in DRC: Five recommendations for successful community engagement in emergencies. World Bank Blogs 2023. (<https://blogs.worldbank.org/en/dev4peace/social-response-ebola-drc-five-recommendations-successful-community-engagement>, accessed 26 March 2024).
640. World Bank. Fighting Epidemics in Complex Emergencies: The Ebola Crisis Response and Recovery through Safety Nets and Community Engagement in the Democratic Republic of the Congo. 2020 (<https://www.worldbank.org/en/results/2020/12/14/fighting-epidemics-in-complex-emergencies-the-ebola-crisis-response-and-recovery-through-safety-nets-and-community-engagement-in-the-democratic-republic-of-the-congo>, accessed 12 December 2024).
641. World Bank. Building a Safety Net in Times of Crisis in the Democratic Republic of the Congo. 2021 (<https://www.worldbank.org/en/news/feature/2021/07/06/building-a-safety-net-in-times-of-crisis-in-democratic-republic-of-the-congo>, accessed 18 December 2024).
642. Ziersch A, Due C. A mixed methods systematic review of studies examining the relationship between housing and health for people from refugee and asylum seeking backgrounds. *Soc Sci Med* 1982 2018;213:199–219.
643. Brake T-M, Dudek V, Sauzet O, Razum O. Psychosocial Attributes of Housing and Their Relationship with Health Among Refugee and Asylum-Seeking Populations in High-Income Countries: Systematic Review. *Public Health Rev* 2023;44:1605602.
644. Egli-Gany D, Aftab W, Hawkes S, Abu-Raddad L, Buse K, Rabbani F et al. The social and structural determinants of sexual and reproductive health and rights in migrants and refugees: a systematic review of reviews. *East Mediterr Health J Rev Sante Mediterr Orient Al-Majallah Al-Sihhiyah Li-Sharq Al-Mutawassit* 2021;27:1203–1213.
645. Koseoglu Ornek O, Waibel J, Wullinger P, Weinmann T. Precarious employment and migrant workers' mental health: a systematic review of quantitative and qualitative studies. *Scand J Work Environ Health* 2022;48:327–350.
646. Hargreaves S, Rustage K, Nellums LB, McAlpine A, Pocock N, Devakumar D et al. Occupational health outcomes among international migrant workers: a systematic review and meta-analysis. *Lancet Glob Health* 2019;7:e872–e882.
647. Wali N, Chen W, Rawal LB, Amanullah ASM, Renzaho AMN. Integrating human rights approaches into public health practices and policies to address health needs amongst Rohingya refugees in Bangladesh: a systematic review and meta-ethnographic analysis. *Arch Public Health Arch Belg Sante Publique* 2018;76:59.
648. Nowak AC, Namer Y, Hornberg C. Health Care for Refugees in Europe: A Scoping Review. *Int J Environ Res Public Health* 2022;19:1278.

649. Kaur H, Saad A, Magwood O, Alkhateeb Q, Mathew C, Khalaf G et al. Understanding the health and housing experiences of refugees and other migrant populations experiencing homelessness or vulnerable housing: a systematic review using GRADE-CERQual. *CMAJ Open* 2021;9:E681-E692.
650. Kusuma YS, Babu BV. Migration and health: A systematic review on health and health care of internal migrants in India. *Int J Health Plann Manage* 2018;33:775-793.
651. Batista R, Pottie K, Bouchard L, Ng E, Tanuseputro P, Tugwell P. Primary Health Care Models Addressing Health Equity for Immigrants: A Systematic Scoping Review. *J Immigr Minor Health* 2018;20:214-230.
652. Global research agenda on health, migration and displacement: strengthening research and translating research priorities into policy and practice. Geneva, World Health Organization, 2023 (<https://iris.who.int/handle/10665/373659>, accessed 25 July 2024). License: CC BY-NC-SA 3.0 IGO.
653. Larrea-Schiavon S, Vázquez-Quesada LM, Bartlett LR, Lam-Cervantes N, Sripad P, Vieitez I et al. Interventions to Improve the Reproductive Health of Undocumented Female Migrants and Refugees in Protracted Situations: A Systematic Review. *Glob Health Sci Pract* 2022;10:1-15.
654. Kamali M, Munyuzangabo M, Siddiqui FJ, Gaffey MF, Meteke S, Als D et al. Delivering mental health and psychosocial support interventions to women and children in conflict settings: a systematic review. *BMJ Glob Health* 2020;5:e002014.
655. Ekezie W, Adaji EE, Murray RL. Essential healthcare services provided to conflict-affected internally displaced populations in low and middle-income countries: A systematic review. *Health Promot Perspect* 2020;10:24-37. (<https://doi.org/10.15171/hpp.2020.06>).
656. Naing C, Whittaker MA, Tanner M. Inter-sectoral approaches for the prevention and control of malaria among the mobile and migrant populations: a scoping review. *Malar J* 2018;17:430.
657. Diaz E, Ortiz-Barreda G, Ben-Shlomo Y, Holdsworth M, Salami B, Rammohan A et al. Interventions to improve immigrant health. A scoping review. *Eur J Public Health* 2017;27:433-439.
658. Aung E, Blondell SJ, Durham J. Interventions for Increasing HIV Testing Uptake in Migrants: A Systematic Review of Evidence. *AIDS Behav* 2017;21:2844-2859.
659. Martin CA, Gowda U, Smith BJ, Renzaho AMN. Systematic Review of the Effect of Lifestyle Interventions on the Components of the Metabolic Syndrome in South Asian Migrants. *J Immigr Minor Health* 2018;20:231-244.
660. Kip A, Priebe S, Holling H, Morina N. Psychological interventions for posttraumatic stress disorder and depression in refugees: A meta-analysis of randomized controlled trials. *Clin Psychol Psychother* 2020;27:489-503.
661. Tovar A, Renzaho AMN, Guerrero AD, Mena N, Ayala GX. A Systematic Review of Obesity Prevention Intervention Studies among Immigrant Populations in the US. *Curr Obes Rep* 2014;3:206-222.
662. Shommu NS, Ahmed S, Rumana N, Barron GRS, McBrien KA, Turin TC. What is the scope of improving immigrant and ethnic minority healthcare using community navigators: A systematic scoping review. *Int J Equity Health* 2016;15:6.
663. Crocker-Buque T, Mindra G, Duncan R, Mounier-Jack S. Immunization, urbanization and slums – a systematic review of factors and interventions. *BMC Public Health* 2017;17:556.
664. Fox S, Kramer E, Agrawal P, Aniyizhai A. Refugee and Migrant Health Literacy Interventions in High-Income Countries: A Systematic Review. *J Immigr Minor Health* 2022;24:207-236.
665. Njue C, Sharmin S, Dawson A. Models of Maternal Healthcare for African refugee women in High-Income Countries: A Systematic Review. *Midwifery* 2022;104:103187.
666. Yadee J, Bangpan M, Thavorn K, Welch V, Tugwell P, Chaiyakunapruk N. Assessing evidence of interventions addressing inequity among migrant populations: a two-stage systematic review. *Int J Equity Health* 2019;18:64.
667. Sieck CJ, Sheon A, Ancker JS, Castek J, Callahan B, Siefer A. Digital inclusion as a social determinant of health. *NPJ Digit Med* 2021;4:52.
668. Romanello M, McGushin A, Napoli CD, Drummond P, Hughes N, Jamart L et al. The 2021 report of the Lancet Countdown on health and climate change: code red for a healthy future. *Lancet* 2021;398:1619-1662.
669. European Commission. The EU as an Issuer: The NextGenerationEU Transformation. Luxembourg: Publications Office of the European Union. 2022 (<https://ec.europa.eu/newsroom/budget/items/759825/en#:~:text=The%20policy%20brief%20describes%20in%20detail%20the%20Commission's,communication%20with%20markets%20and%20structured%20relationships%20with%20banks>, accessed 22 March 2025).

670. United Nations. Widening Digital Gap between Developed, Developing States Threatening to Exclude World's Poorest from Next Industrial Revolution, Speakers Tell Second Committee [website]. UN Meeting Coverage and Press Release, 2023 (<https://press.un.org/en/2023/gaef3587.doc.htm>, accessed 27 February 2025).
671. Buse K, Tomson G, Kuruvilla S et al. Tackling the politics of intersectoral action for the health of people and planet. *BMJ* 2022;376.
672. World Health Organization. Health benefits far outweigh the costs of meeting climate change goals [website]. 2018 (<https://www.who.int/news/item/05-12-2018-health-benefits-far-outweigh-the-costs-of-meeting-climate-change-goals>, accessed 3 June 2024).
673. International Labour Office, editor. Working on a warmer planet: the impact of heat stress on labour productivity and decent work. Geneva, International Labour Office. 2019. (<https://www.ilo.org/publications/working-warmer-planet-effect-heat-stress-productivity-and-decent-work#:~:text=This%20report%20shows%20the%20impact%20of%20heat%20stress,health%20for%20the%20most%20vulnerable%20groups%20of%20workers>, accessed 22 March 2025).
674. UN Women. Explainer: How gender inequality and climate change are interconnected [website]. 2022 (<https://www.unwomen.org/en/news-stories/explainer/2022/02/explainer-how-gender-inequality-and-climate-change-are-interconnected>, accessed 2 May 2024).
675. van Daalen KR, Kallesøe SS, Davey F, Dada S, Jung L, Singh L et al. Extreme events and gender-based violence: a mixed-methods systematic review. *Lancet Planet Health* 2022;6:e504–e523.
676. United Nations Framework Convention on Climate Change (UNFCCC). The Paris Agreement [website]. 2015 (<https://unfccc.int/process-and-meetings/the-paris-agreement>, accessed 20 May 2024).
677. UNFCCC. COP27 Reaches Breakthrough Agreement on New “Loss and Damage” Fund for Vulnerable Countries [website]. 2022 (<https://unfccc.int/news/cop27-reaches-breakthrough-agreement-on-new-loss-and-damage-fund-for-vulnerable-countries>, accessed 2 May 2024).
678. International Energy Agency. SDG7: Data and Projections, Access to electricity. Paris, 2023 (<https://www.iea.org/reports/sdg7-data-and-projections/access-to-electricity>).
679. Mqadi L, Malgas L. The Kuyasa Case Study: An effort towards climate justice and energy poverty alleviation (crediting suppressed demand for energy services under the CDM). 2004. (https://www.eji.snre.umich.edu/EJCC/PRESENTATIONS/LWANDLE%20MQADI_paper.pdf, accessed 18 December 2024).
680. Kuyasa CDM Project. SAEDF Vision: South African Export Development Fund (“SAEDF”) [website]. 2009 (<https://kuyasacdm.co.za/vision.php>, accessed 18 December 2024).
681. SouthSouthNorth (SSN). (<https://southsouthnorth.org/>, accessed 18 December 2024).
682. World Bank Energy Sector Management Assistance Program. Good Practices in City Energy Efficiency: Cape Town-Kuyasa Settlement, South Africa – Low-Income Energy Efficiency Housing Project. 2012 (https://www.esmap.org/sites/default/files/esmap-files/Kuyasa_EECI_Housing_FinalCaseStudy_Africa.pdf, accessed 18 December 2024).
683. World Health Organization. Health in the green economy: health co-benefits of climate change mitigation – housing sector. 2011 (<https://iris.who.int/handle/10665/44609>, accessed 18 December 2024).
684. Vivienne Walsh, Carl Wesselink, Andrew Janisch. Improving urban living conditions in Kuyasa (Cape Town) through energy-efficiency retrofits: the impact on low-income communities. *SouthSouthNorth*. 2011 (https://southsouthnorth.org/wp-content/uploads/2022/01/FINAL-Version-Kuyasa-report_27_07_2011-.pdf).
685. Houghton A, Castillo-Salgado C. Health Co-Benefits of Green Building Design Strategies and Community Resilience to Urban Flooding: A Systematic Review of the Evidence. *Int J Environ Res Public Health* 2017;14:1519.
686. Wang C, Wang J, Norbäck D. A Systematic Review of Associations between Energy Use, Fuel Poverty, Energy Efficiency Improvements and Health. *Int J Environ Res Public Health* 2022;19:7393. (<https://doi.org/10.3390/ijerph19127393>).
687. Beck M, Rivers N, Yonezawa H. A rural myth? Sources and implications of the perceived unfairness of carbon taxes in rural communities. *Ecol Econ* 2016;124:124–34.
688. Sovacool B K. Fuel poverty, affordability, and energy justice in England: Policy insights from the Warm Front Program. *Energy* 2015;93:361–371.
689. Glennon R. The Unfolding Tragedy of Climate Change in Bangladesh. *Sci Am Blog Netw*. 2017. (<https://robertglennon.net/the-unfolding-tragedy-of-climate-change-in-bangladesh/>).
690. Pethick J, Orford J. Rapid rise in Effective Sea-Level in Southwest Bangladesh: Its causes and contemporary rates. *Glob Planet Change* 2013;111:237–245.

691. The World Bank. Shelter from the Storm: Protecting Bangladesh's Coastal Communities from Natural Disasters [website]. 2022 (<https://www.worldbank.org/en/results/2022/08/29/shelter-from-the-storm-protecting-bangladesh-s-coastal-communities-from-natural-disasters>).
692. Begum T. Peace of mind at last: the Bangladeshi villagers digging their way out of the floods. *The Guardian*, 4 May 2023. (<https://www.theguardian.com/global-development/2023/may/04/bangladeshi-villagers-digging-their-way-out-of-the-floods-plinth-settlements>).
693. World Bank. Lighting Up Rural Communities in Bangladesh: The Second Rural Electrification and Renewable Energy Development Project. 2021 (<https://www.worldbank.org/en/results/2021/03/13/lighting-up-rural-communities-in-bangladesh-the-second-rural-electrification-and-renewable-energy-development-project>, accessed 18 December 2024).
694. United Nations. Sustainable Development Goal 7: Affordable and Clean Energy [website]. (<https://www.un.org/sustainabledevelopment/energy/>, accessed 18 December 2024).
695. Grameen Shakti. [website]. (<https://www.gshakti.org/>, accessed 18 December 2024).
696. Khan, Prof. Dr. Shahidul Islam; Rabbi, Md. Fazley. An Innovative Financing Mechanism: Creating Access to Renewable Energy for Rural People of Bangladesh. *Grameen Shakti*, 2013 (https://www.gshakti.org/data/researchAndPub/upload_c7eeda235547928ec2a19e29fa650e36.pdf).
697. Grameen Shakti. Solar Home System [website]. (<https://www.gshakti.org/what-we-do/keyprograms/solarhomesystem>).
698. Grameen Shakti. Biogas [website]. (<https://www.gshakti.org/what-we-do/keyprograms/biogas>, accessed 18 December 2024).
699. Grameen Shakti. Improved Cook Stove [website]. (<https://www.gshakti.org/what-we-do/keyprograms/improvedcookstove>, accessed 18 December 2024).
700. Grameen Shakti. Consultancy & Knowledge Practice [website]. (<https://www.gshakti.org/what-we-do/consultancy-and-knowledge-practice/services-we-offer>, accessed 18 December 2024).
701. International Energy Agency. Energy system of Bangladesh [website]. (<https://www.iea.org/countries/bangladesh>, accessed 18 December 2024).
702. The International Bank for Reconstruction and Development / The World Bank. Improved Cookstoves and Better Health in Bangladesh: Lessons from Household Energy and Sanitation Programs. The World Bank. 2010 (<https://openknowledge.worldbank.org/server/api/core/bitstreams/baf982c6-e4e7-5e85-a3cb-808bd154bc51/content>, accessed 22 March 2025).
703. Zaman SB, Hossain N, Khandker S. Impact of Improved Cooking Stove on Maternal Health in Rural Bangladesh: A Quasi-Experimental Study. *J Med Res Innov* 2017;1:1-9.
704. Berthaud A, Delescluse A, Deligiorgis D, Kumar K, Mane S, Miyamoto S et al. Integrating Gender in Energy Provision Case Study of Bangladesh. Washington (DC), World Bank Group, 2004 (<https://openknowledge.worldbank.org/server/api/core/bitstreams/b05b871d-0c9e-58f6-b4a6-552a78d8bf71/content>, accessed 18 December 2024).
705. Salah Uddin G, Abdullah-Al-Baki C, Donghyun P, Ahmed A, Shu T. Social benefits of solar energy: Evidence from Bangladesh. *Oeconomia Copernic* 2023;14:861-897.
706. Saudia Afrin. How Grameen Shakti is providing a sustainable solution to energy needs in rural Bangladesh. *Dhaka Tribune*, 25 August 2021. (<https://www.dhakatribune.com/science-technology-environment/climate-change/256374/how-grameen-shakti-is-providing-a-sustainable>, accessed 18 December 2024).
707. World Economic Forum. 7 ways to bolster food security in the face of climate change [website]. 2023 (<https://www.weforum.org/agenda/2023/09/7-ways-to-boost-food-security-in-the-face-of-climate-change/>, accessed 21 May 2024).
708. Kesse-Guyot E, Fouillet H, Baudry J, Dussiot A, Langevin B, Allès B et al. Halving food-related greenhouse gas emissions can be achieved by redistributing meat consumption: Progressive optimization results of the NutriNet-Santé cohort. *Sci Total Environ* 2021;789:147901.
709. Red and processed meat in the context of health and the environment: many shades of red and green: information brief. Geneva, World Health Organization, 2023 (<https://iris.who.int/handle/10665/370775>, accessed 5 August 2024). License: CC BY-NC-SA 3.0 IGO.
710. Aerts R, Honnay O, Van Nieuwenhuysse A. Biodiversity and human health: mechanisms and evidence of the positive health effects of diversity in nature and green spaces. *Br Med Bull* 2018;127:5-22.
711. World Health Organization. Biodiversity [website]. 2025 (<https://www.who.int/news-room/fact-sheets/detail/biodiversity>, accessed 5 March 2025).

712. Kuyah S, Buleti S, Dimobe K, Nkurunziza L, Moussa S, Muthuri C et al. Farmer-Managed Natural Regeneration in Africa: Evidence for Climate Change Mitigation and Adaptation in Drylands. In: Dagar JC, Gupta SR, Sileshi GW, editors. *Agroforestry for Sustainable Intensification of Agriculture in Asia and Africa*. Sustainability Sciences in Asia and Africa. Singapore: Springer Nature Singapore; 2023:53–88 (https://doi.org/10.1007/978-981-19-4602-8_3).
713. Tongul H, Hobson M. Scaling up an integrated watershed management approach through social protection programmes in Ethiopia: the MERET and PSNP schemes. *Hunger Nutrition Climate Justice* 2013. Dublin, 2013 (<https://www.mrfcj.org/wp-content/uploads/2015/09/2013-04-16-Ethiopia-MERET.pdf>, accessed 4 April 2024).
714. Richards P. Ebola: How a People's Science Helped End an Epidemic. London, Zed books, 2016.
715. Richards P, Mokuwa GA, Vandt A, Mayhew SH. Re-analysing Ebola spread in Sierra Leone: The importance of local social dynamics. *PLoS ONE* 2020;15.
716. Maat H et al. Towards Sustainable Community-Based Systems for Infectious Disease and Disaster Response; Lessons from Local Initiatives in Four African Countries. *Sustainability* 2021;13:10083.
717. United Nations Development Programme. Solar for Health [website]. (<https://www.undp.org/energy/our-flagship-initiatives/solar-for-health>, accessed 18 December 2024).
718. Green Policy Platform. Solar for Health in Zimbabwe. (<https://www.greenpolicyplatform.org/sites/default/files/downloads/best-practices/Solar%20for%20Health%20in%20Zimbabwe.pdf>, accessed 18 December 2024).
719. Solar for Health Presentation. 2018. (<https://www.undp-capacitydevelopment-health.org/files/UNDP-Solar-For-Health-Presentation-October-2018-%20reduced.pdf>, accessed 18 December 2024).
720. United Nations Development Programme. Solar for Health: 5 Ways Solar Power Can Make Universal Healthcare a Reality [website]. 2018 (<https://stories.undp.org/solar-for-health>).
721. Intergovernmental Panel on Climate Change. Global Warming of 1.5°C. An IPCC Special Report on impacts of global warming of 1.5°C above pre-industrial levels and related global greenhouse gas emission pathways, in the context of strengthening the global response to the threat of climate change, sustainable development, and efforts to eradicate poverty, 1st ed. Cambridge University Press, 2022 (<https://doi.org/10.1017/9781009157940>).
722. Koea JB. Indigenous trauma: A New Zealand perspective. *Injury* 2008;39:S11–S18. ([https://doi.org/10.1016/S0020-1383\(08\)70024-X](https://doi.org/10.1016/S0020-1383(08)70024-X)).
723. United Nations Department of Economic and Social Affairs. Health [website]. (<https://www.un.org/development/desa/indigenouspeoples/mandated-areas/health.html>, accessed 18 December 2024).
724. Clarkson C, Jacobs Z, Marwick B, Fullagar R, Wallis L, Smith M et al. Human occupation of northern Australia by 65,000 years ago. *Nature* 2017;547:306–310.
725. Karanja W. Land and Healing: A Decolonizing Inquiry for Centering Land as the Site of Indigenous Medicine and Healing. In: Wane NN, Todorova MS, Todd KL, editors. *Decolonizing the Spirit in Education and Beyond*. Cham, Springer International Publishing; 2019:45–61 (https://doi.org/10.1007/978-3-030-25320-2_4).
726. Yellowknives Dene First Nation Wellness Division, Lines L-A, Jardine CG. Connection to the land as a youth-identified social determinant of Indigenous Peoples' health. *BMC Public Health* 2019;19:176.
727. Redvers J. "The land is a healer": Perspectives on land-based healing from Indigenous practitioners in northern Canada. *Int J Indig Health* 2020;15:90–107.
728. "Yotti" Kingsley J, Townsend M, Phillips R, Aldous D. "If the land is healthy ... it makes the people healthy": The relationship between caring for Country and health for the Yorta Yorta Nation, Boonwurrung and Bangerang Tribes. *Health Place* 2009;15:291–299.
729. Jubinville D, Smylie J, Wolfe S, Bourgeois C, Berry NS, Rotondi M et al. Relationships to land as a determinant of wellness for Indigenous women, two-spirit, trans, and gender diverse people of reproductive age in Toronto, Canada. *Can J Public Health* 2024;115:253–262.
730. *Mabo v Queensland (No 2)* [1992] HCA 23; (1992) 175 CLR 1, 1992:1 (<https://aiatsis.gov.au/ntpd-resource/742>).
731. Weir JK, Stacey C, Youngeotob K. The Benefits of Caring for Country. Canberra, Australian Institute of Aboriginal and Torres Strait Islander Studies (AIATSIS), 2011 (https://aiatsis.gov.au/sites/default/files/research_pub/benefits-cfc_0_3.pdf).
732. Gracey M, King M. Indigenous health part 1: determinants and disease patterns. *Lancet* 2009;374:65–75.
733. King M, Smith A, Gracey M. Indigenous health part 2: the underlying causes of the health gap. *Lancet* 2009;374:76–85.

734. Australian Human Rights Commission. Close the Gap: Indigenous Health Campaign [website]. 2022 (<https://humanrights.gov.au/our-work/aboriginal-and-torres-strait-islander-social-justice/projects/close-gap-indigenous-health>, accessed 18 December 2024).
735. United Nations High Commissioner for Refugees. UNHCR Iran: New Arrivals from Afghanistan – End November 2022. 2023. (<https://data.unhcr.org/en/documents/details/98461>, accessed 18 December 2024).
736. World Health Organization Regional Office for the Eastern Mediterranean. WHO Completes Major Japan-Funded Project to Enhance Universal Health Coverage in Islamic Republic of Iran [website]. 2024 (<https://www.emro.who.int/iran/news/who-completes-major-japan-funded-project-to-enhance-universal-health-coverage-in-islamic-republic-of-iran.html>, accessed 18 December 2024).
737. International Telecommunication Union. Measuring digital development. Facts and figures 2021. ITU, 2021 (<https://www.itu.int/itu-d/reports/statistics/facts-figures-2021>, accessed 23 October 2023).
738. UN High-Level Panel on Digital Cooperation. The age of digital interdependence: report of the UN Secretary-General's High-Level Panel on Digital Cooperation. 2019. (<https://digitallibrary.un.org/record/3865925>, accessed 23 October 2023).
739. Stängle S, Benedetti FD, Hediger H, Bonmarin M, Loeser M, Fringer A. Use of digital technologies to combat loneliness and social isolation: a cross-sectional study in Swiss outpatient care during COVID-19 pandemic. *BMC Nurs* 2022;21:181.
740. OECD Going Digital Toolkit. Share of individuals who live in households with income in the lowest quintile who use the Internet | Society Indicators [website]. (<https://goingdigital.oecd.org/en/indicator/51>, accessed 5 June 2024).
741. International Telecommunication Union. Measuring digital development: Facts and Figures 2023 [website]. (<https://www.itu.int:443/en/ITU-D/Statistics/Pages/facts/default.aspx>, accessed 5 June 2024).
742. Ahmed AI, Soliman A, Han Y, Qureshi WT, Al-Mallah MH. Racial disparities of internet access in the United States: a multiyear analysis of the national behavioral risk factors surveillance system. *Eur Heart J* 2021;42:ehab724.3084.
743. Thomas J, McCosker A, Parkinson S, Et Al. Measuring Australia's digital divide: the Australian digital inclusion index 2023. Melbourne: ARC Centre of Excellence for Automated Decision-Making and Society, RMIT University, Swinburne University of Technology, and Telstra; 2023 (<https://doi.org/10.25916/528S-NY91>).
744. OECD. Bridging the Digital Gender Divide: Include, upskill, innovate. Organization for Economic Cooperation and Development. 2018 (<https://www.enterprise-development.org/bridging-the-digital-gender-divide-include-upskill-innovate-oecd-2018/>, accessed 5 June 2024).
745. Cho M, Kim KM. Effect of digital divide on people with disabilities during the COVID-19 pandemic. *Disabil Health J* 2022;15:101214.
746. Bonsaksen T, Ruffolo M, Price D, Leung J, Thygesen H, Lamph G et al. Associations between social media use and loneliness in a cross-national population: do motives for social media use matter? *Health Psychol Behav Med* 2023;11:2158089.
747. Greenbaum J, Kaplan D, Young J, Council on Child Abuse and Neglect, Council on Immigrant Child and Family Health. Exploitation, Labor and Sex Trafficking of Children and Adolescents: Health Care Needs of Patients. *Pediatrics* 2022;151:e2022060416.
748. Brown M, Osman S. Addiction, Gambling and Gaming: Chasing the Digital Dragon. In: Kelly C, Lynes A, Hoffin K, editors. *Video Games Crime and Next-Gen Deviance*. Emerald Publishing Limited; 2020:91-111 (<https://doi.org/10.1108/978-1-83867-447-220201006>).
749. Raybould JN, Larkin M, Tunney RJ. Is there a health inequality in gambling related harms? A systematic review. *BMC Public Health* 2021;21:305.
750. Eloundou T, Manning S, Mishkin P, Rock D. GPTs are GPTs: An Early Look at the Labor Market Impact Potential of Large Language Models. 2023. (<https://doi.org/10.48550/arXiv.2303.10130>).
751. Acypreste RD, Paraná E. Artificial Intelligence and employment: a systematic review. *Braz J Polit Econ* 2022;42:1014-1032.
752. Frank, M, Autor, D, Bessen, J, Brynjolfsson, E, Cebrian, M, Deming, D et al. Toward understanding the impact of artificial intelligence on labor. *Proc Natl Acad Sci* 2019;116:6531-6539.
753. Obermayer N, Csizmadia T, Banász Z. Companies on Thin Ice Due to Digital Transformation: The Role of Digital Skills and Human Characteristics. *Int Multidiscip J Soc Sci* 2022;11:88-118.
754. Charlwood A, Guenole N. Can HR adapt to the paradoxes of artificial intelligence? *Hum Resour Manag J* 2022;32:729-742.
755. David Rotman. ChatGPT is about to revolutionize the economy. We need to decide what that looks like. [website]. 2023 (<https://www.technologyreview.com/2023/03/25/1070275/chatgpt-revolutionize-economy-decide-what-looks-like/>, accessed 2 May 2024).

756. Koivusalo M, Svyndarenko A, Mbare B, Perkiö M. Disruptive (dis)engagement: platformisation as a global social policy. *Humanit Soc Sci Commun* 2024;11:584.
757. Klinova K, Korinek A. Unleashing possibilities, ignoring risks: Why we need tools to manage AI's impact on jobs. *Brookings*, 2023 (<https://www.brookings.edu/articles/unleashing-possibilities-ignoring-risks-why-we-need-tools-to-manage-ais-impact-on-jobs/>, accessed 4 April 2024).
758. Datta N, Rong C, Singh S, Stinshoff C, Iacob N, Nigatu NS et al. Working Without Borders: The Promise and Peril of Online Gig Work. Washington (DC): World Bank, 2023 (<https://doi.org/10.1596/40066>).
759. Rodriguez F, Sarraju A, Turakhia MP. The Gig Economy Worker – A New Social Determinant of Health? *JAMA Cardiol* 2022;7:125–126.
760. Greiss J, Murphy D. Fair Work Legislation Amendment (Closing Loopholes) Bill 2023. 2023 (https://www.aph.gov.au/Parliamentary_Business/Bills_Legislation/bd/bd2324a/24bd017, accessed 18 December 2024).
761. López Mourello E. Work on delivery platforms in Argentina: Analysis and policy recommendations. Buenos Aires, ILO Country Office for Argentina, 2020.
762. Asenjo Cruz A, Coddou Mc Manus A, Dhir RK. The platform economy and transformations in the world of work: the case of delivery platform workers in Santiago, Chile. *International Labour Organization*, 2023 ILO working paper / *International Labour Organization*, No. 100 (November 2023); (<https://doi.org/10.54394/JCVG2192>).
763. Seattle Office of Labor Standards. App-Based Worker Minimum Payment Ordinance [website]. 2022 (<https://www.seattle.gov/laborstandards/ordinances/app-based-worker-ordinances/app-based-worker-minimum-payment-ordinance>).
764. Seattle Office of Labor Standards. Gig Worker Paid Sick and Safe Time Ordinance [website]. 2020 (<https://www.seattle.gov/laborstandards/ordinances/covid-19-gig-worker-protections/gig-worker-paid-sick-and-safe-time-ordinance>).
765. Borges do Nascimento IJ, Pizarro AB, Almeida JM, Azzopardi-Muscat N, Gonçalves MA, Björklund M et al. Infodemics and health misinformation: a systematic review of reviews. *Bull World Health Organ* 2022;100:544–561.
766. Vyas DA, Eisenstein LG, Jones DS. Hidden in Plain Sight – Reconsidering the Use of Race Correction in Clinical Algorithms. *N Engl J Med* 2020;383:874–882.
767. HR NEWS. 3 Blockchain Use Cases That Benefit Government Systems. 2024. (<https://hrnews.co.uk/3-blockchain-use-cases-that-benefit-government-systems/>, accessed 25 September 2024).
768. Martinho-Truswell E. How AI Could Help the Public Sector. *Harv Bus Rev* 2018. (<https://hbr.org/2018/01/how-ai-could-help-the-public-sector>, accessed 21 May 2024).
769. Global Government Forum. Big but not scary: how to use big data to shape government policy and delivery [website]. 2024 (<https://www.globalgovernmentforum.com/big-but-not-scary-how-to-use-big-data-to-shape-government-policy-and-delivery/>, accessed 21 May 2024).
770. UN Secretary General. Roadmap for Digital Cooperation. United Nations, 2020 (https://www.un.org/en/content/digital-cooperation-roadmap/assets/pdf/Roadmap_for_Digital_Cooperation_EN.pdf).
771. UN General Assembly. Global Digital Compact. (<https://www.un.org/techenvoy/global-digital-compact>, accessed 25 September 2024).
772. Ratzan SC, Larson HJ, Batista C, Kuzmanovic A, Kalb PE, Rabin KH et al. The Quality Health Information for All Commission: reinventing health communication for the digital era. *Nat Med* 2025;31:22–23.
773. Purnat TD, Clark J. Oversimplified efforts to counter health misinformation are missing the mark. *BMJ* 2025;r393.
774. Graham G, Goren N, Sounderajah V, DeSalvo K. Information is a determinant of health. *Nat Med* 2024;30:927–928.
775. Local 2030 – Localizing the SDGs [website]. 2017 (<https://www.local2030.org/local-action>, accessed 16 October 2023).
776. Ansell C, Sørensen E, Torfing J. The Key Role of Local Governance in Achieving the SDGs. Co-Creation for Sustainability. Leeds: Emerald Publishing Limited; 2022:9–22 (<https://doi.org/10.1108/978-1-80043-798-220220002>).
777. World Health Organization. Policy briefs for a strategic approach for urban health [website]. (<https://www.who.int/teams/social-determinants-of-health/urban-health/policy-briefs>, accessed 27 September 2024).
778. Ministry of Health and Care Services, Norway. Public Health Act (Lov om folkehelsearbeid) [website]. 2011 (<https://lovdata.no/dokument/NL/lov/2011-06-24-29>, accessed 12 March 2024).
779. Buck D. The English Local Government Public Health Reforms: An Independent Assessment. The King's Fund, 2020 (https://assets.kingsfund.org.uk/f/256914/x/9a5f33319f/english_local_government_public_health_reforms_independent_assessment_2020.pdf, accessed 13 March 2024).

780. De Leeuw E. Evidence for Healthy Cities: reflections on practice, method and theory. *Health Promot Int* 2009;24:i19–i36.
781. World Health Organization. How to develop and sustain healthy cities in 20 steps. Copenhagen, World Health Organization Regional Office for Europe, 2022 (<https://www.who.int/europe/publications/item/WHO-EURO-2022-6646-46412-67222>, accessed 12 March 2024).
782. Pearce M. Coventry: a Marmot City [website]. Last Modified: 2015-04-01 publisher: Coventry City Council, 2024 (<https://www.coventry.gov.uk/coventry-marmot-city-1/coventry-marmot-city>, accessed 12 March 2024).
783. Paipa A de. Sistematización de la estrategia HUMANO: red de tejido social [website]. 2019 (<https://es.scribd.com/document/433047541/paipa>, accessed 23 October 2023).
784. World Bank. People-centric model is transforming Chile's social protection system [website]. 2024 (<https://www.worldbank.org/en/news/feature/2024/07/19/modelo-centrado-en-las-personas-est-transformando-el-sistema-de-proteccion-social-en-chile>, accessed 18 December 2024).
785. Case Compass. Chile (GSL). ([https://www.case-compass.org/files/Chile%20\(GSL\).pdf](https://www.case-compass.org/files/Chile%20(GSL).pdf), accessed 18 December 2024).
786. Schultz S, Zorbas C, Peeters A, Yoong S, Backholer K. Strengthening local government policies to address health inequities: perspectives from Australian local government stakeholders. *Int J Equity Health* 2023;22:119.
787. Muchomba FM, Teitler J, Kruse L, Reichman NE. Municipality-Level Variation in Severe Maternal Morbidity and Association with Municipal Expenditures in New Jersey. *JAMA Netw Open* 2021;4:e2135161.
788. Fahy K, Alexiou A, Daras K, Mason K, Bennett D, Taylor-Robinson D et al. Mental health impact of cuts to local government spending on cultural, environmental and planning services in England: a longitudinal ecological study. *BMC Public Health* 2023;23:1441.
789. Rural Health Advocacy Project. Rural-proofing for health: guidelines. Johannesburg, 2015 (<https://www.hst.org.za/publications/NonHST%20Publications/2015-01-13-RHAP-Rural-Proofing-Guideline-A4-Email-1.pdf>, accessed 12 March 2024).
790. Rural Health Information Hub Toolkit. Social Determinants of Health in Rural Communities Toolkit [website]. 2002 (<https://www.ruralhealthinfo.org/toolkits/sdoh>, accessed 12 March 2024).
791. Xeuatvongsa A, Hachiya M, Miyano S, Mizoue T, Kitamura T. Determination of factors affecting the vaccination status of children aged 12–35 months in Lao People's Democratic Republic. *Heliyon* 2017;3:e00265.
792. Lao Statistics Bureau. Lao Social Indicator Survey II 2017, Survey Findings Report. Vientiane, Lao PDR, Lao Statistics Bureau and UNICEF, 2018 (<https://www.unicef.org/laos/media/306/file/LSIS2017ENG.pdf>).
793. Arsenault C, Gage A, Kim MK, Kapoor NR, Akweongo P, Amponsah F et al. COVID-19 and resilience of healthcare systems in ten countries. *Nat Med* 2022;28:1314–1324.
794. World Bank. Household Welfare Monitoring in the Lao PDR [website]. (<https://www.worldbank.org/en/country/lao/brief/monitoring-the-impact-of-covid-19-in-lao-pdr>).
795. Duysburgh E, Kerstens B, Diaz M, Fardhdiani V, Reyes KAV, Phommachanh K et al. Newborn care in Indonesia, Lao People's Democratic Republic and the Philippines: a comprehensive needs assessment. *BMC Pediatr* 2014;14:46.
796. World Health Organization. Building sustainable primary health care in the Lao People's Democratic Republic through community engagement and trust building [website]. 2023 (<https://www.who.int/publications/i/item/WPR-2023-DSI-001>, accessed 5 December 2024).
797. World Health Organization. Evaluating Community Engagement in Public Health: The Case of The Connect Initiative in Lao PDR. Summary Report of the Key Operational and Evaluation Insights from the Village-based Data Collection. 2024.
798. World Health Organization. Addressing Social Determinants of Health in Lao PDR: Local solutions to promote health equity. 2024.
799. World Health Organization Regional Office for the Western Pacific. 39,000 protected against COVID-19 ahead of National Games [website]. 2023 (<https://www.who.int/westernpacific/news-room/feature-stories/item/39-000-protected-against-covid-19-ahead-of-national-games>, accessed 5 December 2024).
800. Phrasisombath K, Kubota S, Elliott EM, Horiuchi S, Ounaphom P, Phimmachak L et al. Reaching the unreached through building trust: a mixed-method study on COVID-19 vaccination in rural Lao PDR. *BMJ Glob Health* 2024;9:e014680.
801. World Health Organization. Integrating Health in Urban and Territorial Planning: A Sourcebook, 1st ed. Geneva, UN-Habitat and World Health Organization, 2020 (<https://iris.who.int/handle/10665/331678>). License: CC BY-NC-SA 3.0 IGO.
802. Utzinger J, Tozan Y, Singer BH. Efficacy and cost-effectiveness of environmental management for malaria control. *Trop Med Int Health TM IH* 2001;6:677–687.

803. Thomson H, Thomas S, Sellstrom E, Petticrew M. Housing improvements for health and associated socio-economic outcomes. *Cochrane Database Syst Rev* 2013;CD008657.
804. Howden-Chapman P, Fyfe C, Nathan K, Keall M, Riggs L, Pierse N. The Effects of Housing on Health and Well-Being in Aotearoa New Zealand. *N Z Popul Rev* 2021;47:16–32.
805. Housing First [website]. (<https://www.hud.govt.nz/our-work/housing-first>, accessed 18 December 2024).
806. Ombler J, Jiang T, Fraser B, Nelson J, McMinn C, Hawkes K et al. Five-Year Post-Housing Outcomes for a Housing First Cohort in Aotearoa, New Zealand. *Int J Homelessness* 2024;4:109–125.
807. Health New Zealand, Te Whatu Ora. Healthy Homes [website]. 2025 (<https://www.tewhatauora.govt.nz/health-services-and-programmes/healthy-homes>, accessed 18 December 2024).
808. Pierse N, Johnson E, Thakurta AG, Chisholm E. Healthy Homes Initiative: Five-year outcomes evaluation. 2024 (<https://www.tewhatauora.govt.nz/publications/healthy-homes-initiative-five-year-outcomes-evaluation>, accessed 18 December 2024).
809. Nieuwenhuijsen MJ. Urban and transport planning pathways to carbon neutral, liveable and healthy cities; A review of the current evidence. *Environ Int* 2020;140:105661.
810. Stevenson M, Thompson J, de Sá TH, Ewing R, Mohan D, McClure R et al. Land use, transport, and population health: estimating the health benefits of compact cities. *Lancet* 2016;388:2925–2935.
811. Wolf KL, Lam ST, McKeen JK, Richardson GRA, van den Bosch M, Bardekjian AC. Urban Trees and Human Health: A Scoping Review. *Int J Environ Res Public Health* 2020;17:4371.
812. Ambient Air Pollution and Health in Accra, Ghana. Geneva, World Health Organization. 2021 (<https://iris.who.int/handle/10665/340678>). License: CC BY-NC-SA 3.0 IGO.
813. World Health Organization. Urban Health Initiative [website]. (<https://www.who.int/initiatives/urban-health-initiative>, accessed 18 December 2024).
814. World Health Organization. Accra Pilot Project – Urban Health Initiative [website]. (<https://www.who.int/initiatives/urban-health-initiative/pilot-projects/accra>, accessed 18 December 2024).
815. World Health Organization. Urban Health Initiative: Accra story [website]. 2021 (<https://www.who.int/news-room/facts-in-pictures/detail/urban-health-initiative-accra-story>, accessed 18 December 2024).
816. Verlinghieri E, Schwanen T. Transport and mobility justice: Evolving discussions. *J Transp Geogr* 2020;87:102798.
817. Child Friendly Cities Initiative. What is the Child Friendly Cities Initiative? [website]. (<https://www.childfriendlycities.org/what-is-the-child-friendly-cities-initiative>, accessed 27 September 2024).
818. Minh A, Muhajarine N, Janus M, Brownell M, Guhn M. A review of neighborhood effects and early child development: How, where, and for whom, do neighborhoods matter? *Health Place* 2017;46:155–174.
819. Christian H, Zubrick SR, Foster S, Giles-Corti B, Bull F, Wood L et al. The influence of the neighborhood physical environment on early child health and development: A review and call for research. *Health Place* 2015;33:25–36.
820. Gascon M, Vrijheid M, Nieuwenhuijsen MJ. The Built Environment and Child Health: An Overview of Current Evidence. *Curr Environ Health Rep* 2016;3:250–257.
821. Watanatada, Patrin. Urban95: creating cities for the youngest people. The Hague, Netherlands, Bernard van Leer Foundation, 2018 (<https://vanleerfoundation.org/wp-content/uploads/2018/06/3.5-Urban95.pdf>, accessed 6 March 2025).
822. Chatterjee S, Dutta C. Healthy public spaces for children within neighbourhoods through urban transformations. *Cities Health* 2024;8:800–815.
823. Decade of Healthy Ageing. What is the UN Decade of Healthy Ageing? [website]. 2022 (<https://www.decadeofhealthyageing.org/about/about-us/what-is-the-decade>, accessed 24 April 2024).
824. World Health Organization. National programmes for age-friendly cities and communities. Geneva, World Health Organization. [website] 2024 (<https://www.who.int/teams/social-determinants-of-health/demographic-change-and-healthy-ageing/age-friendly-environments/national-programmes-afcc>, accessed 24 April 2024).
825. Friel S. Governance, regulation and health equity. In: Drahos P, editor. *Regulatory Theory. Foundations and applications*. ANU Press; 2017:573–590 Foundations and applications; (<https://www.jstor.org/stable/j.ctt1q1crtm.45>, accessed 17 October 2023).
826. World Health Organization. Social participation for universal health coverage, health and well-being. Geneva, World Health Organization, 2024 Seventy-Seventh World Health Assembly; (https://apps.who.int/gb/ebwha/pdf_files/WHA77/A77_ACONF3-en.pdf, accessed 24 July 2024).

827. de Andrade LOM, Filho AP, Solar O, Rígoli F, de Salazar LM, Serrate PC-F et al. Social determinants of health, universal health coverage, and sustainable development: case studies from Latin American countries. *Lancet* 2015;385:1343-1351.
828. World Health Organization. Civil society and health: contributions and potential. Copenhagen, World Health Organization, 2017 (<https://iris.who.int/handle/10665/326290>, accessed 22 April 2024).
829. Blas E, Gilson L, Kelly MP, Labonté R, Lapitan J, Muntaner C et al. Addressing social determinants of health inequities: what can the state and civil society do? *Lancet* 2008;372:1684-1689.
830. Health 2020: a European policy framework and strategy for the 21st century. World Health Organization. Regional Office for Europe, 2013 (<https://iris.who.int/handle/10665/326386>, accessed 29 September 2023).
831. World Health Organization. WHO Civil Society Commission [website]. 2024 (<https://www.who.int/about/collaboration/civil-society-and-ngo-engagement/civil-society-commission>, accessed 22 April 2024).
832. Robert Wood Johnson Foundation. Culture of Health Sentinel Community Insights Brief Report: Grassroots Organization Engagement and Influence on Community Power. 2022 (<https://www.rwjf.org/en/about-rwjf/how-we-work/learning-and-evaluation/sentinel-communities.html>, accessed 20 March 2024).
833. Shannon G, Basu P, Peters LER, Clark-Ginsberg A, Herrera Delgado TM, Gope R et al. Think global, act local: using a translocal approach to understand community-based organisations' responses to planetary health crises during COVID-19. *Lancet Planet Health* 2023;7:e850-e858. ([https://doi.org/10.1016/S2542-5196\(23\)00193-6](https://doi.org/10.1016/S2542-5196(23)00193-6)).
834. Health Foundation. Let's #MakeHealthEqual. Health Equals. (<https://healthequals.org.uk/make-health-equal/>, accessed 23 October 2024).
835. Garcia S, Carrigan C, Wiekping P. Global Civil Society Response to the COVID-19 Crisis. *Voluntas* 2023;34:1310-1321.
836. Adhikari A, Alejandria MCP, Bradlow B, Heller P, Levine AC, Lopez-Portillo MF et al. Public Action and the Pandemic: The Role of Civil Society in Shaping State Responses. Providence, Watson Institute for International & Public Affairs, Brown University Center for Human Rights & Humanitarian Studies, 2022 (<https://home.watson.brown.edu/news/2022-06-16/public-action-and-pandemic-role-civil-society-shaping-state-responses>, accessed 20 March 2024).
837. Carnegie Endowment for International Peace. Global Civil Society in the Shadow of Coronavirus. Carnegie Endowment for International Peace, 2020 (https://carnegieendowment.org/files/Youngs-Coronavirus_Civil_Society_final.pdf).
838. Curty Pereira R, Elliott SJ, Llaguno Cárdenas P. Stepping into the Void: Lessons Learned from Civil Society Organizations during COVID-19 in Rio de Janeiro. *Int J Environ Res Public Health* 2023;20:5507.
839. Doğan E, Genç HD. Early-Responding Civil Society and a Late Coming State: Findings from Turkey during the Pandemic. *Nonprofit Policy Forum* 2021;12:127-146.
840. Desmaison B, Jaime K, Córdova P, Alarcón L, Gallardo L. Collective Infrastructures of Care: Ollas Comunes Defying Food Insecurity During the COVID-19 Pandemic. *Urbanisation* 2022;7:46-65.
841. Coll KM, Flores J, Jiménez M, López N, Lee AL, Carrillo M et al. It's a Matter of Trust: How Thirty Years of History Prepared a Community-Based Organization to Respond to the COVID-19 Pandemic. *Soc Sci* 2023;12.
842. Plasencia G, Gupta R, Kaalund K, Martinez-Bianchi V, Gonzalez-Guarda R, Thoumi A. Systemic Racism Affecting Latinx Population Health During the COVID-19 Pandemic and Beyond: Perspectives of Latinx Community Health Workers and Community-Based Organization Leaders. *Health Equity* 2023;7:715-721.
843. Voice, agency, empowerment – Handbook on social participation for universal health coverage. Geneva, World Health Organization. 2021 (<https://iris.who.int/handle/10665/342704>, accessed 17 April 2024). License: CC BY-NC-SA 3.0 IGO.
844. Participation as a driver of health equity. Copenhagen, World Health Organization, 2019 (<https://www.who.int/europe/publications/i/item/9789289054126>, accessed 17 April 2024).
845. Bodini C, Sanders D, Sengupta A. The contribution of civil society engagement to the achievement of health for all. *People's Health Movement*, 2018.
846. Hernández A, Hurtig A-K, Goicolea I, San Sebastián M, Jerez F, Hernández-Rodríguez F et al. Building collective power in citizen-led initiatives for health accountability in Guatemala: the role of networks. *BMC Health Serv Res* 2020;20:416.
847. Flores W, Hernández A. Health Accountability for Indigenous Populations: Confronting Power through Adaptive Action Cycles. *IDS Bull* 2018;49. (<https://doi.org/10.19088/1968-2018.133>).

848. Romdhane HB, Tlili F, Skhiri A, Zaman S, Phillimore P. Health system challenges of NCDs in Tunisia. *Int J Public Health* 2015;60:39–46.
849. Ben Mesmia H, Rajan D, Bouhafa Chtioui R, Koch K, Jaouadi I, Aboutaleb H et al. The Tunisian experience of participatory health governance: the Societal Dialogue for Health (a qualitative study). *Health Res Policy Syst* 2023;21:84.
850. Friel S. Governance, regulation and health equity. In: Drahoš P, editor. *Regulatory Theory. Foundations and applications*. ANU Press; 2017:573–590. (<https://www.jstor.org/stable/j.ctt1q1crtm.45>, accessed 20 March 2024).
851. Jeong E, Hagose M, Jung H, Ki M, Flahault A. Understanding South Korea's Response to the COVID-19 Outbreak: A Real-Time Analysis. *Int J Environ Res Public Health* 2020;17:9571.
852. OHCHR. Report of the High Commissioner for Human Rights on Civil Society [website]. 2016 (<https://www.ohchr.org/en/calls-for-input/report-high-commissioner-human-rights-civil-society>, accessed 7 December 2023).
853. O'Mara-Eves A, Brunton G, McDaid D, Oliver S, Kavanagh J, Jamal F et al. Community engagement to reduce inequalities in health: a systematic review, meta-analysis and economic analysis. Southampton (UK), NIHR Journals Library, 2013 Public Health Research; (<http://www.ncbi.nlm.nih.gov/books/NBK262817/>, accessed 29 September 2023).
854. Cyril S, Smith BJ, Possamai-Inesedy A, Renzaho AMN. Exploring the role of community engagement in improving the health of disadvantaged populations: a systematic review. *Glob Health Action* 2015;8:29842.
855. O'Mara-Eves A, Brunton G, Oliver S, Kavanagh J, Jamal F, Thomas J. The effectiveness of community engagement in public health interventions for disadvantaged groups: a meta-analysis. *BMC Public Health* 2015;15:129.
856. Farnsworth SK, Böse K, Fajobi O, Souza PP, Peniston A, Davidson LL et al. Community engagement to enhance child survival and early development in low- and middle-income countries: an evidence review. *J Health Commun* 2014;19 Suppl 1:67–88.
857. Jain M, Shisler S, Lane C, Bagai A, Brown E, Engelbert M et al. Use of community engagement interventions to improve child immunisation in low- and middle-income countries: A systematic review and meta-analysis. *Campbell Syst Rev* 2022;18:e1253.
858. Adeagbo M, Olukotun M, Musa S, Alaazi D, Allen U, Renzaho AMN et al. Improving COVID-19 Vaccine Uptake among Black Populations: A Systematic Review of Strategies. *Int J Environ Res Public Health* 2022;19:11971.
859. Kamal A, Hodson A, Pearce JM. A Rapid Systematic Review of Factors Influencing COVID-19 Vaccination Uptake in Minority Ethnic Groups in the UK. *Vaccines* 2021;9:1121.
860. Ha BTT, Quang LN, Thanh PQ, Duc DM, Mirzoev T, Bui TMA. Community engagement in the prevention and control of COVID-19: Insights from Vietnam. *PLOS ONE* 2021;16:e0254432. (<https://doi.org/10.1371/journal.pone.0254432>).
861. Loewenson R, Colvin CJ, Szabzon F, Das S, Khanna R, Coelho VSP et al. Beyond command and control: A rapid review of meaningful community-engaged responses to COVID-19. *Glob Public Health* 2021;16:1439–1453. (<https://doi.org/10.1080/17441692.2021.1900316>).
862. Oliver J, Larsen S, Stinear TP, Hoffmann A, Crouch S, Gibney KB. Reducing mosquito-borne disease transmission to humans: A systematic review of cluster randomised controlled studies that assess interventions other than non-targeted insecticide. *PLoS Negl Trop Dis* 2021;15:e0009601.
863. Munyuzangabo M, Khalifa DS, Gaffey MF, Kamali M, Siddiqui FJ, Meteke S et al. Delivery of sexual and reproductive health interventions in conflict settings: a systematic review. *BMJ Glob Health* 2020;5:e002206.
864. Banna J, Bersamin A. Community involvement in design, implementation and evaluation of nutrition interventions to reduce chronic diseases in indigenous populations in the U.S.: a systematic review. *Int J Equity Health* 2018;17:116.
865. Boelsen-Robinson T, Peeters A, Beauchamp A, Chung A, Gearon E, Backholer K. A systematic review of the effectiveness of whole-of-community interventions by socioeconomic position. *Obes Rev Off J Int Assoc Study Obes* 2015;16:806–816.
866. Gonçalves S. The Effects of Participatory Budgeting on Municipal Expenditures and Infant Mortality in Brazil. *World Dev* 2014;53:94–110. (<https://doi.org/10.1016/j.worlddev.2013.01.009>).
867. Milton B, Attree P, French B, Povall S, Whitehead M, Popay J. The impact of community engagement on health and social outcomes: a systematic review. *Community Dev J* 2012;47:316–334. (<https://doi.org/10.1093/cdj/bsr043>).
868. Wallerstein N, Oetzel JG, Sanchez-Youngman S, Boursaw B, Dickson E, Kastelic S et al. Engage for Equity: A Long-Term Study of Community-Based Participatory Research and Community-Engaged Research Practices and Outcomes. *Health Educ Behav* 2020; 47:380–390. (<https://doi.org/10.1177/1090198119897075>).

869. Kelly M. Cognitive biases in public health and how economics and sociology can help overcome them. *Public Health* 2019;169:163–172.
870. What Works Wellbeing. Tackling loneliness interventions [website]. 2023 (<https://whatworkswellbeing.org/resources/tackling-loneliness-interventions/>, accessed 12 March 2024).
871. Zagic D, Wuthrich VM, Rapee RM, Wolters N. Interventions to improve social connections: a systematic review and meta-analysis. *Soc Psychiatry Psychiatr Epidemiol* 2022;57: 885–906.
872. Bessaha ML, Sabbath EL, Morris Z, Malik S, Scheinfeld L, Saragossi J. A Systematic Review of Loneliness Interventions Among Non-Elderly Adults. *Clin Soc Work J* 2020;48:110–125.
873. Growing Old Before Becoming Rich: Challenges of An Aging Population in Sri Lanka [Sri Lanka]. Asian Development Bank, 2019 (<https://www.adb.org/publications/growing-old-becoming-rich-sri-lanka>, accessed 12 March 2024).
874. Hsueh Y-C, Batchelor R, Liebmann M, Dhanani A, Vaughan L, Fett A-K et al. A Systematic Review of Studies Describing the Effectiveness, Acceptability, and Potential Harms of Place-Based Interventions to Address Loneliness and Mental Health Problems. *Int J Environ Res Public Health* 2022;19:4766.
875. Bagnall A, South J, Southby K, Martino SD, Pilkington G, Mitchell B et al. A systematic review of interventions in community infrastructure (places and spaces) to boost social relations and community wellbeing. Society for Social Medicine & Population Health annual conference, 2019.
876. Younes SR, Marques B, McIntosh J. Public Spaces for Older People: A Review of the Relationship between Public Space to Quality of Life. *Sustainability* 2024;16. (<https://doi.org/10.3390/su16114583>).
877. Levasseur M, G  n  reux M, Bruneau J-F, Vanasse A, Chabot   , Beaulac C et al. Importance of proximity to resources, social support, transportation and neighborhood security for mobility and social participation in older adults: results from a scoping study. *BMC Public Health* 2015;15:503.
878. Bowling A, Stafford M. How do objective and subjective assessments of neighbourhood influence social and physical functioning in older age? Findings from a British survey of ageing. *Soc Sci Med* 1982 2007;64:2533–2549.
879. Reinhard E, Courtin E, van Lenthe FJ, Avendano M. Public transport policy, social engagement and mental health in older age: a quasi-experimental evaluation of free bus passes in England. *J Epidemiol Community Health* 2018;72:361–368.
880. Chibanda D, Weiss HA, Verhey R, Simms V, Munjoma R, Rusakaniko S et al. Effect of a Primary Care-Based Psychological Intervention on Symptoms of Common Mental Disorders in Zimbabwe: A Randomized Clinical Trial. *JAMA* 2016;316:2618.
881. Centre for Global Mental Health. The Friendship Bench [website]. (<https://www.centreforglobalmentalhealth.org/the-friendship-bench>, accessed 12 December 2024).
882. Federal Government of Germany. German Federal Government's Strategy to Counter Loneliness. Berlin, Federal Ministry for Family Affairs, Senior Citizens, Women and Youth. 2023 (<https://www.bmfsfj.de/resource/blob/248208/5c257556f6e203e27e096f46edf3e8ce/strategie-gegen-einsamkeit-englisch-german-federal-governments-strategy-to-counter-loneliness-data.pdf>, accessed 18 December 2024).
883. OECD. Universal Health Coverage and Health Outcomes: Final Report for the G7 Health Ministerial meeting. Paris, Organization for Economic Cooperation and Development. 2016 (<https://www.oecd.org/health/health-systems/G7-Health-Ministerial-meeting.htm>, accessed 10 April 2024).
884. World Health Organization, International Bank for Reconstruction and Development / The World Bank. Tracking Universal Health Coverage: 2023 Global monitoring report: Executive summary. 2023 (<https://www.who.int/publications-detail-redirect/9789240080393>, accessed 23 November 2023).
885. Endalamaw A, Gilks CF, Ambaw F, Assefa Y. Universality of universal health coverage: A scoping review. *PLOS ONE* 2022;17:e0269507.
886. Paul E, Deville C, Bodson O, Sambi  ni NE, Thiam I, Bourgeois M et al. How is equity approached in universal health coverage? An analysis of global and country policy documents in Benin and Senegal. *Int J Equity Health* 2019;18:195.
887. Making fair choices on the path to universal health coverage. Final report of the WHO Consultative Group on Equity and Universal Health Coverage. Geneva, World Health Organization, 2014 (<https://iris.who.int/handle/10665/112671>, accessed 9 April 2024).
888. Asante A, Price J, Hayen A, Jan S, Wiseman V. Equity in Health Care Financing in Low- and Middle-Income Countries: A Systematic Review of Evidence from Studies Using Benefit and Financing Incidence Analyses. *PLOS ONE* 2016;11:e0152866.

889. Health Systems Research Institute. Thailand's Universal Coverage Scheme: Achievements and Challenges. An independent assessment of the first 10 years (2001-2010) [website]. 2012 (https://www.social-protection.org/gimi/gess/ShowResource.action;jsessionid=hzf4mPd2iAZ7dv81aeYWN55OwLxL9fjSWmU5-_hffdrtkac6SJ0q!-491661851?id=28441, accessed 24 October 2023).
890. Hanson K, Brikci N, Erlangga D, Alebachew A, Allegri MD, Balabanova D et al. The Lancet Global Health Commission on financing primary health care: putting people at the centre. *Lancet Glob Health* 2022;10:e715-e772. ([https://doi.org/10.1016/S2214-109X\(22\)00005-5](https://doi.org/10.1016/S2214-109X(22)00005-5)).
891. Darvishi B, Behzadifar M, Ghanbari MK, Ehsanzadeh SJ, Bakhtiari A, Behzadifar M et al. Financial Protection Indexes and the Iranian Health Transformation Plan: A Systematic Review. *Yale J Biol Med* 2021;94:465-476.
892. Odonkor SNNT, Koranteng F, Appiah-Danquah M, Dini L. Do national health insurance schemes guarantee financial risk protection in the drive towards Universal Health Coverage in West Africa? A systematic review of observational studies. *PLOS Glob Public Health* 2023;3:e0001286.
893. Anjorin SS, Ayorinde AA, Abba MS, Mensah D, Okolie EA, Uthman OA et al. Equity of national publicly funded health insurance schemes under the universal health coverage agenda: a systematic review of studies conducted in Africa. *J Public Health Oxf Engl* 2022;44:900-909.
894. Datta BK, Husain MJ, Fatehin S. The crowding out effect of out-of-pocket medication expenses of two major non-communicable diseases in Pakistan. *Int Health* 2020;12:50-59.
895. World Health Organization. 1 in 10 medical products in developing countries is substandard or falsified [website]. 2017 (<https://www.who.int/news/item/28-11-2017-1-in-10-medical-products-in-developing-countries-is-substandard-or-falsified>, accessed 9 April 2024).
896. Kruk ME, Porignon D, Rockers PC, Van Lerberghe W. The contribution of primary care to health and health systems in low- and middle-income countries: a critical review of major primary care initiatives. *Soc Sci Med* 1982 2010;70:904-911.
897. Starfield B, Shi L, Macinko J. Contribution of Primary Care to Health Systems and Health. *Milbank Q* 2005;83:457-502.
898. Building the economic case for primary health care: a scoping review. Geneva, Switzerland, World Health Organization, 2018 Technical series on primary health care. 2018 (<https://iris.who.int/handle/10665/326293>). License: CC BY-NC-SA 3.0 IGO.
899. Bitton A, Fifield J, Ratcliffe H, Karlage A, Wang H, Veillard JH et al. Primary healthcare system performance in low-income and middle-income countries: a scoping review of the evidence from 2010 to 2017. *BMJ Glob Health* 2019;4:e001551. (<https://doi.org/10.1136/bmjgh-2019-001551>).
900. Kringos DS, Boerma W, van der Zee J, Groenewegen P. Europe's strong primary care systems are linked to better population health but also to higher health spending. *Health Aff Proj Hope* 2013;32:686-694.
901. Stenberg K, Hanssen O, Bertram M, Brindley C, Meshreky A, Barkley S et al. Guide posts for investment in primary health care and projected resource needs in 67 low-income and middle-income countries: a modelling study. *Lancet Glob Health* 2019;7:e1500-e1510.
902. World Health Organization and WONCA Working Party on Mental Health. What is primary care mental health? *Ment Health Fam Med* 2008;5:9-13.
903. Allen LN, Smith RW, Simmons-Jones F, Roberts N, Honney R, Currie J. Addressing social determinants of noncommunicable diseases in primary care: a systematic review. *Bull World Health Organ* 2020;98:754-765B. (<http://dx.doi.org/10.2471/BLT.19.248278>).
904. Sumriddetchkajorn K, Shimazaki K, Ono T, Kusaba T, Sato K, Kobayashi N. Universal health coverage and primary care, Thailand. *Bull World Health Organ* 2019;97:415-422.
905. International Labour Organization. ILO-Thailand collaboration helps strengthen Thailand's social security system [website]. 2024 (<https://www.ilo.org/resource/news/ilo-thailand-collaboration-helps-strengthen-thailand%E2%80%99s-social-security>, accessed 18 December 2024).
906. World Health Organization. UHC Service Coverage Index (SDG 3.8.1) [website]. (<https://www.who.int/data/gho/data/indicators/indicator-details/GHO/uhc-index-of-service-coverage>).
907. World Health Organization. Population with household expenditures on health > 10% of total household expenditure or income (%) [website]. ([https://www.who.int/data/gho/data/indicators/indicator-details/GHO/population-with-household-expenditures-on-health-greater-than-10-of-total-household-expenditure-or-income-\(sdg-3-8-2\)-\(-\)](https://www.who.int/data/gho/data/indicators/indicator-details/GHO/population-with-household-expenditures-on-health-greater-than-10-of-total-household-expenditure-or-income-(sdg-3-8-2)-(-))).
908. National Legislative Assembly of Thailand. Primary Health Care System Act B.E. 2562., 2019 (https://www.ratchakitcha.soc.go.th/DATA/PDF/2562/A/056/T_0165.PDF, accessed 7 January 2025).

909. Selvarajah S, Corona Maioli S, Deivanayagam TA, de Moraes Sato P, Devakumar D, Kim S-S et al. Racism, xenophobia, and discrimination: mapping pathways to health outcomes. *Lancet* 2022;400:2109–2124. ([https://doi.org/10.1016/S0140-6736\(22\)02484-9](https://doi.org/10.1016/S0140-6736(22)02484-9)).
910. Firdous T, Darwin Z, Hassan SM. Muslim women's experiences of maternity services in the UK: qualitative systematic review and thematic synthesis. *BMC Pregnancy Childbirth* 2020;20:115. (<https://doi.org/10.1186/s12884-020-2811-8>).
911. Horrill TC, Linton J, Lavoie JG, Martin D, Wiens A, Schultz ASH. Access to cancer care among Indigenous peoples in Canada: A scoping review. *Soc Sci Med* 1982 2019;238:112495.
912. Castro A, Savage V, Kaufman H. Assessing equitable care for Indigenous and Afrodescendant women in Latin America. *Rev Panam Salud Publica Pan Am J Public Health* 2015;38:96–109.
913. Samari G, Alcalá HE, Sharif MZ. Islamophobia, Health, and Public Health: A Systematic Literature Review. *Am J Public Health* 2018;108:e1–e9.
914. Kita S, Minatani M, Hikita N, Matsuzaki M, Shiraishi M, Haruna M. A Systematic Review of the Physical, Mental, Social, and Economic Problems of Immigrant Women in the Perinatal Period in Japan. *J Immigr Minor Health* 2015;17:1863–1881.
915. Daeem R, Mansbach-Kleinfeld I, Farbstein I, Apter A, Elias R, Ifrah A et al. Barriers to help-seeking in Israeli Arab minority adolescents with mental health problems: results from the Galilee study. *Isr J Health Policy Res* 2019;8:45. (<https://doi.org/10.1186/s13584-019-0315-7>).
916. Ben J, Cormack D, Harris R, Paradies Y. Racism and health service utilisation: A systematic review and meta-analysis. *PLOS ONE* 2017;12:e0189900. (<https://doi.org/10.1371/journal.pone.0189900>).
917. FitzGerald C, Hurst S. Implicit bias in healthcare professionals: a systematic review. *BMC Med Ethics* 2017;18:19. (<https://doi.org/10.1186/s12910-017-0179-8>).
918. Watson HL, Downe S. Discrimination against childbearing Romani women in maternity care in Europe: a mixed-methods systematic review. *Reprod Health* 2017;14:1. (<https://doi.org/10.1186/s12978-016-0263-4>).
919. Meghani SH, Byun E, Gallagher RM. Time to take stock: a meta-analysis and systematic review of analgesic treatment disparities for pain in the United States. *Pain Med Malden Mass* 2012;13:150–174.
920. Breaking barriers: Towards more gender-responsive and equitable health systems. Geneva, Switzerland, World Health Organization, 2019 (<https://www.who.int/publications/m/item/breaking-barriers-towards-more-gender-responsive-and-equitable-health-systems>, accessed 10 April 2024).
921. Barr D, P Garnett G, Mayer KH, Morrison M. Key populations are the future of the African HIV/AIDS pandemic. *J Int AIDS Soc* 2021;24:e25750.
922. United Nations. High-Level Meeting on AIDS, 8–10 June 2011, New York [website]. (<https://www.un.org/en/conferences/hiv-aids/newyork2011>).
923. Williams DR, Cooper LA. Reducing Racial Inequities in Health: Using What We Already Know to Take Action. *Int J Environ Res Public Health* 2019;16:606.
924. Pittman P, Chen C, Erikson C, Salsberg E, Luo Q, Vichare A et al. Health Workforce for Health Equity. *Med Care* 2021;59:S405–S408.
925. Browne AJ, Varcoe C, Lavoie J, Smye V, Wong ST, Krause M et al. Enhancing health care equity with Indigenous populations: evidence-based strategies from an ethnographic study. *BMC Health Serv Res* 2016;16:544.
926. Banks L, Kelly M. Cultural safety and the Nursing Council of New Zealand. In: Weppa D, editor. *Cultural Safety in Aotearoa New Zealand*. Cambridge University Press; 2015 (<https://doi.org/10.1017/CBO9781316151136>).
927. WHO guideline on health policy and system support to optimize community health worker programmes: Selected highlights. Geneva, World Health Organization. 2018 (<https://iris.who.int/handle/10665/275501>, accessed 10 April 2024). License: CC BY-NC-SA 3.0 IGO.
928. Ahmed S, Chase LE, Wagnild J, Akhter N, Sturridge S, Clarke A et al. Community health workers and health equity in low- and middle-income countries: systematic review and recommendations for policy and practice. *Int J Equity Health* 2022;21:49.
929. Wadge H, Bhatti Y, Carter A, Harris M, Parston G, Darzi A. *Brazil's Family Health Strategy: Using Community Health Care Workers to Provide Primary Care*. New York: The Commonwealth Fund; 2016 (<https://doi.org/10.26099/s4dx-cm12>).
930. Lavoie JG, Forget EL, Prakash T, Dahl M, Martens P, O'Neil JD. Have investments in on-reserve health services and initiatives promoting community control improved First Nations' health in Manitoba? *Soc Sci Med* 1982 2010;71:717–724.

931. Hernández A, Ruano AL, Marchal B, San Sebastián M, Flores W. Engaging with complexity to improve the health of indigenous people: a call for the use of systems thinking to tackle health inequity. *Int J Equity Health* 2017;16:26.
932. Dudgeon P, Bray A, Walker R. Mitigating the impacts of racism on Indigenous wellbeing through human rights, legislative and health policy reform. *Med J Aust* 2023;218:203–205.
933. Khoury P. Beyond the Biomedical Paradigm: The Formation and Development of Indigenous Community-Controlled Health Organizations in Australia. *Int J Health Serv* 2015;45:471–494.
934. Poirier BF, Hedges J, Soares G, Jamieson LM. Aboriginal Community Controlled Health Services: An Act of Resistance against Australia's Neoliberal Ideologies. *Int J Environ Res Public Health* 2022;19:10058.
935. Foley G. Redfern Aboriginal Medical Service 1971–1991: Twenty Years of Community Service. Redfern, New South Wales, Aboriginal Medical Service Co-operative, 1991.
936. Foley G. Black Power in Redfern 1968–1972. *There Goes the Neighbourhood: Redfern and the Politics of Urban Space*. Performance Space; 2009:125.
937. National Aboriginal Community Controlled Health Organisation (NACCHO). About NACCHO [website]. (<https://www.naccho.org.au/about-us/>, accessed 24 November 2024).
938. Pearson O, Schwartzkopff K, Dawson A, Hagger C, Karagi A, Davy C et al. Aboriginal community controlled health organisations address health equity through action on the social determinants of health of Aboriginal and Torres Strait Islander peoples in Australia. *BMC Public Health* 2020;20:1859.
939. Freeman T, Baum F, Lawless A, Labonté R, Sanders D, Boffa J et al. Case Study of an Aboriginal Community-Controlled Health Service in Australia: Universal, Rights-Based, Publicly Funded Comprehensive Primary Health Care in Action. *Health Hum Rights* 2016;18:93–108.
940. Freeman T, Baum FE, Jolley GM, Lawless A, Edwards T, Javanparast S et al. Service providers' views of community participation at six Australian primary healthcare services: scope for empowerment and challenges to implementation. *Int J Health Plann Manage* 2016;31. (<https://doi.org/10.1002/hpm.2253>).
941. Gleeson D, Townsend B, Tenni BF, Phillips T. Global inequities in access to COVID-19 health products and technologies: A political economy analysis. *Health Place* 2023;83:103051. (<https://doi.org/10.1016/j.healthplace.2023.103051>).
942. Watkinson RE, Williams R, Gillibrand S, Sanders C, Sutton M. Ethnic inequalities in COVID-19 vaccine uptake and comparison to seasonal influenza vaccine uptake in Greater Manchester, UK: A cohort study. *PLOS Med* 2022;19:e1003932. (<https://doi.org/10.1371/journal.pmed.1003932>).
943. Omino M, Kahumbu J. Left on Our Own: COVID-19, TRIPS-Plus Free Trade Agreements, and the Doha Declaration on TRIPS and Public Health. The South Centre, 2020 (https://www.southcentre.int/wp-content/uploads/2022/11/RP170_COVID-19-TRIPS-Plus-FTAs-and-the-Doha-Declaration-on-TRIPS-and-Public-Health_EN.pdf, accessed 10 April 2024).
944. Torreele E, Sherwin H. Health equity requires transformational change: Financial incentives based on worn-out market thinking will not deliver. *PLOS Glob Public Health* 2024;4:e0003147.
945. Swaminathan S, Pécoul B, Abdullah H, Christou C, Gray G, IJsselmuiden C et al. Reboot biomedical R&D in the global public interest. *Nature* 2022;602:207–210.
946. Drugs for Neglected Diseases initiative. Pro-access policies for intellectual property and licensing [website]. (<https://dndi.org/advocacy/pro-access-policies-intellectual-property-licensing/>, accessed 10 April 2024).
947. Coalition for Epidemic Preparedness Innovations. CEPI Equitable Access Policy [website]. 2018 (https://static.cepi.net/downloads/2023-12/Equitable-Access-Policy_0.pdf, accessed 10 April 2024).
948. African Union Development Agency. African Medicines Agency (AMA) [website]. 2022 (<https://www.nepad.org/microsite/african-medicines-agency-ama>, accessed 10 April 2024).
949. Wouters OJ, Shadlen KC, Salcher-Konrad M, Pollard AJ, Larson HJ, Teerawattananon Y et al. Challenges in ensuring global access to COVID-19 vaccines: production, affordability, allocation, and deployment. *Lancet* 2021;397:1023–1034.
950. Kumraj G, Pathak S, Shah S, Majumder P, Jain J, Bhati D et al. Capacity Building for Vaccine Manufacturing Across Developing Countries: The Way Forward. *Hum Vaccines Immunother* 2022;18:2020529.
951. WHO Council on the Economics of Health for All. The mRNA Vaccine Technology Transfer Hub: a pilot for transformative change for the common good? World Health Organization, 2023 (https://cdn.who.int/media/docs/default-source/council-on-the-economics-of-health-for-all/who-council-eh4a_casestudy-mrnahub.pdf?sfvrsn=df0e0628_3&download=true, accessed 19 December 2024).

952. Mangold KA, Bartell TR, Doobay-Persaud AA, Adler MD, Sheehan KM. Expert Consensus on Inclusion of the Social Determinants of Health in Undergraduate Medical Education Curricula. *Acad Med J Assoc Am Med Coll* 2019;94:1355–1360.
953. Doobay-Persaud A, Adler MD, Bartell TR, Sheneman NE, Martinez MD, Mangold KA et al. Teaching the Social Determinants of Health in Undergraduate Medical Education: a Scoping Review. *J Gen Intern Med* 2019;34:720–730.
954. Tomas S. Doctors for Health Equity: The role of the World Medical Association, national medical associations and doctors in addressing the social determinants of health and health equity. Institute of Health Equity, 2016 (<https://www.instituteofhealthequity.org/resources-reports/doctors-for-health-equity-world-medical-association-report/doctors-for-health-equity-wma-full-report-pdf.pdf>, accessed 5 March 2025).
955. Rosa WE, Hannaway CJ, McArdle C, McManus MF, Alharahsheh ST, Marmot M. Nurses for Health Equity: Guidelines for Tackling the Social Determinants of Health. WISH, 2021 (<https://wish.org.qa/wp-content/uploads/2024/01/Nurses-for-Health-Equity.pdf>).
956. The World Medical Association. Declaration of Oslo on Social Determinants of Health [website]. 2011 (<https://www.wma.net/policies-post/wma-declaration-of-oslo-on-social-determinants-of-health/>, accessed 1 December 2023).
957. World Health Organization. Health in All Policies Framework for Country Action. 2024 (<https://www.afro.who.int/publications/health-all-policies-framework-country-action>, accessed 15 April 2024).
958. Integrating the social determinants of health into health workforce education and training. Geneva, World Health Organization, 2023 (<https://iris.who.int/handle/10665/373710>, accessed 10 April 2024). License: CC BY-NC-SA 3.0 IGO.
959. Woolley T, Halili SD, Siega-Sur J-L, Cristobal FL, Reeve C, Ross SJ et al. Socially accountable medical education strengthens community health services. *Med Educ* 2018;52:391–403.
960. Halili Jr S “Ben”, Cristobal F, Woolley T, Ross SJ, Reeve C, Neusy A-J. Addressing health workforce inequities in the Mindanao regions of the Philippines: Tracer study of graduates from a socially-accountable, community-engaged medical school and graduates from a conventional medical school. *Med Teach* 2017;39:859–865.
961. World Health Organization. Multi-Country Initiative [website]. 2024 (<https://www.who.int/initiatives/action-on-the-social-determinants-of-health-for-advancing-equity/about>, accessed 20 May 2024).
962. Boniol M, Kunjumen T, Nair TS, Siyam A, Campbell J, Diallo K. The global health workforce stock and distribution in 2020 and 2030: a threat to equity and “universal” health coverage? *BMJ Glob Health* 2022;7:e009316.
963. Khan N, Gilliar W, Bamrah JS, Dave S. Post-COVID-19: can digital solutions lead to a more equitable global healthcare workforce? *BJPsych Int* 2023;20:18–23.
964. World Health Organization. Fair share for health and care: gender and the undervaluation of health and care work [website]. 2024 (<https://www.who.int/publications-detail-redirect/9789240082854>, accessed 6 June 2024).
965. World Health Organization. Closing the leadership gap: gender equity and leadership in the global health and care workforce [website]. 2021 (<https://www.who.int/publications-detail-redirect/9789240025905>, accessed 18 October 2023).
966. Naqvi H, Williams RD, Chinembiri O, Rodger S. Workforce and workplace racism in health systems: organisations are diverse but not inclusive. *Lancet Lond Engl* 2022;400:2023–2026.
967. World Health Organization. Keeping the promise to the women who will deliver universal health coverage [website]. 2019 (<https://www.who.int/news-room/commentaries/detail/uhc-day-2019-keeping-the-promise-to-the-women-who-will-deliver-uhc>, accessed 11 April 2024).
968. Downs JA, Reif LK, Hokororo A, Fitzgerald DW. Increasing Women in Leadership in Global Health. *Acad Med* 2014;89:1103.
969. World Health Organization. Value gender and equity in the global health workforce [website]. 2024 (<https://www.who.int/activities/value-gender-and-equity-in-the-global-health-workforce>, accessed 11 April 2024).
970. Beyeler L. Why you should care about unpaid care work [website]. (<https://oecd-development-matters.org/2019/03/18/why-you-should-care-about-unpaid-care-work/>, accessed 11 April 2024).
971. Hajek A, Kretzler B, König H-H. Informal Caregiving, Loneliness and Social Isolation: A Systematic Review. *Int J Environ Res Public Health* 2021;18:12101.
972. Lambrias A, Ervin J, Taouk Y, King T. A systematic review comparing cardiovascular disease among informal carers and non-carers. *Int J Cardiol Cardiovasc Risk Prev* 2023;16:200174.
973. Ervin J, Taouk Y, Fleitas Alfonso L, Peasgood T, King T. Longitudinal association between informal unpaid caregiving and mental health amongst working age adults in high-income OECD countries: A systematic review. *EClinicalMedicine* 2022;53:101711.

974. Gérain P, Zech E. Do informal caregivers experience more burnout? A meta-analytic study. *Psychol Health Med* 2021;26:145–161.
975. Alves LC de S, Monteiro DQ, Bento SR, Hayashi VD, Pelegrini LN de C, Vale FAC. Burnout syndrome in informal caregivers of older adults with dementia: A systematic review. *Dement Neuropsychol* 2019;13:415–421.
976. Bom J, Bakx P, Schut F, van Doorslaer E. The Impact of Informal Caregiving for Older Adults on the Health of Various Types of Caregivers: A Systematic Review. *The Gerontologist* 2019;59:e629–e642.
977. Rodríguez-Sánchez B, Aranda-Reneo I, Oliva-Moreno J, López-Bastida J. Assessing the Effect of Including Social Costs in Economic Evaluations of Diabetes-Related Interventions: A Systematic Review. *Clin Outcomes Res CEOR* 2021;13:307–334.
978. Krol M, Papenburg J, van Exel J. Does including informal care in economic evaluations matter? A systematic review of inclusion and impact of informal care in cost-effectiveness studies. *Pharmacoeconomics* 2015;33:123–135.
979. Rodríguez-Sánchez B, Daugbjerg S, Peña-Longobardo LM, Oliva-Moreno J, Aranda-Reneo I, Cicchetti A et al. Does the inclusion of societal costs change the economic evaluations recommendations? A systematic review for multiple sclerosis disease. *Eur J Health Econ HEPAC Health Econ Prev Care* 2023;24:247–277.
980. Peña-Longobardo LM, Rodríguez-Sánchez B, Oliva-Moreno J, Aranda-Reneo I, López-Bastida J. How relevant are social costs in economic evaluations? The case of Alzheimer's disease. *Eur J Health Econ HEPAC Health Econ Prev Care* 2019;20:1207–1236.
981. Engel L, Ajdukovic M, Bucholtz J, McCaffrey N. Valuation of Informal Care Provided to People Living with Dementia: A Systematic Literature Review. *Value Health J Int Soc Pharmacoeconomics Outcomes Res* 2021;24:1863–1870.
982. Marešová P, Dolejš J, Kuca K. Call for a Uniform Strategy of Collecting Alzheimer's Disease Costs: A Review and Meta-Analysis. *J Alzheimers Dis JAD* 2018;63:227–238.
983. Trapero-Bertran M, Oliva-Moreno J. Economic impact of HIV/AIDS: a systematic review in five European countries. *Health Econ Rev* 2014;4:15.
984. Leniz J, Yi D, Yorganci E, Williamson LE, Suji T, Cripps R et al. Exploring costs, cost components, and associated factors among people with dementia approaching the end of life: A systematic review. *Alzheimers Dement N Y N* 2021;7:e12198.
985. Michalowsky B, Kaczynski A, Hoffmann W. The economic and social burden of dementia diseases in Germany – A meta-analysis. *Bundesgesundheitsblatt Gesundheitsforschung Gesundheitsschutz* 2019;62:981–992.
986. Oliva-Moreno J, Trapero-Bertran M, Peña-Longobardo LM, Del Pozo-Rubio R. The Valuation of Informal Care in Cost-of-Illness Studies: A Systematic Review. *Pharmacoeconomics* 2017;35:331–345.
987. Soledad Salvador. The National Integrated Care System in Uruguay: An Opportunity for the Economic Empowerment of Women. UN Women, 2019 (<https://lac.unwomen.org/sites/default/files/Field%20Office%20Americas/Documentos/Publicaciones/2019/10/SNIC%20web%20INGLES.pdf>, accessed 18 December 2024).
988. Government of Uruguay. Ley N° 19.353: Creación del Sistema Nacional Integrado de Cuidados (SNIC) [website]. 2015 (<https://www.impo.com.uy/bases/leyes/19353-2015>).
989. Junta Nacional de Cuidados. Plan Nacional de Cuidados 2016-2020. Government of Uruguay, 2015 (https://www.gub.uy/sistema-cuidados/sites/sistema-cuidados/files/2020-01/plan-nacional-de-cuidados-2016-2020_0.pdf).
990. Comité Consultivo de Cuidados [website]. 2022 (<https://www.gub.uy/sistema-cuidados/politicas-y-gestion/comite-consultivo-cuidados>, accessed 7 January 2025).
991. Amarante V, Barro P, Colacce M. Patrones de uso del tiempo de varones y mujeres en Uruguay. 2007-2022. Instituto de Economía, Facultad de Ciencias Económicas y Administración, Universidad de la República, Uruguay, 2024 Serie Documentos de Trabajo; (https://iecon.fcea.udelar.edu.uy/images/2024/DT/DT_11-24.pdf).
992. Sara Thomas. Doctors for Health Equity – World Medical Association Report. Institute of Health Equity, 2016 (<https://www.instituteofhealthequity.org/resources-reports/doctors-for-health-equity-world-medical-association-report>, accessed 24 April 2024).
993. Ministry of Health, Spain. Social and family context: factors that should be included in the Digital Health History (Condicionantes sociales y del contexto familiar que sería recomendable incluir en la Historia de Salud Digital). 2023 (<https://www.sanidad.gob.es/areas/promocionPrevencion/entornosSaludables/saludComunitaria/documentosTecnicos/docs/condicionantesSocialesycontextofamiliar.pdf>, accessed 24 April 2024).

994. Baum F. Health Sectors as Actors and Advocates for Health and Health Equity. In: Baum F, editor. *Governing for Health: Advancing Health and Equity through Policy and Advocacy*. New York: Oxford University Press; 2019:37-58 (<https://doi.org/10.1093/oso/9780190258948.003.0004>).
995. Allen M, Malhotra AM, Allwood D. Acting on the social determinants of health to reduce health inequalities: innovative approaches by provider trusts. UCL Partners, 2023 (<https://uclpartners.com/news-item/trusts-develop-innovative-approaches-to-tackling-social-determinants-of-health/>, accessed 18 April 2024).
996. Ministry of Health and Care Services, Norway. The Norwegian Public Health Act [website]. 2012 (<https://health-inequalities.eu/jwddb/the-norwegian-public-health-act/>, accessed 4 April 2024).
997. Ministère de la Santé et des Services sociaux Québec. Plan d'action interministériel 2022-2025 de la Politique gouvernementale de prévention en santé – Publications du ministère de la Santé et des Services sociaux [website]. 2024 (<https://publications.msss.gouv.qc.ca/msss/document-003411/>, accessed 4 April 2024).
998. Government of South Australia. Public Health Act 2011 [website]. (<https://www.sahealth.sa.gov.au/wps/wcm/connect/Public+Content/SA+Health+Internet/About+us/Legislation/Public+Health+Act/Public+Health+Act>, accessed 4 April 2024).
999. Government of South Australia. Evolution of the Health in All Policies approach [website]. Last Modified: 18 May 2023 (<https://www.sahealth.sa.gov.au/wps/wcm/connect/Public+Content/SA+Health+Internet/About+us/About+SA+Health/Health+in+All+Policies/Evolution+of+the+Health+in+All+Policies+approach>, accessed 4 April 2024).
1000. Van Eyk H, Harris E, Baum F, Delany-Crowe T, Lawless A, MacDougall C. Health in All Policies in South Australia—Did It Promote and Enact an Equity Perspective? *Int J Environ Res Public Health* 2017;14:1288. (<https://doi.org/10.3390/ijerph14111288>).
1001. Lillefjell M, Hope S, Anthun KS, Hermansen E, Vik JT, Sund ER et al. Planning for Health Equity: How Municipal Strategic Documents and Project Plans Reflect Intentions Instructed by the Norwegian Public Health Act. *Societies* 2023;13:74.
1002. Fosse E. Norwegian policies to reduce social inequalities in health: Developments from 1987 to 2021. *Scand J Public Health* 2022;50:882-886.
1003. Tallarek Née Grimm MJ, Helgesen MK, Fosse E. Reducing social inequities in health in Norway: Concerted action at state and local levels? *Health Policy* 2013;113:228-235.
1004. Gov.co. Plan Decenal Salud Pública 2022-2031 [website]. 2024 (<https://www.minsalud.gov.co/plandecenal/Paginas/PDSP-2022-2031.aspx>, accessed 4 April 2024).
1005. Rivillas JC, Colonia FD. Reducing causes of inequity: policies focused on social determinants of health during generational transitions in Colombia. *Glob Health Action* 2017;10:1349238.
1006. Shiffman J, Shawar YR. Framing and the formation of global health priorities. *Lancet* 2022;399:1977-1990.
1007. Katikireddi SV, Bond L, Hilton S. Changing Policy Framing as a Deliberate Strategy for Public Health Advocacy: A Qualitative Policy Case Study of Minimum Unit Pricing of Alcohol. *Milbank Q* 2014;92:250-283.
1008. Patchett AD, Yeatman HR, Johnson KM. Obesity framing for health policy development in Australia, France and Switzerland. *Health Promot Int* 2014;dau066.
1009. Colombini M, Mayhew SH, Hawkins B, Bista M, Joshi SK, Schei B et al. Agenda setting and framing of gender-based violence in Nepal: how it became a health issue. *Health Policy Plan* 2016;31:493-503.
1010. Townsend B, Schram A, Baum F, Labonté R, Friel S. How does policy framing enable or constrain inclusion of social determinants of health and health equity on trade policy agendas? *Crit Public Health* 2020;30:115-126.
1011. Fosse E, Helgesen MK. Policies to address the social determinants of health in the Nordic countries. Stockholm | Helsinki, Nordic Welfare Centre (<https://nordicwelfare.org/en/publikationer/social-determinants-of-health/>, accessed 1 April 2024).
1012. Lal A, Abdalla SM, Chattu VK, Erondur NA, Lee T-L, Singh S et al. Pandemic preparedness and response: exploring the role of universal health coverage within the global health security architecture. *Lancet Glob Health* 2022;10:e1675-e1683.
1013. Preparación y respuesta ante emergencias sensibles a poblaciones migrantes en Chile: perspectivas postpandémicas [website]. 2024 (<https://www.medwave.cl/enfoques/actualidad/2788.html>, accessed 8 April 2024).
1014. Maciel ELN, Sanchez MN, Cruz AMD, Cravo Neto DB, Lima NVT. Brazil's Pivotal Moment in Public Health: Establishing the Interministerial Committee (CIEDDS) for the Elimination of Tuberculosis and Socially Determined Diseases. *Rev Soc Bras Med Trop* 2024;57:e00601-2024.
1015. Economic Commission for Latin America and the Caribbean. Bolsa Familia (2003-2021) [website]. (<https://dds.cepal.org/bpsnc/programme?id=6>).

1016. Pescarini JM, Williamson E, Nery JS, Ramond A, Ichihara MY, Fiaccone RL et al. Effect of a conditional cash transfer programme on leprosy treatment adherence and cure in patients from the nationwide 100 Million Brazilian Cohort: a quasi-experimental study. *Lancet Infect Dis* 2020;20:618–627.
1017. Luz PM, Struchiner CJ. Robust evidence of Bolsa Família Programme's benefits. *Lancet HIV* 2022;9:e669–e670.
1018. Bennett S, Glandon D, Rasanathan K. Governing multisectoral action for health in low-income and middle-income countries: unpacking the problem and rising to the challenge. *BMJ Glob Health* 2018;3:e000880. (<https://doi.org/10.1136/bmjgh-2018-000880>).
1019. Rasanathan K, Bennett S, Atkins V, Beschel R, Carrasquilla G, Charles J et al. Governing multisectoral action for health in low- and middle-income countries. *PLOS Med* 2017;14:e1002285.
1020. World Health Organization and the Government of South Australia. The Adelaide Statement on Health in All Policies: moving towards a shared governance for health and well-being. *Health Promot Int* 2010;25:258–260.
1021. El-Jardali F, Fadlallah R, Daher N. Multi-sectoral collaborations in selected countries of the Eastern Mediterranean region: assessment, enablers and missed opportunities from the COVID-19 pandemic response. *Health Res Policy Syst* 2024;22:14.
1022. World health statistics 2017: monitoring health for the SDGs, sustainable development goals. Geneva, World Health Organization, 2017 (<https://iris.who.int/handle/10665/255336>, accessed 9 January 2025). License: CC BY-NC-SA 3.0 IGO.
1023. Reitsma MB, Claypool AL, Vargo J, Shete PB, McCorvie R, Wheeler WH et al. Racial/Ethnic Disparities In COVID-19 Exposure Risk, Testing, and Cases at the Subcounty Level In California. *Health Aff (Millwood)* 2021;40:870–878.
1024. California Department of Public Health. Health in All Policies [website]. (<https://www.cdph.ca.gov/Programs/OHE/Pages/HIAP.aspx#:~:text=The%20Health%20in%20All%20Policies,Office%20of%20the%20Attorney%20General>, accessed 19 December 2024).
1025. Glesmann C. Over 15 years of creating innovative solutions to gun violence: The City of Richmond, California, Office of Neighborhood Safety. Evident Change, 2023.
1026. World Health Organization. Health inequality monitoring: harnessing data to advance health equity. Geneva, World Health Organization, 2024 (<https://iris.who.int/handle/10665/379703>). License: CC BY-NC-SA 3.0 IGO).
1027. World Health Organization. Health Equity Assessment Toolkit [website]. (https://www.who.int/data/inequality-monitor/assessment_toolkit, accessed 25 July 2024).
1028. World Health Organization. Health Equity Policy Tool: a framework to track policies for increasing health equity in the WHO European Region. Copenhagen, World Health Organization Regional Office for Europe, 2019 (<https://iris.who.int/handle/10665/346027>, accessed 18 April 2024).
1029. Public Health England. Health Equity Assessment Tool [website]. 2021 (<https://www.gov.uk/government/publications/health-equity-assessment-tool-heat>, accessed 18 April 2024).
1030. O'Flaherty M, Lloyd-Williams F, Capewell S, Boland A, Maden M, Collins B et al. Modelling tool to support decision-making in the NHS Health Check programme: workshops, systematic review and co-production with users. *Health Technol Assess* 2021;25:1–234.
1031. Centre for Addiction and Mental Health. Health Equity Impact Assessment [website]. 2024 (<https://www.camh.ca/en/professionals/professionals--projects/heia>, accessed 18 April 2024).
1032. World Health Organization. Primary Health Care Country Case Study Compendium [website]. (<https://www.who.int/teams/primary-health-care/evidence-and-innovation/primary-health-care-case-study-compendium>, accessed 30 July 2024).
1033. Schönbach J-K, Bolte G, Czwikla G, Manz K, Mensing M, Muellmann S et al. Equity impacts of interventions to increase physical activity among older adults: a quantitative health impact assessment. *Int J Behav Nutr Phys Act* 2020;17:103. (<https://doi.org/10.1186/s12966-020-00999-4>).
1034. Bender AM, Sørensen J, Diderichsen F, Brønnum-Hansen H. A health inequality impact assessment from reduction in overweight and obesity. *BMC Public Health* 2020;20:1823. (<https://doi.org/10.1186/s12889-020-09831-x>).
1035. Barros A, Boutayeb A, Brown C, Dean HD, Di Ruggiero E, Ferrelli RM et al. Towards a global monitoring system for implementing the Rio Political Declaration on Social Determinants of Health: developing a core set of indicators for government action on the social determinants of health to improve health equity. *Int J Equity Health* 2018;17:136.
1036. World Health Organization. Health inequality monitoring: harnessing data to advance health equity. 2024 (<https://iris.who.int/handle/10665/379703>, accessed 23 March 2025). CC BY-NC-SA 3.0 IGO.
1037. World Health Organization. Health Inequality Data Repository [website]. (<https://www.who.int/data/inequality-monitor/data>, accessed 25 July 2024).

1038. Biermann O, Mwoka M, Ettman CK, Abdalla SM, Shawky S, Ambuko J et al. Data, Social Determinants, and Better Decision-making for Health: the 3-D Commission. *J Urban Health* 2021;98:4-14. (<https://doi.org/10.1007/s11524-021-00556-9>).
1039. Harari L, Lee C. Intersectionality in quantitative health disparities research: A systematic review of challenges and limitations in empirical studies. *Soc Sci Med* 1982 2021;277:113876.
1040. Bauer GR, Churchill SM, Mahendran M, Walwyn C, Lizotte D, Villa-Rueda AA. Intersectionality in quantitative research: A systematic review of its emergence and applications of theory and methods. *SSM – Popul Health* 2021;14:100798.
1041. Kilpeläinen K, Parikka S, Koponen P, Koskinen S, Rotko T, Koskela T et al. Finnish experiences of health monitoring: local, regional, and national data sources for policy evaluation. *Glob Health Action* 2016;9:28824. (<https://doi.org/10.3402/gha.v9.28824>).
1042. Valle AM. The Mexican experience in monitoring and evaluation of public policies addressing social determinants of health. *Glob Health Action* 2016;9:29030. (<https://doi.org/10.3402/gha.v9.29030>).
1043. Rivillas JC, Colonia FD. Reducing causes of inequity: policies focused on social determinants of health during generational transitions in Colombia. *Glob Health Action* 2017;10.
1044. World Health Organization. Webinar: Monitoring Social Determinants of Health Equity [website]. 2024 (<https://www.who.int/news-room/events/detail/2024/11/13/default-calendar/webinar-monitoring-social-determinants-of-health-equity>, accessed 7 January 2025).
1045. Posayanonda T. Multisectoral Collaboration and HiAP Approach towards Health Equity: Thailand Experience. 2023. (https://research.tuni.fi/uploads/2023/09/9be72ab7-3-sept62023_pillar_panel_tipicha_case_thailand.pdf, accessed 7 January 2025).
1046. Intawong K. Global health [website]. 2023 (<https://prezi.com/p/l4p3n73gro2q/global-health/>, accessed 7 January 2025).
1047. Urban Indian Health Institute. Data Dashboard. Urban Indian Health Inst. (<https://www.uihi.org/urban-indian-health/data-dashboard/>, accessed 18 June 2024).
1048. Torres I, Thapa B, Robbins G, Koya SF, Abdalla SM, Arah OA et al. Data Sources for Understanding the Social Determinants of Health: Examples from Two Middle-Income Countries: the 3-D Commission. *J Urban Health Bull N Y Acad Med* 2021;98:31-40. (<https://doi.org/10.1007/s11524-021-00558-7>).
1049. Patra BG, Sharma MM, Vekaria V, Adekanattu P, Patterson OV, Glicksberg B et al. Extracting social determinants of health from electronic health records using natural language processing: a systematic review. *J Am Med Inform Assoc* 2021;28:2716-2727. (<https://doi.org/10.1093/jamia/ocab170>).
1050. Lindenfeld Z, Pagán JA, Chang JE. Utilizing Publicly Available Community Data to Address Social Determinants of Health: A Compendium of Data Sources. *Inq J Health Care Organ Provis Financ* 2023;60:00469580231152318. (<https://doi.org/10.1177/00469580231152318>).
1051. Edjoc R. Key health inequalities in Canada: a national portrait. Ottawa, Public Health Agency of Canada, 2018.
1052. Miszkurka M. Monitoring health inequalities to inform action – the Pan-Canadian Health Inequalities Reporting Initiative. *Eur J Public Health* 2020;30:ckaa165.484.
1053. Public Health Agency of Canada, Pan-Canadian Public Health Network, Statistics Canada, Canadian Institute for Health Information. Pan-Canadian Health Inequalities Data Tool. 2022. (https://health-infobase.canada.ca/health-inequalities/docs/Architecture_Map_GenPop_En_2022.pdf).

ANNEX 1:

Declaration of Interests

The expert groups' terms of reference outlined roles such as evidence review, oversight of methodology and formulation of recommendations. All members completed and signed WHO's Declaration of Interests form, reviewed independently by WHO's ethics team, with follow-ups conducted to address any declared conflicts. No relevant conflicts were identified.

ANNEX 2:

Methodology

1. Introduction

The methodology for this report ensures transparency, reproducibility and alignment with WHO's ethical and quality standards. This section details the processes for evidence synthesis, stakeholder engagement, validation and conflict-of-interest management, addressing key requirements outlined in WHO's publication checklist and operational frameworks.

2. Formation of the expert groups

At the commencement of developing the report, two ad hoc expert groups (Policy Expert Group, and Scientific Expert Group) were established, comprising global experts in health equity, social determinants and public policy serving in their individual capacity (see Annex 1). The expert groups were convened to provide strategic and technical guidance on development of the report.

Members were selected through a transparent process. A list of possible members was developed based on recommendations put forward by the Department of Social Determinants of Health, the Division for UHC/Healthier Populations and by all six WHO regional offices. The final selection of members was made by the Department of Social Determinants and the Division of UHC/Healthier Populations, based on the following criteria:

- ▶ expertise in health equity, social determinants and policy analysis
- ▶ regional representation and gender diversity, to ensure inclusive representation
- ▶ independence from conflicts of interest.

The expert groups were convened by the Secretariat at regular intervals over the period 2021-2024 to provide public policy strategic and technical guidance, reviewing evidence, shaping the methodology and contributing to recommendations.

3. Stakeholder engagement and consultations

A participatory approach was adopted to ensure diverse and meaningful stakeholder engagement. This process included:

- ▶ virtual consultations and written feedback from Member States, WHO regional offices, civil society and subject-matter experts;
- ▶ inclusion of voices from low- and middle-income countries and populations experiencing marginalization.

As part of the iterative refinement process, a draft summary of the report and its proposed recommendations was presented to the 154th WHO Executive Board and the 77th World Health Assembly. Member States provided overwhelmingly positive feedback, which was systematically incorporated into the final report.

4. Evidence synthesis and analysis

The methodology for evidence synthesis incorporated a rigorous approach to data collection, analysis and interpretation:

- ▶ Systematic reviews of peer-reviewed literature, WHO reports and grey literature formed the evidence base.
- ▶ Inclusion criteria prioritized relevance to social determinants of health equity; geographic diversity and methodological rigour.
- ▶ Exclusion criteria focused on outdated evidence and studies with significant methodological limitations.

Ten scoping reviews and background papers commissioned by the WHO Secretariat were used to identify the latest evidence on the social determinants of health equity, progress in tackling them and best practice:

1. *Interventions on the social determinants of health: a rapid review of published literature from 2014 until today.* Jessica Allen, Antia Castedo, Angela Donkin, Lorenzo Lionello and Jamaica Noferini. Institute for Health Equity, University College London, United Kingdom.
2. *Towards a World Safer from Health Emergencies: required action on social determinants of health.* Royal Institution for the Advancement of Learning / McGill University, Canada.
3. *Interests, ideas and institutions: traversing political economy challenges to reduce health inequities through action on the social determinants of health.* Sharon Friel, Australian National University, Australia.

4. *The case for action on social determinants of health and health equity.* Salma Abdalla, Boston University.
5. *Investing in equitable human capital: the economic case for action on social determinants of health and health equity and the costs of inaction.* Pooja Yerramilli, Seed Global Health, United States of America. A synthesis of this paper has been published here: Pooja Yerramilli, Mickey Chopra, Kumanan Rasanathan. The cost of inaction on health equity and its social determinants. *BMJ Global Health* 2024;9:e012690.
6. *Progress and obstacles on social determinants of health over the last decade.* Miriam Vandenburg, Flinders University, Australia, with Fran Baum, Sue Gibbons, Sharon Friel, Ashley Schram, Hailay Abrha Gesesew. A synthesis of this paper has been published here: Miriam van den Berg, Joanne Flavel, Ashley Schram, Sharon Friel, Hailay Abrha Gesesew, Fran Baum. Social, cultural and political conditions for advancing health equity: examples from eight country case studies (2011-2021). *BMJ Global Health* 2024;9:e015694.
7. *Key policies and interventions for action on social determinants of health and health equity by country context.* Laura Frost, Beth Pratt, Global Health Insights, United States of America.
8. *Analyses, evidence, and key recommendations on commercial determinants of health: inputs for the World Report on Social Determinants of Health Equity.* Jennifer Lacy-Nichols, Alexandra Jones, Kent Buse, The George Institute, Australia.
9. *Insufficient progress on social determinants of health over the last decade: explanations and prospects for overcoming them in the future.* Andrew Cassels, GH Associates, United Kingdom.
10. *Healthy climate, healthy people: a framework for supporting faster, fairer joint action on climate change and health equity.* Kirsty Wild and Alistair Woodward. The University of Auckland, New Zealand. A synthesis of this paper has been published here: Kirsty Wild, Kumanan Rasanathan, Sudhvir Singh, Alistair Woodward. Balancing short-term and long-term climate-health equity strategies: a framework for supporting faster, fairer joint action on climate change and health equity. *BMJ Global Health* 2025;9:e016094.

Data analysis utilized frameworks such as PROGRESS-Plus to examine disparities across socioeconomic and demographic dimensions. Evidence was synthesized to identify key patterns, determinants and actionable recommendations.

5. Country example selection and validation

Country examples were included in the report to illustrate diverse contexts, innovative policies and practical applications of actions addressing social determinants of health, to improve health equity in countries across the world. Country examples were selected through a transparent process. A list of possible country examples was developed based on recommendations put forward by the Department of Social Determinants of Health, the Division for UHC/Healthier Populations and by all six WHO regional offices. The final selection of country examples was made by the Department of Social Determinants, based on the following criteria:

1. **Geographic diversity:** To ensure representativeness of diverse settings, there are no more than three examples from any one country and there is a distribution of examples across all six WHO regions in the report.
2. **Policy/intervention overview:** Each country example must provide a clear explanation of the policy/intervention, including its rationale (theory) for addressing social determinants of health and its focus on health equity, gaps, gradients or services for marginalized populations.
3. **Implementation details:** The selected examples must describe when, where and how the policy/intervention was implemented, including enabling factors, the roles of government or social sectors, and community involvement.
4. **Health sector role:** Each example must outline the health sector's role (lead or supportive) and specify if WHO was involved.
5. **Results and outcomes:** The selected examples must present preliminary results or outcomes, ideally focusing on health services or status improvements for marginalized groups, or reductions in health inequalities. Where these are unavailable, the examples include social determinants-related indicators.
6. **Credible references:** Each country example provides peer-reviewed sources or official reports from government or UN entities, if possible.

Drafting and validation of the country examples was done through consultations with WHO headquarters, and regional and country offices, as well as government officials and other representatives from the featured countries, to ensure contextual relevance and accuracy.

6. Conflict of interest management, and ethical and quality standards

All contributors, including expert group members and external reviewers, signed WHO's Disclosure of Interests forms. The forms were reviewed

independently to assess potential conflicts. Contributors with declared conflicts were excluded from decision-making on related sections of the report. Adherence to WHO's conflict-of-interest policies ensured transparency and high ethical standards throughout the process. Emphasis was also placed on transparent alignment with WHO's quality standards.

7. Validation and review process

The report underwent iterative review and refinement:

- ▶ Drafts were reviewed by expert group members, WHO leadership and external experts.
- ▶ Feedback was systematically incorporated, to ensure alignment with WHO's ethical guidelines and scientific rigour.
- ▶ Peer reviews and consultations ensured the findings were actionable and relevant to global and regional contexts.

8. Response to feedback from the 154th Executive Board and 77th World Health Assembly

In response to the Assembly's Resolution 74.16, a draft summary of the report and its proposed recommendations was presented to the 154th WHO Executive Board and the 77th World Health Assembly. The positive feedback provided by Member States informed the iterative refinement and finalization of the report, ensuring its alignment with Member State priorities.

9. Limitations

While efforts were made to ensure comprehensive coverage, certain limitations persist, including:

- ▶ gaps in data availability, particularly in low-resource settings
- ▶ variability in contextual relevance and implementation of country examples.

10. Strengths

This methodology reflects WHO's commitment to evidence-based policy development, inclusivity and ethical integrity. This comprehensive approach, informed by WHO's operational frameworks and ethical guidelines, ensures that the report reflects diverse perspectives and provides actionable, evidence-based recommendations.



Equity and Health Unit
Department of Social Determinants of Health
World Health Organization
20, Avenue Appia
1211 Geneva 27
Switzerland

Email: equityandhealth@who.int
Website: <https://www.who.int/health-topics/social-determinants-of-health>

